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## Chapter 23: GAUSS' LAW

1. A total charge of $6.3 \times 10^{-8}$ C is distributed uniformly throughout a 2.7-cm radius sphere. The volume charge density is:
   A. $3.7 \times 10^{-7}$ C/m$^3$
   B. $6.9 \times 10^{-6}$ C/m$^3$
   C. $6.9 \times 10^{-6}$ C/m$^2$
   D. $2.5 \times 10^{-4}$ C/m$^3$
   E. $7.6 \times 10^{-4}$ C/m$^3$
   ans: E

2. Charge is placed on the surface of a 2.7-cm radius isolated conducting sphere. The surface charge density is uniform and has the value $6.9 \times 10^{-6}$ C/m$^2$. The total charge on the sphere is:
   A. $5.6 \times 10^{-10}$ C
   B. $2.1 \times 10^{-8}$ C
   C. $4.7 \times 10^{-8}$ C
   D. $6.3 \times 10^{-8}$ C
   E. $9.5 \times 10^{-3}$ C
   ans: D

3. A spherical shell has an inner radius of 3.7 cm and an outer radius of 4.5 cm. If charge is distributed uniformly throughout the shell with a volume density of $6.1 \times 10^{-4}$ C/m$^3$ the total charge is:
   A. $1.0 \times 10^{-7}$ C
   B. $1.3 \times 10^{-7}$ C
   C. $2.0 \times 10^{-7}$ C
   D. $2.3 \times 10^{-7}$ C
   E. $4.0 \times 10^{-7}$ C
   ans: A

4. A cylinder has a radius of 2.1 cm and a length of 8.8 cm. Total charge $6.1 \times 10^{-7}$ C is distributed uniformly throughout. The volume charge density is:
   A. $5.3 \times 10^{-5}$ C/m$^3$
   B. $5.3 \times 10^{-5}$ C/m$^2$
   C. $8.5 \times 10^{-4}$ C/m$^3$
   D. $5.0 \times 10^{-3}$ C/m$^3$
   E. $6.3 \times 10^{-2}$ C/m$^3$
   ans: D

5. When a piece of paper is held with one face perpendicular to a uniform electric field the flux through it is $25\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$. When the paper is turned 25° with respect to the field the flux through it is:

   A. 0
   B. $12\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$
   C. $21\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$
   D. $23\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$
   E. $25\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$

   ans: D

6. The flux of the electric field $(24\,\mathrm{N/C})\,\hat{\mathrm{i}} + (30\,\mathrm{N/C})\,\hat{\mathrm{j}} + (16\,\mathrm{N/C})\,\hat{\mathrm{k}}$ through a $2.0\,\mathrm{m}^2$ portion of the $yz$ plane is:

   A. $32\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$
   B. $34\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$
   C. $42\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$
   D. $48\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$
   E. $60\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$

   ans: D

7. Consider Gauss's law: $\oint \vec{E}\cdot d\vec{A} = q/\epsilon_0$. Which of the following is true?

   A. $\vec{E}$ must be the electric field due to the enclosed charge
   B. If $q = 0$, then $\vec{E} = 0$ everywhere on the Gaussian surface
   C. If the three particles inside have charges of $+q$, $+q$, and $-2q$, then the integral is zero
   D. on the surface $\vec{E}$ is everywhere parallel to $d\vec{A}$
   E. If a charge is placed outside the surface, then it cannot affect $\vec{E}$ at any point on the surface

   ans: C

8. A charged point particle is placed at the center of a spherical Gaussian surface. The electric flux $\Phi_E$ is changed if:

   A. the sphere is replaced by a cube of the same volume
   B. the sphere is replaced by a cube of one-tenth the volume
   C. the point charge is moved off center (but still inside the original sphere)
   D. the point charge is moved to just outside the sphere
   E. a second point charge is placed just outside the sphere

   ans: D

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9. Choose the INCORRECT statement:

   A. Gauss' law can be derived from Coulomb's law
   B. Gauss' law states that the net number of lines crossing any closed surface in an outward direction is proportional to the net charge enclosed within the surface
   C. Coulomb's law can be derived from Gauss' law and symmetry
   D. Gauss' law applies to a closed surface of any shape
   E. According to Gauss' law, if a closed surface encloses no charge, then the electric field must vanish everywhere on the surface

   ans: E

10. The outer surface of the cardboard center of a paper towel roll:
    A. is a possible Gaussian surface
    B. cannot be a Gaussian surface because it encloses no charge
    C. cannot be a Gaussian surface since it is an insulator
    D. cannot be a Gaussian surface because it is not a closed surface
    E. none of the above

    ans: D

11. A physics instructor in an anteroom charges an electrostatic generator to $25\,\mu$C, then carries it into the lecture hall. The net electric flux in $\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$ through the lecture hall walls is:
    A. 0
    B. $25 \times 10^{-6}$
    C. $2.2 \times 10^{5}$
    D. $2.8 \times 10^{6}$
    E. can not tell unless the lecture hall dimensions are given

    ans: D

12. A point particle with charge $q$ is placed inside the cube but not at its center. The electric flux through any one side of the cube:
    A. is zero
    B. is $q/\epsilon_0$
    C. is $q/4\epsilon_0$
    D. is $q/6\epsilon_0$
    E. cannot be computed using Gauss' law

    ans: E

13. A particle with charge 5.0-$\mu$C is placed at the corner of a cube. The total electric flux in $\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$ through all sides of the cube is:
    A. 0
    B. $7.1 \times 10^{4}$
    C. $9.4 \times 10^{4}$
    D. $1.4 \times 10^{5}$
    E. $5.6 \times 10^{5}$

    ans: E

14. A point particle with charge $q$ is at the center of a Gaussian surface in the form of a cube. The electric flux through any one face of the cube is:
    A. $q/\epsilon_0$
    B. $q/4\pi\epsilon_0$
    C. $q/3\epsilon_0$
    D. $q/6\epsilon_0$
    E. $q/12\epsilon_0$

    ans: D

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15. The table below gives the electric flux in $\mathrm{N \cdot m^2/C}$ through the ends and round surfaces of four Gaussian surfaces in the form of cylinders. Rank the cylinders according to the charge inside, from the most negative to the most positive.

| | left end | right end | rounded surface |
|---|---|---|---|
| cylinder 1: | $+2 \times 10^{-9}$ | $+4 \times 10^{-9}$ | $-6 \times 10^{-9}$ |
| cylinder 2: | $+3 \times 10^{-9}$ | $-2 \times 10^{-9}$ | $+6 \times 10^{-9}$ |
| cylinder 3: | $-2 \times 10^{-9}$ | $-5 \times 10^{-9}$ | $+3 \times 10^{-9}$ |
| cylinder 4: | $+2 \times 10^{-9}$ | $-5 \times 10^{-9}$ | $-3 \times 10^{-9}$ |

A. 1, 2, 3, 4
B. 4, 3, 2, 1
C. 3, 4, 2, 1
D. 3, 1, 4, 2
E. 4, 3, 1, 2

ans: E

16. A conducting sphere of radius 0.01 m has a charge of $1.0 \times 10^{-9}$ C deposited on it. The magnitude of the electric field in N/C just outside the surface of the sphere is:

A. 0
B. 450
C. 900
D. 4500
E. 90,000

ans: C

17. A round wastepaper basket with a 0.15-m radius opening is in a uniform electric field of 300 N/C, perpendicular to the opening. The total flux through the sides and bottom, in $\mathrm{N \cdot m^2}$ C, is:

A. 0
B. 4.2
C. 21
D. 280
E. can not tell without knowing the areas of the sides and bottom

ans: C

18. 10 C of charge are placed on a spherical conducting shell. A particle with a charge of $-3$ C is placed at the center of the cavity. The net charge on the inner surface of the shell is:

A. $-7$ C
B. $-3$ C
C. 0 C
D. $+3$ C
E. $+7$ C

ans: D

19. 10 C of charge are placed on a spherical conducting shell. A particle with a charge of $-3$ C is placed at the center of the cavity. The net charge on the outer surface of the shell is:

A. $-7$ C
B. $-3$ C
C. 0 C
D. $+3$ C
E. $+7$ C

ans: E

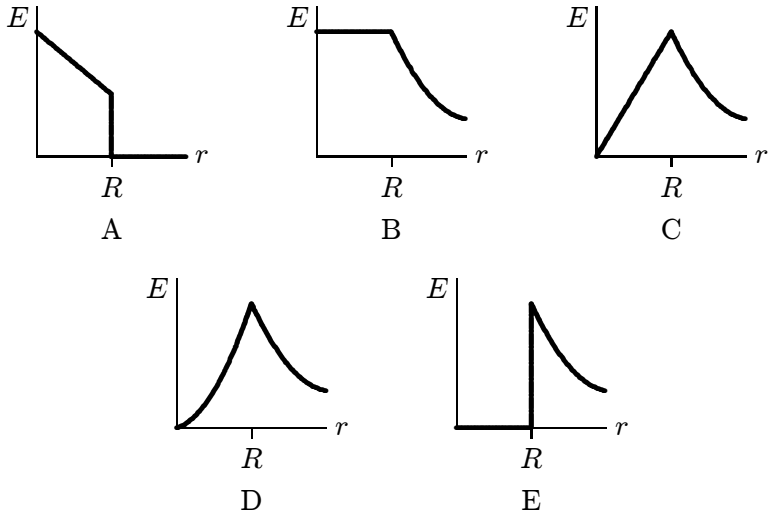
20. A 30-N/C uniform electric field points perpendicularly toward the left face of a large neutral conducting sheet. The surface charge density in $\mathrm{C/m^2}$ on the left and right faces, respectively, are:

A. $-2.7 \times 10^{-9}\ \mathrm{C/m^2}$; $+2.7 \times 10^{-9}\ \mathrm{C/m^2}$
B. $+2.7 \times 10^{-9}\ \mathrm{C/m^2}$; $-2.7 \times 10^{-9}\ \mathrm{C/m^2}$
C. $-5.3 \times 10^{-9}\ \mathrm{C/m^2}$; $+5.3 \times 10^{-9}\ \mathrm{C/m^2}$
D. $+5.3 \times 10^{-9}\ \mathrm{C/m^2}$; $-5.3 \times 10^{-9}\ \mathrm{C/m^2}$
E. 0; 0

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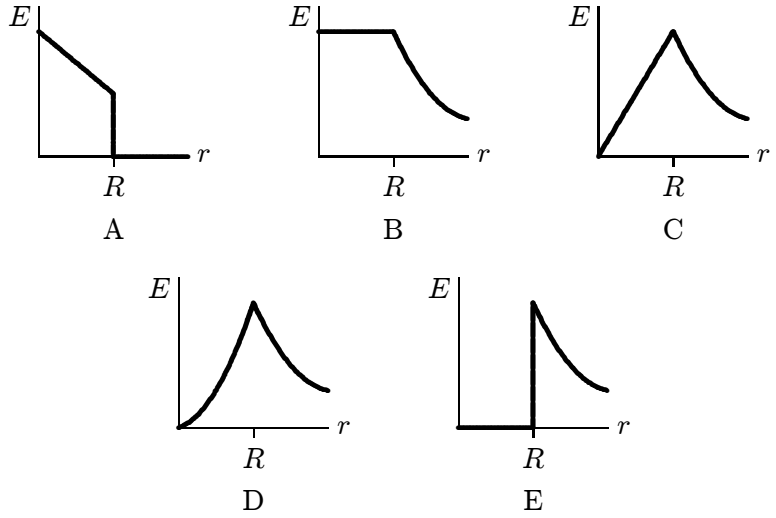
ans: A

21. A solid insulating sphere of radius $R$ contains positive charge that is distributed with a volume charge density that does not depend on angle but does increase with distance from the sphere center. Which of the graphs below might give the magnitude $E$ of the electric field as a function of the distance $r$ from the center of the sphere?




ans: D

22. Which of the following graphs represents the magnitude of the electric field as a function of the distance from the center of a solid charged conducting sphere of radius $R$?




ans: E

23. Charge $Q$ is distributed uniformly throughout an insulating sphere of radius $R$. The magnitude of the electric field at a point $R/2$ from the center is:

A. $Q/4\pi\epsilon_0 R^2$
B. $Q/\pi\epsilon_0 R^2$
C. $3Q/4\pi\epsilon_0 R^2$
D. $Q/8\pi\epsilon_0 R^2$
E. none of these

ans: D

24. Positive charge $Q$ is distributed uniformly throughout an insulating sphere of radius $R$, centered at the origin. A particle with positive charge $Q$ is placed at $x = 2R$ on the $x$ axis. The magnitude of the electric field at $x = R/2$ on the $x$ axis is:

A. $Q/4\pi\epsilon_0 R^2$
B. $Q/8\pi\epsilon_0 R^2$
C. $Q/72\pi\epsilon_0 R^2$
D. $17Q/72\pi\epsilon_0 R^2$
E. none of these

ans: C

25. Charge $Q$ is distributed uniformly throughout a spherical insulating shell. The net electric flux in $\mathrm{N}\cdot\mathrm{m}^2/\mathrm{C}$ through the inner surface of the shell is:

A. 0
B. $Q/\epsilon_0$
C. $2Q/\epsilon_0$
D. $Q/4\pi\epsilon_0$
E. $Q/2\pi\epsilon_0$

ans: A

26. Charge $Q$ is distributed uniformly throughout a spherical insulating shell. The net electric flux in $\mathrm{N \cdot m^2/C}$ through the outer surface of the shell is:

A. 0
B. $Q/\epsilon_0$
C. $2Q/\epsilon_0$
D. $Q/4\epsilon_0$
E. $Q/2\pi\epsilon_0$

ans: B

27. A 3.5-cm radius hemisphere contains a total charge of $6.6 \times 10^{-7}\,\mathrm{C}$. The flux through the rounded portion of the surface is $9.8 \times 10^4\,\mathrm{N \cdot m^2/C}$. The flux through the flat base is:

A. 0
B. $+2.3 \times 10^4\,\mathrm{N \cdot m^2/C}$
C. $-2.3 \times 10^4\,\mathrm{N \cdot m^2/C}$
D. $-9.8 \times 10^4\,\mathrm{N \cdot m^2/C}$
E. $+9.8 \times 10^4\,\mathrm{N \cdot m^2/C}$

ans: C

28. Charge is distributed uniformly along a long straight wire. The electric field 2 cm from the wire is 20 N/C. The electric field 4 cm from the wire is:

A. 120 N/C
B. 80 N/C
C. 40 N/C
D. 10 N/C
E. 5 N/C

ans: D

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29. Positive charge $Q$ is placed on a conducting spherical shell with inner radius $R_1$ and outer radius $R_2$. A particle with charge $q$ is placed at the center of the cavity. The magnitude of the electric field at a point in the cavity, a distance $r$ from the center, is:

A. zero
B. $Q/4\pi\epsilon_0 R_1^2$
C. $q/4\pi\epsilon_0 r^2$
D. $(q+Q)/4\pi\epsilon_0 r^2$
E. $(q+Q)/4\pi\epsilon_0(R_1^2 - r^2)$

ans: C

30. Positive charge $Q$ is placed on a conducting spherical shell with inner radius $R_1$ and outer radius $R_2$. A point charge $q$ is placed at the center of the cavity. The magnitude of the electric field at a point outside the shell, a distance $r$ from the center, is:

A. zero
B. $Q/4\pi\epsilon_0 r^2$
C. $q/4\pi\epsilon_0 r^2$
D. $(q+Q)/4\pi\epsilon_0 r^2$
E. $(q+Q)/4\pi\epsilon_0(R_1^2 - r^2)$

ans: D

31. Positive charge $Q$ is placed on a conducting spherical shell with inner radius $R_1$ and outer radius $R_2$. A point charge $q$ is placed at the center of the cavity. The magnitude of the electric field produced by the charge on the inner surface at a point in the interior of the conductor, a distance $r$ from the center, is:

A. 0
B. $Q/4v\pi\epsilon_0 R_1^2$
C. $Q/4\pi\epsilon_0 R_2^2$
D. $q/4\pi\epsilon_0 r^2$
E. $Q/4\pi\epsilon_0 r^2$

ans: D

32. A long line of charge with $\lambda_\ell$ charge per unit length runs along the cylindrical axis of a cylindrical shell which carries a charge per unit length of $\lambda_c$. The charge per unit length on the inner and outer surfaces of the shell, respectively are:

A. $\lambda_\ell$ and $\lambda_c$
B. $-\lambda_\ell$ and $\lambda_c + \lambda_\ell$
C. $-\lambda_\ell$ and $\lambda_c - \lambda_c$
D. $\lambda_\ell + \lambda_c$ and $\lambda_c - \lambda_\ell$
E. $\lambda_\ell - \lambda_c$ and $\lambda_c + \lambda_\ell$

ans: B

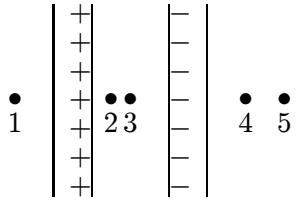
33. Charge is distributed uniformly on the surface of a large flat plate. The electric field 2 cm from the plate is 30 N/C. The electric field 4 cm from the plate is:

A. 120 N/C
B. 80 N/C
C. 30 N/C
D. 15 N/C
E. 7.5 N/C

ans: C

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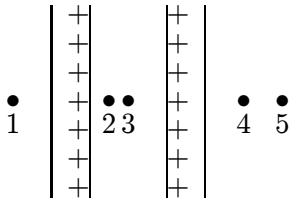
34. Two large insulating parallel plates carry charge of equal magnitude, one positive and the other negative, that is distributed uniformly over their inner surfaces. Rank the points 1 through 5 according to the magnitude of the electric field at the points, least to greatest.




A. 1, 2, 3, 4, 5
B. 2, then 1, 3, and 4 tied, then 5
C. 1, 4, and 5 tie, then 2 and 3 tie
D. 2 and 3 tie, then 1 and 4 tie, then 5
E. 2 and 3 tie, then 1, 4, and 5 tie

ans: C

35. Two large parallel plates carry positive charge of equal magnitude that is distributed uniformly over their inner surfaces. Rank the points 1 through 5 according to the magnitude of the electric field at the points, least to greatest.




A. 1, 2, 3, 4, 5
B. 5, 4, 3, 2, 1
C. 1, 4, and 5 tie, then 2 and 3 tie
D. 2 and 3 tie, then 1 and 4 tie, then 5
E. 2 and 3 tie, then 1, 4, and 5 tie

ans: E

36. A particle with charge $Q$ is placed outside a large neutral conducting sheet. At any point in the interior of the sheet the electric field produced by charges on the surface is directed:

A. toward the surface
B. away from the surface
C. toward $Q$
D. away from $Q$
E. none of the above

ans: C

37. A hollow conductor is positively charged. A small uncharged metal ball is lowered by a silk thread through a small opening in the top of the conductor and allowed to touch its inner surface. After the ball is removed, it will have:

A. a positive charge
B. a negative charge
C. no appreciable charge
D. a charge whose sign depends on what part of the inner surface it touched
E. a charge whose sign depends on where the small hole is located in the conductor

ans: C

38. A spherical conducting shell has charge $Q$. A particle with charge $q$ is placed at the center of the cavity. The charge on the inner surface of the shell and the charge on the outer surface of the shell, respectively, are:

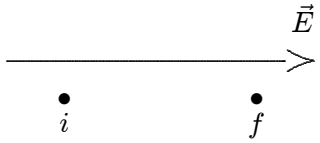
A. 0, $Q$
B. $q$, $Q - q$
C. $Q$, 0
D. $-q$, $Q + q$
E. $-q$, 0

ans: D

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## Chapter 24: ELECTRIC POTENTIAL

1. An electron moves from point $i$ to point $f$, in the direction of a uniform electric field. During this displacement:




   A. the work done by the field is positive and the potential energy of the electron-field system increases
   B. the work done by the field is negative and the potential energy of the electron-field system increases
   C. the work done by the field is positive and the potential energy of the electron-field system decreases
   D. the work done by the field is negative and the potential energy of the electron-field system decreases
   E. the work done by the field is positive and the potential energy of the electron-field system does not change

   ans: B

2. A particle with a charge of $5.5 \times 10^{-8}\,\mathrm{C}$ is 3.5 cm from a particle with a charge of $-2.3 \times 10^{-8}\,\mathrm{C}$. The potential energy of this two-particle system, relative to the potential energy at infinite separation, is:

   A. $3.2 \times 10^{-4}\,\mathrm{J}$
   B. $-3.2 \times 10^{-4}\,\mathrm{J}$
   C. $9.3 \times 10^{-3}\,\mathrm{J}$
   D. $-9.3 \times 10^{-3}\,\mathrm{J}$
   E. zero

   ans: B

3. A particle with a charge of $5.5 \times 10^{-8}\,\mathrm{C}$ is fixed at the origin. A particle with a charge of $-2.3 \times 10^{-8}\,\mathrm{C}$ is moved from $x = 3.5\,\mathrm{cm}$ on the $x$ axis to $y = 4.3\,\mathrm{cm}$ on the $y$ axis. The change in potential energy of the two-particle system is:

   A. $3.1 \times 10^{-3}\,\mathrm{J}$
   B. $-3.1 \times 10^{-3}\,\mathrm{J}$
   C. $6.0 \times 10^{-5}\,\mathrm{J}$
   D. $-6.0 \times 10^{-5}\,\mathrm{J}$
   E. 0

   ans: C

4. A particle with a charge of $5.5 \times 10^{-8}$ C charge is fixed at the origin. A particle with a charge of $-2.3 \times 10^{-8}$ C charge is moved from $x = 3.5$ cm on the $x$ axis to $y = 3.5$ cm on the $y$ axis. The change in the potential energy of the two-particle system is:

   A. $3.2 \times 10^{-4}$ J
   B. $-3.2 \times 10^{-4}$ J
   C. $9.3 \times 10^{-3}$ J
   D. $-9.3 \times 10^{-3}$ J
   E. 0

   ans: E

5. Three particles lie on the $x$ axis: particle 1, with a charge of $1 \times 10^{-8}$ C is at $x = 1$ cm, particle 2, with a charge of $2 \times 10^{-8}$ C, is at $x = 2$ cm, and particle 3, with a charge of $-3 \times 10^{-8}$ C, is at $x = 3$ cm. The potential energy of this arrangement, relative to the potential energy for infinite separation, is:

   A. $+4.9 \times 10^{-4}$ J
   B. $-4.9 \times 10^{-4}$ J
   C. $+8.5 \times 10^{-4}$ J
   D. $-8.5 \times 10^{-4}$ J
   E. zero

   ans: B

6. Two identical particles, each with charge $q$, are placed on the $x$ axis, one at the origin and the other at $x = 5$ cm. A third particle, with charge $-q$, is placed on the $x$ axis so the potential energy of the three-particle system is the same as the potential energy at infinite separation. Its $x$ coordinate is:

   A. 13 cm
   B. 2.5 cm
   C. 7.5 cm
   D. 10 cm
   E. $-5$ cm

   ans: A

7. Choose the correct statement:

   A. A proton tends to go from a region of low potential to a region of high potential
   B. The potential of a negatively charged conductor must be negative
   C. If $\vec{E} = 0$ at a point P then $V$ must be zero at P
   D. If $V = 0$ at a point P then $\vec{E}$ must be zero at P
   E. None of the above are correct

   ans: E

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8. If 500 J of work are required to carry a charged particle between two points with a potential difference of 20 V, the magnitude of the charge on the particle is:

   A. 0.040 C
   B. 12.5 C
   C. 20 C
   D. cannot be computed unless the path is given
   E. none of these

   ans: B

9. The potential difference between two points is 100 V. If a particle with a charge of 2 C is transported from one of these points to the other, the magnitude of the work done is:

A. 200 J
B. 100 J
C. 50 J
D. 100 J
E. 2 J

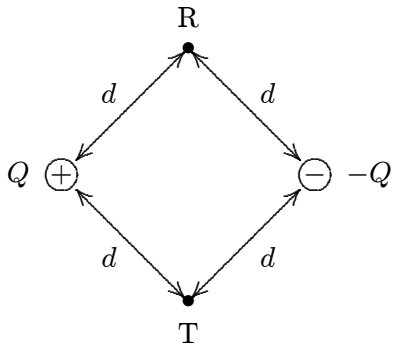
ans: A

10. During a lightning discharge, 30 C of charge move through a potential difference of $1.0 \times 10^8$ V in $2.0 \times 10^{-2}$ s. The energy released by this lightning bolt is:

A. $1.5 \times 10^{11}$ J
B. $3.0 \times 10^9$ J
C. $6.0 \times 10^7$ J
D. $3.3 \times 10^6$ J
E. 1500 J

ans: B

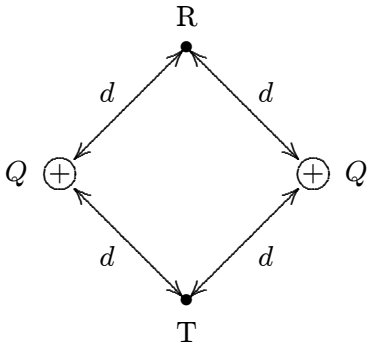
11. Points R and T are each a distance $d$ from each of two particles with charges of equal magnitudes and opposite signs as shown. If $k = 1/4\pi\epsilon_0$, the work required to move a particle with a negative charge $q$ from R to T is:




A. 0
B. $kqQ/d^2$
C. $kqQ/d$
D. $kqQ/(\sqrt{2}d)$
E. $kQq/(2d)$

ans: A

12. Points R and T are each a distance $d$ from each of two particles with equal positive charges as shown. If $k = 1/4\pi\epsilon_0$, the work required to move a particle with charge $q$ from R to T is:

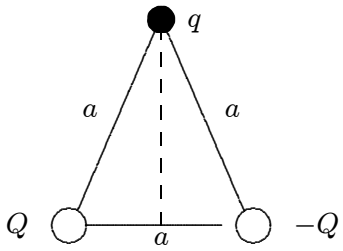



A. 0
B. $kQq/d^2$
C. $kQq/d$
D. $kQq/(\sqrt{2}d)$
E. $kQq/(2d)$

ans: A

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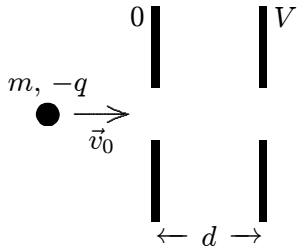
13. Two particle with charges $Q$ and $-Q$ are fixed at the vertices of an equilateral triangle with sides of length $a$. If $k = 1/4\pi\epsilon_0$, the work required to move a particle with charge $q$ from the other vertex to the center of the line joining the fixed particles is:




A. 0
B. $kQq/a$
C. $kQq/a^2$
D. $2kQq/a$
E. $\sqrt{2}kQq/a$

ans: A

14. A particle with mass $m$ and charge $-q$ is projected with speed $v_0$ into the region between two parallel plates as shown. The potential difference between the two plates is $V$ and their separation is $d$. The change in kinetic energy of the particle as it traverses this region is:




A. $-qV/d$
B. $2qV/mv_0^2$
C. $qV$
D. $mv_0^2/2$
E. none of these

ans: C

15. An electron is accelerated from rest through a potential difference $V$. Its final speed is proportional to:

A. $V$
B. $V^2$
C. $\sqrt{V}$
D. $1/V$
E. $1/\sqrt{V}$

ans: C

16. In separate experiments, four different particles each start from far away with the same speed and impinge directly on a gold nucleus. The masses and charges of the particles are

particle 1: mass $m_0$, charge $q_0$
particle 2: mass $2m_0$, charge $2q_0$
particle 3: mass $2m_0$, charge $q_0/2$
particle 4: mass $m_0/2$, charge $2q_0$

Rank the particles according to the distance of closest approach to the gold nucleus, from smallest to largest.

A. 1, 2, 3, 4
B. 4, 3, 2, 1
C. 3, 1 and 2 tie, then 4
D. 4, 1 and 2 tie, then 1
E. 1 and 2 tie, then 3, 4

ans: C

17. Two large parallel conducting plates are separated by a distance $d$, placed in a vacuum, and connected to a source of potential difference $V$. An oxygen ion, with charge $2e$, starts from rest on the surface of one plate and accelerates to the other. If $e$ denotes the magnitude of the electron charge, the final kinetic energy of this ion is:

A. $eV/2$
B. $eV/d$
C. $eVd$
D. $Vd/e$
E. $2eV$

ans: E

18. An electron volt is :

A. the force acting on an electron in a field of 1 N/C
B. the force required to move an electron 1 meter
C. the energy gained by an electron in moving through a potential difference of 1 volt
D. the energy needed to move an electron through 1 meter in any electric field
E. the work done when 1 coulomb of charge is moved through a potential difference of 1 volt.

ans: C

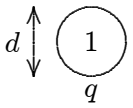
19. An electron has charge $-e$ and mass $m_e$. A proton has charge $e$ and mass $1840m_e$. A "proton volt" is equal to:

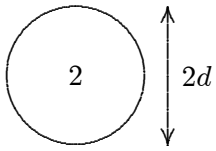
A. 1 eV
B. 1840 eV
C. $(1/1840)$ eV
D. $\sqrt{1840}$ eV
E. $(1/\sqrt{1840})$ eV

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ans: A

20. Two conducting spheres, one having twice the diameter of the other, are separated by a distance large compared to their diameters. The smaller sphere (1) has charge $q$ and the larger sphere (2) is uncharged. If the spheres are then connected by a long thin wire:




A. 1 and 2 have the same potential
B. 2 has twice the potential of 1
C. 2 has half the potential of 1
D. 1 and 2 have the same charge
E. all of the charge is dissipated

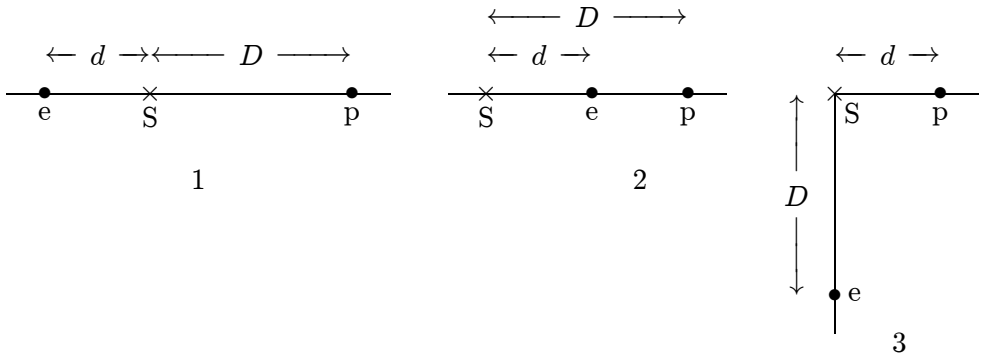
ans: A

21. Two conducting spheres are far apart. The smaller sphere carries a total charge $Q$. The larger sphere has a radius that is twice that of the smaller and is neutral. After the two spheres are connected by a conducting wire, the charges on the smaller and larger spheres, respectively, are:

A. $Q/2$ and $Q/2$
B. $Q/3$ and $2Q/3$
C. $2Q/3$ and $Q/3$
D. zero and $Q$
E. $2Q$ and $-Q$

ans: B

22. Three possible configurations for an electron e and a proton p are shown below. Take the zero of potential to be at infinity and rank the three configurations according to the potential at S, from most negative to most positive.




A. 1, 2, 3
B. 3, 2, 1
C. 2, 3, 1
D. 1 and 2 tie, then 3
E. 1 and 3 tie, then 2

ans: D

23. A conducting sphere with radius $R$ is charged until the magnitude of the electric field just outside its surface is $E$. The electric potential of the sphere, relative to the potential far away, is:

A. zero
B. $E/R$
C. $E/R^2$
D. $ER$
E. $ER^2$

ans: D

24. A 5-cm radius conducting sphere has a surface charge density of $2 \times 10^{-6}\,\mathrm{C/m^2}$ on its surface. Its electric potential, relative to the potential far away, is:

A. $1.1 \times 10^4\,\mathrm{V}$
B. $2.2 \times 10^4\,\mathrm{V}$
C. $2.3 \times 10^5\,\mathrm{V}$
D. $3.6 \times 10^5\,\mathrm{V}$
E. $7.2 \times 10^6\,\mathrm{V}$

ans: A

25. A hollow metal sphere is charged to a potential $V$. The potential at its center is:

A. $V$
B. 0
C. $-V$
D. $2V$
E. $\pi V$

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ans: A

26. Positive charge is distributed uniformly throughout a non-conducting sphere. The highest electric potential occurs:

A. at the center
B. at the surface
C. halfway between the center and surface
D. just outside the surface
E. far from the sphere

ans: A

27. A total charge of $7 \times 10^{-8}\,\mathrm{C}$ is uniformly distributed throughout a non-conducting sphere with a radius of 5 cm. The electric potential at the surface, relative to the potential far away, is about:

A. $-1.3 \times 10^4\,\mathrm{V}$
B. $1.3 \times 10^4\,\mathrm{V}$
C. $7.0 \times 10^5\,\mathrm{V}$
D. $-6.3 \times 10^4\,\mathrm{V}$
E. 0

ans: B

28. Eight identical spherical raindrops are each at a potential $V$, relative to the potential far away. They coalesce to make one spherical raindrop whose potential is:

A. $V/8$
B. $V/2$
C. $2V$
D. $4V$
E. $8V$

ans: D

29. A metal sphere carries a charge of $5 \times 10^{-9}$ C and is at a potential of 400 V, relative to the potential far away. The potential at the center of the sphere is:

A. 400 V
B. $-400$ V
C. $2 \times 10^{-6}$ V
D. 0
E. none of these

ans: A

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30. A 5-cm radius isolated conducting sphere is charged so its potential is +100 V, relative to the potential far away. The charge density on its surface is:

A. $+2.2 \times 10^{-7}$ C/m$^2$
B. $-2.2 \times 10^{-7}$ C/m$^2$
C. $+3.5 \times 10^{-7}$ C/m$^2$
D. $-3.5 \times 10^{-7}$ C/m$^2$
E. $+1.8 \times 10^{-8}$ C/m$^2$

ans: E

31. A conducting sphere has charge $Q$ and its electric potential is $V$, relative to the potential far away. If the charge is doubled to $2Q$, the potential is:

A. $V$
B. $2V$
C. $4V$
D. $V/2$
E. $V/4$

ans: B

32. The potential difference between the ends of a 2-meter stick that is parallel to a uniform electric field is 400 V. The magnitude of the electric field is:

A. zero
B. 100 V/m
C. 200 V/m
D. 400 V/m
E. 800 V/m

ans: E

33. In a certain region of space the electric potential increases uniformly from east to west and does not vary in any other direction. The electric field:

A. points east and varies with position
B. points east and does not vary with position
C. points west and varies with position
D. points west and does not vary with position
E. points north and does not vary with position

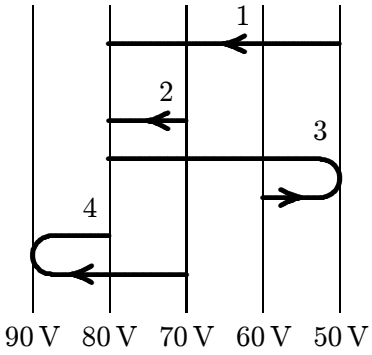
ans: B

34. If the electric field is in the positive $x$ direction and has a magnitude given by $E = Cx^2$, where $C$ is a constant, then the electric potential is given by $V =$:

A. $2Cx$
B. $-2Cx$
C. $Cx^3/3$
D. $-Cx^3/3$
E. $-3Cx^3$

ans: D

35. An electron goes from one equipotential surface to another along one of the four paths shown below. Rank the paths according to the work done by the electric field, from least to greatest.




A. 1, 2, 3, 4
B. 4, 3, 2, 1
C. 1, 3, 4 and 2 tie
D. 4 and 2 tie, then 3, then 1
E. 4, 3, 1, 2

ans: D

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36. The work required to carry a particle with a charge of 6.0 C from a 5.0-V equipotential surface to a 6.0-V equipotential surface and back again to the 5.0-V surface is:

A. 0
B. $1.2 \times 10^{-5}$ J
C. $3.0 \times 10^{-5}$ J
D. $6.0 \times 10^{-5}$ J
E. $6.0 \times 10^{-6}$ J

ans: A

37. The equipotential surfaces associated with a charged point particles are:

A. radially outward from the particle
B. vertical planes
C. horizontal planes
D. concentric spheres centered at the particle
E. concentric cylinders with the particle on the axis.

ans: D

38. The electric field in a region around the origin is given by $\vec{E} = C(x\,\hat{\mathrm{i}} + y\,\hat{\mathrm{j}})$, where $C$ is a constant. The equipotential surfaces in that region are:

A. concentric cylinders with axes along the $z$ axis
B. concentric cylinders with axes along the $x$ axis
C. concentric spheres centered at the origin
D. planes parallel to the $xy$ plane
E. planes parallel to the $yz$ plane

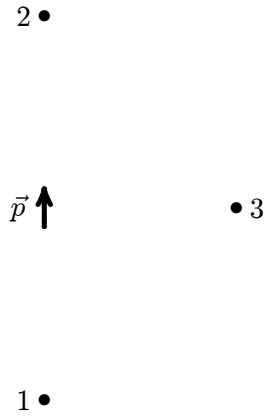
ans: A

39. The electric potential in a certain region of space is given by $V = -7.5x^2 + 3x$, where $V$ is in volts and $x$ is in meters. In this region the equipotential surfaces are:

A. planes parallel to the $x$ axis
B. planes parallel to the $yz$ plane
C. concentric spheres centered at the origin
D. concentric cylinders with the $x$ axis as the cylinder axis
E. unknown unless the charge is given

ans: B

40. In the diagram, the points 1, 2, and 3 are all the same very large distance from a dipole. Rank the points according to the values of the electric potential at them, from the most negative to the most positive.




A. 1, 2, 3
B. 3, 2, 1
C. 2, 3, 1
D. 1, 3, 2
E. 1 and 2 tie, then 3

ans: D

41. A particle with charge $q$ is to be brought from far away to a point near an electric dipole. No work is done if the final position of the particle is on:

A. the line through the charges of the dipole
B. a line that is perpendicular to the dipole moment
C. a line that makes an angle of 45° with the dipole moment
D. a line that makes an angle of 30° with the dipole moment
E. none of the above

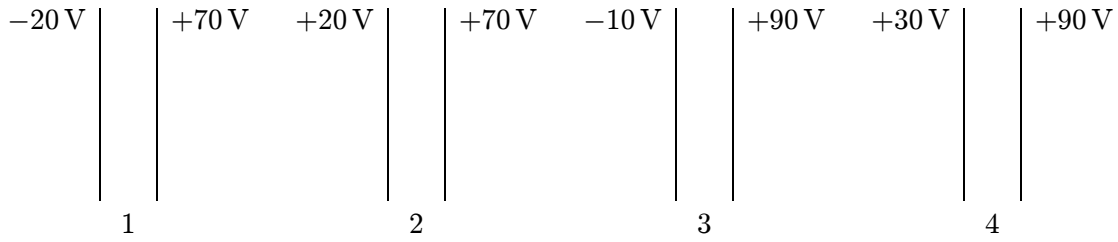
ans: B

42. Equipotential surfaces associated with an electric dipole are:

A. spheres centered on the dipole
B. cylinders with axes along the dipole moment
C. planes perpendicular to the dipole moment
D. planes parallel to the dipole moment
E. none of the above

ans: E

43. The diagram shows four pairs of large parallel conducting plates. The value of the electric potential is given for each plate. Rank the pairs according to the magnitude of the electric field between the plates, least to greatest.




A. 1, 2, 3, 4
B. 4, 3, 2, 1
C. 2, 3, 1, 4
D. 2, 4, 1, 3
E. 3, 2, 4, 1

ans: D

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# Chapter 25: CAPACITANCE

1. The units of capacitance are equivalent to:
   A. J/C
   B. V/C
   C. $J^2/C$
   D. C/J
   E. $C^2/J$

   ans: E

2. A farad is the same as a:
   A. J/V
   B. V/J
   C. C/V
   D. V/C
   E. N/C

   ans: C

3. A capacitor C "has a charge Q". The actual charges on its plates are:
   A. $Q$, $Q$
   B. $Q/2$, $Q/2$
   C. $Q$, $-Q$
   D. $Q/2$, $-Q/2$
   E. $Q$, 0

   ans: C

4. Each plate of a capacitor stores a charge of magnitude 1 mC when a 100-V potential difference is applied. The capacitance is:
   A. $5\,\mu$F
   B. $10\,\mu$F
   C. $50\,\mu$F
   D. $100\,\mu$F
   E. none of these

   ans: B

5. To charge a 1-F capacitor with 2 C requires a potential difference of:
   A. 2 V
   B. 0.2 V
   C. 5 V
   D. 0.5 V
   E. none of these

   ans: A

6. The capacitance of a parallel-plate capacitor with plate area $A$ and plate separation $d$ is given by:

   A. $\epsilon_0 d/A$
   B. $\epsilon_0 d/2A$
   C. $\epsilon_0 A/d$
   D. $\epsilon_0 A/2d$
   E. $Ad/\epsilon_0$

   ans: C

7. The capacitance of a parallel-plate capacitor is:

   A. proportional to the plate area
   B. proportional to the charge stored
   C. independent of any material inserted between the plates
   D. proportional to the potential difference of the plates
   E. proportional to the plate separation

   ans: A

8. The plate areas and plate separations of five parallel plate capacitors are

   capacitor 1: area $A_0$, separation $d_0$
   capacitor 2: area $2A_0$, separation $2d_0$
   capacitor 3: area $2A_0$, separation $d_0/2$
   capacitor 4: area $A_0/2$, separation $2d_0$
   capacitor 5: area $A_0$, separation $d_0/2$

   Rank these according to their capacitances, least to greatest.

   A. 1, 2, 3, 4, 5
   B. 5, 4, 3, 2, 1
   C. 5, 3 and 4 tie, then 1, 2
   D. 4, 1 and 2 tie, then 5, 3
   E. 3, 5, 1 and 2 tie, 1, 4

   ans: D

9. The capacitance of a parallel-plate capacitor can be increased by:

   A. increasing the charge
   B. decreasing the charge
   C. increasing the plate separation
   D. decreasing the plate separation
   E. decreasing the plate area

   ans: D

10. If both the plate area and the plate separation of a parallel-plate capacitor are doubled, the capacitance is:

    A. doubled
    B. halved
    C. unchanged
    D. tripled
    E. quadrupled

    ans: C

11. If the plate area of an isolated charged parallel-plate capacitor is doubled:
    A. the electric field is doubled
    B. the potential difference is halved
    C. the charge on each plate is halved
    D. the surface charge density on each plate is doubled
    E. none of the above

    ans: B

12. If the plate separation of an isolated charged parallel-plate capacitor is doubled:
    A. the electric field is doubled
    B. the potential difference is halved
    C. the charge on each plate is halved
    D. the surface charge density on each plate is doubled
    E. none of the above

    ans: E

13. Pulling the plates of an isolated charged capacitor apart:
    A. increases the capacitance
    B. increases the potential difference
    C. does not affect the potential difference
    D. decreases the potential difference
    E. does not affect the capacitance

    ans: B

14. If the charge on a parallel-plate capacitor is doubled:
    A. the capacitance is halved
    B. the capacitance is doubled
    C. the electric field is halved
    D. the electric field is doubled
    E. the surface charge density is not changed on either plate

    ans: D

15. A parallel-plate capacitor has a plate area of $0.2\,\mathrm{m}^2$ and a plate separation of $0.1\,\mathrm{mm}$. To obtain an electric field of $2.0 \times 10^6\,\mathrm{V/m}$ between the plates, the magnitude of the charge on each plate should be:
    A. $8.9 \times 10^{-7}\,\mathrm{C}$
    B. $1.8 \times 10^{-6}\,\mathrm{C}$
    C. $3.5 \times 10^{-6}\,\mathrm{C}$
    D. $7.1 \times 10^{-6}\,\mathrm{C}$
    E. $1.4 \times 10^{-5}\,\mathrm{C}$

    ans: D

16. A parallel-plate capacitor has a plate area of $0.2\,\mathrm{m}^2$ and a plate separation of 0.1 mm. If the charge on each plate has a magnitude of $4 \times 10^{-6}\,\mathrm{C}$ the potential difference across the plates is approximately:

A. 0
B. $4 \times 10^{-2}\,\mathrm{V}$
C. $1 \times 10^{2}\,\mathrm{V}$
D. $2 \times 10^{2}\,\mathrm{V}$
E. $4 \times 10^{8}\,\mathrm{V}$

ans: D

17. The capacitance of a spherical capacitor with inner radius $a$ and outer radius $b$ is proportional to:

A. $a/b$
B. $b - a$
C. $b^2 - a^2$
D. $ab/(b - a)$
E. $ab/(b^2 - a^2)$

ans: D

18. The capacitance of a single isolated spherical conductor with radius $R$ is proportional to:

A. $R$
B. $R^2$
C. $1/R$
D. $1/R^2$
E. none of these

ans: A

19. Two conducting spheres have radii of $R_1$ and $R_2$, with $R_1$ greater than $R_2$. If they are far apart the capacitance is proportional to:

A. $R_1 R_2/(R_1 - R_2)$
B. $R_1^2 - R_2^2$
C. $(R_1 - R_2)/R_1 R_2$
D. $R_1^2 + R_2^2$
E. none of these

ans: A

20. The capacitance of a cylindrical capacitor can be increased by:

A. decreasing both the radius of the inner cylinder and the length
B. increasing both the radius of the inner cylinder and the length
C. increasing the radius of the outer cylindrical shell and decreasing the length
D. decreasing the radius of the inner cylinder and increasing the radius of the outer cylindrical shell
E. only by decreasing the length

ans: B

21. A battery is used to charge a series combination of two identical capacitors. If the potential difference across the battery terminals is $V$ and total charge $Q$ flows through the battery during the charging process then the charge on the positive plate of each capacitor and the potential difference across each capacitor are:

A. $Q/2$ and $V/2$, respectively
B. $Q$ and $V$, respectively
C. $Q/2$ and $V$, respectively
D. $Q$ and $V/2$, respectively
E. $Q$ and $2V$, respectively

ans: D

22. A battery is used to charge a parallel combination of two identical capacitors. If the potential difference across the battery terminals is $V$ and total charge $Q$ flows through the battery during the charging process then the charge on the positive plate of each capacitor and the potential difference across each capacitor are:

A. $Q/2$ and $V/2$, respectively
B. $Q$ and $V$, respectively
C. $Q/2$ and $V$, respectively
D. $Q$ and $V/2$, respectively
E. $Q$ and $2V$, respectively

ans: C

23. A 2-$\mu$F and a 1-$\mu$F capacitor are connected in series and a potential difference is applied across the combination. The 2-$\mu$F capacitor has:

A. twice the charge of the 1-$\mu$F capacitor
B. half the charge of the 1-$\mu$F capacitor
C. twice the potential difference of the 1-$\mu$F capacitor
D. half the potential difference of the 1-$\mu$F capacitor
E. none of the above

ans: D

24. A 2-$\mu$F and a 1-$\mu$F capacitor are connected in parallel and a potential difference is applied across the combination. The 2-$\mu$F capacitor has:

A. twice the charge of the 1-$\mu$F capacitor
B. half the charge of the 1-$\mu$F capacitor
C. twice the potential difference of the 1-$\mu$F capacitor
D. half the potential difference of the 1-$\mu$F capacitor
E. none of the above

ans: A

25. Let $Q$ denote charge, $V$ denote potential difference, and $U$ denote stored energy. Of these quantities, capacitors in series must have the same:

A. $Q$ only
B. $V$ only
C. $U$ only
D. $Q$ and $U$ only
E. $V$ and $U$ only

ans: A

26. Let $Q$ denote charge, $V$ denote potential difference, and $U$ denote stored energy. Of these quantities, capacitors in parallel must have the same:

A. $Q$ only
B. $V$ only
C. $U$ only
D. $Q$ and $U$ only
E. $V$ and $U$ only

ans: B

27. Capacitors $C_1$ and $C_2$ are connected in parallel. The equivalent capacitance is given by:

A. $C_1C_2/(C_1 + C_2)$
B. $(C_1 + C_2)/C_1C_2$
C. $1/(C_1 + C_2)$
D. $C_1/C_2$
E. $C_1 + C_2$

ans: E

28. Capacitors $C_1$ and $C_2$ are connected in series. The equivalent capacitance is given by:

A. $C_1C_2/(C_1 + C_2)$
B. $(C_1 + C_2)/C_1C_2$
C. $1/(C_1 + C_2)$
D. $C_1/C_2$
E. $C_1 + C_2$

ans: A

29. Capacitors $C_1$ and $C_2$ are connected in series and a potential difference is applied to the combination. If the capacitor that is equivalent to the combination has the same potential difference, then the charge on the equivalent capacitor is the same as:

A. the charge on $C_1$
B. the sum of the charges on $C_1$ and $C_2$
C. the difference of the charges on $C_1$ and $C_2$
D. the product of the charges on $C_1$ and $C_2$
E. none of the above

ans: A

30. Capacitors $C_1$ and $C_2$ are connected in parallel and a potential difference is applied to the combination. If the capacitor that is equivalent to the combination has the same potential difference, then the charge on the equivalent capacitor is the same as:

A. the charge on $C_1$
B. the sum of the charges on $C_1$ and $C_2$
C. the difference of the charges on $C_1$ and $C_2$
D. the product of the charges on $C_1$ and $C_2$
E. none of the above

ans: B

31. Two identical capacitors are connected in series and two, each identical to the first, are connected in parallel. The equivalent capacitance of the series connection is _____ the equivalent capacitance of parallel connection.

A. twice
B. four times
C. half
D. one-fourth
E. the same as

ans: D

32. Two identical capacitors, each with capacitance $C$, are connected in parallel and the combination is connected in series to a third identical capacitor. The equivalent capacitance of this arrangement is:

A. $2C/3$
B. $C$
C. $3C/2$
D. $2C$
E. $3C$

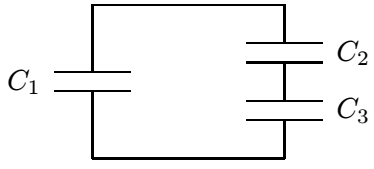
ans: A

33. A 2-$\mu$F and a 1-$\mu$F capacitor are connected in series and charged from a battery. They store charges $P$ and $Q$, respectively. When disconnected and charged separately using the same battery, they have charges $R$ and $S$, respectively. Then:

A. $R > S > Q = P$
B. $P > Q > R = S$
C. $R > P = Q > S$
D. $R = P > S = Q$
E. $R > P > S = Q$

ans: A

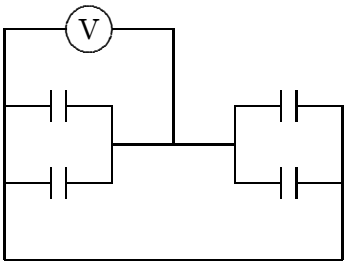
34. Capacitor $C_1$ is connected alone to a battery and charged until the magnitude of the charge on each plate is $4.0 \times 10^{-8}$ C. Then it is removed from the battery and connected to two other capacitors $C_2$ and $C_3$, as shown. The charge on the positive plate of $C_1$ is then $1.0 \times 10^{-8}$ C. The charges on the positive plates of $C_2$ and $C_3$ are:




A. $q_2 = 3.0 \times 10^{-8}$ C and $q_3 = 3.0 \times 10^{-8}$ C
B. $q_2 = 2.0 \times 10^{-8}$ C and $q_3 = 2.0 \times 10^{-8}$ C
C. $q_2 = 5.0 \times 10^{-8}$ C and $q_3 = 1.0 \times 10^{-8}$ C
D. $q_2 = 3.0 \times 10^{-8}$ C and $q_3 = 1.0 \times 10^{-8}$ C
E. $q_2 = 1.0 \times 10^{-8}$ C and $q_3 = 3.0 \times 10^{-8}$ C

ans: A

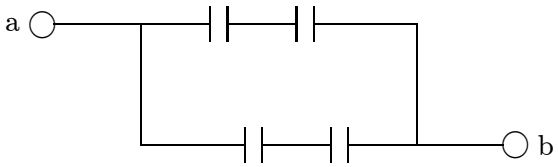
35. Each of the four capacitors shown is 500 $\mu$F. The voltmeter reads 1000 V. The magnitude of the charge, in coulombs, on each capacitor plate is:




A. 0.2
B. 0.5
C. 20
D. 50
E. none of these

ans: B

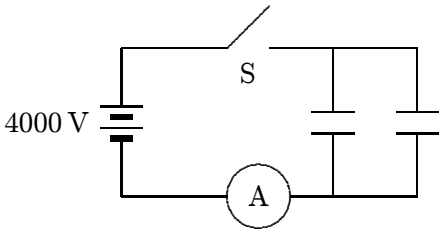
36. The diagram shows four 6-$\mu$F capacitors. The capacitance between points a and b is:




A. 3 $\mu$F
B. 4 $\mu$F
C. 6 $\mu$F
D. 9 $\mu$F
E. 1 $\mu$F

ans: C

37. Each of the two 25-$\mu$F capacitors shown is initially uncharged. How many coulombs of charge pass through the ammeter A after the switch S is closed?




A. 0.10
B. 0.20
C. 10
D. 0.05
E. none of these

ans: B

38. A 20-F capacitor is charged to 200 V. Its stored energy is:

A. 4000 J
B. 4 J
C. 0.4 J
D. 2000 J
E. 0.1 J

ans: C

39. A charged capacitor stores 10 C at 40 V. Its stored energy is:

A. 400 J
B. 4 J
C. 0.2 J
D. 2.5 J
E. 200 J

ans: E

40. A 2-$\mu$F and a 1-$\mu$F capacitor are connected in series and charged by a battery. They store energies $P$ and $Q$, respectively. When disconnected and charged separately using the same battery, they store energies $R$ and $S$, respectively. Then:

A. $R > P > S > Q$
B. $P > Q > R > S$
C. $R > P > Q > S$
D. $P > R > S > Q$
E. $R > S > Q > P$

ans: E

41. The quantity $(1/2)\epsilon_0 E^2$ has the significance of:

A. energy/farad
B. energy/coulomb
C. energy
D. energy/volume
E. energy/volt

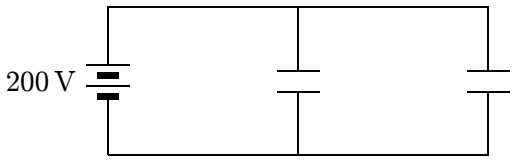
ans: D

42. Capacitors A and B are identical. Capacitor A is charged so it stores 4 J of energy and capacitor B is uncharged. The capacitors are then connected in parallel. The total stored energy in the capacitors is now:

A. 16 J
B. 8 J
C. 4 J
D. 2 J
E. 1 J

ans: D

43. To store a total of 0.040 J of energy in the two identical capacitors shown, each should have a capacitance of:




A. $0.10\,\mu$F
B. $0.50\,\mu$F$0.10\,\mu$F
C. $1.0\,\mu$J
D. $1.5\,\mu$F
E. $2.0\,\mu$F

ans: C

44. A battery is used to charge a parallel-plate capacitor, after which it is disconnected. Then the plates are pulled apart to twice their original separation. This process will double the:

A. capacitance
B. surface charge density on each plate
C. stored energy
D. electric field between the two places
E. charge on each plate

ans: C

45. A parallel-plate capacitor has a plate area of $0.3\,\text{m}^2$ and a plate separation of 0.1 mm. If the charge on each plate has a magnitude of $5 \times 10^{-6}$ C then the force exerted by one plate on the other has a magnitude of about:

A. 0
B. 5 N
C. 9 N
D. $1 \times 10^4$ N
E. $9 \times 10^5$ N

ans: B

46. A certain capacitor has a capacitance of $5.0\,\mu$F. After it is charged to $5.0\,\mu$C and isolated, the plates are brought closer together so its capacitance becomes $10\,\mu$F. The work done by the agent is about:

A. zero
B. $1.25 \times 10^{-6}$ J
C. $-1.25 \times 10^{-6}$ J
D. $8.3 \times 10^{-7}$ J
E. $-8.3 \times 10^{-7}$ J

ans: C

47. A dielectric slab is slowly inserted between the plates of a parallel plate capacitor, while the potential difference between the plates is held constant by a battery. As it is being inserted:

A. the capacitance, the potential difference between the plates, and the charge on the positive plate all increase
B. the capacitance, the potential difference between the plates, and the charge on the positive plate all decrease
C. the potential difference between the plates increases, the charge on the positive plate decreases, and the capacitance remains the same
D. the capacitance and the charge on the positive plate decrease but the potential difference between the plates remains the same
E. the capacitance and the charge on the positive plate increase but the potential difference between the plates remains the same

ans: E

48. An air-filled parallel-plate capacitor has a capacitance of 1 pF. The plate separation is then doubled and a wax dielectric is inserted, completely filling the space between the plates. As a result, the capacitance becomes 2 pF. The dielectric constant of the wax is:

A. 0.25
B. 0.5
C. 2.0
D. 4.0
E. 8.0

ans: D

49. One of materials listed below is to be placed between two identical metal sheets, with no, air gap, to form a parallel-plate capacitor. Which produces the greatest capacitance?

A. material of thickness 0.1 mm and dielectric constant 2
B. material of thickness 0.2 mm and dielectric constant 3
C. material of thickness 0.3 mm and dielectric constant 2
D. material of thickness 0.4 mm and dielectric constant 8
E. material of thickness 0.5 mm and dielectric constant 11

ans: E

50. Two capacitors are identical except that one is filled with air and the other with oil. Both capacitors carry the same charge. The ratio of the electric fields $E_{\text{air}}/E_{\text{oil}}$ is:

A. between 0 and 1
B. 0
C. 1
D. between 1 and infinity
E. infinite

ans: D

51. A parallel-plate capacitor, with air dielectric, is charged by a battery, after which the battery is disconnected. A slab of glass dielectric is then slowly inserted between the plates. As it is being inserted:
    A. a force repels the glass out of the capacitor
    B. a force attracts the glass into the capacitor
    C. no force acts on the glass
    D. a net charge appears on the glass
    E. the glass makes the plates repel each other

    ans: B

52. Two parallel-plate capacitors with the same plate separation but different capacitance are connected in parallel to a battery. Both capacitors are filled with air. The quantity that is NOT the same for both capacitors when they are fully charged is:
    A. potential difference
    B. energy density
    C. electric field between the plates
    D. charge on the positive plate
    E. dielectric constant

    ans: D

53. Two parallel-plate capacitors with the same plate area but different capacitance are connected in parallel to a battery. Both capacitors are filled with air. The quantity that is the same for both capacitors when they are fully charged is:
    A. potential difference
    B. energy density
    C. electric field between the plates
    D. charge on the positive plate
    E. plate separation

    ans: A

54. Two parallel-plate capacitors with different plate separation but the same capacitance are connected in series to a battery. Both capacitors are filled with air. The quantity that is NOT the same for both capacitors when they are fully charged is:
    A. potential difference
    B. stored energy
    C. electric field between the plates
    D. charge on the positive plate
    E. dielectric constant

    ans: C

55. Two parallel-plate capacitors with different capacitance but the same plate separation are connected in series to a battery. Both capacitors are filled with air. The quantity that is the same for both capacitors when they are fully charged is:
    A. potential difference
    B. stored energy
    C. energy density
    D. electric field between the plates
    E. charge on the positive plate

    ans: E

## Chapter 26: CURRENT AND RESISTANCE

1. A car battery is rated at $80\,\mathrm{A}\cdot\mathrm{h}$. An ampere-hour is a unit of:
   A. power
   B. energy
   C. current
   D. charge
   E. force

   ans: D

2. Current has units:
   A. kilowatt·hour
   B. coulomb/second
   C. coulomb
   D. volt
   E. ohm

   ans: B

3. Current has units:
   A. kilowatt·hour
   B. ampere
   C. coulomb
   D. volt
   E. ohm

   ans: B

4. The units of resistivity are:
   A. ohm
   B. ohm·meter
   C. ohm/meter
   D. ohm/meter$^2$
   E. none of these

   ans: B

5. The rate at which electrical energy is used may be measured in:
   A. watt/second
   B. watt·second
   C. watt
   D. joule·second
   E. kilowatt·hour

   ans: C

6. Energy may be measured in:
   A. kilowatt
   B. joule·second
   C. watt
   D. watt·second
   E. volt/ohm

   ans: D

7. Which one of the following quantities is correctly matched to its unit?
   A. Power — kW·h
   B. Energy — kW
   C. Potential difference — J/C
   D. Current — A/s
   E. Resistance — V/C

   ans: C

8. Current is a measure of:
   A. force that moves a charge past a point
   B. resistance to the movement of a charge past a point
   C. energy used to move a charge past a point
   D. amount of charge that moves past a point per unit time
   E. speed with which a charge moves past a point

   ans: D

9. A 60-watt light bulb carries a current of 0.5 A. The total charge passing through it in one hour is:
   A. 120 C
   B. 3600 C
   C. 3000 C
   D. 2400 C
   E. 1800 C

   ans: E

10. A 10-ohm resistor has a constant current. If 1200 C of charge flow through it in 4 minutes what is the value of the current?
   A. 3.0 A
   B. 5.0 A
   C. 11 A
   D. 15 A
   E. 20 A

   ans: D

11. Conduction electrons move to the right in a certain wire. This indicates that:
    A. the current density and electric field both point right
    B. the current density and electric field both point left
    C. the current density points right and the electric field points left
    D. the current density points left and the electric field points right
    E. the current density points left but the direction of the electric field is unknown

    ans: B

12. Two wires made of different materials have the same uniform current density. They carry the same current only if:
    A. their lengths are the same
    B. their cross-sectional areas are the same
    C. both their lengths and cross-sectional areas are the same
    D. the potential differences across them are the same
    E. the electric fields in them are the same

    ans: B

13. A wire with a length of 150 m and a radius of 0.15 mm carries a current with a uniform current density of $2.8 \times 10^7\ \mathrm{A/m^2}$. The current is:
    A. $0.63\ \mathrm{A^2}$
    B. 2.0 A
    C. $5.9\ \mathrm{A^2}$
    D. 296 A
    E. $400\ \mathrm{A^2}$

    ans: B

14. In a conductor carrying a current we expect the electron drift speed to be:
    A. much greater than the average electron speed
    B. much less than the average electron speed
    C. about the same as the average electron speed
    D. less than the average electron speed at low temperature and greater than the average electron speed at high temperature
    E. less than the average electron speed at high temperature and greater than the average electron speed at low temperature

    ans: B

15. Two substances are identical except that the electron mean free time for substance A is twice the electron mean free time for substance B. If the same electric field exists in both substances the electron drift speed in A is:
    A. the same as in B
    B. twice that in B
    C. half that in B
    D. four times that in B
    E. one-fourth that in B

    ans: B

16. The current is zero in a conductor when no potential difference is applied because:
    A. the electrons are not moving
    B. the electrons are not moving fast enough
    C. for every electron with a given velocity there is another with a velocity of equal magnitude and opposite direction.
    D. equal numbers of electrons and protons are moving together
    E. otherwise Ohm's law would not be valid

    ans: C

17. The current density is the same in two wires. Wire A has twice the free-electron concentration of wire B. The drift speed of electrons in A is:
    A. twice that of electrons in B
    B. four times that of electrons in B
    C. half that of electrons in B
    D. one-fourth that of electrons in B
    E. the same as that of electrons in B

    ans: C

18. Copper contains $8.4\times10^{28}$ free electrons/$\mathrm{m}^3$. A copper wire of cross-sectional area $7.4\times10^{-7}\,\mathrm{m}^2$ carries a current of 1 A. The electron drift speed is approximately:
    A. $3\times10^{8}\,\mathrm{m/s}$
    B. $10^{3}\,\mathrm{m/s}$
    C. $1\,\mathrm{m/s}$
    D. $10^{-4}\,\mathrm{m/s}$
    E. $10^{-23}\,\mathrm{m/s}$

    ans: D

19. If $\vec{J}$ is the current density and $d\vec{A}$ is a vector element of area then the integral $\int \vec{J}\cdot d\vec{A}$ over an area represents:
    A. the electric flux through the area
    B. the average current density at the position of the area
    C. the resistance of the area
    D. the resistivity of the area
    E. the current through the area

    ans: E

20. If the potential difference across a resistor is doubled:
    A. only the current is doubled
    B. only the current is halved
    C. only the resistance is doubled
    D. only the resistance is halved
    E. both the current and resistance are doubled

    ans: A

21. Five cylindrical wires are made of the same material. Their lengths and radii are

wire 1: length $\ell$, radius $r$
wire 2: length $\ell/4$, radius $r/2$
wire 3: length $\ell/2$, radius $r/2$
wire 4: length $\ell$, radius $r/2$
wire 5: length $5\ell$, radius $2r$

Rank the wires according to their resistances, least to greatest.

A. 1, 2, 3, 4, 5
B. 5, 4, 3, 2, 1
C. 1 and 2 tie, then 5, 3, 4
D. 1, 3, 4, 2, 5
E. 1, 2, 4, 3, 5

ans: C

22. Of the following, the copper conductor that has the least resistance is:

A. thin, long and hot
B. thick, short and cool
C. thick, long and hot
D. thin, short and cool
E. thin, short and hot

ans: B

23. A cylindrical copper rod has resistance $R$. It is reformed to twice its original length with no change of volume. Its new resistance is:

A. $R$
B. $2R$
C. $4R$
D. $8R$
E. $R/2$

ans: C

24. The resistance of a rod does NOT depend on:

A. its temperature
B. its material
C. its length
D. its conductivity
E. the shape of its (fixed) cross-sectional area

ans: E

25. A certain wire has resistance $R$. Another wire, of the same material, has half the length and half the diameter of the first wire. The resistance of the second wire is:

A. $R/4$
B. $R/2$
C. $R$
D. $2R$
E. $4R$

ans: D

26. A nichrome wire is 1 m long and $1 \times 10^{-6}\,\mathrm{m}^2$ in cross-sectional area. When connected to a potential difference of 2 V, a current of 4 A exists in the wire. The resistivity of this nichrome is:

A. $10^{-7}\,\Omega \cdot \mathrm{m}$
B. $2 \times 10^{-7}\,\Omega \cdot \mathrm{m}$
C. $4 \times 10^{-7}\,\Omega \cdot \mathrm{m}$
D. $5 \times 10^{-7}\,\Omega \cdot \mathrm{m}$
E. $8 \times 10^{-7}\,\Omega \cdot \mathrm{m}$

ans: D

27. Two conductors are made of the same material and have the same length. Conductor A is a solid wire of diameter 1 m. Conductor B is a hollow tube of inside diameter 1 m and outside diameter 2 m. The ratio of their resistance, $R_A/R_B$, is:

A. 1
B. $\sqrt{2}$
C. 2
D. 3
E. 4

ans: D

28. Conductivity is:

A. the same as resistivity, it is just more convenient to use for good conductors
B. expressed in $\Omega^{-1}$
C. equal to 1/resistance
D. expressed in $(\Omega \cdot \mathrm{m})^{-1}$
E. not a meaningful quantity for an insulator

ans: D

29. A certain sample carries a current of 4 A when the potential difference is 2 V and a current of 10 A when the potential difference is 4 V. This sample:

A. obeys Ohm's law
B. has a resistance of $0.5\,\Omega$ at 1 V
C. has a resistance of $2.5\,\Omega$ at 1 V
D. has a resistance of $2.5\,\Omega$ at 2 V
E. does not have a resistance

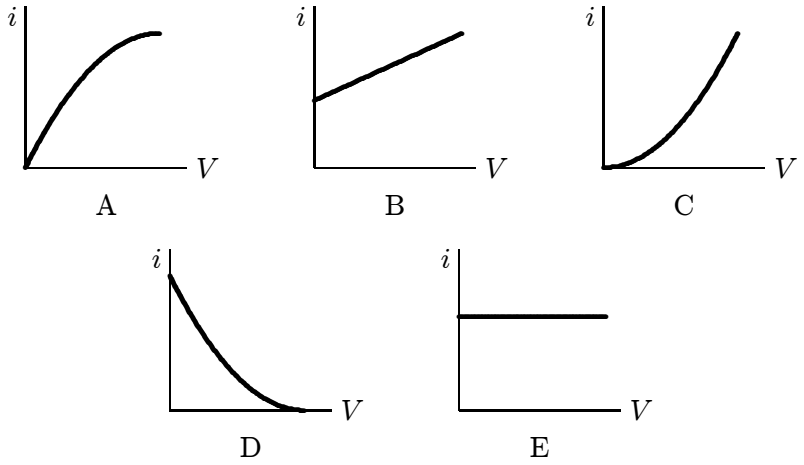
ans: B

30. A current of 0.5 A exists in a 60-ohm lamp. The applied potential difference is:

A. 15 V
B. 30 V
C. 60 V
D. 120 V
E. none of these

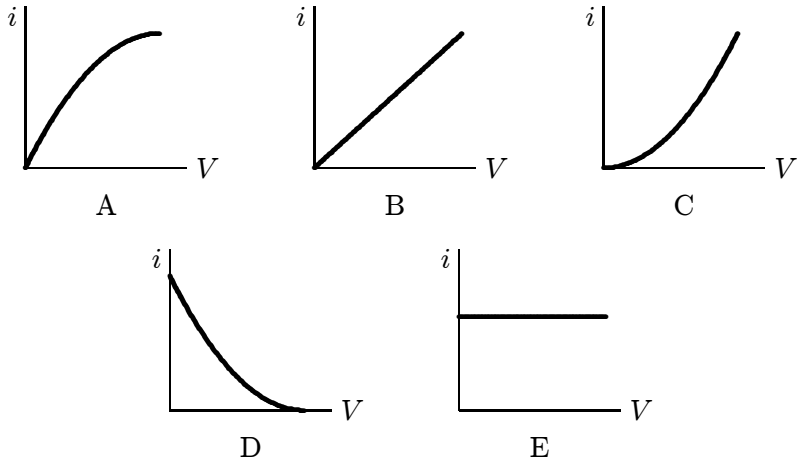
ans: B

31. Which of the following graphs best represents the current-voltage relationship of an incandescent light bulb?




ans: A

32. Which of the following graphs best represents the current-voltage relationship for a device that obeys Ohm's law?




ans: B

33. Two wires are made of the same material and have the same length but different radii. They are joined end-to-end and a potential difference is maintained across the combination. Of the following the quantity that is the same for both wires is:

A. potential difference
B. current
C. current density
D. electric field
E. conduction electron drift speed

ans: B

34. For an ohmic substance the resistivity is the proportionality constant for:
    A. current and potential difference
    B. current and electric field
    C. current density and potential difference
    D. current density and electric field
    E. potential difference and electric field

    ans: D

35. For an ohmic resistor, resistance is the proportionality constant for:
    A. potential difference and electric field
    B. current and electric field
    C. current and length
    D. current and cross-sectional area
    E. current and potential difference

    ans: E

36. For an ohmic substance, the resistivity depends on:
    A. the electric field
    B. the potential difference
    C. the current density
    D. the electron mean free time
    E. the cross-sectional area of the sample

    ans: D

37. For a cylindrical resistor made of ohmic material, the resistance does NOT depend on:
    A. the current
    B. the length
    C. the cross-sectional area
    D. the resistivity
    E. the electron drift velocity

    ans: A

38. For an ohmic substance, the electron drift velocity is proportional to:
    A. the cross-sectional area of the sample
    B. the length of the sample
    C. the mass of an electron
    D. the electric field in the sample
    E. none of the above

    ans: D

39. You wish to triple the rate of energy dissipation in a heating device. To do this you could triple:

A. the potential difference keeping the resistance the same
B. the current keeping the resistance the same
C. the resistance keeping the potential difference the same
D. the resistance keeping the current the same
E. both the potential difference and current

ans: D

40. A student kept her 60-watt, 120-volt study lamp turned on from 2:00 PM until 2:00 AM. How many coulombs of charge went through it?

A. 150
B. 3,600
C. 7,200
D. 18,000
E. 21,600

ans: E

41. A flat iron is marked "120 V, 600 W". In normal use, the current in it is:

A. 2 A
B. 4 A
C. 5 A
D. 7.2 A
E. 0.2 A

ans: C

42. An certain resistor dissipates 0.5 W when connected to a 3 V potential difference. When connected to a 1 V potential difference, this resistor will dissipate:

A. 0.5 W
B. 0.167 W
C. 1.5 W
D. 0.056 W
E. none of these

ans: D

43. An ordinary light bulb is marked "60 W, 120 V". Its resistance is:

A. 60 Ω
B. 120 Ω
C. 180 Ω
D. 240 Ω
E. 15 Ω

ans: D

44. The mechanical equivalent of heat is 1 cal = 4.18 J. The specific heat of water is 1 cal/g · K. An electric immersion water heater, rated at 400 W, should heat a kilogram of water from 10° C to 30° C in about:

A. 3.5 min
B. 1 min
C. 15 min
D. 45 min
E. 15 s

ans: A

45. It is better to send 10,000 kW of electric power long distances at 10,000 V rather than at 220 V because:

A. there is less heating in the transmission wires
B. the resistance of the wires is less at high voltages
C. more current is transmitted at high voltages
D. the insulation is more effective at high voltages
E. the $iR$ drop along the wires is greater at high voltage

ans: A

46. Suppose the electric company charges 10 cents per kW·h. How much does it cost to use a 125 W lamp 4 hours a day for 30 days?

A. $1.20
B. $1.50
C. $1.80
D. $7.20
E. none of these

ans: B

47. A certain x-ray tube requires a current of 7 mA at a voltage of 80 kV. The rate of energy dissipation (in watts) is:

A. 560
B. 5600
C. 26
D. 11.4
E. 87.5

ans: A

48. The mechanical equivalent of heat is 1 cal = 4.18 J. A heating coil, connected to a 120-V source, provides 60,000 calories in 10 minutes. The current in the coil is:

A. 0.83 A
B. 2 A
C. 3.5 A
D. 20 A
E. 50 A

ans: C

49. You buy a "75 W" light bulb. The label means that:
    A. no matter how you use the bulb, the power will be 75 W
    B. the bulb was filled with 75 W at the factory
    C. the actual power dissipated will be much higher than 75 W since most of the power appears as heat
    D. the bulb is expected to burn out after you use up its 75 W
    E. none of the above

    ans: E

50. A current of 0.3 A is passed through a lamp for 2 minutes using a 6-V power supply. The energy dissipated by this lamp during the 2 minutes is:
    A. 1.8 J
    B. 12 J
    C. 20 J
    D. 36 J
    E. 216 J

    ans: E

## Chapter 27: CIRCUITS

1. "The sum of the currents into a junction equals the sum of the currents out of the junction" is a consequence of:

   A. Newton's third law
   B. Ohm's law
   C. Newton's second law
   D. conservation of energy
   E. conservation of charge

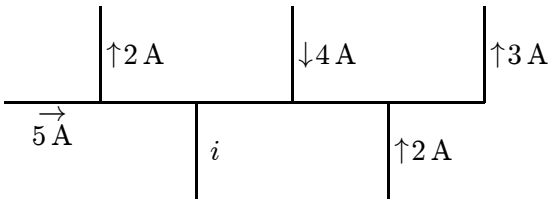
   ans: E

2. "The sum of the emf's and potential differences around a closed loop equals zero" is a consequence of:

   A. Newton's third law
   B. Ohm's law
   C. Newton's second law
   D. conservation of energy
   E. conservation of charge

   ans: D

3. A portion of a circuit is shown, with the values of the currents given for some branches. What is the direction and value of the current $i$?




   A. ↓, 6 A
   B. ↑, 6 A
   C. ↓, 4 A
   D. ↑, 4 A
   E. ↓, 2 A

   ans: A

4. Four wires meet at a junction. The first carries 4 A into the junction, the second carries 5 A out of the junction, and the third carries 2 A out of the junction. The fourth carries:

   A. 7 A out of the junction
   B. 7 A into the junction
   C. 3 A out of the junction
   D. 3 A into the junction
   E. 1 A into the junction

   ans: D

5. In the context of the loop and junctions rules for electrical circuits a junction is:
   A. where a wire is connected to a resistor
   B. where a wire is connected to a battery
   C. where only two wires are joined
   D. where three or more wires are joined
   E. where a wire is bent

   ans: D

6. For any circuit the number of independent equations containing emf's, resistances, and currents equals:
   A. the number of junctions
   B. the number of junctions minus 1
   C. the number of branches
   D. the number of branches minus 1
   E. the number of closed loops

   ans: C

7. If a circuit has $L$ closed loops, $B$ branches, and $J$ junctions the number of independent loop equations is:
   A. $B - J + 1$
   B. $B - J$
   C. $B$
   D. $L$
   E. $L - J$

   ans: A

8. A battery is connected across a series combination of two identical resistors. If the potential difference across the terminals is $V$ and the current in the battery is $i$, then:
   A. the potential difference across each resistor is $V$ and the current in each resistor is $i$
   B. the potential difference across each resistor is $V/2$ and the current in each resistor is $i/2$
   C. the potential difference across each resistor is $V$ and the current in each resistor is $i/2$
   D. the potential difference across each resistor is $V/2$ and the current in each resistor is $i$
   E. none of the above are true

   ans: D

9. A battery is connected across a parallel combination of two identical resistors. If the potential difference across the terminals is $V$ and the current in the battery is $i$, then:
   A. the potential difference across each resistor is $V$ and the current in each resistor is $i$
   B. the potential difference across each resistor is $V/2$ and the current in each resistor is $i/2$
   C. the potential difference across each resistor is $V$ and the current in each resistor is $i/2$
   D. the potential difference across each resistor is $V/2$ and the current in each resistor is $i$
   E. none of the above are true

   ans: C

10. A total resistance of 3.0 Ω is to be produced by combining an unknown resistor $R$ with a 12 Ω resistor. What is the value of $R$ and how is it to be connected to the 12 Ω resistor?

A. 4.0 Ω, parallel
B. 4.0 Ω, series
C. 2.4 Ω, parallel
D. 2.4 Ω, series
E. 9.0 Ω, series

ans: A

11. By using only two resistors, $R_1$ and $R_2$, a student is able to obtain resistances of 3 Ω, 4 Ω, 12 Ω, and 16 Ω. The values of $R_1$ and $R_2$ (in ohms) are:

A. 3, 4
B. 2, 12
C. 3, 16
D. 4, 12
E. 4, 16

ans: D

12. Four 20-Ω resistors are connected in parallel and the combination is connected to a 20-V emf device. The current in the device is:

A. 0.25 A
B. 1.0 A
C. 4.0 A
D. 5.0 A
E. 100 A

ans: C

13. Four 20-Ω resistors are connected in parallel and the combination is connected to a 20-V emf device. The current in any one of the resistors is:

A. 0.25 A
B. 1.0 A
C. 4.0 A
D. 5.0 A
E. 100 A

ans: B

14. Four 20-Ω resistors are connected in series and the combination is connected to a 20-V emf device. The current in any one of the resistors is:

A. 0.25 A
B. 1.0 A
C. 4.0 A
D. 5.0 A
E. 100 A

ans: A

15. Four 20-Ω resistors are connected in series and the combination is connected to a 20-V emf device. The potential difference across any one of the resistors is:

A. 1 V
B. 4 V
C. 5 V
D. 20 V
E. 80 V

ans: C

16. Nine identical wires, each of diameter $d$ and length $L$, are connected in parallel. The combination has the same resistance as a single similar wire of length $L$ but whose diameter is:

A. $3d$
B. $9d$
C. $d/3$
D. $d/9$
E. $d/81$

ans: A

17. Nine identical wires, each of diameter $d$ and length $L$, are connected in series. The combination has the same resistance as a single similar wire of length $L$ but whose diameter is:

A. $3d$
B. $9d$
C. $d/3$
D. $d/9$
E. $d/81$

ans: C

18. Two wires made of the same material have the same lengths but different diameters. They are connected in parallel to a battery. The quantity that is NOT the same for the wires is:

A. the end-to-end potential difference
B. the current
C. the current density
D. the electric field
E. the electron drift velocity

ans: B

19. Two wires made of the same material have the same lengths but different diameters. They are connected in series to a battery. The quantity that is the same for the wires is:

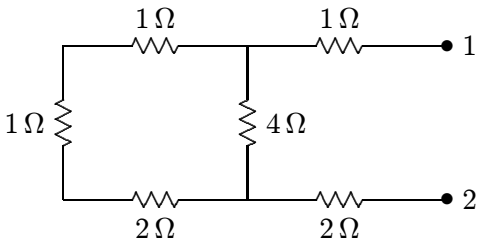
A. the end-to-end potential difference
B. the current
C. the current density
D. the electric field
E. the electron drift velocity

ans: B



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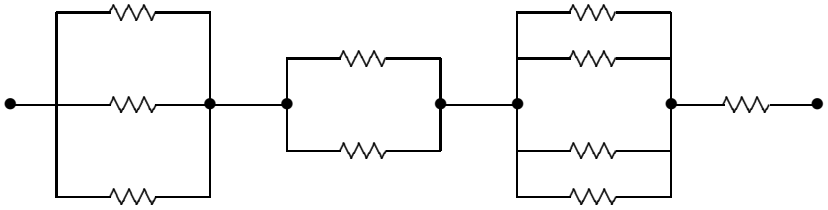
20. The equivalent resistance between points 1 and 2 of the circuit shown is:




A. 3 Ω
B. 4 Ω
C. 5 Ω
D. 6 Ω
E. 7 Ω

ans: C

21. Each of the resistors in the diagram has a resistance of 12 Ω. The resistance of the entire circuit is:



A. 5.76 Ω
B. 25 Ω
C. 48 Ω
D. 120 Ω
E. none of these

ans: B

22. The resistance of resistor 1 is twice the resistance of resistor 2. The two are connected in parallel and a potential difference is maintained across the combination. Then:

A. the current in 1 is twice that in 2
B. the current in 1 is half that in 2
C. the potential difference across 1 is twice that across 2
D. the potential difference across 1 is half that across 2
E. none of the above are true

ans: B

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23. The resistance of resistor 1 is twice the resistance of resistor 2. The two are connected in series and a potential difference is maintained across the combination. Then:

A. the current in 1 is twice that in 2
B. the current in 1 is half that in 2
C. the potential difference across 1 is twice that across 2
D. the potential difference across 1 is half that across 2
E. none of the above are true

ans: C

24. Resistor 1 has twice the resistance of resistor 2. The two are connected in series and a potential difference is maintained across the combination. The rate of thermal energy generation in 1 is:

A. the same as that in 2
B. twice that in 2
C. half that in 2
D. four times that in 2
E. one-fourth that in 2

ans: B

25. Resistor 1 has twice the resistance of resistor 2. The two are connected in parallel and a potential difference is maintained across the combination. The rate of thermal energy generation in 1 is:

A. the same as that in 2
B. twice that in 2
C. half that in 2
D. four times that in 2
E. one-fourth that in 2

ans: C

26. The emf of a battery is equal to its terminal potential difference:

A. under all conditions
B. only when the battery is being charged
C. only when a large current is in the battery
D. only when there is no current in the battery
E. under no conditions

ans: D

27. The terminal potential difference of a battery is less than its emf:

A. under all conditions
B. only when the battery is being charged
C. only when the battery is being discharged
D. only when there is no current in the battery
E. under no conditions

ans: C

28. A battery has an emf of 9 V and an internal resistance of 2 Ω. If the potential difference across its terminals is greater than 9 V:

A. it must be connected across a large external resistance
B. it must be connected across a small external resistance
C. the current must be out of the positive terminal
D. the current must be out of the negative terminal
E. the current must be zero

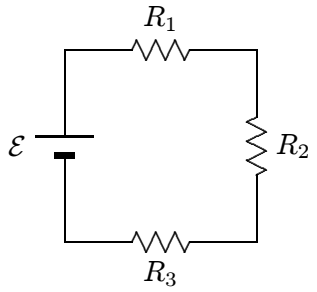
ans: D

29. A battery with an emf of 24 V is connected to a 6-Ω resistor. As a result, current of 3 A exists in the resistor. The terminal potential difference of the battery is:

A. 0
B. 6 V
C. 12 V
D. 18 V
E. 24 V

ans: D

30. In the diagram $R_1 > R_2 > R_3$. Rank the three resistors according to the current in them, least to greatest.




A. 1, 2, 3
B. 3, 2, 1
C. 1, 3, 2
D. 3, 1, 3
E. All are the same

ans: E

31. Resistances of 2.0 Ω, 4.0 Ω, and 6.0 Ω and a 24-V emf device are all in parallel. The current in the 2.0-Ω resistor is:

A. 12 A
B. 4.0 A
C. 2.4 A
D. 2.0 A
E. 0.50 A

ans: A

32. Resistances of 2.0 Ω, 4.0 Ω, and 6.0 Ω and a 24-V emf device are all in series. The potential difference across the 2.0-Ω resistor is:

A. 4 V
B. 8 V
C. 12 V
D. 24 V
E. 48 V

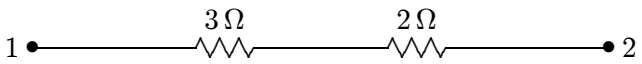
ans: A

33. A battery with an emf of 12 V and an internal resistance of 1 Ω is used to charge a battery with an emf of 10 V and an internal resistance of 1 Ω. The current in the circuit is:

A. 1 A
B. 2 A
C. 4 A
D. 11 A
E. 22 A

ans: A

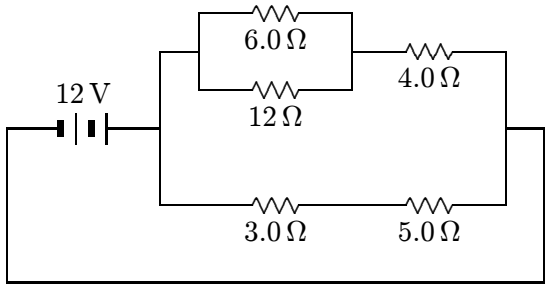
34. In the diagram, the current in the 3-Ω resistor is 4 A. The potential difference between points 1 and 2 is:




A. 0.75 V
B. 0.8 V
C. 1.25 V
D. 12 V
E. 20 V

ans: E

35. The current in the 5.0-Ω resistor in the circuit shown is:




A. 0.42 A
B. 0.67 A
C. 1.5 A
D. 2.4 A
E. 3.0 A

ans: C

36. A 3-Ω and a 1.5-Ω resistor are wired in parallel and the combination is wired in series to a 4-Ω resistor and a 10-V emf device. The current in the 3-Ω resistor is:

A. 0.33 A
B. 0.67 A
C. 2.0 A
D. 3.3 A
E. 6.7 A

ans: B

37. A 3-Ω and a 1.5-Ω resistor are wired in parallel and the combination is wired in series to a 4-Ω resistor and a 10-V emf device. The potential difference across the 3-Ω resistor is:

A. 2.0 V
B. 6.0 V
C. 8.0 V
D. 10 V
E. 12 V

ans: A

38. Two identical batteries, each with an emf of 18 V and an internal resistance of 1 Ω, are wired in parallel by connecting their positive terminals together and connecting their negative terminals together. The combination is then wired across a 4-Ω resistor. The current in the 4-Ω resistor is:

A. 1.0 A
B. 2.0 A
C. 4.0 A
D. 3.6 A
E. 7.2 A

ans: C

39. Two identical batteries, each with an emf of 18 V and an internal resistance of 1 Ω, are wired in parallel by connecting their positive terminals together and connecting their negative terminals together. The combination is then wired across a 4-Ω resistor. The current in each battery is:

A. 1.0 A
B. 2.0 A
C. 4.0 A
D. 3.6 A
E. 7.2 A

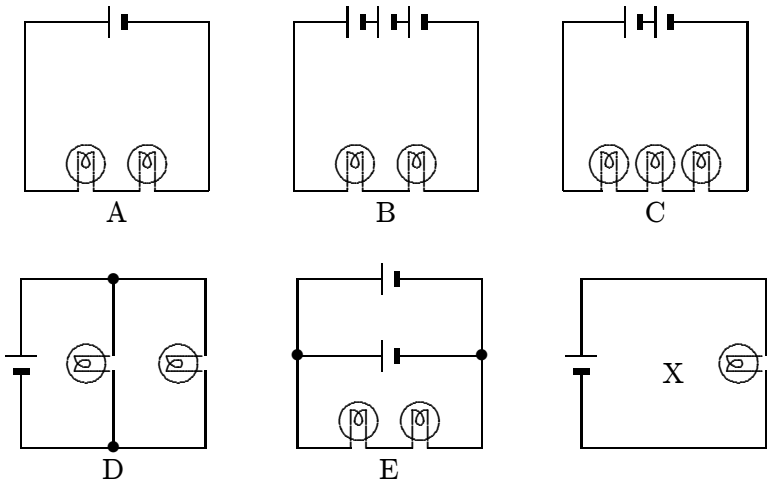
ans: B

40. Two identical batteries, each with an emf of 18 V and an internal resistance of 1 Ω, are wired in parallel by connecting their positive terminals together and connecting their negative terminals together. The combination is then wired across a 4-Ω resistor. The potential difference across the 4-Ω resistor is:

A. 4.0 V
B. 8.0 V
C. 14 V
D. 16 V
E. 29 V

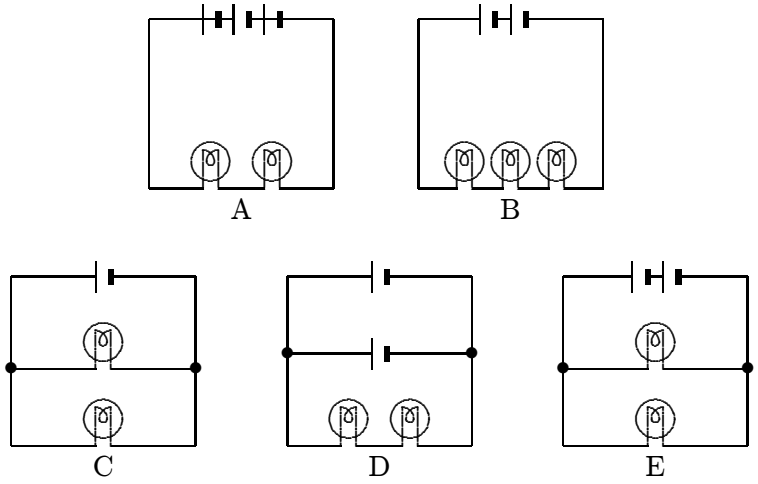
ans: D

41. In the diagrams, all light bulbs are identical and all emf devices are identical. In which circuit (A, B, C, D, E) will the bulbs glow with the same brightness as in circuit X?




ans: D

42. In the diagrams, all light bulbs are identical and all emf devices are identical. In which circuit (A, B, C, D, E) will the bulbs be dimmest?




ans: D

43. A 120-V power line is protected by a 15-A fuse. What is the maximum number of "120 V, 500 W" light bulbs that can be operated at full brightness from this line?

A. 1
B. 2
C. 3
D. 4
E. 5

ans: C

44. Two 110-V light bulbs, one "25 W" and the other "100 W", are connected in series to a 110 V source. Then:

A. the current in the 100-W bulb is greater than that in the 25-W bulb
B. the current in the 100-W bulb is less than that in the 25-W bulb
C. both bulbs will light with equal brightness
D. each bulb will have a potential difference of 55 V
E. none of the above

ans: E

45. A resistor with resistance $R_1$ and a resistor with resistance $R_2$ are connected in parallel to an ideal battery with emf $\mathcal{E}$. The rate of thermal energy generation in the resistor with resistance $R_1$ is:

A. $\mathcal{E}^2/R_1$
B. $\mathcal{E}^2 R_1/(R_1 + R_2)^2$
C. $\mathcal{E}^2/(R_1 + R_2)$
D. $\mathcal{E}^2/R_2$
E. $\mathcal{E}^2 R_1/R_2^2$

ans: A

46. In an antique automobile, a 6-V battery supplies a total of 48 W to two identical headlights in parallel. The resistance (in ohms) of each bulb is:

A. 0.67
B. 1.5
C. 3
D. 4
E. 8

ans: B

47. Resistor 1 has twice the resistance of resistor 2. They are connected in parallel to a battery. The ratio of the thermal energy generation rate in 1 to that in 2 is:

A. 1 : 4
B. 1 : 2
C. 1 : 1
D. 2 : 1
E. 4 : 1

ans: B

48. A series circuit consists of a battery with internal resistance $r$ and an external resistor $R$. If these two resistances are equal ($r = R$) then the thermal energy generated per unit time by the internal resistance $r$ is:

A. the same as by $R$
B. half that by $R$
C. twice that by $R$
D. one-third that by $R$
E. unknown unless the emf is given

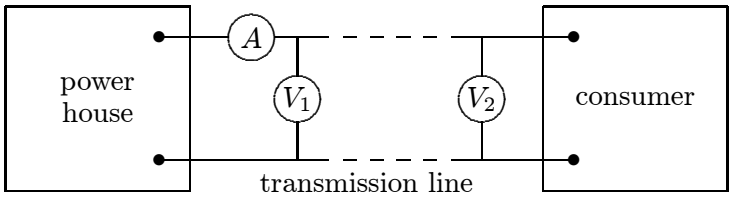
ans: A

49. The positive terminals of two batteries with emf's of $\mathcal{E}_1$ and $\mathcal{E}_2$, respectively, are connected together. Here $\mathcal{E}_2 > \mathcal{E}_1$. The circuit is completed by connecting the negative terminals. If each battery has an internal resistance $r$, the rate with which electrical energy is converted to chemical energy in the smaller battery is:

A. $\mathcal{E}_1^2/r$
B. $\mathcal{E}_1^2/2r$
C. $(\mathcal{E}_2 - \mathcal{E}_1)\mathcal{E}_1/r$
D. $(\mathcal{E}_2 - \mathcal{E}_1)\mathcal{E}_1/2r$
E. $\mathcal{E}_2^2/2r$

ans: D

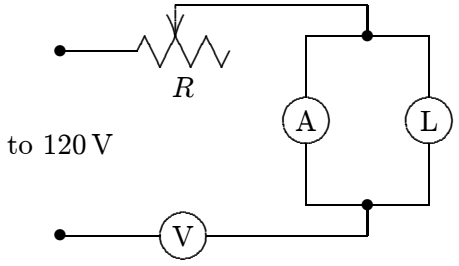
50. In the figure, voltmeter $V_1$ reads 600 V, voltmeter $V_2$ reads 580 V, and ammeter A reads 100 A. The power wasted in the transmission line connecting the power house to the consumer is:




A. 1 kW
B. 2 kW
C. 58 kW
D. 59 kW
E. 60 kW

ans: B

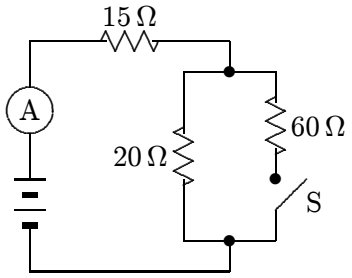
51. The circuit shown was wired for the purpose of measuring the resistance of the lamp L. Inspection shows that:




A. voltmeter V and rheostat R should be interchanged
B. the circuit is satisfactory
C. the ammeter A should be in parallel with R, not L
D. the meters, V and A, should be interchanged
E. L and V should be interchanged

ans: D

52. When switch S is open, the ammeter in the circuit shown reads 2.0 A. When S is closed, the ammeter reading:




A. increases slightly
B. remains the same
C. decreases slightly
D. doubles
E. halves

ans: A

53. A certain galvanometer has a resistance of 100 Ω and requires 1 mA for full scale deflection. To make this into a voltmeter reading 1 V full scale, connect a resistance of:

A. 1000 Ω in parallel
B. 900 Ω in series
C. 1000 Ω in series
D. 10 Ω in parallel
E. 0.1 Ω in series

ans: B

54. To make a galvanometer into an ammeter, connect:

A. a high resistance in parallel
B. a high resistance in series
C. a low resistance in series
D. a low resistance in parallel
E. a source of emf in series

ans: D

55. A certain voltmeter has an internal resistance of $10,000\,\Omega$ and a range from 0 to 100 V. To give it a range from 0 to 1000 V, one should connect:

A. $100,000\,\Omega$ in series
B. $100,000\,\Omega$ in parallel
C. $1000\,\Omega$ in series
D. $1000\,\Omega$ in parallel
E. $90,000\,\Omega$ in series

ans: E

56. A certain ammeter has an internal resistance of $1\,\Omega$ and a range from 0 to 50 mA. To make its range from 0 to 5 A, use:

A. a series resistance of $99\,\Omega$
B. an extremely large (say $10^6\ \Omega$ ) series resistance
C. a resistance of $99\,\Omega$ in parallel
D. a resistance of $1/99\,\Omega$ in parallel
E. a resistance of $1/1000\,\Omega$ in parallel

ans: D

57. A galvanometer has an internal resistance of $12\,\Omega$ and requires 0.01 A for full scale deflection. To convert it to a voltmeter reading 3 V full scale, one must use a series resistance of:

A. $102\,\Omega$
B. $288\,\Omega$
C. $300\,\Omega$
D. $360\,\Omega$
E. $412\,\Omega$

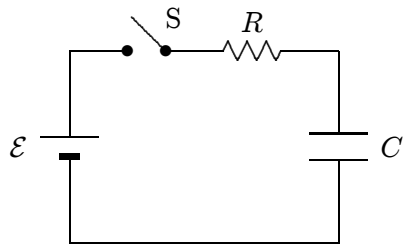
ans: B

58. A certain voltmeter has an internal resistance of $10,000\,\Omega$ and a range from 0 to 12 V. To extend its range to 120 V, use a series resistance of:

A. $1,111\,\Omega$
B. $90,000\,\Omega$
C. $100,000\,\Omega$
D. $108,000\,\Omega$
E. $120,000\,\Omega$

ans: B

59. Four circuits have the form shown in the diagram. The capacitor is initially uncharged and the switch S is open.




The values of the emf $\mathcal{E}$, resistance $R$, and capacitance $C$ for each of the circuits are

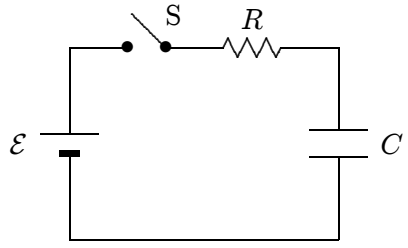
circuit 1: $\mathcal{E} = 18\,\text{V}$, $R = 3\,\Omega$, $C = 1\,\mu\text{F}$
circuit 2: $\mathcal{E} = 18\,\text{V}$, $R = 6\,\Omega$, $C = 9\,\mu\text{F}$
circuit 3: $\mathcal{E} = 12\,\text{V}$, $R = 1\,\Omega$, $C = 7\,\mu\text{F}$
circuit 4: $\mathcal{E} = 10\,\text{V}$, $R = 5\,\Omega$, $C = 7\,\mu\text{F}$

Rank the circuits according to the current just after switch S is closed least to greatest.

A. 1, 2, 3, 4
B. 4, 3, 2, 1
C. 4, 2, 3, 1
D. 4, 2, 1, 3
E. 3, 1, 2, 4

ans: D

60. Four circuits have the form shown in the diagram. The capacitor is initially uncharged and the switch S is open.




The values of the emf $\mathcal{E}$, resistance $R$, and capacitance $C$ for each of the circuits are

circuit 1: $\mathcal{E} = 18\,\text{V}$, $R = 3\,\Omega$, $C = 1\,\mu\text{F}$
circuit 2: $\mathcal{E} = 18\,\text{V}$, $R = 6\,\Omega$, $C = 9\,\mu\text{F}$
circuit 3: $\mathcal{E} = 12\,\text{V}$, $R = 1\,\Omega$, $C = 7\,\mu\text{F}$
circuit 4: $\mathcal{E} = 10\,\text{V}$, $R = 5\,\Omega$, $C = 7\,\mu\text{F}$

Rank the circuits according to the time after switch S is closed for the capacitors to reach half their final charges, least to greatest.

A. 1, 2, 3, 4
B. 4, 3, 2, 1
C. 1, 3, 4, 2
D. 1 and 2 tied, then 4, 3
E. 4, 3, then 1 and 2 tied

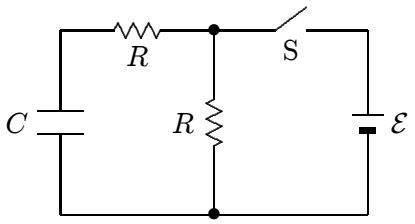
ans: C

61. The time constant $RC$ has units of:

A. second/farad
B. second/ohm
C. 1/second
D. second/watt
E. none of these

ans: E

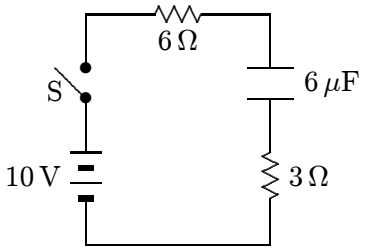
62. In the circuit shown, both resistors have the same value $R$. Suppose switch S is initially closed. When it is then opened, the circuit has a time constant $\tau_a$. Conversely, suppose S is initially open. When it is then closed, the circuit has a time constant $\tau_b$. The ratio $\tau_a/\tau_b$ is:




A. 1
B. 2
C. 0.5
D. 0.667
E. 1.5

ans: B

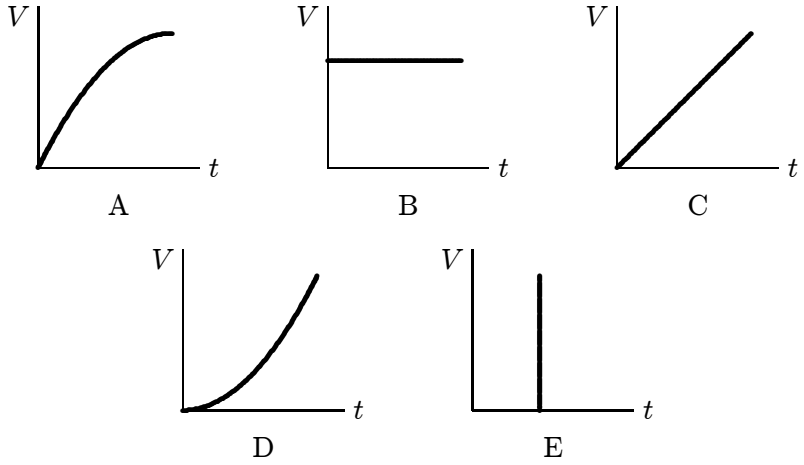
63. In the circuit shown, the capacitor is initially uncharged. At time $t = 0$, switch S is closed. If $\tau$ denotes the time constant, the approximate current through the $3\,\Omega$ resistor when $t = \tau/10$ is:




A. 0.38 A
B. 0.50 A
C. 0.75 A
D. 1.0 A
E. 1.5 A

ans: D

64. Suppose the current charging a capacitor is kept constant. Which graph below correctly gives the potential difference $V$ across the capacitor as a function of time?




ans: C

65. A charged capacitor is being discharged through a resistor. At the end of one time constant the charge has been reduced by $(1 - 1/e) = 63\%$ of its initial value. At the end of two time constants the charge has been reduced by what percent of its initial value?

A. 82%
B. 86%
C. 100%
D. Between 90% and 100%
E. Need to know more data to answer the question

ans: B

66. An initially uncharged capacitor C is connected in series with resistor R. This combination is then connected to a battery of emf $V_0$. Sufficient time elapses so that a steady state is reached. Which of the following statements is NOT true?

A. The time constant is independent of $V_0$
B. The final charge on C is independent of $R$
C. The total thermal energy generated by R is independent of $R$
D. The total thermal energy generated by R is independent of $V_0$
E. The initial current (just after the battery was connected) is independent of $C$

ans: C

67. A certain capacitor, in series with a resistor, is being charged. At the end of 10 ms its charge is half the final value. The time constant for the process is about:

A. 0.43 ms
B. 2.3 ms
C. 6.9 ms
D. 10 ms
E. 14 ms

ans: E

68. A certain capacitor, in series with a 720-Ω resistor, is being charged. At the end of 10 ms its charge is half the final value. The capacitance is about:

A. 9.6 $\mu$F
B. 14 $\mu$F
C. 20 $\mu$F
D. 7.2 F
E. 10 F

ans: C

69. In the capacitor discharge formula $q = q_0 e^{-t/RC}$ the symbol $t$ represents:

A. the time constant
B. the time it takes for C to lose the fraction $1/e$ of its initial charge
C. the time it takes for C to lose the fraction $(1 - 1/e)$ of its initial charge
D. the time it takes for C to lose essentially all of its initial charge
E. none of the above

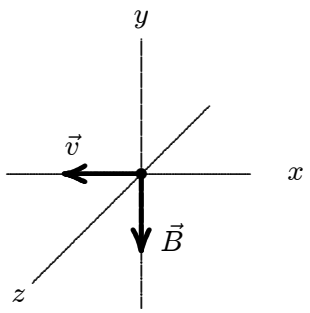
ans: E

## Chapter 28: MAGNETIC FIELDS

1. Units of a magnetic field might be:
   A. C·m/s
   B. C·s/m
   C. C/kg
   D. kg/C·s
   E. N/C·m
   ans: D

2. In the formula $\vec{F} = q\vec{v} \times \vec{B}$:
   A. $\vec{F}$ must be perpendicular to $\vec{v}$ but not necessarily to $\vec{B}$
   B. $\vec{F}$ must be perpendicular to $\vec{B}$ but not necessarily to $\vec{v}$
   C. $\vec{v}$ must be perpendicular to $\vec{B}$ but not necessarily to $\vec{F}$
   D. all three vectors must be mutually perpendicular
   E. $\vec{F}$ must be perpendicular to both $\vec{v}$ and $\vec{B}$
   ans: E

3. An electron moves in the negative $x$ direction, through a uniform magnetic field in the negative $y$ direction. The magnetic force on the electron is:




   A. in the negative $x$ direction
   B. in the positive $y$ direction
   C. in the negative $y$ direction
   D. in the positive $z$ direction
   E. in the negative $z$ direction
   ans: E

4. At any point the magnetic field lines are in the direction of:
   A. the magnetic force on a moving positive charge
   B. the magnetic force on a moving negative charge
   C. the velocity of a moving positive charge
   D. the velocity of a moving negative charge
   E. none of the above
   ans: E

5. The magnetic force on a charged particle is in the direction of its velocity if:
   A. it is moving in the direction of the field
   B. it is moving opposite to the direction of the field
   C. it is moving perpendicular to the field
   D. it is moving in some other direction
   E. never

   ans: E

6. A magnetic field exerts a force on a charged particle:
   A. always
   B. never
   C. if the particle is moving across the field lines
   D. if the particle is moving along the field lines
   E. if the particle is at rest

   ans: C

7. The direction of the magnetic field in a certain region of space is determined by firing a test charge into the region with its velocity in various directions in different trials. The field direction is:
   A. one of the directions of the velocity when the magnetic force is zero
   B. the direction of the velocity when the magnetic force is a maximum
   C. the direction of the magnetic force
   D. perpendicular to the velocity when the magnetic force is zero
   E. none of the above

   ans: A

8. An electron is moving north in a region where the magnetic field is south. The magnetic force exerted on the electron is:
   A. zero
   B. up
   C. down
   D. east
   E. west

   ans: A

9. A magnetic field CANNOT:
   A. exert a force on a charged particle
   B. change the velocity of a charged particle
   C. change the momentum of a charged particle
   D. change the kinetic energy of a charged particle
   E. change the trajectory of a charged particle

   ans: D

10. A proton (charge $e$), traveling perpendicular to a magnetic field, experiences the same force as an alpha particle (charge $2e$) which is also traveling perpendicular to the same field. The ratio of their speeds, $v_{\text{proton}}/v_{\text{alpha}}$, is:

A. 0.5
B. 1
C. 2
D. 4
E. 8

ans: C

11. A hydrogen atom that has lost its electron is moving east in a region where the magnetic field is directed from south to north. It will be deflected:

A. up
B. down
C. north
D. south
E. not at all

ans: A

12. A beam of electrons is sent horizontally down the axis of a tube to strike a fluorescent screen at the end of the tube. On the way, the electrons encounter a magnetic field directed vertically downward. The spot on the screen will therefore be deflected:

A. upward
B. downward
C. to the right as seen from the electron source
D. to the left as seen from the electron source
E. not at all

ans: C

13. An electron (charge $= -1.6 \times 10^{-19}$ C) is moving at $3 \times 10^5$ m/s in the positive $x$ direction. A magnetic field of 0.8 T is in the positive $z$ direction. The magnetic force on the electron is:

A. 0
B. $4 \times 10^{-14}$ N, in the positive $z$ direction
C. $4 \times 10^{-14}$ N, in the negative $z$ direction
D. $4 \times 10^{-14}$ N, in the positive $y$ direction
E. $4 \times 10^{-14}$ N, in the negative $y$ direction

ans: D

14. At one instant an electron (charge $= -1.6 \times 10^{-19}$ C) is moving in the $xy$ plane, the components of its velocity being $v_x = 5 \times 10^5$ m/s and $v_y = 3 \times 10^5$ m/s. A magnetic field of 0.8 T is in the positive $x$ direction. At that instant the magnitude of the magnetic force on the electron is:

A. 0
B. $2.6 \times 10^{-14}$ N
C. $3.8 \times 10^{-14}$ N
D. $6.4 \times 10^{-14}$ N
E. $1.0 \times 10^{-13}$ N

ans: C

15. At one instant an electron (charge $= -1.6 \times 10^{-19}$ C) is moving in the $xy$ plane, the components of its velocity being $v_x = 5 \times 10^5$ m/s and $v_y = 3 \times 10^5$ m/s. A magnetic field of 0.8 T is in the positive $x$ direction. At that instant the magnitude of the magnetic force on the electron is:

A. 0
B. $3.8 \times 10^{-14}$ N
C. $5.1 \times 10^{-14}$ N
D. $6.4 \times 10^{-14}$ N
E. $7.5 \times 10^{-14}$ N

ans: B

16. An electron travels due north through a vacuum in a region of uniform magnetic field $\vec{B}$ that is also directed due north. It will:

A. be unaffected by the field
B. speed up
C. slow down
D. follow a right-handed corkscrew path
E. follow a left-handed corkscrew path

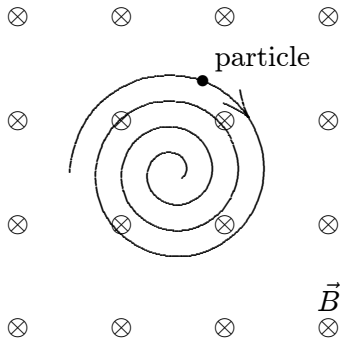
ans: A

17. At one instant an electron is moving in the positive $x$ direction along the $x$ axis in a region where there is a uniform magnetic field in the positive $z$ direction. When viewed from a point on the positive $z$ axis, it subsequent motion is:

A. straight ahead
B. counterclockwise around a circle in the $xy$ plane
C. clockwise around a circle in the $xy$ plane
D. in the positive $z$ direction
E. in the negative $z$ direction

ans: B

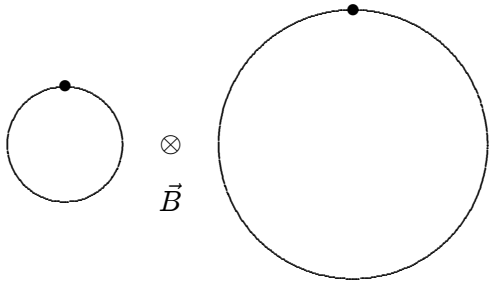
18. A uniform magnetic field is directed into the page. A charged particle, moving in the plane of the page, follows a clockwise spiral of decreasing radius as shown. A reasonable explanation is:




A. the charge is positive and slowing down
B. the charge is negative and slowing down
C. the charge is positive and speeding up
D. the charge is negative and speeding up
E. none of the above

ans: B

19. An electron and a proton each travel with equal speeds around circular orbits in the same uniform magnetic field, as shown in the diagram (not to scale). The field is into the page on the diagram. Because the electron is less massive than the proton and because the electron is negatively charged and the proton is positively charged:




A. the electron travels clockwise around the smaller circle and the proton travels counterclockwise around the larger circle
B. the electron travels counterclockwise around the smaller circle and the proton travels clockwise around the larger circle
C. the electron travels clockwise around the larger circle and the proton travels counterclockwise around the smaller circle
D. the electron travels counterclockwise around the larger circle and the proton travels clockwise around the smaller circle
E. the electron travels counterclockwise around the smaller circle and the proton travels counterclockwise around the larger circle

ans: A

20. An electron is launched with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$. The angle $\theta$ between $\vec{v}$ and $\vec{B}$ is between 0 and 90°. As a result, the electron follows a helix, its velocity vector $\vec{v}$ returning to its initial value in a time interval of:

A. $2\pi m/eB$
B. $2\pi mv/e$B
C. $2\pi mv \sin\theta/eB$
D. $2\pi mv \cos\theta/eB$
E. none of these

ans: A

21. An electron and a proton are both initially moving with the same speed and in the same direction at 90° to the same uniform magnetic field. They experience magnetic forces, which are initially:

A. identical
B. equal in magnitude but opposite in direction
C. in the same direction and differing in magnitude by a factor of 1840
D. in opposite directions and differing in magnitude by a factor of 1840
E. equal in magnitude but perpendicular to each other.

ans: B

22. An electron enters a region of uniform perpendicular $\vec{E}$ and $\vec{B}$ fields. It is observed that the velocity $\vec{v}$ of the electron is unaffected. A possible explanation is:

A. $\vec{v}$ is parallel to $\vec{E}$ and has magnitude $E/B$
B. $\vec{v}$ is parallel to $\vec{B}$
C. $\vec{v}$ is perpendicular to both $\vec{E}$ and $\vec{B}$ and has magnitude $B/E$
D. $\vec{v}$ is perpendicular to both $\vec{E}$ and $\vec{B}$ and has magnitude $E/B$
E. the given situation is impossible

ans: D

23. A charged particle is projected into a region of uniform, parallel, $\vec{E}$ and $\vec{B}$ fields. The force on the particle is:

A. zero
B. at some angle $< 90°$ with the field lines
C. along the field lines
D. perpendicular to the field lines
E. unknown (need to know the sign of the charge)

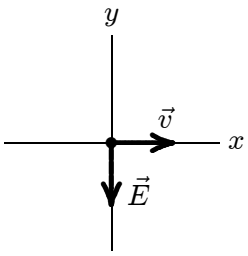
ans: B

24. A uniform magnetic field is in the positive $z$ direction. A positively charged particle is moving in the positive $x$ direction through the field. The net force on the particle can be made zero by applying an electric field in what direction?

A. Positive $y$
B. Negative $y$
C. Positive $x$
D. Negative $x$
E. Positive $z$

ans: B

25. An electron is traveling in the positive $x$ direction. A uniform electric field $\vec{E}$ is in the negative $y$ direction. If a uniform magnetic field with the appropriate magnitude and direction also exists in the region, the total force on the electron will be zero. The appropriate direction for the magnetic field is:




A. the positive $y$ direction
B. the negative $y$ direction
C. into the page
D. out of the page
E. the negative $x$ direction

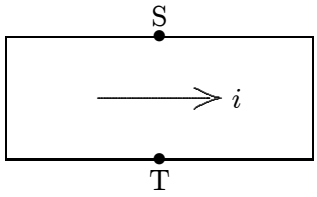
ans: C

26. An ion with a charge of $+3.2\times10^{-19}$ C is in a region where a uniform electric field of $5\times10^{4}$ V/m is perpendicular to a uniform magnetic field of 0.8 T. If its acceleration is zero then its speed must be:

A. 0
B. $1.6 \times 10^4$ m/s
C. $4.0 \times 10^4$ m/s
D. $6.3 \times 10^4$ m/s
E. any value but 0

ans: D

27. The current is from left to right in the conductor shown. The magnetic field is into the page and point S is at a higher potential than point T. The charge carriers are:




A. positive
B. negative
C. neutral
D. absent
E. moving near the speed of light

ans: A

28. Electrons (mass $m$, charge $-e$) are accelerated from rest through a potential difference $V$ and are then deflected by a magnetic field $\vec{B}$ that is perpendicular to their velocity. The radius of the resulting electron trajectory is:

A. $(\sqrt{2eV/m})/B$
B. $B\sqrt{2eV}/m$
C. $(\sqrt{2mV/e})/B$
D. $B\sqrt{2mV}/e$
E. none of these

ans: C

29. In a certain mass spectrometer, an ion beam passes through a velocity filter consisting of mutually perpendicular fields $\vec{E}$ and $\vec{B}$. The beam then enters a region of another magnetic field $\vec{B}'$ perpendicular to the beam. The radius of curvature of the resulting ion beam is proportional to:

A. $EB'/B$
B. $EB/B'$
C. $BB'/E$
D. $B/EB'$
E. $E/BB'$

ans: E

30. A cyclotron operates with a given magnetic field and at a given frequency. If $R$ denotes the radius of the final orbit, the final particle energy is proportional to:

A. $1/R$
B. $R$
C. $R^2$
D. $R^3$
E. $R^4$

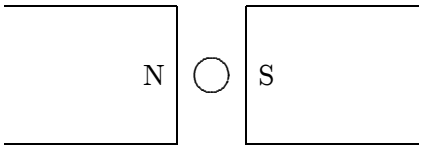
ans: C

31. J. J. Thomson's experiment, involving the motion of an electron beam in mutually perpendicular $\vec{E}$ and $\vec{B}$ fields, gave the value of:

A. mass of an electron
B. charge of an electron
C. Earth's magnetic field
D. charge/mass ratio for electrons
E. Avogadro's number

ans: D

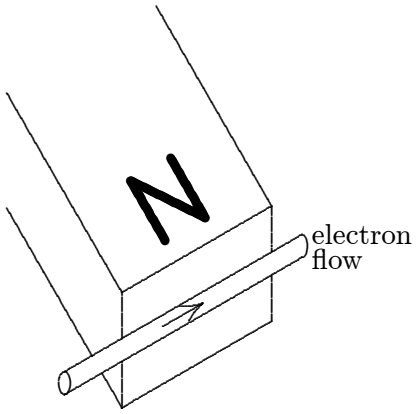
32. The diagram shows a straight wire carrying a flow of electrons into the page. The wire is between the poles of a permanent magnet. The direction of the magnetic force exerted on the wire is:




A. $\uparrow$
B. $\downarrow$
C. $\leftarrow$
D. $\rightarrow$
E. into the page

ans: A

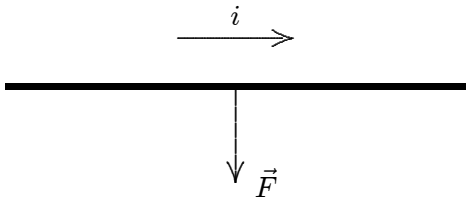
33. The figure shows the motion of electrons in a wire that is near the N pole of a magnet. The wire will be pushed:




A. toward the magnet
B. away from the magnet
C. downwards
D. upwards
E. along its length

ans: D

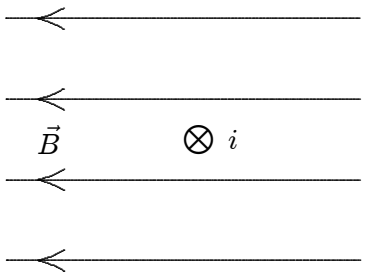
34. The diagram shows a straight wire carrying current $i$ in a uniform magnetic field. The magnetic force on the wire is indicated by an arrow but the magnetic field is not shown. Of the following possibilities, the direction of the magnetic field is:




A. opposite the direction of the current
B. opposite the direction of $\vec{F}$
C. in the direction of $\vec{F}$
D. into the page
E. out of the page

ans: E

35. The figure shows a uniform magnetic field $\vec{B}$ directed to the left and a wire carrying a current into the page. The magnetic force acting on the wire is:




A. toward the top of the page
B. toward the bottom of the page
C. toward the left
D. toward the right
E. zero

ans: A

36. A loop of wire carrying a current of 2.0 A is in the shape of a right triangle with two equal sides, each 15 cm long. A 0.7 T uniform magnetic field is parallel to the hypotenuse. The resultant magnetic force on the two equal sides has a magnitude of:

A. 0
B. 0.21 N
C. 0.30 N
D. 0.41 N
E. 0.51 N

ans: A

37. A loop of wire carrying a current of 2.0 A is in the shape of a right triangle with two equal sides, each 15 cm long. A 0.7 T uniform magnetic field is in the plane of the triangle and is perpendicular to the hypotenuse. The magnetic force on either of the two equal sides has a magnitude of:

A. zero
B. 0.105 N
C. 0.15 N
D. 0.21 N
E. 0.25 N

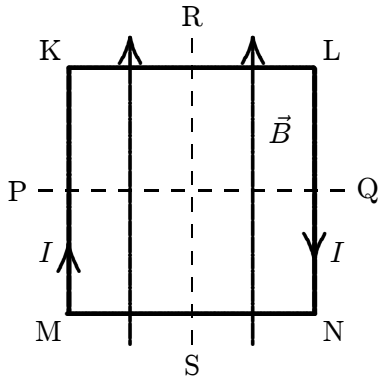
ans: C

38. A current is clockwise around the outside edge of this page and a uniform magnetic field is directed parallel to the page, from left to right. If the magnetic force is the only force acting on the page, the page will turn so the right edge:

A. moves toward you
B. moves away from you
C. moves to your right
D. moves to your left
E. does not move

ans: A

39. A square loop of wire lies in the plane of the page and carries a current I as shown. There is a uniform magnetic field $\vec{B}$ parallel to the side MK as indicated. The loop will tend to rotate:




A. about PQ with KL coming out of the page
B. about PQ with KL going into the page
C. about RS with MK coming out of the page
D. about RS with MK going into the page
E. about an axis perpendicular to the page.

ans: A

40. The units of magnetic dipole moment are:

A. ampere
B. ampere·meter
C. ampere·meter$^2$
D. ampere/meter
E. ampere/meter$^2$

ans: C

41. You are facing a loop of wire which carries a clockwise current of 3.0 A and which surrounds an area of $5.8 \times 10^{-2}\,\mathrm{m}^2$. The magnetic dipole moment of the loop is:

A. $3.0\,\mathrm{A}\cdot\mathrm{m}^2$, away from you
B. $3.0\,\mathrm{A}\cdot\mathrm{m}^2$, toward you
C. $0.17\,\mathrm{A}\cdot\mathrm{m}^2$, away from you
D. $0.17\,\mathrm{A}\cdot\mathrm{m}^2$, toward you
E. $0.17\,\mathrm{A}\cdot\mathrm{m}^2$, left to right

ans: C

42. The magnetic torque exerted on a flat current-carrying loop of wire by a uniform magnetic field $\vec{B}$ is:

A. maximum when the plane of the loop is perpendicular to $\vec{B}$
B. maximum when the plane of the loop is parallel to $\vec{B}$
C. dependent on the shape of the loop for a fixed loop area
D. independent of the orientation of the loop
E. such as to rotate the loop around the magnetic field lines

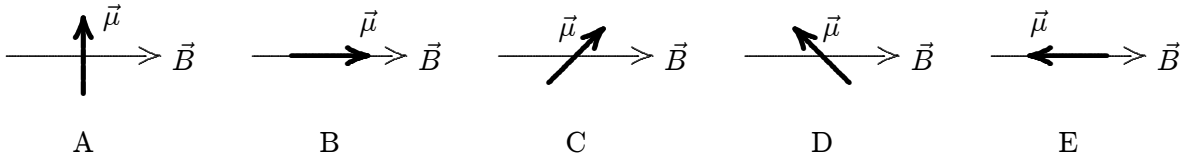
ans: B

43. A circular loop of wire with a radius of 20 cm lies in the $xy$ plane and carries a current of 2 A, counterclockwise when viewed from a point on the positive $z$ axis. Its magnetic dipole moment is:

A. $0.25\,\text{A} \cdot \text{m}^2$, in the positive $z$ direction
B. $0.25\,\text{A} \cdot \text{m}^2$, in the negative $z$ direction
C. $2.5\,\text{A} \cdot \text{m}^2$, in the positive $z$ direction
D. $2.5\,\text{A} \cdot \text{m}^2$, in the negative $z$ direction
E. $0.25\,\text{A} \cdot \text{m}^2$, in the $xy$ plane

ans: A

44. The diagrams show five possible orientations of a magnetic dipole $\vec{\mu}$ in a uniform magnetic field $\vec{B}$. For which of these does the magnetic torque on the dipole have the greatest magnitude?




ans: A

45. The magnetic dipole moment of a current-carrying loop of wire is in the positive $z$ direction. If a uniform magnetic field is in the positive $x$ direction the magnetic torque on the loop is:

A. 0
B. in the positive $y$ direction
C. in the negative $y$ direction
D. in the positive $z$ direction
E. in the negative $z$ direction

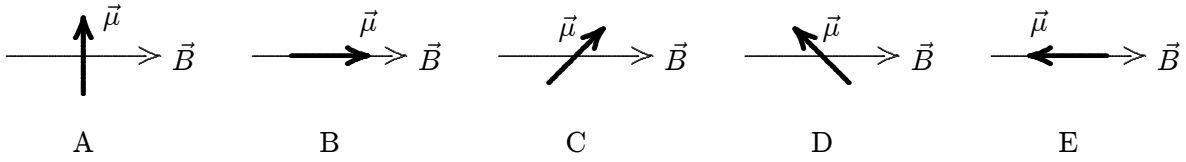
ans: B

46. For a loop of current-carrying wire in a uniform magnetic field the potential energy is a minimum if the magnetic dipole moment of the loop is:

A. in the same direction as the field
B. in the direction opposite to that of the field
C. perpendicular to the field
D. at an angle of 45° to the field
E. none of the above

ans: A

47. The diagrams show five possible orientations of a magnetic dipole $\vec{\mu}$ in a uniform magnetic field $\vec{B}$. For which of these is the potential energy the greatest?




ans: E

48. A loop of current-carrying wire has a magnetic dipole moment of $5 \times 10^{-4}\,\mathrm{A \cdot m^2}$. The moment initially is aligned with a 0.5-T magnetic field. To rotate the loop so its dipole moment is perpendicular to the field and hold it in that orientation, you must do work of:

A. 0
B. $2.5 \times 10^{-4}\,\mathrm{J}$
C. $-2.5 \times 10^{-4}\,\mathrm{J}$
D. $1.0 \times 10^{-3}\,\mathrm{J}$
E. $-1.0 \times 10^{-3}\,\mathrm{J}$

ans: B

## Chapter 29: MAGNETIC FIELDS DUE TO CURRENTS

1. Suitable units for $\mu_0$ are:

   A. tesla
   B. newton/ampere$^2$
   C. weber/meter
   D. kilogram·ampere/meter
   E. tesla·meter/ampere

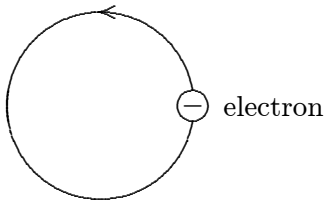
   ans: E

2. A "coulomb" is:

   A. one ampere per second
   B. the quantity of charge that will exert a force of 1 N on a similar charge at a distance of 1 m
   C. the amount of current in each of two long parallel wires, separated by 1 m, that produces a force of $2 \times 10^{-7}$ N/m
   D. the amount of charge that flows past a point in one second when the current is 1 A
   E. an abbreviation for a certain combination of kilogram, meter and second

   ans: D

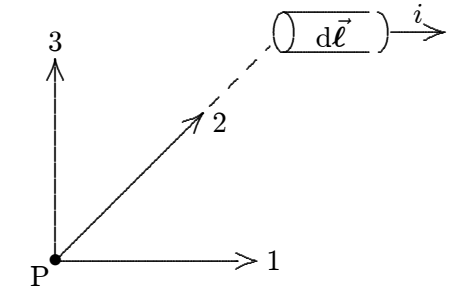
3. Electrons are going around a circle in a counterclockwise direction as shown. At the center of the circle they produce a magnetic field that is:




   A. into the page
   B. out of the page
   C. to the left
   D. to the right
   E. zero

   ans: A

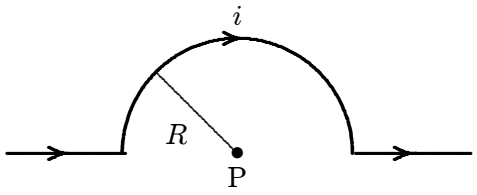
4. In the figure, the current element $i\,\mathrm{d}\vec{\ell}$, the point P, and the three vectors (1, 2, 3) are all in the plane of the page. The direction of $\mathrm{d}\vec{B}$, due to this current element, at the point P is:




A. in the direction marked "1"
B. in the direction marked "2"
C. in the direction marked "3"
D. out of the page
E. into the page

ans: E

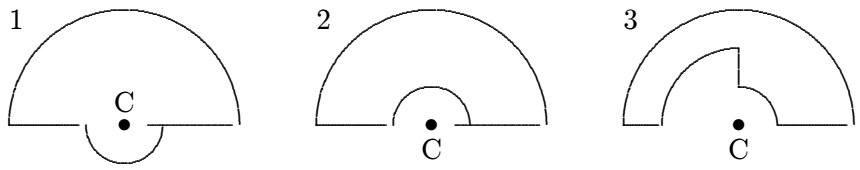
5. The magnitude of the magnetic field at point P, at the center of the semicircle shown, is given by:




A. $2\mu_0 i/R$
B. $\mu_0 i/R$
C. $\mu_0 i/4\pi R$
D. $\mu_0 i/2R$
E. $\mu_0 i/4R$

ans: E

6. The diagrams show three circuits consisting of concentric circular arcs (either half or quarter circles of radii $r$, $2r$, and $3r$) and radial lengths. The circuits carry the same current. Rank them according to the magnitudes of the magnetic fields they produce at C, least to greatest.




A. 1, 2, 3
B. 3, 2, 1
C. 1, 3, 2
D. 2, 3, 1
E. 2, 1, 3

ans: B

7. Lines of the magnetic field produced by a long straight wire carrying a current are:

A. in the direction of the current
B. opposite to the direction of the current
C. radially outward from the wire
D. radially inward toward the wire
E. circles that are concentric with the wire

ans: E

8. In an overhead straight wire, the current is north. The magnetic field due to this current, at our point of observation, is:

A. east
B. up
C. north
D. down
E. west

ans: E

9. A wire carrying a large current $i$ from east to west is placed over an ordinary magnetic compass. The end of the compass needle marked "N" will point:

A. north
B. south
C. east
D. west
E. the compass will act as an electric motor, hence the needle will keep rotating

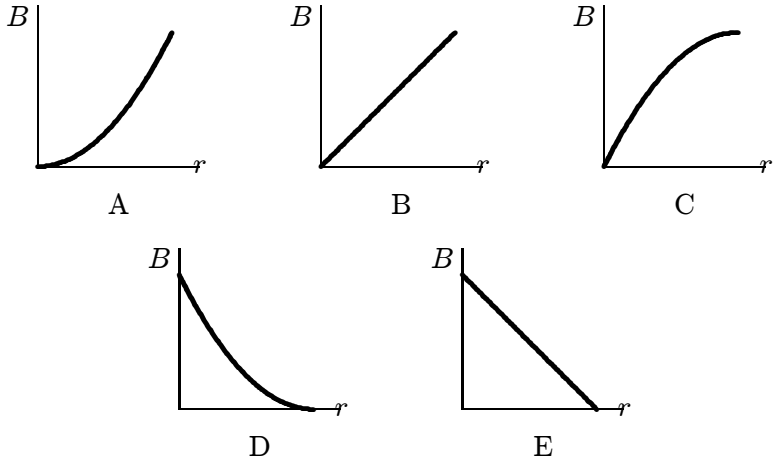
ans: B

10. The magnetic field outside a long straight current-carrying wire depends on the distance $R$ from the wire axis according to:

A. $R$
B. $1/R$
C. $1/R^2$
D. $1/R^3$
E. $1/R^{3/2}$

ans: B

11. Which graph correctly gives the magnitude of the magnetic field outside an infinitely long straight current-carrying wire as a function of the distance $r$ from the wire?




ans: D

12. The magnetic field a distance 2 cm from a long straight current-carrying wire is $2.0 \times 10^{-5}$ T. The current in the wire is:

A. 0.16 A
B. 1.0 A
C. 2.0 A
D. 4.0 A
E. 25 A

ans: C

13. Two long parallel straight wires carry equal currents in opposite directions. At a point midway between the wires, the magnetic field they produce is:

A. zero
B. non-zero and along a line connecting the wires
C. non-zero and parallel to the wires
D. non-zero and perpendicular to the plane of the two wires
E. none of the above

ans: D

14. Two long straight wires are parallel and carry current in the same direction. The currents are 8.0 and 12 A and the wires are separated by 0.40 cm. The magnetic field in tesla at a point midway between the wires is:

A. 0
B. $4.0 \times 10^{-4}$
C. $8.0 \times 10^{-4}$
D. $12 \times 10^{-4}$
E. $20 \times 10^{-4}$

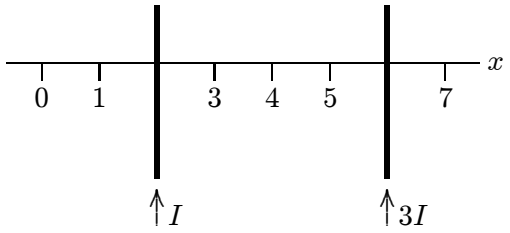
ans: B

15. Two long straight wires are parallel and carry current in opposite directions. The currents are 8.0 and 12 A and the wires are separated by 0.40 cm. The magnetic field in tesla at a point midway between the wires is:

A. 0
B. $4.0 \times 10^{-4}$
C. $8.0 \times 10^{-4}$
D. $12 \times 10^{-4}$
E. $20 \times 10^{-4}$

ans: E

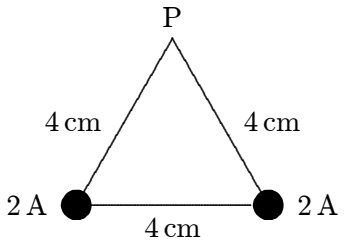
16. Two long straight current-carrying parallel wires cross the $x$ axis and carry currents $I$ and $3I$ in the same direction, as shown. At what value of $x$ is the net magnetic field zero?




A. 0
B. 1
C. 3
D. 5
E. 7

ans: C

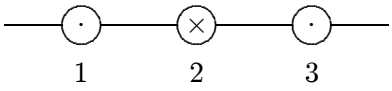
17. Two long straight wires pierce the plane of the paper at vertices of an equilateral triangle as shown below. They each carry 2 A, out of the paper. The magnetic field at the third vertex (P) has magnitude (in T):




A. $1.0 \times 10^{-5}$
B. $1.7 \times 10^{-5}$
C. $2.0 \times 10^{-5}$
D. $5.0 \times 10^{-6}$
E. $8.7 \times 10^{-6}$

ans: B

18. The diagram shows three equally spaced wires that are perpendicular to the page. The currents are all equal, two being out of the page and one being into the page. Rank the wires according to the magnitudes of the magnetic forces on them, from least to greatest.




A. 1, 2, 3
B. 2, 1 and 3 tie
C. 2 and 3 tie, then 1
D. 1 and 3 tie, then 2
E. 3, 2, 1

ans: B

19. Two parallel wires carrying equal currents of 10 A attract each other with a force of 1 mN. If both currents are doubled, the force of attraction will be:

A. 1 mN
B. 4 mN
C. 0.5 mN
D. 0.25 mN
E. 2 mN

ans: B

20. Two parallel long wires carry the same current and repel each other with a force $F$ per unit length. If both these currents are doubled and the wire separation tripled, the force per unit length becomes:

A. $2F/9$
B. $4F/9$
C. $2F/3$
D. $4F/3$
E. $6F$

ans: D

21. Two parallel wires, 4 cm apart, carry currents of 2 A and 4 A respectively, in the same direction. The force per unit length in N/m of one wire on the other is:

A. $1 \times 10^{-3}$, repulsive
B. $1 \times 10^{-3}$, attractive
C. $4 \times 10^{-5}$, repulsive
D. $4 \times 10^{-5}$, attractive
E. none of these

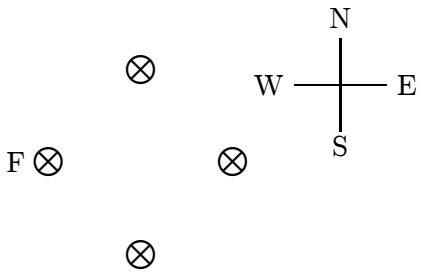
ans: D

22. Two parallel wires, 4 cm apart, carry currents of 2 A and 4 A respectively, in opposite directions. The force per unit length in N/m of one wire on the other is:

A. $1 \times 10^{-3}$, repulsive
B. $1 \times 10^{-3}$, attractive
C. $4 \times 10^{-5}$, repulsive
D. $4 \times 10^{-5}$, attractive
E. none of these

ans: C

23. Four long straight wires carry equal currents into the page as shown. The magnetic force exerted on wire F is:




A. north
B. east
C. south
D. west
E. zero

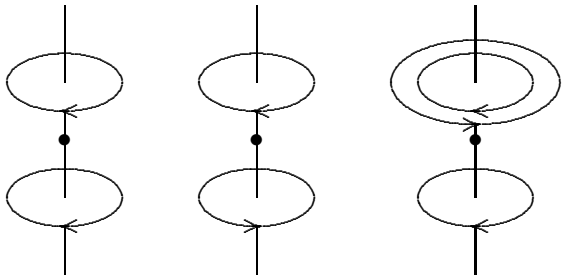
ans: B

24. A constant current is sent through a helical coil. The coil:

A. tends to get shorter
B. tends to get longer
C. tends to rotate about its axis
D. produces zero magnetic field at its center
E. none of the above

ans: A

25. The diagram shows three arrangements of circular loops, centered on vertical axes and carrying identical currents in the directions indicated. Rank the arrangements according to the magnitudes of the magnetic fields at the midpoints between the loops on the central axes.




A. 1, 2, 3
B. 2, 1, 3
C. 2, 3, 1
D. 3, 2, 1
E. 3, 1, 2

ans: C

26. Helmholtz coils are commonly used in the laboratory because the magnetic field between them:

A. can be varied more easily than the fields of other current arrangements
B. is especially strong
C. nearly cancels Earth's magnetic field
D. is parallel to the plane of the coils
E. is nearly uniform

ans: E

27. If the radius of a pair of Helmholtz coils is $R$ then the distance between the coils is:

A. $R/4$
B. $R/2$
C. $R$
D. $2R$
E. $4R$

ans: C

28. If $R$ is the distance from a magnetic dipole, then the magnetic field it produces is proportional to:

A. $R$
B. $1/R$
C. $R^2$
D. $1/R^2$
E. $1/R^3$

ans: E

29. A square loop of current-carrying wire with edge length $a$ is in the $xy$ plane, the origin being at its center. Along which of the following lines can a charge move without experiencing a magnetic force?

A. $x = 0$, $y = a/2$
B. $x = a/2$, $y = a/2$
C. $x = a/2$, $y = 0$
D. $x = 0$, $y = 0$
E. $x = 0$, $z = 0$

ans: D

30. In Ampere's law, $\oint \vec{B} \cdot d\vec{s} = \mu_0 i$, the integration must be over any:

A. surface
B. closed surface
C. path
D. closed path
E. closed path that surrounds all the current producing $\vec{B}$

ans: D

31. In Ampere's law, $\oint \vec{B} \cdot d\vec{s} = \mu_0 i$, the symbol $d\vec{s}$ is:

A. an infinitesimal piece of the wire that carries current $i$
B. in the direction of $\vec{B}$
C. perpendicular to $\vec{B}$
D. a vector whose magnitude is the length of the wire that carries current $i$
E. none of the above

ans: E

32. In Ampere's law, $\oint \vec{B} \cdot d\vec{s} = \mu_0 i$, the direction of the integration around the path:

A. must be clockwise
B. must be counterclockwise
C. must be such as to follow the magnetic field lines
D. must be along the wire in the direction of the current
E. none of the above

ans: E

33. A long straight wire carrying a 3.0 A current enters a room through a window 1.5 m high and 1.0 m wide. The path integral $\oint \vec{B} \cdot d\vec{s}$ around the window frame has the value (in T·m):

A. 0.20
B. $2.5 \times 10^{-7}$
C. $3.0 \times 10^{-7}$
D. $3.8 \times 10^{-6}$
E. none of these

ans: D

34. Two long straight wires enter a room through a door. One carries a current of 3.0 A into the room while the other carries a current of 5.0 A out. The magnitude of the path integral $\oint \vec{B} \cdot d\vec{s}$ around the door frame is:

A. $2.5 \times 10^{-6}\ \mathrm{T \cdot m}$
B. $3.8 \times 10^{-6}\ \mathrm{T \cdot m}$
C. $6.3 \times 10^{-6}\ \mathrm{T \cdot m}$
D. $1.0 \times 10^{-5}\ \mathrm{T \cdot m}$
E. none of these

ans: A

35. If the magnetic field $\vec{B}$ is uniform over the area bounded by a circle with radius $R$, the net current through the circle is:

A. 0
B. $2\pi RB/\mu_0$
C. $\pi R^2 B/\mu_0$
D. $RB/2\mu_0$
E. $2RB/\mu_0$

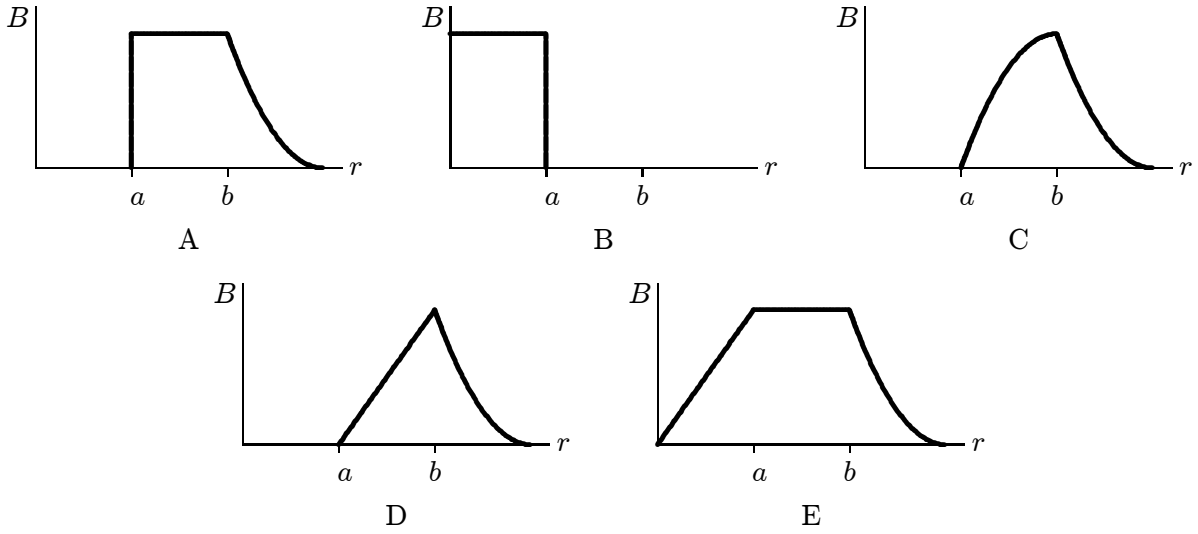
ans: A

36. The magnetic field at any point is given by $\vec{B} = A\vec{r} \times \hat{\mathrm{k}}$, where $\vec{r}$ is the position vector of the point and $A$ is a constant. The net current through a circle of radius $R$, in the $xy$ plane and centered at the origin is given by:

A. $\pi AR^2/\mu_0$
B. $2\pi AR/\mu_0$
C. $4\pi AR^3/3\mu_0$
D. $2\pi AR^2/\mu_0$
E. $\pi AR^2/2\mu_0$

ans: D

37. A hollow cylindrical conductor (inner radius = $a$, outer radius = $b$) carries a current $i$ uniformly spread over its cross section. Which graph below correctly gives $B$ as a function of the distance $r$ from the center of the cylinder?




ans: C

38. A long straight cylindrical shell carries current $i$ parallel to its axis and uniformly distributed over its cross section. The magnitude of the magnetic field is greatest:

A. at the inner surface of the shell
B. at the outer surface of the shell
C. inside the shell near the middle
D. in hollow region near the inner surface of the shell
E. near the center of the hollow region

ans: B

39. A long straight cylindrical shell has inner radius $R_i$ and outer radius $R_o$. It carries current $i$, uniformly distributed over its cross section. A wire is parallel to the cylinder axis, in the hollow region ($r < R_i$). The magnetic field is zero everywhere outside the shell ($r > R_o$). We conclude that the wire:

A. is on the cylinder axis and carries current $i$ in the same direction as the current in the shell
B. may be anywhere in the hollow region but must be carrying current $i$ in the direction opposite to that of the current in the shell
C. may be anywhere in the hollow region but must be carrying current $i$ in the same direction as the current in the shell
D. is on the cylinder axis and carries current $i$ in the direction opposite to that of the current in the shell
E. does not carry any current

ans: D

40. A long straight cylindrical shell has inner radius $R_i$ and outer radius $R_o$. It carries a current $i$, uniformly distributed over its cross section. A wire is parallel to the cylinder axis, in the hollow region ($r < R_i$). The magnetic field is zero everywhere in the hollow region. We conclude that the wire:

A. is on the cylinder axis and carries current $i$ in the same direction as the current in the shell
B. may be anywhere in the hollow region but must be carrying current $i$ in the direction opposite to that of the current in the shell
C. may be anywhere in the hollow region but must be carrying current $i$ in the same direction as the current in the shell
D. is on the cylinder axis and carries current $i$ in the direction opposite to that of the current in the shell
E. does not carry any current

ans: E

41. The magnetic field $B$ inside a long ideal solenoid is independent of:

A. the current
B. the core material
C. the spacing of the windings
D. the cross-sectional area of the solenoid
E. the direction of the current

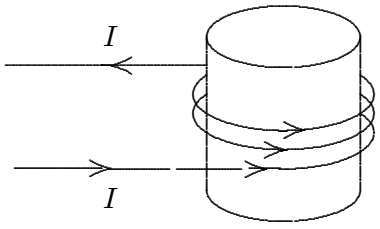
ans: D

42. Two long ideal solenoids (with radii 20 mm and 30 mm, respectively) have the same number of turns of wire per unit length. The smaller solenoid is mounted inside the larger, along a common axis. The magnetic field within the inner solenoid is zero. The current in the inner solenoid must be:

A. two-thirds the current in the outer solenoid
B. one-third the current in the outer solenoid
C. twice the current in the outer solenoid
D. half the current in the outer solenoid
E. the same as the current in the outer solenoid

ans: E

43. Magnetic field lines inside the solenoid shown are:




A. clockwise circles as one looks down the axis from the top of the page
B. counterclockwise circles as one looks down the axis from the top of the page
C. toward the top of the page
D. toward the bottom of the page
E. in no direction since $B = 0$

ans: C

44. Solenoid 2 has twice the radius and six times the number of turns per unit length as solenoid 1. The ratio of the magnetic field in the interior of 2 to that in the interior of 1 is:

A. 2
B. 4
C. 6
D. 1
E. 1/3

ans: C

45. A solenoid is 3.0 cm long and has a radius of 0.50 cm. It is wrapped with 500 turns of wire carrying a current of 2.0 A. The magnetic field at the center of the solenoid is:

A. $9.9 \times 10^{-8}$ T
B. $1.3 \times 10^{-3}$ T
C. $4.2 \times 10^{-2}$ T
D. 16 T
E. 20 T

ans: C

46. A toroid with a square cross section carries current $i$. The magnetic field has its largest magnitude:

A. at the center of the hole
B. just inside the toroid at its inner surface
C. just inside the toroid at its outer surface
D. at any point inside (the field is uniform)
E. none of the above

ans: B

47. A toroid has a square cross section with the length of an edge equal to the radius of the inner surface. The ratio of the magnitude of the magnetic field at the inner surface to the magnitude of the field at the outer surface is:

A. 1/4
B. 1/2
C. 1
D. 2
E. 4

ans: D

## Chapter 30: INDUCTION AND INDUCTANCE

1. The normal to a certain 1-$m^2$ area makes an angle of 60° with a uniform magnetic field. The magnetic flux through this area is the same as the flux through a second area that is perpendicular to the field if the second area is:

   A. 0.866 $m^2$
   B. 1.15 $m^2$
   C. 0.5 $m^2$
   D. 2 $m^2$
   E. 1 $m^2$

   ans: C

2. Suppose this page is perpendicular to a uniform magnetic field and the magnetic flux through it is 5 Wb. If the page is turned by 30° around an edge the flux through it will be:

   A. 2.5 Wb
   B. 4.3 Wb
   C. 5 Wb
   D. 5.8 Wb
   E. 10 Wb

   ans: B

3. A 2-T uniform magnetic field makes an angle of 30° with the $z$ axis. The magnetic flux through a 3-$m^2$ portion of the $xy$ plane is:

   A. 2.0 Wb
   B. 3.0 Wb
   C. 5.2 Wb
   D. 6.0 Wb
   E. 12 Wb

   ans: C

4. A uniform magnetic field makes an angle of 30° with the $z$ axis. If the magnetic flux through a 1-$m^2$ portion of the $xy$ plane is 5 Wb then the magnetic flux through a 2-$m^2$ portion of the same plane is:

   A. 2.5 Wb
   B. 4.3 Wb
   C. 5 Wb
   D. 5.8 Wb
   E. 10 Wb

   ans: E

5. 1 weber is the same as:

   A. 1 V/s
   B. 1 T/s
   C. 1 T/m
   D. 1 T · $m^2$
   E. 1 T/$m^2$

   ans: D

6. 1 weber is the same as:
   A. $1\,\mathrm{V}\cdot\mathrm{s}$
   B. $1\,\mathrm{T}\cdot\mathrm{s}$
   C. $1\,\mathrm{T/m}$
   D. $1\,\mathrm{V/s}$
   E. $1\,\mathrm{T/m^2}$

   ans: A

7. The units of motional emf are:
   A. volt/second
   B. volt·meter/second
   C. volt/tesla
   D. tesla/second
   E. tesla·meter$^2$/second

   ans: E

8. Faraday's law states that an induced emf is proportional to:
   A. the rate of change of the magnetic field
   B. the rate of change of the electric field
   C. the rate of change of the magnetic flux
   D. the rate of change of the electric flux
   E. zero

   ans: C

9. The emf that appears in Faraday's law is:
   A. around a conducting circuit
   B. around the boundary of the surface used to compute the magnetic flux
   C. throughout the surface used to compute the magnetic flux
   D. perpendicular to the surface used to compute the magnetic flux
   E. none of the above

   ans: B

10. If the magnetic flux through a certain region is changing with time:
    A. energy must be dissipated as heat
    B. an electric field must exist at the boundary
    C. a current must flow around the boundary
    D. an emf must exist around the boundary
    E. a magnetic field must exist at the boundary

    ans: D

11. A square loop of wire lies in the plane of the page. A decreasing magnetic field is directed into the page. The induced current in the loop is:

A. counterclockwise
B. clockwise
C. zero
D. up the left edge and from right to left along the top edge
E. through the middle of the page

ans: B

12. As an externally generated magnetic field through a certain conducting loop increases in magnitude, the field produced at points inside the loop by the current induced in the loop must be:

A. increasing in magnitude
B. decreasing in magnitude
C. in the same direction as the applied field
D. directed opposite to the applied field
E. perpendicular to the applied field

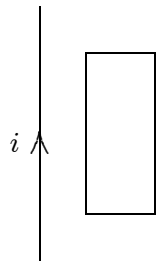
ans: D

13. At any instant of time the total magnetic flux through a stationary conducting loop is less in magnitude than the flux associated with an externally applied field. This might occur because:

A. the applied field is normal to the loop and increasing in magnitude
B. the applied field is normal to the loop and decreasing in magnitude
C. the applied field is parallel to the plane of the loop and increasing in magnitude
D. the applied field is parallel to the plane of the loop and decreasing in magnitude
E. the applied field is tangent to the loop

ans: A

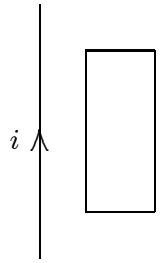
14. A long straight wire is in the plane of a rectangular conducting loop. The straight wire carries a constant current $i$, as shown. While the wire is being moved toward the rectangle the current in the rectangle is:




A. zero
B. clockwise
C. counterclockwise
D. clockwise in the left side and counterclockwise in the right side
E. counterclockwise in the left side and clockwise in the right side

ans: C

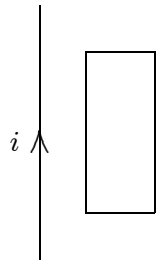
15. A long straight wire is in the plane of a rectangular conducting loop. The straight wire carries an increasing current in the direction shown. The current in the rectangle is:




A. zero
B. clockwise
C. counterclockwise
D. clockwise in the left side and counterclockwise in the right side
E. counterclockwise in the left side and clockwise in the right side

ans: C

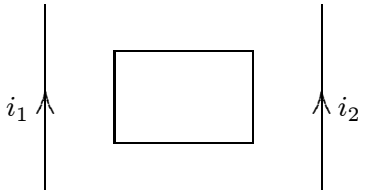
16. A long straight wire is in the plane of a rectangular conducting loop. The straight wire initially carries a constant current $i$ in the direction shown. While the current $i$ is being shut off, the current in the rectangle is:




A. zero
B. clockwise
C. counterclockwise
D. clockwise in the left side and counterclockwise in the right side
E. counterclockwise in the left side and clockwise in the right side

ans: B

17. A rectangular loop of wire is placed midway between two long straight parallel conductors as shown. The conductors carry currents $i_1$ and $i_2$, as indicated. If $i_1$ is increasing and $i_2$ is constant, then the induced current in the loop is:




A. zero
B. clockwise
C. counterclockwise
D. depends on $i_1 - i_2$
E. depends on $i_1 + i_2$

ans: C

18. You push a permanent magnet with its north pole away from you toward a loop of conducting wire in front of you. Before the north pole enters the loop the current in the loop is:

A. zero
B. clockwise
C. counterclockwise
D. to your left
E. to your right

ans: C

19. A vertical bar magnet is dropped through the center of a horizontal loop of wire, with its north pole leading. At the instant when the midpoint of the magnet is in the plane of the loop, the induced current at point P, viewed from above, is:

A. maximum and clockwise
B. maximum and counterclockwise
C. not maximum but clockwise
D. not maximum but counterclockwise
E. essentially zero

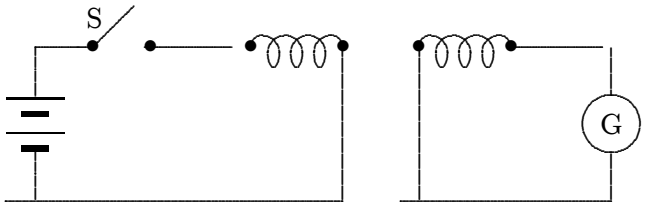
ans: E

20. A circular loop of wire rotates about a diameter in a magnetic field that is perpendicular to the axis of rotation. Looking in the direction of the field at the loop the induced current is:

A. always clockwise
B. always counterclockwise
C. clockwise in the lower half of the loop and counterclockwise in the upper half
D. clockwise in the upper half of the loop and counterclockwise in the lower half
E. sometimes clockwise and sometimes counterclockwise

ans: E

21. In the experiment shown:




A. there is a steady reading in G as long as S is closed
B. a motional emf is generated when S is closed
C. the current in the battery goes through G
D. there is a current in G just after S is opened or closed
E. since the two loops are not connected, the current in G is always zero

ans: D

22. The emf developed in a coil X due to the current in a neighboring coil Y is proportional to the:

A. magnetic field in X
B. rate of change of magnetic field in X
C. resistance of X
D. thickness of the wire in X
E. current in Y

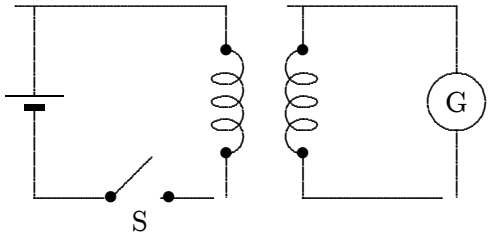
ans: B

23. One hundred turns of insulated copper wire are wrapped around an iron core of cross-sectional area $0.100\,\mathrm{m}^2$. The circuit is completed by connecting the coil to a 10-$\Omega$ resistor. As the magnetic field along the coil axis changes from 1.00 T in one direction to 1.00 T in the other direction, the total charge that flows through the resistor is:

A. $10^{-2}\,\mathrm{C}$
B. $2 \times 10^{-2}\,\mathrm{C}$
C. $1\,\mathrm{C}$
D. $2\,\mathrm{C}$
E. $0.20\,\mathrm{C}$

ans: D

24. In the circuit shown, there will be a non-zero reading in galvanometer G:




A. only just after S is closed
B. only just after S is opened
C. only while S is kept closed
D. never
E. only just after S is opened or closed

ans: E

25. A magnet moves inside a coil. Consider the following factors:

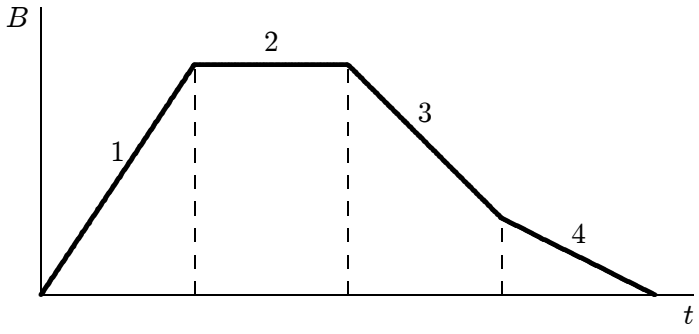
I. strength of the magnet
II. number of turns in the coil
III. speed at which the magnet moves

Which can affect the emf induced in the coil?

A. I only
B. II only
C. III only
D. I and II only
E. I, II, III

ans: E

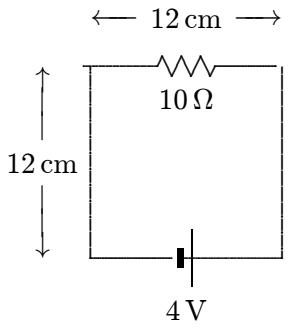
26. The graph shows the magnitude $B$ of a uniform magnetic field that is perpendicular to the plane of a conducting loop. Rank the five regions indicated on the graph according to the magnitude of the emf induced in the loop, from least to greatest.




A. 1, 2, 3, 4
B. 2, 4, 3, 1
C. 4, 3, 1, 2
D. 1, 3, 4, 2
E. 4, 3, 2, 1

ans: B

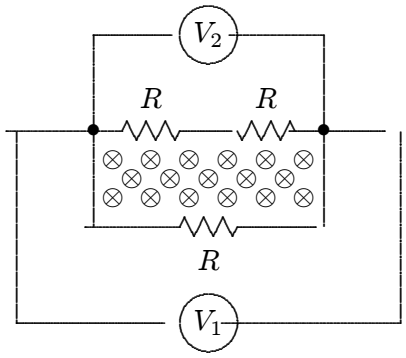
27. The circuit shown is in a uniform magnetic field that is into the page. The current in the circuit is 0.20 A. At what rate is the magnitude of the magnetic field changing? Is it increasing or decreasing?:




A. zero
B. 140 T/s, decreasing
C. 140 T/s, increasing
D. 420 T/s, decreasing
E. 420 T/s, increasing

ans: B

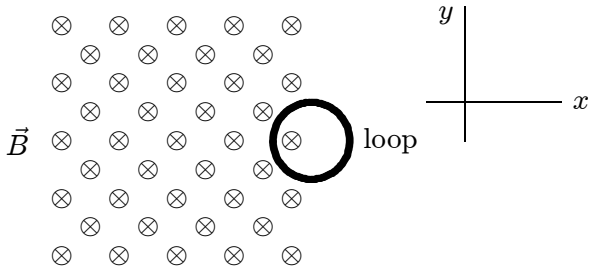
28. A changing magnetic field pierces the interior of a circuit containing three identical resistors. Two voltmeters are connected to the same points, as shown. $V_1$ reads 1 mV. $V_2$ reads:




A. 0
B. 1/3 mV
C. 1/2 mV
D. 1 mV
E. 2 mV

ans: E

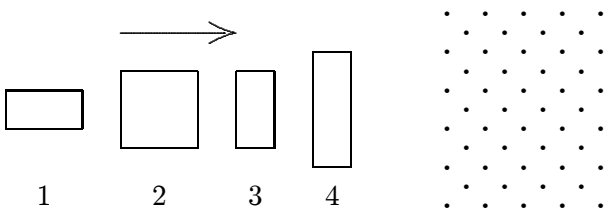
29. A circular loop of wire is positioned half in and half out of a square region of constant uniform magnetic field directed into the page, as shown. To induce a clockwise current in this loop:




A. move it in $+x$ direction
B. move it in $+y$ direction
C. move it in $-y$ direction
D. move it in $-x$ direction
E. increase the strength of the magnetic field

ans: A

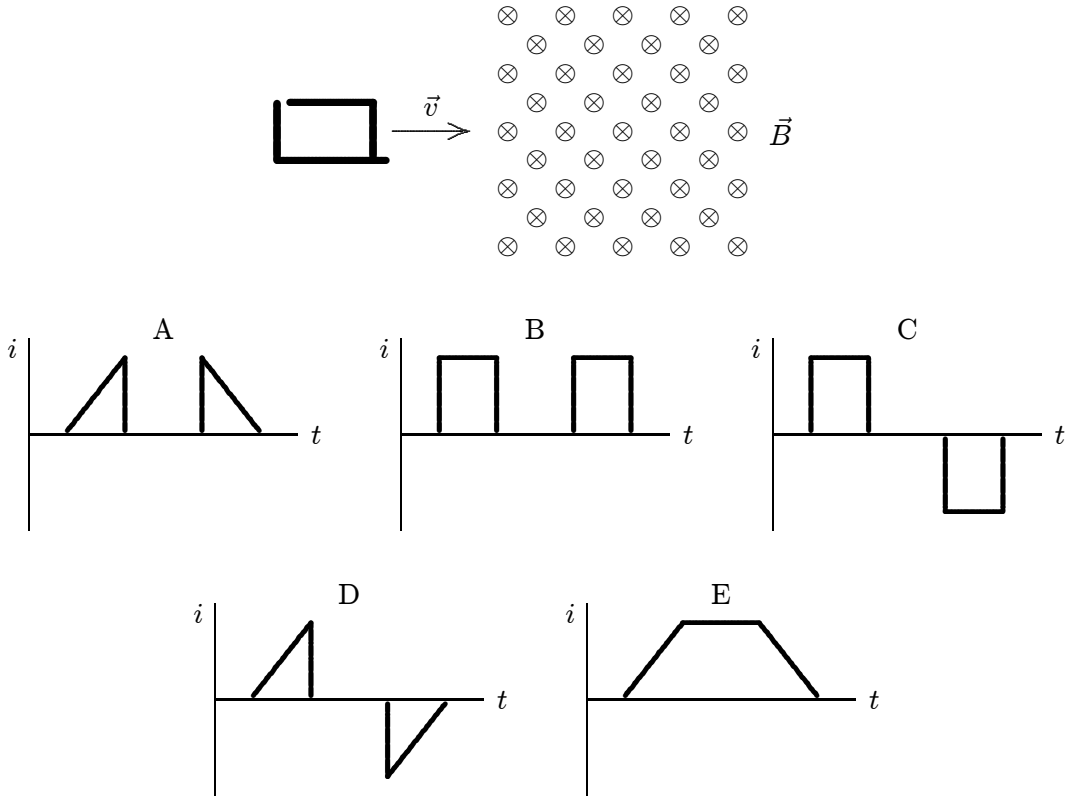
30. The four wire loops shown have edge lengths of either $L$, $2L$, or $3L$. They will move with the same speed into a region of uniform magnetic field $\vec{B}$, directed out of the page. Rank them according to the maximum magnitude of the induced emf, least to greatest.




A. 1 and 2 tie, then 3 and 4 tie
B. 3 and 4 tie, then 1 and 2 tie
C. 4, 2, 3, 1
D. 1, then 2 and 3 tie, then 4
E. 1, 2, 3, 4

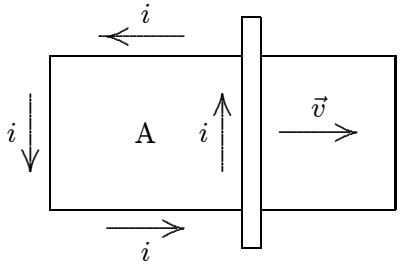
ans: D

31. A square loop of wire moves with a constant speed $v$ from a field-free region into a region of constant uniform magnetic field, as shown. Which of the five graphs correctly shows the induced current $i$ in the loop as a function of time $t$?




ans: C

32. The figure shows a bar moving to the right on two conducting rails. To make an induced current $i$ in the direction indicated, a constant magnetic field in region A should be in what direction?




A. Right
B. Left
C. Into the page
D. Out of the page
E. Impossible; this cannot be done with a constant magnetic field

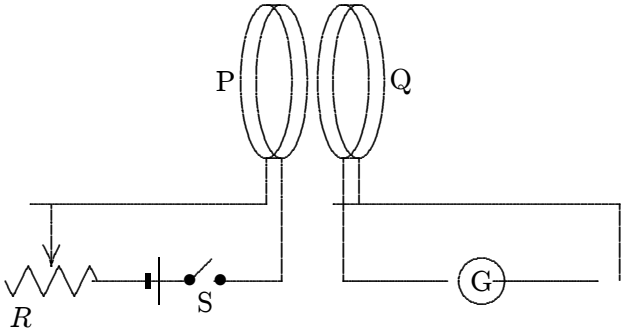
ans: C

33. A car travels northward at 75 km/h along a straight road in a region where Earth's magnetic field has a vertical component of $0.50 \times 10^{-4}$ T. The emf induced between the left and right side, separated by 1.7 m, is:

A. 0
B. 1.8 mV
C. 3.6 mV
D. 6.4 mV
E. 13 mV

ans: B

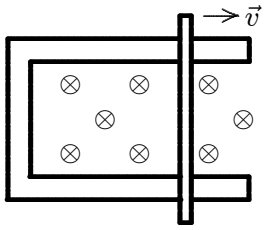
34. Coils P and Q each have a large number of turns of insulated wire. When switch S is closed, the pointer of galvanometer G is deflected toward the left. With S now closed, to make the pointer of G deflect toward the right one could:




A. move the slide of the rheostat R quickly to the right
B. move coil P toward coil Q
C. move coil Q toward coil P
D. open S
E. do none of the above

ans: D

35. A rod lies across frictionless rails in a constant uniform magnetic field $\vec{B}$, as shown. The rod moves to the right with speed $v$. In order for the emf around the circuit to be zero, the magnitude of the magnetic field should:




A. not change
B. increase linearly with time
C. decrease linearly with time
D. increase quadratically with time
E. decrease quadratically with time

ans: C

36. A rectangular loop of wire has area A. It is placed perpendicular to a uniform magnetic field $B$ and then spun around one of its sides at frequency $f$. The maximum induced emf is:

A. $BAf$
B. $BAf$
C. $2BAf$
D. $2\pi BAf$
E. $4\pi BAf$

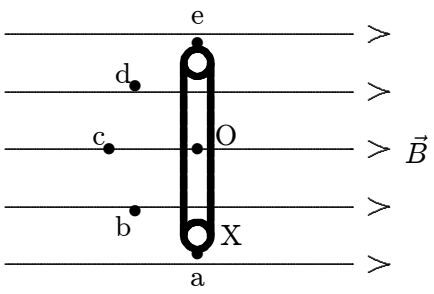
ans: D

37. A rectangular loop of wire is placed perpendicular to a uniform magnetic field and then spun around one of its sides at frequency $f$. The induced emf is a maximum when:

A. the flux is zero
B. the flux is a maximum
C. the flux is half its maximum value
D. the derivative of the flux with respect to time is zero
E. none of the above

ans: A

38. The diagram shows a circular loop of wire that rotates at a steady rate about a diameter O that is perpendicular to a uniform magnetic field. The maximum induced emf occurs when the point X on the loop passes:




A. a
B. b
C. c
D. d
E. e

ans: C

39. A copper hoop is held in a vertical east-west plane in a uniform magnetic field whose field lines run along the north-south direction. The largest induced emf is produced when the hoop is:

A. rotated about a north-south axis
B. rotated about an east-west axis
C. moved rapidly, without rotation, toward the east
D. moved rapidly, without rotation, toward the south
E. moved rapidly, without rotation, toward the northwest

ans: B

40. A 10-turn conducting loop with a radius of 3.0 cm spins at 60 revolutions per second in a magnetic field of 0.50 T. The maximum emf generated is:

A. 0.014 V
B. 0.53 V
C. 5.3 V
D. 18 V
E. 180 V

ans: C

41. A single loop of wire with a radius of 7.5 cm rotates about a diameter in a uniform magnetic field of 1.6 T. To produce a maximum emf of 1.0 V, it should rotate at:

A. 0
B. 2.7 rad/s
C. 5.6 rad/s
D. 35 rad/s
E. 71 rad/s

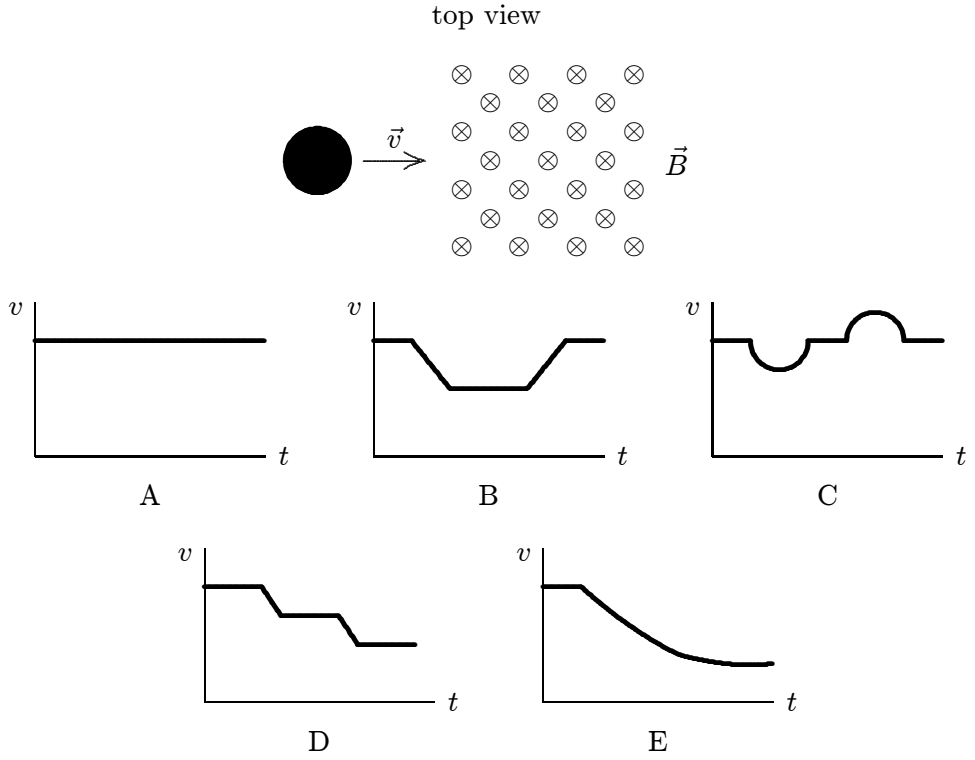
ans: D

42. A merry-go-round has an area of $300\,\mathrm{m}^2$ and spins at 2 rpm about a vertical axis at a place where Earth's magnetic field is vertical and has a magnitude of $5 \times 10^{-5}$ T. The emf around the rim is:

A. 0
B. 0.5 mV
C. 3.1 mV
D. 15 mV
E. 30 mV

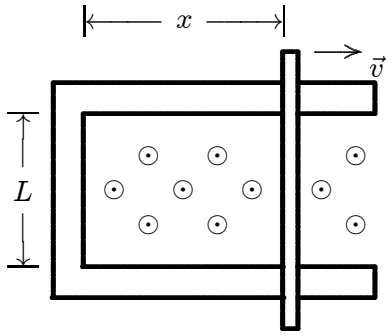
ans: A

43. A copper penny slides on a horizontal frictionless table. There is a square region of constant uniform magnetic field perpendicular to the table, as shown. Which graph correctly shows the speed $v$ of the penny as a function of time $t$?




ans: D

44. A rod with resistance $R$ lies across frictionless conducting rails in a constant uniform magnetic field $\vec{B}$, as shown. Assume the rails have negligible resistance. The magnitude of the force that must be applied by a person to pull the rod to the right at constant speed $v$ is:




A. 0
B. $BLv$
C. $BLv/R$
D. $B^2L^2v/R$
E. $B^2Lxv/R$

ans: D

45. A rod of length $L$ and electrical resistance $R$ moves through a constant uniform magnetic field $\vec{B}$, perpendicular to the rod. The force that must be applied by a person to keep the rod moving with constant velocity $\vec{v}$ is:

A. 0
B. $BLv$
C. $BLv/R$
D. $B^2L^2v/R$
E. $B^2L^2v^2/R$

ans: A

46. As a loop of wire with a resistance of $10\,\Omega$ moves in a constant non-uniform magnetic field, it loses kinetic energy at a uniform rate of 4.0 mJ/s. The induced current in the loop:

A. is 0
B. is 2 mA
C. is 2.8 mA
D. is 20 mA
E. cannot be calculated from the given data

ans: D

47. As a loop of wire with a resistance of $10\,\Omega$ moves in a non-uniform magnetic field, it loses kinetic energy at a uniform rate of 5 mJ/s. The induced emf in the loop:

A. is 0
B. is 0.2 V
C. is 0.28 V
D. is 2 V
E. cannot be calculated from the given data

ans: B

48. An electric field is associated with every:

A. magnetic field
B. time-dependent magnetic field
C. time-dependent magnetic flux
D. object moving in a magnetic field
E. conductor moving in a magnetic field

ans: B

49. A cylindrical region of radius R = 3.0 cm contains a uniform magnetic field parallel to its axis. If the electric field induced at a point $R/2$ from the cylinder axis is $4.5 \times 10^{-3}$ V/m the magnitude of the magnetic field must be changing at the rate:

A. 0
B. 0.30 T/s
C. 0.60 T/s
D. 1.2 T/s
E. 2.4 T/s

ans: C

50. A cylindrical region of radius $R$ contains a uniform magnetic field parallel to its axis. The field is zero outside the cylinder. If the magnitude of the field is changing at the rate $dB/dt$, the electric field induced at a point $2R$ from the cylinder axis is:

A. zero
B. $2R\,dB/dt$
C. $R\,dB/dt$
D. $(R/2)\,dB/dt$
E. $(R/4)\,dB/dt$

ans: E

51. A cylindrical region of radius $R$ contains a uniform magnetic field, parallel to its axis, with magnitude that is changing linearly with time. If $r$ is the radial distance from the cylinder axis, the magnitude of the induced electric field inside the cylinder is proportional to:

A. $R$
B. $r$
C. $r^2$
D. $1/r$
E. $1/r^2$

ans: B

52. A cylindrical region of radius $R$ contains a uniform magnetic field, parallel to its axis, with magnitude that is changing linearly with time. If $r$ is the radial distance from the cylinder axis, the magnitude of the induced electric field outside the cylinder is proportional to:

A. $R$
B. $r$
C. $r^2$
D. $1/r$
E. $1/r^2$

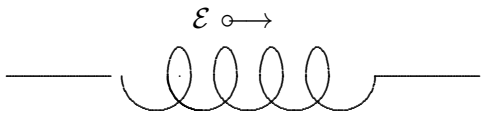
ans: D

53. The unit "henry" is equivalent to:

A. volt·second/ampere
B. volt/second
C. ohm
D. ampere·volt/second
E. ampere·second/volt

ans: A

54. The diagram shows an inductor that is part of a circuit. The direction of the emf induced in the inductor is indicated. Which of the following is possible?




A. The current is constant and rightward
B. The current is constant and leftward
C. The current is increasing and rightward
D. The current is increasing and leftward
E. None of the above

ans: D

55. A 10-turn ideal solenoid has an inductance of 3.5 mH. When the solenoid carries a current of 2.0 A the magnetic flux through each turn is:

A. 0
B. $3.5 \times 10^{-4}$ wb
C. $7.0 \times 10^{-4}$ wb
D. $7.0 \times 10^{-3}$ wb
E. $7.0 \times 10^{-2}$ wb

ans : C

56. A 10-turn ideal solenoid has an inductance of 4.0 mH. To generate an emf of 2.0 V the current should change at a rate of:

A. zero
B. 5.0 A/s
C. 50 A/s
D. 250 A/s
E. 500 A/s

ans: E

57. A long narrow solenoid has length $\ell$ and a total of $N$ turns, each of which has cross-sectional area $A$. Its inductance is:

A. $\mu_0 N^2 A\ell$
B. $\mu_0 N^2 A/\ell$
C. $\mu_0 N A/\ell$
D. $\mu_0 N^2 \ell/A$
E. none of these

ans: B

58. A flat coil of wire, having 5 turns, has an inductance $L$. The inductance of a similar coil having 20 turns is:

A. $4L$
B. $L/4$
C. $16L$
D. $L/16$
E. $L$

ans: C

59. An inductance $L$, resistance $R$, and ideal battery of emf $\mathcal{E}$ are wired in series. A switch in the circuit is closed at time 0, at which time the current is zero. At any later time $t$ the current $i$ is given by:

A. $(\mathcal{E}/R)(1 - e^{-Lt/R})$
B. $(\mathcal{E}/R)e^{-Lt/R}$
C. $(\mathcal{E}/R)(1 + e^{-Rt/L})$
D. $(\mathcal{E}/R)e^{-Rt/L}$
E. $(\mathcal{E}/R)(1 - e^{-Rt/L})$

ans: E

60. An inductance $L$, resistance R, and ideal battery of emf $\mathcal{E}$ are wired in series. A switch in the circuit is closed at time 0, at which time the current is zero. At any later time $t$ the emf of the inductor is given by:

A. $\mathcal{E}(1 - e^{-Lt/R})$
B. $\mathcal{E}e^{-Lt/R}$
C. $\mathcal{E}(1 + e^{-Rt/L})$
D. $\mathcal{E}e^{-Rt/L}$
E. $\mathcal{E}(1 - e^{-Rt/L})$

ans: D

61. An inductance $L$, resistance $R$, and ideal battery of emf $\mathcal{E}$ are wired in series. A switch in the circuit is closed at time 0, at which time the current is zero. At any later time $t$ the potential difference across the resistor is given by:

A. $\mathcal{E}(1 - e^{-Lt/R})$
B. $\mathcal{E}e^{-Lt/R}$
C. $\mathcal{E}(1 + e^{-Rt/L})$
D. $\mathcal{E}e^{-Rt/L}$
E. $\mathcal{E}(1 - e^{-Rt/L})$

ans: E

62. An 8.0-mH inductor and a 2.0-Ω resistor are wired in series to an ideal battery. A switch in the circuit is closed at time 0, at which time the current is zero. The current reaches half its final value at time:

A. 2.8 ms
B. 4.0 ms
C. 3 s
D. 170 s
E. 250 s

ans: A

63. An 8.0-mH inductor and a 2.0-Ω resistor are wired in series to a 20-V ideal battery. A switch in the circuit is closed at time 0, at which time the current is zero. After a long time the current in the resistor and the current in the inductor are:

A. 0, 0
B. 10 A, 10 A
C. 2.5 A, 2.5 A
D. 10 A, 2.5 A
E. 10 A, 0

ans: B

64. An 8.0-mH inductor and a 2.0-Ω resistor are wired in series to a 20-V ideal battery. A switch in the circuit is closed at time 0, at which time the current is zero. Immediately after the switch is thrown the potential differences across the inductor and resistor are:

A. 0, 20 V
B. 20 V, 0
C. 10 V, 10 V
D. 16 V, 4 V
E. unknown since the rate of change of the current is not given

ans: B

65. An inductor with inductance $L$ resistor with resistance $R$ are wired in series to an ideal battery with emf $\mathcal{E}$. A switch in the circuit is closed at time 0, at which time the current is zero. A long time after the switch is thrown the potential differences across the inductor and resistor:

A. 0, $\mathcal{E}$
B. $\mathcal{E}$, 0
C. $\mathcal{E}/2$, $\mathcal{E}/2$
D. $(L/R)\mathcal{E}$, $(R/L)\mathcal{E}$
E. cannot be computed unless the rate of change of the current is given

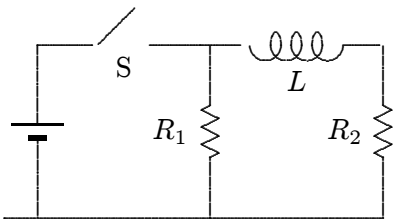
ans: A

66. If both the resistance and the inductance in an $LR$ series circuit are doubled the new inductive time constant will be:

A. twice the old
B. four times the old
C. half the old
D. one-fourth the old
E. unchanged

ans: E

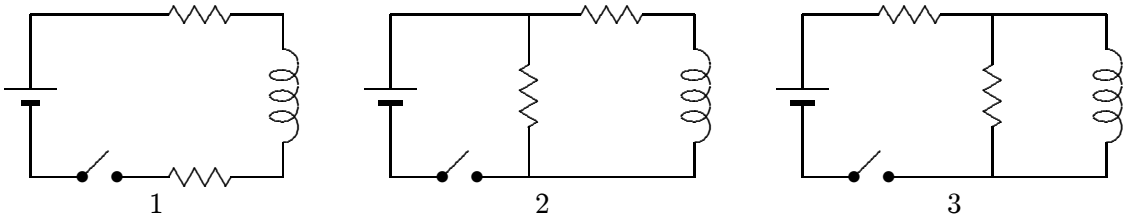
67. When the switch S in the circuit shown is closed, the time constant for the growth of current in $R_2$ is:




A. $L/R_1$
B. $L/R_2$
C. $L/(R_1 + R_2)$
D. $L(R_1 + R_2)/(R_1R_2)$
E. $(L/R_1 + L/R_2)/2$

ans: B

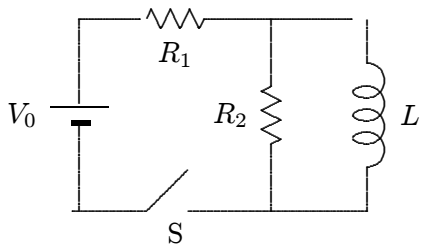
68. The diagrams show three circuits with identical batteries, identical inductors, and identical resistors. Rank them according to the current through the battery just after the switch is closed, from least to greatest.




A. 3, 2, 1
B. 2 and 3 ties, then 1
C. 1, 3, 2
D. 1, 2, 3
E. 3, 1, 2

ans: C

69. Immediately after switch S in the circuit shown is closed, the current through the battery is:




A. 0
B. $V_0/R_1$
C. $V_0/R_2$
D. $V_0/(R_1 + R_2)$
E. $V_0(R_1 + R_2)/(R_1R_2)$

ans: D

70. A 3.5-mH inductor and a 4.5-mH inductor are connected in series. The equivalent inductance is:

A. 2.0 mH
B. 0.51 mH
C. 0.13 mH
D. 1.0 mH
E. 8.0 mH

ans: E

71. A 3.5-mH inductor and a 4.5-mH inductor are connected in series and a time varying current is established in them. When the total emf of the combination is 16 V, the emf of the larger inductor is:

A. 7.0 V
B. 9.0 V
C. 2.3 V
D. 28 V
E. 36 V

ans: B

72. A 3.5-mH inductor and a 4.5-mH inductor are connected in parallel. The equivalent inductance is:

A. 2.0 mH
B. 0.51 mH
C. 0.13 mH
D. 1.0 mH
E. 8.0 mH

ans: A

73. A 3.5-mH inductor and a 4.5-mH inductor are connected in parallel. When the total emf of the combination is 16 V, the rate of change of the current in the larger inductor is:

A. $2.0 \times 10^3$ A/s
B. $3.6 \times 10^3$ A/s
C. $4.6 \times 10^3$ A/s
D. $7.0 \times 10^3$ A/s
E. $8.1 \times 10^3$ A/s

ans: B

74. An inductor with inductance $L$ and an inductor with inductance $2L$ are connected in parallel. When the rate of change of the current in the larger inductor is 1200 A/s the rate of change of the current in the smaller inductor is:

A. 400 A/s
B. 1200 A/s
C. 1600 A/s
D. 2000 A/s
E. 2400 A/s

ans: E

75. The stored energy in an inductor:

A. depends, in sign, upon the direction of the current
B. depends on the rate of change of current
C. is proportional to the square of the inductance
D. has units J/H
E. none of the above

ans: E

76. An inductance $L$ and a resistance $R$ are connected in series to an ideal battery. A switch in the circuit is closed at time 0, at which time the current is zero. The energy stored in the inductor is a maximum:

A. just after the switch is closed
B. at the time $t = L/R$ after the switch is closed
C. at the time $t = L/R$ after the switch is closed
D. at the time $t = 2L/R$ after the switch is closed
E. a long time after the switch is closed

ans: E

77. An inductance $L$ and a resistance $R$ are connected in series to an ideal battery. A switch in the circuit is closed at time 0, at which time the current is zero. The rate of increase of the energy stored in the inductor is a maximum:

A. just after the switch is closed
B. at the time $t = L/R$ after the switch is closed
C. at the time $t = L/R$ after the switch is closed
D. at the time $t = (L/R) \ln 2$ after the switch is closed
E. a long time after the switch is closed

ans: D

78. In each of the following operations, energy is expended. The LEAST percentage of returnable electrical energy will be yielded by:

A. charging a capacitor
B. charging a storage battery
C. sending current through a resistor
D. establishing a current through an inductor
E. moving a conducting rod through a magnetic field

ans: C

79. A current of 10 A in a certain inductor results in a stored energy of 40 J. When the current is changed to 5 A in the opposite direction, the stored energy changes by:

A. 20 J
B. 30 J
C. 40 J
D. 50 J
E. 60 J

ans: B

80. A 6.0-mH inductor is in a series circuit with a resistor and an ideal battery. At the instant the current in the circuit is 5.0 A the energy stored in the inductor is:

A. 0
B. $7.5 \times 10^{-2}$ J
C. $15 \times 10^{-2}$ J
D. $30 \times 10^{-2}$ J
E. unknown since the rate of change of the current is not given

ans: B

81. A 6.0-mH inductor is in a circuit. At the instant the current is 5.0 A and its rate of change is 200 A/s, the rate with which the energy stored in the inductor is increasing is:

A. $7.5 \times 10^{-2}$ W
B. 120 W
C. 240 W
D. 3.0 W
E. 6.0 W

ans: E

82. A 6.0-mH inductor and a 3.0-$\Omega$ resistor are wired in series to a 12-V ideal battery. A switch in the circuit is closed at time 0, at which time the current is zero. 2.0 ms later the energy stored in the inductor is:

A. 0
B. $2.5 \times 10^{-2}$ J
C. $1.9 \times 10^{-2}$ J
D. $3.8 \times 10^{-2}$ J
E. $9.6 \times 10^{-3}$ J

ans: C

83. The quantity $B^2/\mu_0$ has units of:

A. J
B. J/H
C. J/m
D. $J/m^3$
E. $H/m^3$

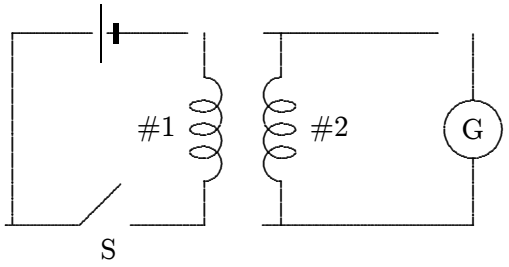
ans: D

84. A 0.20-cm radius cylinder, 3.0 cm long, is wrapped with wire to form an inductor. At the instant the magnetic field in the interior is 5.0 mT the energy stored in the field is about:

A. 0
B. $3.8 \times 10^{-6}$ J
C. $7.5 \times 10^{-6}$ J
D. $7.5 \times 10^{-4}$ J
E. 9.9 J

ans: B

85. In the diagram, assume that all the magnetic field lines generated by coil 1 pass through coil 2. Coil 1 has 100 turns and coil 2 has 400 turns. Then:




A. the power supplied to coil 1 is equal to the power delivered by coil 2
B. the emf around coil 1 will be one-fourth the emf around coil 2
C. the current in coil 1 will be one-fourth the current in coil 2
D. the emfs will be the same in the two coils
E. none of the above

ans: E

## Chapter 31: ELECTROMAGNETIC OSCILLATIONS AND ALTERNATING CURRENT

1. A charged capacitor and an inductor are connected in series. At time $t = 0$ the current is zero, but the capacitor is charged. If $T$ is the period of the resulting oscillations, the next time after $t = 0$ that the current is a maximum is:

   A. $T$
   B. $T/4$
   C. $T/2$
   D. $T$
   E. $2T$

   ans: B

2. A charged capacitor and an inductor are connected in series. At time $t = 0$ the current is zero, but the capacitor is charged. If $T$ is the period of the resulting oscillations, the next time after $t = 0$ that the charge on the capacitor is a maximum is:

   A. $T$
   B. $T/4$
   C. $T/2$
   D. $T$
   E. $2T$

   ans: C

3. A charged capacitor and an inductor are connected in series. At time $t = 0$ the current is zero, but the capacitor is charged. If $T$ is the period of the resulting oscillations, the next time after $t = 0$ that the voltage across the inductor is a maximum is:

   A. $T$
   B. $T/4$
   C. $T/2$
   D. $T$
   E. $2T$

   ans: C

4. A charged capacitor and an inductor are connected in series. At time $t = 0$ the current is zero, but the capacitor is charged. If $T$ is the period of the resulting oscillations, the next time after $t = 0$ that the energy stored in the magnetic field of the inductor is a maximum is:

   A. $T$
   B. $T/4$
   C. $T/2$
   D. $T$
   E. $2T$

   ans: B

5. A charged capacitor and an inductor are connected in series. At time $t = 0$ the current is zero, but the capacitor is charged. If $T$ is the period of the resulting oscillations, the next time after $t = 0$ that the energy stored in the electric field of the capacitor is a maximum is:

   A. $T$
   B. $T/4$
   C. $T/2$
   D. $T$
   E. $2T$

   ans: C

6. A capacitor in an $LC$ oscillator has a maximum potential difference of 15 V and a maximum energy of 360 $\mu$J. At a certain instant the energy in the capacitor is 40 $\mu$J. At that instant what is the potential difference across the capacitor?

   A. zero
   B. 5 V
   C. 10 V
   D. 15 V
   E. 20 V

   ans: B

7. Which of the following has the greatest effect in decreasing the oscillation frequency of an $LC$ circuit? Using instead:

   A. $L/2$ and $C/2$
   B. $L/2$ and $2C$
   C. $2L$ and $C/2$
   D. $2L$ and $2C$
   E. none of these

   ans: D

8. We desire to make an $LC$ circuit that oscillates at 100 Hz using an inductance of 2.5 H. We also need a capacitance of:

   A. 1 F
   B. 1 mF
   C. 1 $\mu$F
   D. 100 $\mu$F
   E. 1 pF

   ans: C

9. An $LC$ circuit consists of a 1-$\mu$F capacitor and a 4 mH inductor. Its oscillation frequency is approximately:

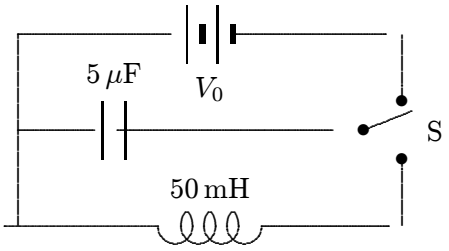
   A. 0.025 Hz
   B. 25 Hz
   C. 60 Hz
   D. 2500 Hz
   E. 15,800 Hz

   ans: D

10. An $LC$ circuit has an oscillation frequency of $10^5$ Hz. If C = 0.1 $\mu$F, then L must be about:
    A. 10 mH
    B. 1 mH
    C. 25 $\mu$H
    D. 2.5 $\mu$H
    E. 1 pH

    ans: C

11. In the circuit shown, switch S is first pushed up to charge the capacitor. When S is then pushed down, the current in the circuit will oscillate at a frequency of:




    A. 318 Hz
    B. 0.01 Hz
    C. 12.500 Hz
    D. 2000 Hz
    E. depends on $V_0$

    ans: A

12. Radio receivers are usually tuned by adjusting the capacitor of an $LC$ circuit. If $C = C_1$ for a frequency of 600 kHz, then for a frequency of 1200 kHz one must adjust $C$ to:
    A. $C_1/2$
    B. $C_1/4$
    C. $2C_1$
    D. $4C_1$
    E. $\sqrt{2}C_1$

    ans: B

13. An $LC$ series circuit with an inductance $L$ and a capacitance $C$ has an oscillation frequency $f$. Two inductors, each with inductance $L$, and two capacitors, each with capacitance $C$, are all wired in series and the circuit is completed. The oscillation frequency is:
    A. $f/4$
    B. $f/2$
    C. $f$
    D. $2f$
    E. $4f$

    ans: C

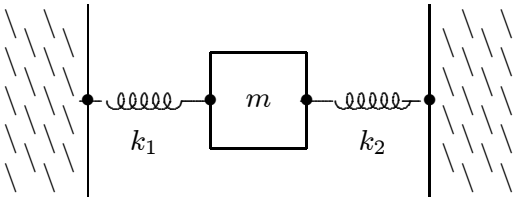
14. The electrical analog of a spring constant $k$ is:

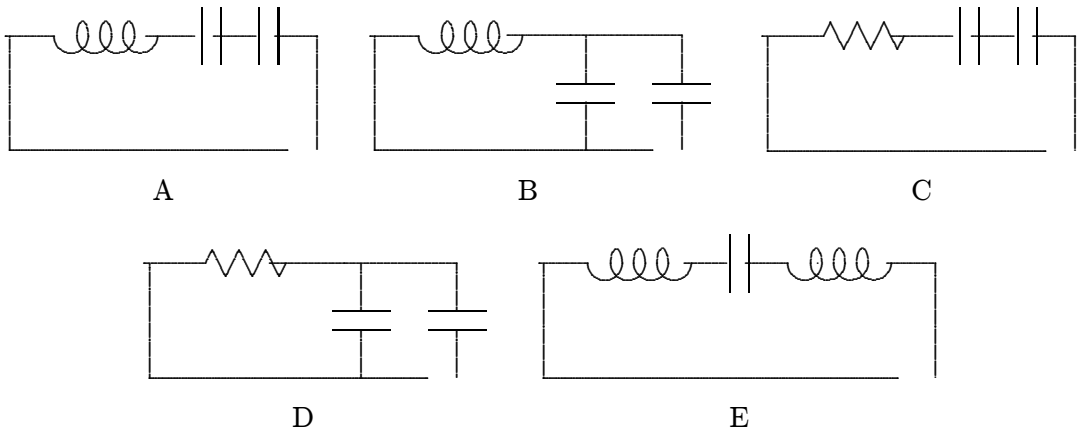
A. $L$
B. $1/L$
C. $C$
D. $1/C$
E. $R$

ans: D

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15. Consider the mechanical system consisting of two springs and a block, as shown. Which one of the five electrical circuits (A, B, C, D, E) is the analog of the mechanical system?




ans: A

16. A 150-g block on the end of a spring with a spring constant of 35 N/m is pulled aside 25 cm and released from rest. In the electrical analog the initial charge on the capacitor is:

A. 0.15 C
B. 6.67 C
C. 0.025 C
D. 40 C
E. 35 C

ans: C

17. A 150-g block on the end of a spring with a spring constant of 35 N/m is pulled aside 25 cm and released from rest. In the electrical analog the maximum charge on the capacitor is 0.25 C. The maximum current in the $LC$ circuit is:

A. 0.38 A
B. 0.025 A
C. 40 A
D. 2.3 A
E. 5.3 A

ans: A

18. A capacitor in an $LC$ oscillator has a maximum potential difference of 15 V and a maximum energy of 360 $\mu$J. At a certain instant the energy in the capacitor is 40 $\mu$J. At that instant what is the potential difference across the capacitor?

A. zero
B. 5 V
C. 10 V
D. 15 V
E. 20 V

ans: B

19. A capacitor in an $LC$ oscillator has a maximum potential difference of 15 V and a maximum energy of 360 $\mu$J. At a certain instant the energy in the capacitor is 40 $\mu$J. At that instant what is the emf induced in the inductor?

A. zero
B. 5 V
C. 10 V
D. 15 V
E. 20 V

ans: C

20. In an oscillating $LC$ circuit, the total stored energy is $U$. The maximum energy stored in the capacitor during one cycle is:

A. $U/2$
B. $U/\sqrt{2}$
C. $U$
D. $U/(2\pi)$
E. $U/\pi$

ans: C

21. In an oscillating $LC$ circuit, the total stored energy is $U$ and the maximum charge on the capacitor is $Q$. When the charge on the capacitor is $Q/2$, the energy stored in the inductor is:

A. $U/2$
B. $U/4$
C. $(4/3)U$
D. $3U/2$
E. $3U/4$

ans: E

22. The total energy in an $LC$ circuit is $5.0 \times 10^{-6}$ J. If C = 15 $\mu$F the charge on the capacitor is:
    A. 0.82 $\mu$C
    B. 8.5 $\mu$C
    C. 12 $\mu$C
    D. 17 $\mu$C
    E. 24 $\mu$C

    ans: C

23. The total energy in an $LC$ circuit is $5.0 \times 10^{-6}$ J. If $L$ = 25 mH the maximum current is:
    A. 10 mA
    B. 14 mA
    C. 20 mA
    D. 28 mA
    E. 40 mA

    ans: C

24. At time $t = 0$ the charge on the 50-$\mu$F capacitor in an $LC$ circuit is 15 $\mu$C and there is no current. If the inductance is 20 mH the maximum current is:
    A. 15 nA
    B. 15 $\mu$A
    C. 6.7 mA
    D. 15 mA
    E. 15 A

    ans: D

25. An $LC$ circuit has an inductance of 20 mH and a capacitance of 5.0 $\mu$F. At time $t = 0$ the charge on the capacitor is 3.0 $\mu$C and the current is 7.0 mA. The total energy is:
    A. $4.1 \times 10^{-7}$ J
    B. $4.9 \times 10^{-7}$ J
    C. $9.0 \times 10^{-7}$ J
    D. $1.4 \times 10^{-6}$ J
    E. $2.8 \times 10^{-6}$ J

    ans: D

26. An $LC$ circuit has a capacitance of 30 $\mu$F and an inductance of 15 mH. At time $t = 0$ the charge on the capacitor is 10 $\mu$C and the current is 20 mA. The maximum charge on the capacitor is:
    A. 8.9 $\mu$C
    B. 10 $\mu$C
    C. 12 $\mu$C
    D. 17 $\mu$C
    E. 24 $\mu$C

    ans: D

27. An $LC$ circuit has an inductance of 15 mH and a capacitance of 10 $\mu$F. At one instant the charge on the capacitor is 25 $\mu$C. At that instant the current is changing at the rate of:

A. 0
B. $1.7 \times 10^{-8}$ A/s
C. $5.9 \times 10^{-3}$ A/s
D. $3.8 \times 10^{-2}$ A/s
E. 170 A/s

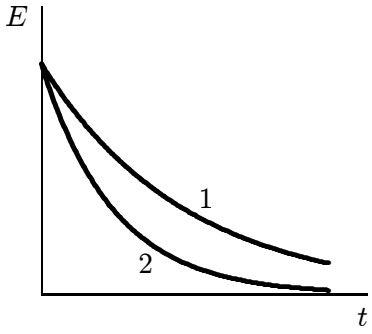
ans: E

28. An $LC$ circuit has a capacitance of 30 $\mu$F and an inductance of 15 mH. At time $t = 0$ the charge on the capacitor is 10 $\mu$C and the current is 20 mA. The maximum current is:

A. 18 mA
B. 20 mA
C. 25 mA
D. 35 mA
E. 42 mA

ans: C

29. The graphs show the total electromagnetic energy in two RLC circuits as functions of time. Which of the following statements might be true?




A. Circuit 1 has a smaller resistance and a larger inductance
B. Circuit 1 has a larger resistance and a smaller inductance
C. Circuit 1 has the same resistance and a larger inductance
D. Circuit 1 has a larger resistance and a larger capacitance
E. Circuit 1 has the same resistance and a smaller capacitance

ans: A

30. An $RLC$ circuit has a resistance of 200 $\Omega$ and an inductance of 15 mH. Its oscillation frequency is 7000 Hz. At time $t = 0$ the current is 25 mA and there is no charge on the capacitor. After five complete cycles the current is:

A. zero
B. $1.8 \times 10^{-6}$ A
C. $2.1 \times 10^{-4}$ A
D. $2.3 \times 10^{-3}$ A
E. $2.5 \times 10^{-2}$ A

ans: C

31. An *RLC* circuit has an inductance of 25 mH and a capacitance of 5.0 $\mu$F. The charge on the capacitor does NOT oscillate but rather decays exponentially to zero. The resistance in the circuit must be:

A. greater than or equal to $20,000\Omega$
B. less than $20,000\Omega$ but greater than $10,000\Omega$
C. less than $10,000\Omega$ but greater than $5,000\Omega$
D. less than $5,000\Omega$ but greater than 0
E. 0

ans: A

32. A series circuit with an inductance of 15 mH, a capacitance of 35 $\mu$F, and a resistance of 5.0 $\Omega$ contains a sinusoidal source of emf with a frequency of 500 Hz. The frequency with which the charge on the capacitor oscillates is:

A. 500 Hz
B. 1.4 kHz
C. greater than 1.4 kHz
D. less than 500 Hz
E. between 500 Hz and 1.4 kHz

ans: A

33. The rapid exponential decay in just a few cycles of the charge on the plates of capacitor in an *RLC* circuit might be due to:

A. a large inductance
B. a large capacitance
C. a small capacitance
D. a large resistance
E. a small resistance

ans: D

34. An *RLC* circuit has a capacitance of 12 $\mu$F, an inductance of 25 mH, and a resistance of $60\Omega$. The current oscillates with an angular frequency of:

A. $1.2 \times 10^3$ rad/s
B. $1.4 \times 10^3$ rad/s
C. $1.8 \times 10^3$ rad/s
D. $2.2 \times 10^3$ rad/s
E. $2.6 \times 10^3$ rad/s

ans: B

35. The angular frequency of a certain *RLC* series circuit is $\omega_0$. A source of sinusoidal emf, with angular frequency $2\omega$, is inserted into the circuit. After transients die out the angular frequency of the current oscillations is:

A. $\omega_0/2$
B. $\omega_0$
C. $2\omega_0$
D. $1.5\omega_0$
E. $3\omega_0$

ans: C

36. The angular frequency of a certain *RLC* series circuit is $\omega_0$. A source of sinusoidal emf, with angular frequency $\omega$, is inserted into the circuit and $\omega$ is varied while the amplitude of the source is held constant. For which of the following values of $\omega$ is the amplitude of the current oscillations the greatest?

A. $\omega_0/5$
B. $\omega_0/2$
C. $\omega_0$
D. $2\omega_0$
E. None of them (they all produce the same current amplitude)

ans: C

37. An *RLC* circuit has a sinusoidal source of emf. The average rate at which the source supplies energy is 5 nW. This must also be:

A. the average rate at which energy is stored in the capacitor
B. the average rate at which energy is stored in the inductor
C. the average rate at which energy is dissipated in the resistor
D. twice the average rate at which energy is stored in the capacitor
E. three times the average rate at which energy is stored in the inductor

ans: C

38. In a purely capacitive circuit the current:

A. leads the voltage by one-fourth of a cycle
B. leads the voltage by one-half of a cycle
C. lags the voltage by one-fourth of a cycle
D. lags the voltage by one-half of a cycle
E. is in phase with the potential difference across the plates

ans: A

39. In a purely resistive circuit the current:

A. leads the voltage by one-fourth of a cycle
B. leads the voltage by one-half of a cycle
C. lags the voltage by one-fourth of a cycle
D. lags the voltage by one-half of a cycle
E. is in phase with the voltage

ans: E

40. In a purely inductive circuit, the current lags the voltage by:

A. zero
B. one-fourth of a cycle
C. one-half of a cycle
D. three-fourths of a cycle
E. one cycle

ans: B

41. A series $RL$ circuit is connected to an emf source of angular frequency $\omega$. The current:
    A. leads the applied emf by $\tan^{-1}(\omega L/R)$
    B. lags the applied emf by $\tan^{-1}(\omega L/R)$
    C. lags the applied emf by $\tan^{-1}(\omega R/L)$
    D. leads the applied emf by $\tan^{-1}(\omega R/L)$
    E. is zero

    ans: B

42. A series $RC$ circuit is connected to an emf source having angular frequency $\omega$. The current:
    A. leads the source emf by $\tan^{-1}(1/\omega CR)$
    B. lags the source emf by $\tan^{-1}(1/\omega CR)$
    C. leads the source emf by $\tan^{-1}(\omega CR)$
    D. lags the source emf by $\tan^{-1}(\omega CR)$
    E. leads the source emf by $\pi/4$

    ans: A

43. In an $RLC$ series circuit, which is connected to a source of emf $\mathcal{E}_m \cos(\omega t)$, the current lags the voltage by 45° if:
    A. $R = 1/\omega C - \omega L$
    B. $R = 1/\omega L - \omega C$
    C. $R = \omega L - 1/\omega C$
    D. $R = \omega C - 1/\omega L$
    E. $\omega L = 1/\omega C$

    ans: C

44. A coil has a resistance of 60 Ω and an impedance of 100 Ω. Its reactance, in ohms, is:
    A. 40
    B. 60
    C. 80
    D. 117
    E. 160

    ans: C

45. The reactance in ohms of a 35-$\mu$F capacitor connected to a 400-Hz generator is:
    A. 0
    B. 0.014
    C. 0.088
    D. 11
    E. 71

    ans: D

46. A 35-$\mu$F capacitor is connected to a source of sinusoidal emf with a frequency of 400 Hz and a maximum emf of 20 V. The maximum current is:

A. 0
B. 0.28 A
C. 1.8 A
D. 230 A
E. 1400 A

ans: C

47. A 45-mH inductor is connected to a source of sinusoidal emf with a frequency of 400 Hz and a maximum emf of 20 V. The maximum current is:

A. 0
B. 0.18 A
C. 1.1 A
D. 360 A
E. 2300 A

ans: B

48. The impedance of an $RLC$ series circuit is definitely increased if:

A. $C$ decreases
B. $L$ increases
C. $L$ decreases
D. $R$ increases
E. $R$ decreases

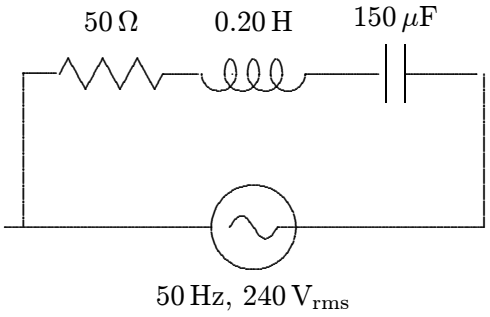
ans: D

49. An $RLC$ series circuit has $R = 4\,\Omega$, $X_C = 3\,\Omega$, and $X_L = 6\,\Omega$. The impedance of this circuit is:

A. $5\,\Omega$
B. $7\,\Omega$
C. $9.8\,\Omega$
D. $13\,\Omega$
E. $7.8\,\Omega$

ans: A

50. The impedance of the circuit shown is:




A. 21 Ω
B. 50 Ω
C. 63 Ω
D. 65 Ω
E. 98 Ω

ans: D

51. An electric motor, under load, has an effective resistance of 30 Ω and an inductive reactance of 40 Ω. When powered by a source with a maximum voltage of 420 V, the maximum current is:

A. 6.0 A
B. 8.4 A
C. 10.5 A
D. 12.0 A
E. 14.0 A

ans: B

52. An *RL* series circuit is connected to an ac generator with a maximum emf of 20 V. If the maximum potential difference across the resistor is 16 V, then the maximum potential difference across the inductor is:

A. 2 V
B. 4 V
C. 12 V
D. 25.6 V
E. 36 V

ans: C

53. When the amplitude of the oscillator in a series *RLC* circuit is doubled:

A. the impedance is doubled
B. the voltage across the capacitor is halved
C. the capacitive reactance is halved
D. the power factor is doubled
E. the current amplitude is doubled

ans: E

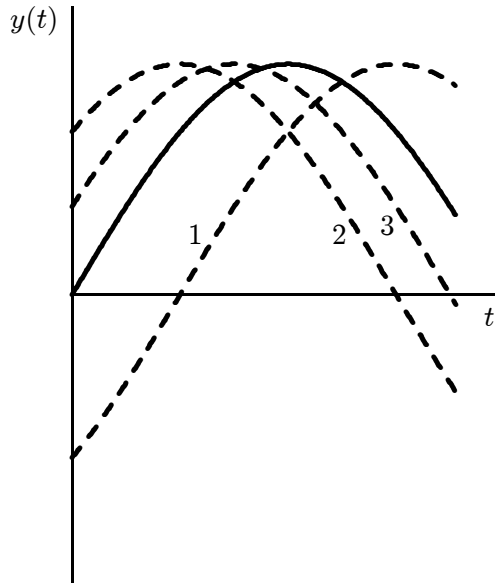
54. When the frequency of the oscillator in a series $RLC$ circuit is doubled:
    A. the capacitive reactance is doubled
    B. the capacitive reactance is halved
    C. the impedance is doubled
    D. the current amplitude is doubled
    E. the current amplitude is halved

    ans: B

55. In an $RLC$ series circuit, the source voltage is leading the current at a given frequency $f$. If $f$ is lowered slightly, then the circuit impedance will:
    A. increase
    B. decrease
    C. remain the same
    D. need to know the amplitude of the source voltage
    E. need to know whether the phase angle is larger or smaller than 45°

    ans: B

56. In the diagram, the function $y(t) = y_m \sin(\omega t)$ is plotted as a solid curve. The other three curves have the form $y(t) = y_m \sin(\omega t + \phi)$, where $\phi$ is between $-\pi/2$ and $+\pi/2$. Rank the curves according to the value of $\phi$, from the most negative to the most positive.




    A. 1, 2, 3
    B. 2, 3, 1
    C. 3, 2, 1
    D. 1, 3, 2
    E. 2, 1, 3

    ans: D

57. An $RLC$ series circuit has $L = 100\,\text{mH}$ and $C = 1\,\mu\text{F}$. It is connected to a 1000-Hz source and the source emf is found to lead the current by 75°. The value of $R$ is:

A. $12.6\,\Omega$
B. $126\,\Omega$
C. $175\,\Omega$
D. $1750\,\Omega$
E. $1810\,\Omega$

ans: B

58. An $RLC$ series circuit is driven by a sinusoidal emf with angular frequency $\omega_d$. If $\omega_d$ is increased without changing the amplitude of the emf the current amplitude increases. If $L$ is the inductance, $C$ is the capacitance, and $R$ is the resistance, this means that:

A. $\omega_d L > 1/\omega_d C$
B. $\omega_d L < 1/\omega_d C$
C. $\omega_d L = 1/\omega_d C$
D. $\omega_d L > R$
E. $\omega_d L < R$

ans: B

59. In a sinusoidally driven series $RLC$ circuit, the inductive reactance is $X_L = 200\,\Omega$, the capacitive reactance is $X_C = 100\,\Omega$, and the resistance is $R = 50\,\Omega$. The current and applied emf would be in phase if:

A. the resistance is increased to $100\,\Omega$, with no other changes
B. the resistance is increased to $200\,\Omega$, with no other changes
C. the inductance is reduced to zero, with no other changes
D. the capacitance is doubled, with no other changes
E. the capacitance is halved, with no other changes

ans: E

60. In a sinusoidally driven series $RLC$ circuit the current lags the applied emf. The rate at which energy is dissipated in the resistor can be increased by:

A. decreasing the capacitance and making no other changes
B. increasing the capacitance and making no other changes
C. increasing the inductance and making no other changes
D. increasing the driving frequency and making no other changes
E. decreasing the amplitude of the driving emf and making no other changes

ans: A

61. An $RLC$ series circuit, connected to a source E, is at resonance. Then:

A. the voltage across $R$ is zero
B. the voltage across $R$ equals the applied voltage
C. the voltage across $C$ is zero
D. the voltage across $L$ equals the applied voltage
E. the applied voltage and current differ in phase by 90°

ans: B

62. An *RLC* series circuit is connected to an oscillator with a maximum emf of 100 V. If the voltage amplitudes $V_R$, $V_L$, and $V_C$ are all equal to each other, then $V_R$ must be:

A. 33 V
B. 50 V
C. 67 V
D. 87 V
E. 100 V

ans: E

63. A resistor, an inductor, and a capacitor are connected in parallel to a sinusoidal source of emf. Which of the following is true?

A. The currents in all branches are in phase.
B. The potential differences across all branches are in phase.
C. The current in the capacitor branch leads the current in the inductor branch by one-fourth of a cycle
D. The potential difference across the capacitor branch leads the potential difference across the inductor branch by one-fourth of a cycle.
E. The current in the capacitor branch lags the current in the inductor branch by one-fourth of a cycle.

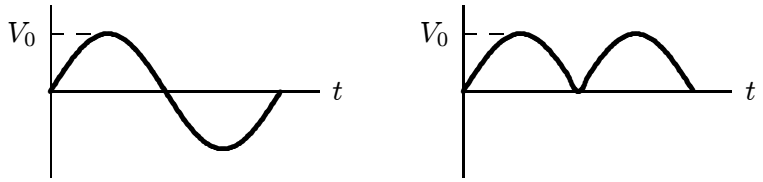
ans: B

64. The rms value of an ac current is:

A. its peak value
B. its average value
C. that steady current that produces the same rate of heating in a resistor as the actual current
D. that steady current that will charge a battery at the same rate as the actual current
E. zero

ans: C

65. The rms value of a sinusoidal voltage is $V_0/\sqrt{2}$, where $V_0$ is the amplitude. What is the rms value of its fully rectified wave? Recall that $V_{\text{rect}}(t) = |V(t)|$.




A. $V_0^2/\sqrt{2}$
B. $V_0^2/2$
C. $\sqrt{2}V_0$
D. $V_0/\sqrt{2}$
E. $V_0/(2\sqrt{2})$

ans: D

66. A sinusoidal voltage $V(t)$ has an rms value of 100 V. Its maximum value is:

A. 100 V
B. 707 V
C. 70.7 V
D. 141 V
E. 200 V

ans: D

67. An ac generator produces 10 V (rms) at 400 rad/s. It is connected to a series $RL$ circuit ($R = 17.3\,\Omega$, $L = 0.025$ H). The rms current is:

A. 0.50 A and leads the emf by 30°
B. 0.71 A and lags the emf by 30°
C. 1.40 A and lags the emf by 60°
D. 0.50 A and lags the emf by 30°
E. 0.58 A and leads the emf by 90°

ans: D

68. An ac generator producing 10 V (rms) at 200 rad/s is connected in series with a 50-Ω resistor, a 400-mH inductor, and a 200-$\mu$F capacitor. The rms current in amperes is:

A. 0.125
B. 0.135
C. 0.18
D. 0.20
E. 0.40

ans: B

69. An ac generator producing 10 V (rms) at 200 rad/s is connected in series with a 50-Ω resistor, a 400-mH inductor, and a 200-$\mu$F capacitor. The rms voltage (in volts) across the resistor is:

A. 2.5
B. 3.4
C. 6.7
D. 10.0
E. 10.8

ans: C

70. An ac generator producing 10 V (rms) at 200 rad/s is connected in series with a 50-Ω resistor, a 400-mH inductor, and a 200-$\mu$F capacitor. The rms voltage (in volts) across the capacitor is:

A. 2.5
B. 3.4
C. 6.7
D. 10.0
E. 10.8

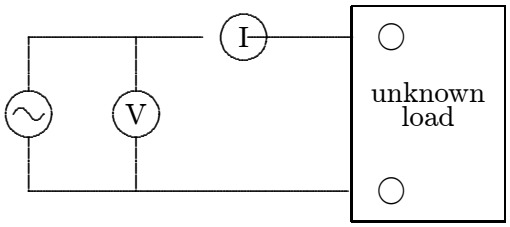
ans: B

71. An ac generator producing 10 V (rms) at 200 rad/s is connected in series with a 50-Ω resistor, a 400-mH inductor, and a 200-$\mu$F capacitor. The rms voltage (in volts) across the inductor is:

A. 2.5
B. 3.4
C. 6.7
D. 10.0
E. 10.8

ans: E

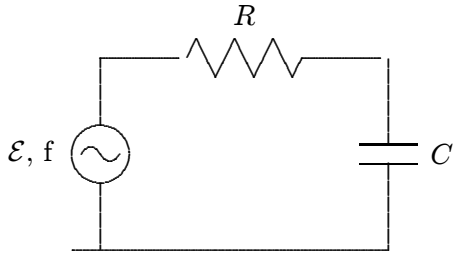
72. The ideal meters shown read rms current and voltage. The average power delivered to the load is:




A. definitely equal to $VI$
B. definitely more than $VI$
C. possibly equal to $VI$ even if the load contains an inductor and a capacitor
D. definitely less than $VI$
E. zero, as is the average of any sine wave

ans: C

73. The average power supplied to the circuit shown passes through a maximum when which one of the following is increased continuously from a very low to a very high value?




A. Source emf $\mathcal{E}$
B. $R$
C. $C$
D. Source frequency $f$
E. None of these

ans: B

74. In a series $RLC$ circuit the rms value of the generator emf is $E$ and the rms value of the current is $i$. The current lags the emf by $\phi$. The average power supplied by the generator is given by:

A. $(iE/2)\cos\phi$
B. $iE$
C. $i^2/Z$
D. $i^2Z$
E. $i^2R$

ans: E

75. The units of the power factor are:

A. ohm
B. watt
C. radian
D. $\text{ohm}^{1/2}$
E. none of these

ans: E

76. A series circuit consists of a 15-$\Omega$ resistor, a 25-mH inductor, and a 35-$\mu$F capacitor. If the frequency is 100 Hz the power factor is:

A. 0
B. 0.20
C. 0.45
D. 0.89
E. 1.0

ans: C

77. The main reason that alternating current replaced direct current for general use is:

A. ac generators do not need slip rings
B. ac voltages may be conveniently transformed
C. electric clocks do not work on dc
D. a given ac current does not heat a power line as much as the same dc current
E. ac minimizes magnetic effects

ans: B

78. A step-down transformer is used to:

A. increase the power
B. decrease the power
C. increase the voltage
D. decrease the voltage
E. change ac to dc

ans: D

79. Iron, rather than copper, is used in the core of transformers because iron:
    A. can withstand a higher temperature
    B. has a greater resistivity
    C. has a very high permeability
    D. makes a good permanent magnet
    E. insulates the primary from the secondary

    ans: C

80. The core of a transformer is made in a laminated form to:
    A. facilitate easy assembly
    B. reduce $i^2R$ losses in the coils
    C. increase the magnetic flux
    D. save weight
    E. prevent eddy currents

    ans: E

81. A generator supplies 100 V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:
    A. 1000 V
    B. 500 V
    C. 250 V
    D. 100 V
    E. 10 V

    ans: A

82. The resistance of the primary coil of a well-designed, 1 : 10 step-down transformer is 1 Ω. With the secondary circuit open, the primary is connected to a 12 V ac generator. The primary current is:
    A. essentially zero
    B. about 12 A
    C. about 120 A
    D. depends on the actual number of turns in the primary coil
    E. depends on the core material

    ans: A

83. The primary of an ideal transformer has 100 turns and the secondary has 600 turns. Then:
    A. the power in the primary circuit is less than that in the secondary circuit
    B. the currents in the two circuits are the same
    C. the voltages in the two circuits are the same
    D. the primary current is six times the secondary current
    E. the frequency in the secondary circuit is six times that in the primary circuit

    ans: D

84. The primary of a 3 : 1 step-up transformer is connected to a source and the secondary is connected to a resistor $R$. The power dissipated by $R$ in this situation is $P$. If $R$ is connected directly to the source it will dissipate a power of:

A. $P/9$
B. $P/3$
C. $P$
D. $3P$
E. $9P$

ans: A

85. In an ideal 1 : 8 step-down transformer, the primary power is 10 kW and the secondary current is 25 A. The primary voltage is:

A. $25,600\,\text{V}$
B. $3200\,\text{V}$
C. $400\,\text{V}$
D. $50\,\text{V}$
E. $6.25\,\text{V}$

ans: B

86. A source with an impedance of $100\,\Omega$ is connected to the primary coil of a transformer and a resistance $R$ is connected to the secondary coil. If the transformer has 500 turns in its primary coil and 100 turns in its secondary coil the greatest power will be dissipated in the resistor if $R =$:

A. 0
B. $0.25\,\Omega$
C. $4.0\,\Omega$
D. $50\,\Omega$
E. $100\,\Omega$

ans: C

## Chapter 32: MAXWELL'S EQUATIONS; MAGNETISM AND MATTER

1. Gauss' law for magnetism:
   A. can be used to find $\vec{B}$ due to given currents provided there is enough symmetry
   B. is false because there are no magnetic poles
   C. can be used with open surfaces because there are no magnetic poles
   D. contradicts Faraday's law because one says $\Phi_B = 0$ and the other says $\mathcal{E} = -d\Phi_B/dt$
   E. none of the above

   ans: E

2. Gauss' law for magnetism tells us:
   A. the net charge in any given volume
   B. that the line integral of a magnetic field around any closed loop must vanish
   C. the magnetic field of a current element
   D. that magnetic monopoles do not exist
   E. charges must be moving to produce magnetic fields

   ans: D

3. The statement that magnetic field lines form closed loops is a direct consequence of:
   A. Faraday's law
   B. Ampere's law
   C. Gauss' law for electricity
   D. Gauss' law for magnetism
   E. the Lorentz force

   ans: D

4. A magnetic field parallel to the $x$ axis with a magnitude that decreases with increasing $x$ but does not change with $y$ and $z$ is impossible according to:
   A. Faraday's law
   B. Ampere's law
   C. Gauss' law for electricity
   D. Gauss' law for magnetism
   E. Newton's second law

   ans: D

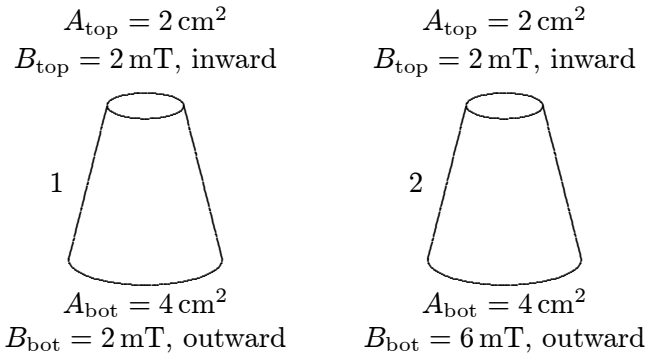
5. According to Gauss' law for magnetism, magnetic field lines:
   A. form closed loops
   B. start at south poles and end at north poles
   C. start at north poles and end at south poles
   D. start at both north and south poles and end at infinity
   E. do not exist

   ans: A

6. The magnetic field lines due to an ordinary bar magnet:
   A. form closed curves
   B. cross one another near the poles
   C. are more numerous near the N pole than near the S pole
   D. do not exist inside the magnet
   E. none of the above

   ans: A

7. Four closed surfaces are shown. The areas $A_{\text{top}}$ and $A_{\text{bot}}$ of the top and bottom faces and the magnitudes $B_{\text{top}}$ and $B_{\text{bot}}$ of the uniform magnetic fields through the top and bottom faces are given. The fields are perpendicular to the faces and are either inward or outward. Rank the surfaces according to the magnitude of the magnetic flux through the curved sides, least to greatest.




   A. 1, 2, 3, 4
   B. 3, 4, 1, 2
   C. 1, 2, 4, 3
   D. 4, 3, 2, 1
   E. 2, 1, 4, 3

   ans: B

8. Consider the four Maxwell equations:

   1. $\oint \vec{E} \cdot \overrightarrow{\mathrm{d}}\, \vec{A} = q/\epsilon_0$
   2. $\oint \vec{B} \cdot \mathrm{d}\vec{A} = 0$
   3. $\oint \vec{E} \cdot \mathrm{d}\vec{s} = -\mathrm{d}\Phi_B/\mathrm{d}t$
   4. $\oint \vec{B} \cdot \mathrm{d}\vec{s} = \mu_0 i + \mu_0 \epsilon_0 \, \mathrm{d}\Phi_E/\mathrm{d}t$

   Which of these must be modified if magnetic poles are discovered?

   A. Only 1
   B. Only 2
   C. Only 2 and 3
   D. Only 3 and 4
   E. Only 2, 3, and 4

   ans: C

9. One of the Maxwell equations begins with $\oint \vec{B} \cdot \mathrm{d}\vec{s} = \ldots$. The symbol "$\mathrm{d}\vec{s}$" means:

   A. an infinitesimal displacement of a charge
   B. an infinitesimal displacement of a magnetic pole
   C. an infinitesimal inductance
   D. an infinitesimal surface area
   E. none of the above

   ans: E

10. One of the Maxwell equations begins with $\oint \vec{E} \cdot \mathrm{d}\vec{s} = \ldots$. The "$\circ$" symbol in the integral sign means:

    A. the same as the subscript in $\mu_0$
    B. integrate clockwise around the path
    C. integrate counterclockwise around the path
    D. integrate around a closed path
    E. integrate over a closed surface

    ans: D

11. One of the Maxwell equations begins with $\oint \vec{B} \cdot \mathrm{d}\vec{A} = \ldots$. The "$\circ$" symbol in the integral sign means:

    A. the same as the subscript in $\mu_0$
    B. integrate clockwise around the path
    C. integrate counterclockwise around the path
    D. integrate around a closed path
    E. integrate over a closed surface

    ans: E

12. One of the crucial facts upon which the Maxwell equations are based is:

    A. the numerical value of the electron charge
    B. charge is quantized
    C. the numerical value of the charge/mass ratio of the electron
    D. there are three types of magnetic materials
    E. none of the above

    ans: E

13. Two of Maxwell's equations contain a path integral on the left side and an area integral on the right. For them:

A. the path must pierce the area
B. the path must be well-separated from the area
C. the path must be along a field line and the area must be perpendicular to the field line
D. the path must be the boundary of the area
E. the path must lie in the area, away from its boundary

ans: D

14. Two of Maxwell's equations contain an integral over a closed surface. For them the infinitesimal vector area $d\vec{A}$ is always:

A. tangent to the surface
B. perpendicular to the surface and pointing outward
C. perpendicular to the surface and pointing inward
D. tangent to a field line
E. perpendicular to a field line

ans: B

15. Two of Maxwell's equations contain a path integral on the left side and an area integral on the right. The directions of the infinitesimal path element $d\vec{s}$ and infinitesimal area element $d\vec{A}$ are:

A. always in the same direction
B. always in opposite directions
C. always perpendicular to each other
D. never perpendicular to each other
E. none of the above

ans: E

16. Two of Maxwell's equations contain a path integral on the left side and an area integral on the right. Suppose the area is the surface of a piece of paper at which you are looking and $d\vec{A}$ is chosen to point toward you. Then, the path integral is:

A. clockwise around the circumference of the paper
B. counterclockwise around the circumference of the paper
C. from left to right
D. from right to left
E. from top to bottom

ans: B

17. Which of the following equations can be used, along with a symmetry argument, to calculate the electric field of a point charge?

A. $\oint \vec{E} \cdot d\vec{A} = q/\epsilon_0$
B. $\oint \vec{B} \cdot d\vec{A} = 0$
C. $\oint \vec{E} \cdot d\vec{s} = -d\Phi_B/dt$
D. $\oint \vec{B} \cdot d\vec{s} = \mu_0 i + \mu_0 \epsilon_0 d\Phi_E/dt$
E. None of these

ans: A

18. Which of the following equations can be used, along with a symmetry argument, to calculate the magnetic field of a long straight wire carrying current?
A. $\oint \vec{E} \cdot d\vec{A} = q/\epsilon_0$
B. $\oint \vec{B} \cdot d\vec{A} = 0$
C. $\oint \vec{E} \cdot d\vec{s} = -d\Phi_B/dt$
D. $\oint \vec{B} \cdot d\vec{s} = \mu_0 i + \mu_0 \epsilon_0 d\Phi_E/dt$
E. None of these

ans: D

19. Which of the following equations can be used to show that magnetic field lines form closed loops?
A. $\oint \vec{E} \cdot d\vec{A} = q/\epsilon_0$
B. $\oint \vec{B} \cdot d\vec{A} = 0$
C. $\oint \vec{E} \cdot d\vec{s} = -d\Phi_B/dt$
D. $\oint \vec{B} \cdot d\vec{s} = \mu_0 i + \mu_0 \epsilon_0 d\Phi_E/dt$
E. None of these

ans: B

20. Which of the following equations, along with a symmetry argument, can be used to calculate the magnetic field produced by a uniform time-varying electric field?
A. $\oint \vec{E} \cdot d\vec{A} = q/\epsilon_0$
B. $\oint \vec{B} \cdot d\vec{A} = 0$
C. $\oint \vec{E} \cdot d\vec{s} = -d\Phi_B/dt$
D. $\oint \vec{B} \cdot d\vec{s} = \mu_0 i + \mu_0 \epsilon_0 d\Phi_E/dt$
E. None of these

ans: D

21. Which of the following equations, along with a symmetry argument, can be used to calculate the electric field produced by a uniform time-varying magnetic field?
A. $\oint \vec{E} \cdot d\vec{A} = q/\epsilon_0$
B. $\oint \vec{B} \cdot d\vec{A} = 0$
C. $\oint \vec{E} \cdot d\vec{s} = -d\Phi_B/dt$
D. $\oint \vec{B} \cdot d\vec{s} = \mu_0 i + \mu_0 \epsilon_0 d\Phi_E/dt$
E. None of these

ans: C

22. Which of the following equations, along with a symmetry argument, can be used to calculate the magnetic field between the plates of a charging parallel plate capacitor with circular plates?
A. $\oint \vec{E} \cdot d\vec{A} = q/\epsilon_0$
B. $\oint \vec{B} \cdot d\vec{A} = 0$
C. $\oint \vec{E} \cdot d\vec{s} = -d\Phi_B/dt$
D. $\oint \vec{B} \cdot d\vec{s} = \mu_0 i + \mu_0 \epsilon_0 d\Phi_E/dt$
E. None of these

ans: D

23. Maxwell's equations, along with an appropriate symmetry argument, can be used to calculate:
    A. the electric force on a given charge
    B. the magnetic force on a given moving charge
    C. the flux of a given electric field
    D. the flux of a given magnetic field
    E. none of these

    ans: E

24. The polarity of an unmarked magnet can be determined using:
    A. a charged glass rod
    B. a compass
    C. an electroscope
    D. another unmarked magnet
    E. iron filings

    ans: B

25. A bar magnet is placed vertically with its S pole up and its N pole down. Its $\vec{B}$ field at its center is:
    A. zero
    B. down
    C. up due to the weight of the magnet
    D. horizontal
    E. slightly below the horizontal

    ans: B

26. A bar magnet is broken in half. Each half is broken in half again, etc. The observation is that each piece has both a north and south pole. This is usually explained by:
    A. Ampere's theory that all magnetic phenomena result from electric currents
    B. our inability to divide the magnet into small enough pieces
    C. Coulomb's law
    D. Lenz' law
    E. conservation of charge.

    ans: A

27. A small bar magnet is suspended horizontally by a string. When placed in a uniform horizontal magnetic field, it will:
    A. translate in the direction of $\vec{B}$
    B. translate in the opposite direction of $\vec{B}$
    C. rotate so as to be at right angles to $\vec{B}$
    D. rotate so as to be vertical
    E. none of the above

    ans: E

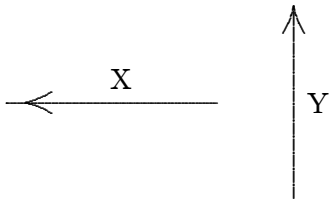
28. Magnetic dipole X is fixed and dipole Y is free to move. Dipole Y will initially:




A. move toward X but not rotate
B. move away from X but not rotate
C. move toward X and rotate
D. move away from X and rotate
E. rotate but not translate

ans: A

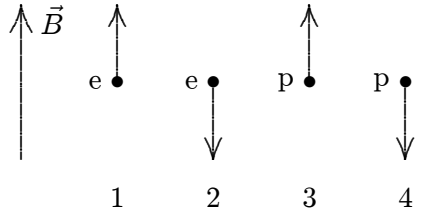
29. Magnetic dipole X is fixed and dipole Y is free to move. Dipole Y will initially:




A. move toward X but not rotate
B. move away from X but not rotate
C. move toward X and rotate
D. move away from X and rotate
E. rotate but not move toward or away from X

ans: E

30. The diagram shows the angular momentum vectors of two electrons and two protons in the same external magnetic field. The field points upward in the diagram. Rank the situations according to the potential energy, least to greatest.




A. 1 and 3 tie, then 2 and 4 tie
B. 2 and 3 tie, then 1 and 4 tie
C. 1 and 2 tie, then 3 and 4 tie
D. 3 and 4 tie, then 1 and 2 tie
E. all tie

ans: B

31. The energy of a magnetic dipole in an external magnetic field is least when:
    A. the dipole moment is parallel to the field
    B. the dipole moment is antiparallel to the field
    C. the dipole moment is perpendicular to the field
    D. none of the above (the same energy is associated with all orientations)
    E. none of the above (no energy is associated with the dipole-field interaction)

    ans: A

32. The magnetic properties of materials stem chiefly from:
    A. particles with north poles
    B. particles with south poles
    C. motions of protons within nuclei
    D. proton spin angular momentum
    E. electron magnetic dipole moments

    ans: E

33. Magnetization is:
    A. the current density in an object
    B. the charge density of moving charges in an object
    C. the magnetic dipole moment of an object
    D. the magnetic dipole moment per unit volume of an object
    E. the magnetic field per unit volume produced by an object

    ans: D

34. The units of magnetization are:
    A. ampere
    B. ampere·meter
    C. ampere·meter$^2$
    D. ampere/meter
    E. ampere/meter$^2$

    ans: D

35. If $\vec{L}$ is the orbital angular momentum of an electron, the magnetic dipole moment associated with its orbital motion:
    A. is in the direction of $\vec{L}$ and has magnitude proportional to $L$
    B. is opposite to the direction of $\vec{L}$ and has magnitude proportional to $L$
    C. is in the direction of $\vec{L}$ and has magnitude proportional to $L^2$
    D. is opposite to the direction of $\vec{L}$ and has magnitude proportional to $L^2$
    E. does not depend on $\vec{L}$

    ans: B

36. If an electron has an orbital angular momentum with magnitude $L$ the magnitude of the orbital contribution to its magnetic dipole moment is given by:

A. $eL/m$
B. $eL/2m$
C. $2eL/m$
D. $mL/e$
E. $mL/2$

ans: B

37. An electron traveling with speed $v$ around a circle of radius $r$ is equivalent to a current of:

A. $evr/2$
B. $ev/r$
C. $ev/2\pi r$
D. $2\pi er/v$
E. $2\pi ev/r$

ans: C

38. The intrinsic magnetic dipole moments of protons and neutrons are much less than that of an electron because:

A. their masses are greater
B. their angular momenta are much less
C. their angular momenta are much greater
D. their charges are much less
E. their radii are much less

ans: A

39. The spin magnetic dipole moment of an electron:

A. is in the same direction as the spin angular momentum
B. is zero
C. has a magnitude that depends on the orbital angular momentum
D. has a magnitude that depends on the applied magnetic field
E. none of the above

ans: E

40. If an electron has zero orbital angular momentum, the magnitude of its magnetic dipole moment equals:

A. zero
B. half the Bohr magneton
C. a Bohr magneton
D. twice a Bohr magneton
E. none of these

ans: C

41. The magnetic dipole moment of an atomic electron is typically:
    A. much less than a Bohr magneton
    B. a few Bohr magnetons
    C. much greater than a Bohr magneton
    D. much greater or much less than a Bohr magneton, depending on the atom
    E. not related to the value of the Bohr magneton

    ans: B

42. The magnitude of the Bohr magneton in J/T is about:
    A. $10^{-15}$
    B. $10^{-19}$
    C. $10^{-23}$
    D. $10^{-27}$
    E. $10^{-31}$

    ans: C

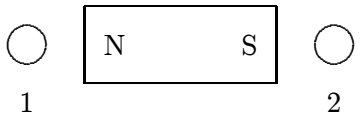
43. The molecular theory of magnetism can explain each of the following EXCEPT:
    A. an N pole attracts a S pole
    B. stroking an iron bar with a magnet will magnetize the bar
    C. when a bar magnet is broken in two, each piece is a bar magnet
    D. heating tends to destroy magnetization
    E. hammering tends to destroy magnetization

    ans: A

44. Lenz' law can explain:
    A. paramagnetism only
    B. diamagnetism only
    C. ferromagnetism only
    D. only two of the three types of magnetism
    E. all three of the types of magnetism

    ans: B

45. The diagram shows two small diamagnetic spheres, one near each end of a bar magnet. Which of the following statements is true?



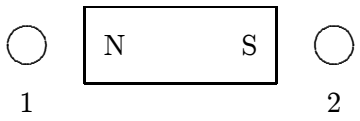

    A. The force on 1 is toward the magnet and the force on 2 is away from the magnet
    B. The force on 1 is away from the magnet and the force on 2 is away from the magnet
    C. The forces on 1 and 2 are both toward the magnet
    D. The forces on 1 and 2 are both away from the magnet
    E. The magnet does not exert a force on either sphere

    ans: D

46. Paramagnetism is closely associated with:
    A. the tendency of electron dipole moments to align with an applied magnetic field
    B. the tendency of electron dipole moments to align opposite to an applied magnetic field
    C. the exchange force between electrons
    D. the force exerted by electron dipole moments on each other
    E. the torque exerted by electron dipole moments on each other

    ans: A

47. The diagram shows two small paramagnetic spheres, one near each end of a bar magnet. Which of the following statements is true?




    A. The force on 1 is toward the magnet and the force on 2 is away from the magnet
    B. The force on 1 is away from the magnet and the force on 2 is away from the magnet
    C. The forces on 1 and 2 are both toward the magnet
    D. The forces on 1 and 2 are both away from the magnet
    E. The magnet does not exert a force on either sphere

    ans: C

48. A paramagnetic substance is placed in a weak magnetic field and its absolute temperature $T$ is increased. As a result, its magnetization:
    A. increases in proportion to $T$
    B. increases in proportion to $T^2$
    C. remains the same
    D. decreases in proportion to $1/T$
    E. decreases in proportion to $1/T^2$

    ans: D

49. A magnetic field $\vec{B}_0$ is applied to a paramagnetic substance. In the interior the magnetic field produced by the magnetic dipoles of the substance is:
    A. greater than $\vec{B}_0$ and in the opposite direction
    B. less than $\vec{B}_0$ and in the opposite direction
    C. greater than $\vec{B}_0$ and in the same direction
    D. less than $\vec{B}_0$ and in the same direction
    E. the same as $\vec{B}_0$

    ans: D

50. A paramagnetic substance, in an external magnetic field, is thermally isolated. The field is then removed. As a result:
    A. the magnetic energy of the magnetic dipoles decreases
    B. the temperature of the substance increases
    C. the magnetization decreases, but only slightly
    D. the magnetization reverses direction
    E. none of the above

    ans: E

51. A magnetic field $\vec{B}_0$ is applied to a diamagnetic substance. In the interior the magnetic field produced by the magnetic dipoles of the substance is:
    A. greater than $\vec{B}_0$ and in the opposite direction
    B. less than $\vec{B}_0$ and in the opposite direction
    C. greater than $\vec{B}_0$ and in the same direction
    D. less than $\vec{B}_0$ and in the same direction
    E. the same as $\vec{B}_0$

    ans: B

52. Ferromagnetism is closely associated with:
    A. the tendency of electron dipole moments to align with an applied magnetic field
    B. the tendency of electron dipole moments to align opposite to an applied magnetic field
    C. the tendency of electron dipole moments to change magnitude in an applied magnetic field
    D. the tendency of electron dipole moments to align with each other
    E. the force exerted by electron dipole moments on each other

    ans: D

53. Of the three chief kinds of magnetic materials (diamagnetic, paramagnetic, and ferromagnetic), which are used to make permanent magnets?
    A. Only diamagnetic
    B. Only ferromagnetic
    C. Only paramagnetic
    D. Only paramagnetic and ferromagnetic
    E. All three

    ans: B

54. When a permanent magnet is strongly heated:
    A. nothing happens
    B. it becomes an induced magnet
    C. it loses its magnetism
    D. its magnetism increases
    E. its polarity reverses

    ans: C

55. Magnetization vectors in neighboring ferromagnetic domains are:
    A. always in opposite directions
    B. always in the same direction
    C. always in different directions
    D. sometimes in different directions and sometimes in the same direction
    E. sometimes in opposite directions and sometimes in the same direction

    ans: C

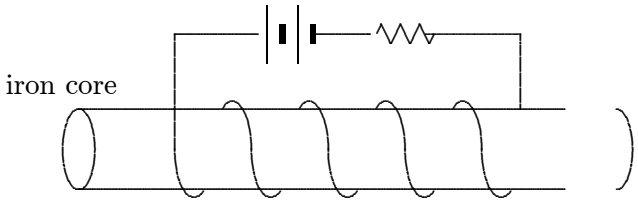
56. The behavior of ferromagnetic domains in an applied magnetic field gives rise to:
    A. hysteresis
    B. ferromagnetism
    C. the Curie law
    D. a lowering of the Curie temperature
    E. Gauss' law for magnetism

    ans: A

57. Because ferromagnets have ferromagnetic domains, the net magnetization:
    A. can never be in the same direction as an applied field
    B. may not vanish when an applied field is reduced to zero
    C. can never vanish
    D. is proportional to any applied magnetic field
    E. is always opposite to the direction of any applied magnetic field

    ans: B

58. The soft iron core in the solenoid shown is removable. Then:




    A. the current will be larger without the core
    B. the current will be larger with the core
    C. one must do work to remove the core
    D. the circuit will do work in expelling the core
    E. the stored energy is the same with or without the core

    ans: C

59. An unmagnetized steel bar is placed inside a solenoid. As the current in the solenoid is slowly increased from zero to some large value, the magnetization of the bar:
    A. increases proportionally with the current
    B. remains zero for awhile and then increases linearly with any further increase in current
    C. increases with increasing current at first but later is much less affected by it
    D. is unaffected by the current
    E. increases quadratically with the current

    ans: C

60. The magnetic field of Earth is roughly the same as that of a magnetic dipole with a dipole moment of about:

A. $10^{17}$ J/T
B. $10^{19}$ J/T
C. $10^{21}$ J/T
D. $10^{23}$ J/T
E. $10^{25}$ J/T

ans: D

61. Of the following places, one would expect that the horizontal component of Earth's magnetic field is largest in:

A. Maine
B. Florida
C. Maryland
D. New York
E. Iowa

ans: B

62. A positively charged ion, due to a cosmic ray, is headed through Earth's atmosphere toward the center of Earth. Due to Earth's magnetic field, the ion will be deflected:

A. south
B. north
C. west
D. east
E. not at all since it is a charge and not a pole

ans: D

63. Maxwell's great contribution to electromagnetic theory was his hypothesis that:

A. work is required to move a magnetic pole through a closed path surrounding a current
B. a time-varying electric flux acts as a current for purposes of producing a magnetic field
C. the speed of light could be determined from simple electrostatic and magnetostatic experiments (finding the values of $\mu_0$ and $\epsilon_0$)
D. the magnetic force on a moving charge particle is perpendicular to both $\vec{v}$ and $\vec{B}$
E. magnetism could be explained in terms of circulating currents in atoms

ans: B

64. Displacement current is:

A. $d\Phi_E/dt$
B. $\epsilon_0 d\Phi_E/dt$
C. $\mu_0 d\Phi_E/dt$
D. $\mu_0\epsilon_0 d\Phi_E/dt$
E. $-d\Phi_B/dt$

ans: B

65. Displacement current exists wherever there is:

A. a magnetic field
B. moving charge
C. a changing magnetic field
D. an electric field
E. a changing electric field

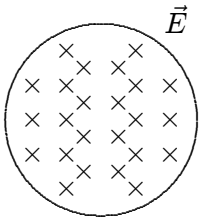
ans: E

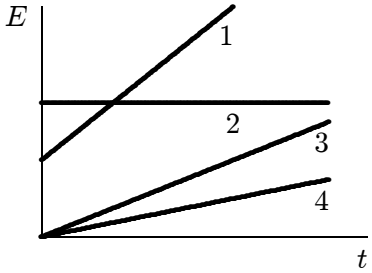
66. Displacement current exists in the region between the plates of a parallel plate capacitor if:

A. the capacitor leaks charge across the plates
B. the capacitor is being discharged
C. the capacitor is fully charged
D. the capacitor is fully discharged
E. none of the above are true

ans: B

67. An electric field exists in the cylindrical region shown and is parallel to the cylinder axis. The magnitude of the field might vary with time according to any of the four graphs shown. Rank the four variations according to the magnitudes of the magnetic field induced at the edge of the region, least to greatest.

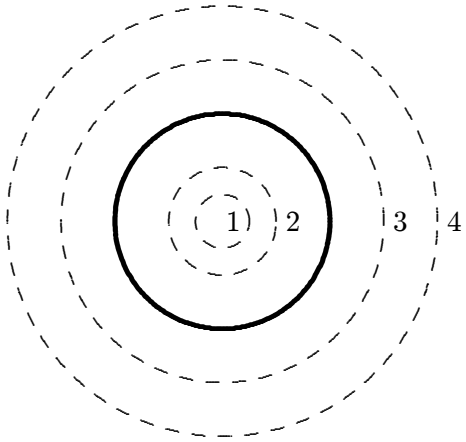



A. 2, 4, 3, 1
B. 3 and 4 tie, then 1, 2
C. 4, 3, 2, 1
D. 4, 3, 1, 2
E. 2, 1, 3, 4

ans: A

68. The diagram shows one plate of a parallel-plate capacitor from within the capacitor. The plate is circular and has radius $R$. The dashed circles are four integration paths and have radii of $r_1 = R/4$, $r_2 = R/2$, $r_3 = 3R/2$, and $r_4 = 2R$. Rank the paths according to the magnitude of $\oint \vec{B} \cdot \mathrm{d}\vec{s}$ around the paths during the discharging of the capacitor, least to greatest.




A. 1, 2 and 3 tie, then 4
B. 1, 2, 3, 4
C. 1, 2, 3 and 4 tie
D. 4, 3, 1, 2
E. all tie

ans: C

69. A 1.2-m radius cylindrical region contains a uniform electric field along the cylinder axis. It is increasing uniformly with time. To obtain a total displacement current of $2.0 \times 10^{-9}$ A through a cross section of the region, the magnitude of the electric field should change at a rate of:

A. $5.0\,\mathrm{V/m \cdot s}$
B. $12\,\mathrm{V/m \cdot s}$
C. $37\,\mathrm{V/m \cdot s}$
D. $50\,\mathrm{V/m \cdot s}$
E. $4.0 \times 10^7\,\mathrm{V/m \cdot s}$

ans: D

70. A current of 1 A is used to charge a parallel plate capacitor with square plates. If the area of each plate is $0.6\,\mathrm{m}^2$ the displacement current through a $0.3\,\mathrm{m}^2$ area wholly between the capacitor plates and parallel to them is:

A. 1 A
B. 2 A
C. 0.7 A
D. 0.5 A
E. 0.25 A

ans: D

71. A 1-$\mu$F capacitor is connected to an emf that is increasing uniformly with time at a rate of 100 V/s. The displacement current between the plates is:

A. 0
B. $1 \times 10^{-8}$ A
C. $1 \times 10^{-6}$ A
D. $1 \times 10^{-4}$ A
E. 100 A

ans: D

72. A magnetic field exists between the plates of a capacitor:

A. always
B. never
C. when the capacitor is fully charged
D. while the capacitor is being charged
E. only when the capacitor is starting to be charged

ans: D

73. Suppose you are looking into one end of a long cylindrical tube in which there is a uniform electric field, pointing away from you. If the magnitude of the field is decreasing with time the direction of the induced magnetic field is:

A. toward you
B. away from you
C. clockwise
D. counterclockwise
E. to your right

ans: D

74. Suppose you are looking into one end of a long cylindrical tube in which there is a uniform electric field, pointing away from you. If the magnitude of the field is decreasing with time the field lines of the induced magnetic field are:

A. circles
B. ellipses
C. straight lines parallel to the electric field
D. straight lines perpendicular to the electric field
E. none of the above

ans: A

75. A cylindrical region contains a uniform electric field that is along the cylinder axis and is changing with time. If $r$ is distance from the cylinder axis the magnitude of the magnetic field within the region is:

A. uniform
B. proportional to $1/r$
C. proportional to $r^2$
D. proportional to $1/r^2$
E. proportional to $r$

ans: E

76. A cylindrical region contains a uniform electric field that is parallel to the axis and is changing with time. If $r$ is distance from the cylinder axis the magnitude of the magnetic field outside the region is:

A. 0
B. proportional to $1/r$
C. proportional to $r^2$
D. proportional to $1/r^2$
E. proportional to $r$

ans: B

77. A 0.70-m radius cylindrical region contains a uniform electric field that is parallel to the axis and is decreasing at the rate $5.0 \times 10^{12}\,\mathrm{V/m \cdot s}$. The magnetic field at a point 0.25 m from the axis has a magnitude of:

A. 0
B. $7.0 \times 10^{-6}\,\mathrm{T}$
C. $2.8 \times 10^{-5}\,\mathrm{T}$
D. $5.4 \times 10^{-5}\,\mathrm{T}$
E. $7.0 \times 10^{-5}\,\mathrm{T}$

ans: B

78. A 0.70-m radius cylindrical region contains a uniform electric field that is parallel to the axis and is decreasing at the rate $5.0 \times 10^{12}\,\mathrm{V/m \cdot s}$. The magnetic field at a point 1.2 m from the axis has a magnitude of:

A. 0
B. $7.0 \times 10^{-6}\,\mathrm{T}$
C. $1.1 \times 10^{-5}\,\mathrm{T}$
D. $2.3 \times 10^{-5}\,\mathrm{T}$
E. $2.8 \times 10^{-5}\,\mathrm{T}$

ans: C

79. A 1-A current is used to charge a parallel plate capacitor. A large square piece of paper is placed between the plates and parallel to them so it sticks out on all sides. The value of the integral $\oint \vec{B} \cdot d\vec{s}$ around the perimeter of the paper is:

A. $2\,\mathrm{T \cdot m}$
B. $4\pi \times 10^{-7}\,\mathrm{T \cdot m}$
C. $8.85 \times 10^{-12}\,\mathrm{T \cdot m}$
D. $10^{-7}\,\mathrm{T \cdot m}$
E. not determined from the given quantities

ans: B

80. A sinusoidal emf is connected to a parallel plate capacitor. The magnetic field between the plates is:

A. 0
B. constant
C. sinusoidal and its amplitude does not depend on the frequency of the source
D. sinusoidal and its amplitude is proportional to the frequency of the source
E. sinusoidal and its amplitude is inversely proportional to the frequency of the source

ans: D

81. An electron is on the $z$ axis moving toward the $xy$ plane but it has not reached that plane yet. At that instant:

A. there is only a true current through the $xy$ plane
B. there is only a displacement current through the $xy$ plane
C. there are both true and displacement currents through the $xy$ plane
D. there is neither a true nor a displacement current through the $xy$ plane
E. none of the above are true

ans: B

## Chapter 33: ELECTROMAGNETIC WAVES

1. Select the correct statement:
   A. ultraviolet light has a longer wavelength than infrared
   B. blue light has a higher frequency than $x$ rays
   C. radio waves have higher frequency than gamma rays
   D. gamma rays have higher frequency than infrared waves
   E. electrons are a type of electromagnetic wave

   ans: D

2. Consider: radio waves (r), visible light (v), infrared light (i), x-rays (x), and ultraviolet light (u). In order of increasing frequency, they are:
   A. r, v, i, x, u
   B. r, i, v, u, x
   C. i, r, v, u, x
   D. i, v, r, u, x
   E. r, i, v, x, u

   ans: B

3. The order of increasing wavelength for blue (b), green (g), red (r), and yellow (y) light is:
   A. r, y, g, b
   B. r, g, y, b
   C. g, y, b, r
   D. b, g, y, r
   E. b, y, g, r

   ans: D

4. Of the following human eyes are most sensitive to:
   A. red light
   B. violet light
   C. blue light
   D. green light
   E. none of these (they are equally sensitive to all colors)

   ans: D

5. Which of the following is NOT true for electromagnetic waves?
   A. they consist of changing electric and magnetic fields
   B. they travel at different speeds in vacuum, depending on their frequency
   C. they transport energy
   D. they transport momentum
   E. they can be reflected

   ans: B

6. The product $\mu_0\epsilon_0$ has the same units as:
   A. $(\text{velocity})^2$
   B. $(\text{velocity})^{1/2}$
   C. $1/\text{velocity}$
   D. $1/\text{velocity}^2$
   E. $1/\text{velocity}^{1/2}$

   ans: D

7. Maxwell's equations predict that the speed of electromagnetic waves in free space is given by:
   A. $\mu_0\epsilon_0$
   B. $(\mu_0\epsilon_0)^{1/2}$
   C. $1/\mu_0\epsilon_0$
   D. $1/(\mu_0\epsilon_0)^{1/2}$
   E. $1/(\mu_0\epsilon_0)^2$

   ans: D

8. Maxwell's equations predict that the speed of light in free space is
   A. an increasing function of frequency
   B. a decreasing function of frequency
   C. independent of frequency
   D. a function of the distance from the source
   E. a function of the size of the source

   ans: C

9. The speed of light in vacuum is about:
   A. 1100 ft/s
   B. $93 \times 10^6$ m/s
   C. $6 \times 10^{23}$ m/s
   D. $3 \times 10^{10}$ cm/s
   E. 186,000 mph

   ans: D

10. The Sun is about $1.5 \times 10^{11}$ m away. The time for light to travel this distance is about:
   A. $4.5 \times 10^{18}$ s
   B. 8 s
   C. 8 min
   D. 8 hr
   E. 8 yr

   ans: C

11. The time for a radar signal to travel to the Moon and back, a one-way distance of about $3.8 \times 10^8$ m, is:

A. 1.3 s
B. 2.5 s
C. 8 s
D. 8 min
E. $1 \times 10^6$ s

ans: B

12. Which of the following types of electromagnetic radiation travels at the greatest speed in vacuum?

A. Radio waves
B. Visible light
C. X rays
D. Gamma rays
E. All of these travel at the same speed

ans: E

13. Radio waves differ from visible light waves in that radio waves:

A. travel slower
B. have a higher frequency
C. travel faster
D. have a lower frequency
E. require a material medium

ans: D

14. Visible light has a frequency of about:

A. $5 \times 10^{18}$ Hz
B. $5 \times 10^{16}$ Hz
C. $5 \times 10^{14}$ Hz
D. $5 \times 10^{12}$ Hz
E. $5 \times 10^{10}$ Hz

ans: C

15. The theoretical upper limit for the frequency of electromagnetic waves is:

A. just slightly greater than that of red light
B. just slightly less than that of blue light
C. the greatest x-ray frequency
D. none of the above (there is no upper limit)
E. none of the above (but there is an upper limit)

ans: D

16. Radio waves of wavelength 3 cm have a frequency of:
    A. 1 MHz
    B. 9 MHz
    C. 100 MHz
    D. 10,000 MHz
    E. 900 MHz

    ans: D

17. Radio waves of wavelength 300 m have a frequency of:
    A. $10^{-3}$ kHz
    B. 500 kHz
    C. 1 MHz
    D. 9 MHz
    E. 108 kHz

    ans: C

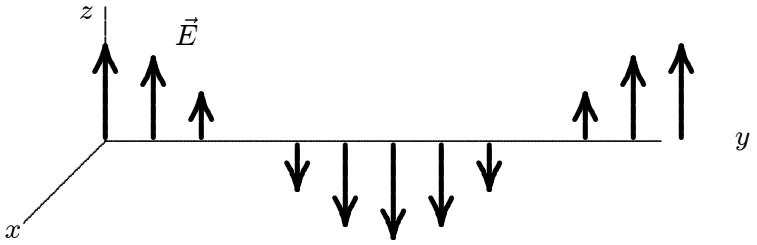
18. If the electric field in a plane electromagnetic wave is given by $E_m \sin[(3 \times 10^6\ \mathrm{m}^{-1})x - \omega t]$, the value of $\omega$ is:
    A. 0.01 rad/s
    B. 10 rad/s
    C. 100 rad/s
    D. $9 \times 10^{14}$ rad/s
    E. $9 \times 10^{16}$ rad/s

    ans: D

19. An electromagnetic wave is generated by:
    A. any moving charge
    B. any accelerating charge
    C. only a charge with changing acceleration
    D. only a charge moving in a circle
    E. only a charge moving in a straight line

    ans: B

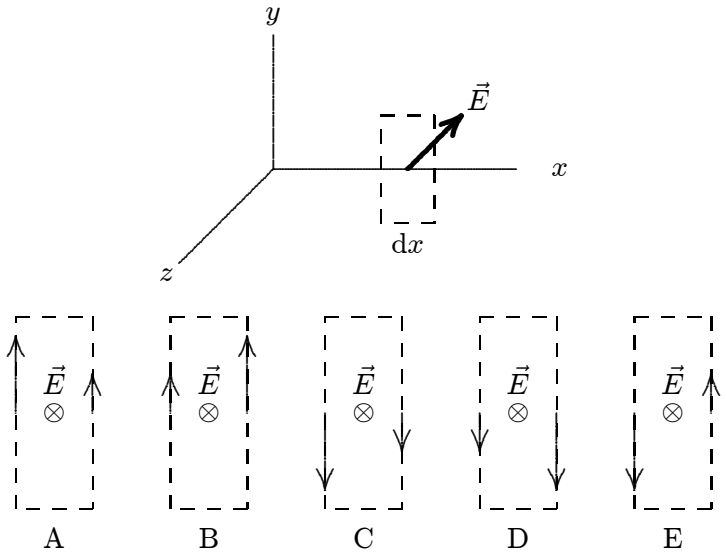
20. The electric field for a plane electromagnetic wave traveling in the $+y$ direction is shown. Consider a point where $\vec{E}$ is in the $+z$ direction. The $\vec{B}$ field is:




A. in the $+x$ direction and in phase with the $\vec{E}$ field
B. in the$-x$ direction and in phase with the $\vec{E}$ field
C. in the $+x$ direction and one-fourth of a cycle out of phase with the $\vec{E}$ field
D. in the $+z$ direction and in phase with the $\vec{E}$ field
E. in the $+z$ direction and one-fourth of a cycle out of phase with the $\vec{E}$ field

ans: A

21. A plane electromagnetic wave is traveling in the positive $x$ direction. At the instant shown the electric field at the extremely narrow dashed rectangle is in the negative $z$ direction and its magnitude is decreasing. Which diagram correctly shows the directions and relative magnitudes of the magnetic field at the edges of the rectangle?




ans: B

22. In a plane electromagnetic wave in vacuum, the ratio $E/B$ of the amplitudes in SI units of the two fields is:

A. the speed of light
B. an increasing function of frequency
C. a decreasing function of frequency
D. $\sqrt{2}$
E. $1/\sqrt{2}$

ans: A

23. If the magnetic field in a plane electromagnetic wave is along the $y$ axis and its component is given by $B_m \sin(kx - \omega t)$, in SI units, then the electric field is along the $z$ axis and its component is given by:

A. $(cB_m)\cos(kx - \omega t)$
B. $-(cB_m/c)\cos(kx - \omega t)$
C. $-(cB_m/c)\sin(kx - \omega t)$
D. $B_m \cos(kx - \omega t)$
E. $(cB_m/c)\sin(kx - \omega t)$

ans: E

24. If the electric field in a plane electromagnetic wave is along the $y$ axis and its component is given by $E_m \sin(kx + \omega t)$, in SI units, then the magnetic field is along the $z$ axis and its component is given by:

A. $(E_m/c)\cos(kx + \omega t)$
B. $-(E_m/c)\cos(kx + \omega t)$
C. $-(E_m/c)\sin(kx + \omega t)$
D. $E_m \cos(kx + \omega t)$
E. $(E_m/c)\sin(kx + \omega t)$

ans: C

25. An electromagnetic wave is traveling in the positive $x$ direction with its electric field along the $z$ axis and its magnetic field along the $y$ axis. The fields are related by:

A. $\partial E/\partial x = \mu_0 \epsilon_0 \partial B/\partial x$
B. $\partial E/\partial x = \mu_0 \epsilon_0 \partial B/\partial t$
C. $\partial B/\partial x = \mu_0 \epsilon_0 \partial E/\partial x$
D. $\partial B/\partial x = \mu_0 \epsilon_0 \partial E/\partial t$
E. $\partial B/\partial x = -\mu_0 \epsilon_0 \partial E/\partial t$

ans: E

26. If the amplitude of the electric field in a plane electromagnetic wave is 100 V/m then the amplitude of the magnetic field is:

A. $3.3 \times 10^{-7}$ T
B. $6.7 \times 10^{-7}$ T
C. 0.27 T
D. $8.0 \times 10^{7}$ T
E. $3.0 \times 10^{9}$ T

ans: A

27. For an electromagnetic wave the direction of the vector $\vec{E} \times \vec{B}$ gives:

A. the direction of the electric field
B. the direction of the magnetic field
C. the direction of wave propagation
D. the direction of the electromagnetic force on a proton
E. the direction of the emf induced by the wave

ans: C

28. The dimensions of $\vec{S} = (1/\mu_0)\vec{E} \times \vec{B}$ are:
    A. $\mathrm{J/m^2}$
    B. J/s
    C. W/s
    D. $\mathrm{W/m^2}$
    E. $\mathrm{J/m^3}$

    ans: D

29. The time-averaged energy in a sinusoidal electromagnetic wave is:
    A. overwhelmingly electrical
    B. slightly more electrical than magnetic
    C. equally divided between the electric and magnetic fields
    D. slightly more magnetic than electrical
    E. overwhelmingly magnetic

    ans: C

30. At a certain point and a certain time the electric field of an electromagnetic wave is in the negative $z$ direction and the magnetic field is in the positive $y$ direction. Which of the following statements is true?
    A. Energy is being transported in the positive $x$ direction but half a cycle later, when the electric field is in the opposite direction, it will be transported in the negative $x$ direction
    B. Energy is being transported in the positive $x$ direction and half a cycle later, when the electric field is in the opposite direction, it will still be transported in the positive $x$ direction
    C. Energy is being transported in the negative $x$ direction but half a cycle later, when the electric field is in the opposite direction, it will be transported in the positive $x$ direction
    D. Energy is being transported in the negative $x$ direction and half a cycle later, when the electric field is in the opposite direction, it will still be transported in the negative $x$ direction
    E. None of the above are true

    ans: B

31. An electromagnetic wave is transporting energy in the negative $y$ direction. At one point and one instant the magnetic field is in the positive $x$ direction. The electric field at that point and instant is:
    A. positive $y$ direction
    B. negative $y$ direction
    C. positive $z$ direction
    D. negative $z$ direction
    E. negative $x$ direction

    ans: D

32. A point source emits electromagnetic energy at a rate of 100 W. The intensity 10 m from the source is:
A. $10\,\mathrm{W/m^2}$
B. $1.6\,\mathrm{W/m^2}$
C. $1\,\mathrm{W/m^2}$
D. $0.024\,\mathrm{W/m^2}$
E. $0.080\,\mathrm{W/m^2}$
ans: E

33. The light intensity 10 m from a point source is $1000\,\mathrm{W/m^2}$. The intensity 100 m from the same source is:
A. $1000\,\mathrm{W/m^2}$
B. $100\,\mathrm{W/m^2}$
C. $10\,\mathrm{W/m^2}$
D. $1\,\mathrm{W/m^2}$
E. $0.1\,\mathrm{W/m^2}$
ans: C

34. When the distance between a point source of light and a light meter is reduced from 6.0 m to 2.0 m, the intensity of illumination at the meter will be the original value multiplied by:
A. 3
B. 9
C. 1/3
D. 1/9
E. 1
ans: A

35. The magnetic field in a sinusoidal light wave has an amplitude of $3.3 \times 10^{-7}$ T. The intensity of the wave is:
A. $1.7 \times 10^{-4}\,\mathrm{W/m^2}$
B. $13\,\mathrm{W/m^2}$
C. $27\,\mathrm{W/m^2}$
D. $1.0 \times 10^{5}\,\mathrm{W/m^2}$
E. $4.0 \times 10^{10}\,\mathrm{W/m^2}$
ans: B

36. A sinusoidal electromagnetic wave with an electric field amplitude of 100 V/m is incident normally on a surface with an area of $1\,\mathrm{cm^2}$ and is completely absorbed. The energy absorbed in 10 s is:
A. 1.3 mJ
B. 13 mJ
C. 27 mJ
D. 130 mJ
E. 270 mJ
ans: B

37. Evidence that electromagnetic waves carry momentum is:
    A. the tail of a comet points away from the Sun
    B. electron flow through a wire generates heat
    C. a charged particle in a magnetic field moves in a circular orbit
    D. heat can be generated by rubbing two sticks together
    E. the Doppler effect

    ans: A

38. Light of uniform intensity shines perpendicularly on a totally absorbing surface, fully illuminating the surface. If the area of the surface is decreased:
    A. the radiation pressure increases and the radiation force increases
    B. the radiation pressure increases and the radiation force decreases
    C. the radiation pressure stays the same and the radiation force increases
    D. the radiation pressure stays the same and the radiation force decreases
    E. the radiation pressure decreases and the radiation force decreases

    ans: D

39. Light with an intensity of $1\,\mathrm{kW/m^2}$ falls normally on a surface and is completely absorbed. The radiation pressure is:
    A. $1\,\mathrm{kPa}$
    B. $3 \times 10^{11}\,\mathrm{Pa}$
    C. $1.7 \times 10^{-6}\,\mathrm{Pa}$
    D. $3.3 \times 10^{-6}\,\mathrm{Pa}$
    E. $6.7 \times 10^{-6}\,\mathrm{Pa}$

    ans: D

40. Light with an intensity of $1\,\mathrm{kW/m^2}$ falls normally on a surface and is completely reflected. The radiation pressure is:
    A. $1\,\mathrm{kPa}$
    B. $3 \times 10^{11}\,\mathrm{Pa}$
    C. $1.7 \times 10^{-6}\,\mathrm{Pa}$
    D. $3.3 \times 10^{-6}\,\mathrm{Pa}$
    E. $6.7 \times 10^{-6}\,\mathrm{Pa}$

    ans: E

41. Light with an intensity of $1\,\mathrm{kW/m^2}$ falls normally on a surface with an area of $1\,\mathrm{cm^2}$ and is completely absorbed. The force of the radiation on the surface is:
    A. $1.0 \times 10^{-4}\,\mathrm{N}$
    B. $3.3 \times 10^{-11}\,\mathrm{N}$
    C. $1.7 \times 10^{-10}\,\mathrm{N}$
    D. $3.3 \times 10^{-10}\,\mathrm{N}$
    E. $6.7 \times 10^{-10}\,\mathrm{N}$

    ans: D

42. Light with an intensity of $1\,\text{kW/m}^2$ falls normally on a surface with an area of $1\,\text{cm}^2$ and is completely reflected. The force of the radiation on the surface is:

A. $1.0 \times 10^{-4}\,\text{N}$
B. $3.3 \times 10^{-11}\,\text{N}$
C. $1.7 \times 10^{-10}\,\text{N}$
D. $3.3 \times 10^{-10}\,\text{N}$
E. $6.7 \times 10^{-10}\,\text{N}$

ans: E

43. A company claims to have developed material that absorbs light energy without a transfer of momentum. Such material is:

A. impossible
B. possible, but very expensive
C. inexpensive and already in common use
D. in use by NASA but is not commercially available
E. a breakthrough in high technology

ans: A

44. Polarization experiments provide evidence that light is:

A. a longitudinal wave
B. a stream of particles
C. a transverse wave
D. some type of wave
E. nearly monochromatic

ans: C

45. A vertical automobile radio antenna is sensitive to electric fields that are polarized:

A. horizontally
B. in circles around the antenna
C. vertically
D. normal to the antenna in the forward direction
E. none of the above

ans: C

46. For linearly polarized light the plane of polarization is:

A. perpendicular to both the direction of polarization and the direction of propagation
B. perpendicular to the direction of polarization and parallel to the direction of propagation
C. parallel to the direction of polarization and perpendicular to the direction of propagation
D. parallel to both the direction of polarization and the direction of propagation
E. none of the above

ans: D

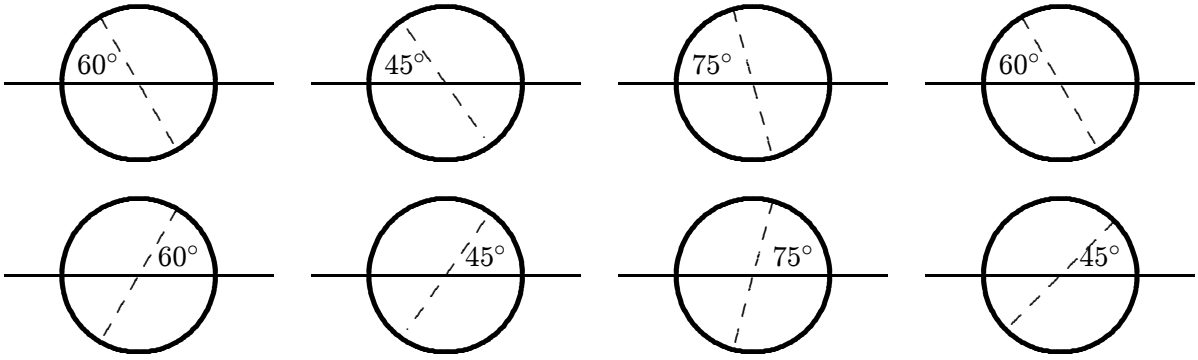
47. Light from any ordinary source (such as a flame) is usually:
    A. unpolarized
    B. plane polarized
    C. circularly polarized
    D. elliptically polarized
    E. monochromatic

    ans: A

48. The electric field in unpolarized light:
    A. has no direction at any time
    B. rotates rapidly
    C. is always parallel to the direction of propagation
    D. changes direction randomly and often
    E. remains along the same line but reverses direction randomly and often

    ans: D

49. The diagrams show four pairs of polarizing sheets, with the polarizing directions indicated by dashed lines. The two sheets of each pair are placed one behind the other and the front sheet is illuminated by unpolarized light. The incident intensity is the same for all pairs of sheets. Rank the pairs according to the intensity of the transmitted light, least to greatest.




    A. 1, 2, 3, 4
    B. 4, 2, 1, 3
    C. 2, 4, 3, 1
    D. 2, 4, 1, 3
    E. 3, 1, 4, 2

    ans: D

50. A clear sheet of polaroid is placed on top of a similar sheet so that their polarizing axes make an angle of 30° with each other. The ratio of the intensity of emerging light to incident unpolarized light is:

A. 1 : 4
B. 1 : 3
C. 1 : 2
D. 3 : 4
E. 3 : 8

ans: E

51. An unpolarized beam of light has intensity $I_0$. It is incident on two ideal polarizing sheets. The angle between the axes of polarization of these sheets is $\theta$. Find $\theta$ if the emerging light has intensity $I_0/4$:

A. $\sin^{-1}(1/2)$
B. $\sin^{-1}(1/\sqrt{5})$
C. $\cos^{-1}(1/2)$
D. $\cos^{-1}(1/\sqrt{2})$
E. $\tan^{-1}(1/4)$

ans: D

52. In a stack of three polarizing sheets the first and third are crossed while the middle one has its axis at 45° to the axes of the other two. The fraction of the intensity of an incident unpolarized beam of light that is transmitted by the stack is:

A. 1/2
B. 1/3
C. 1/4
D. 1/8
E. 0

ans: D

53. Three polarizing sheets are placed in a stack with the polarizing directions of the first and third perpendicular to each other. What angle should the polarizing direction of the middle sheet make with the polarizing direction of the first sheet to obtain maximum transmitted intensity when unpolarized light is incident on the stack?

A. 0
B. 30°
C. 45°
D. 60°
E. 90°

ans: C

54. Three polarizing sheets are placed in a stack with the polarizing directions of the first and third perpendicular to each other. What angle should the polarizing direction of the middle sheet make with the polarizing direction of the first sheet to obtain zero transmitted intensity when unpolarized light is incident on the stack?

A. 0
B. 30°
C. 45°
D. 60°
E. All angles allow light to pass through

ans: A

55. The relation $\theta_{\text{incident}} = \theta_{\text{reflected}}$, which applies as a ray of light strikes an interface between two media, is known as:

A. Faraday's law
B. Snell's law
C. Ampere's law
D. Cole's law
E. none of these

ans: E

56. The relation $n_1 \sin\theta_1 = n_2 \sin\theta_2$, which applies as a ray of light strikes an interface between two media, is known as:

A. Gauss' law
B. Snell's law
C. Faraday's law
D. Cole's law
E. law of sines

ans: B

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57. As used in the laws of reflection and refraction, the "normal" direction is:

A. any convenient direction
B. tangent to the interface
C. along the incident ray
D. perpendicular to the electric field vector of the light
E. perpendicular to the interface

ans: E

58. When an electromagnetic wave meets a reflecting surface, the direction taken by the reflected wave is determined by:

A. the material of the reflecting surface
B. the angle of incidence
C. the index of the medium
D. the intensity of the wave
E. the wavelength

ans: B

59. The index of refraction of a substance is:

A. the speed of light in the substance
B. the angle of refraction
C. the angle of incidence
D. the speed of light in vacuum divided by the speed of light in the substance
E. measured in radians

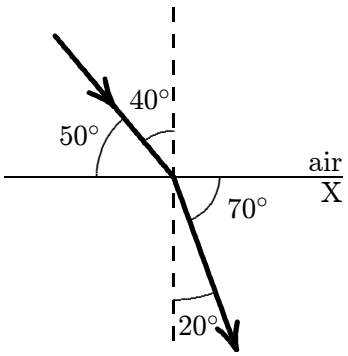
ans: D

60. The units of index of refraction are:

A. m/s
B. s/m
C. radian
D. $m/s^2$
E. none of these

ans: E

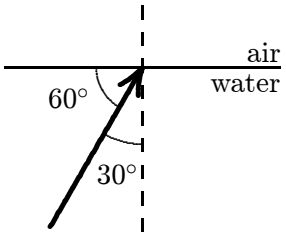
61. The diagram shows the passage of a ray of light from air into a substance X. The index of refraction of X is:




A. 0.53
B. 0.88
C. 1.9
D. 2.2
E. 3.0

ans: C

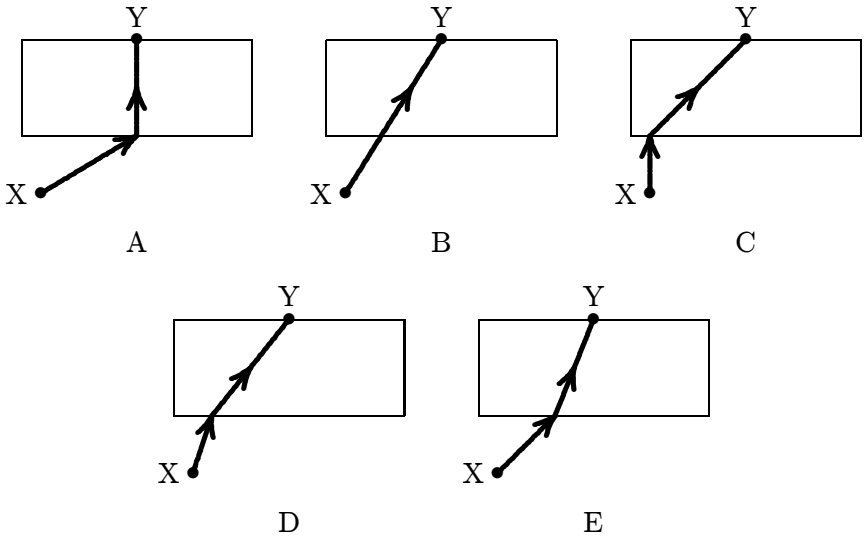
62. If $n_{\text{water}} = 1.33$, what is the angle of refraction for the ray shown?




A. 19°
B. 22°
C. 36°
D. 42°
E. 48°

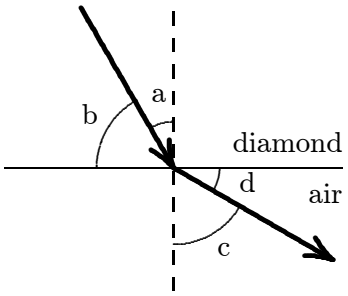
ans: D

63. Which diagram below illustrates the path of a light ray as it travels from a given point X in air to another given point Y in glass?




ans: E

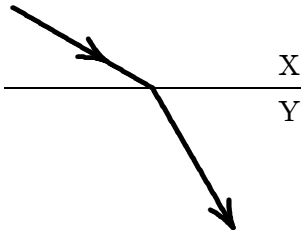
64. The index of refraction for diamond is 2.5. Which of the following is correct for the situation shown?




A. $(\sin a)/(\sin b) = 2.5$
B. $(\sin b)/(\sin d) = 2.5$
C. $(\cos a)/(\cos c) = 2.5$
D. $(\sin c)/(\sin a) = 2.5$
E. $a/c = 2.5$

ans: D

65. When light travels from medium X to medium Y as shown:




A. both the speed and the frequency decrease
B. both the speed and the frequency increase
C. both the speed and the wavelength decrease
D. both the speed and the wavelength increase
E. both the wavelength and the frequency are unchanged

ans: C

66. A ray of light passes obliquely through a plate of glass having parallel faces. The emerging ray:

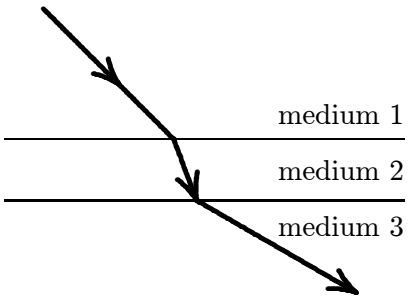
A. is totally internally reflected
B. is bent more toward the normal than the incident ray
C. is bent further away from the normal than the incident ray
D. is parallel to the incident ray but displaced sideways
E. lies on the same straight line as the incident ray

ans: D

67. When light passes from air to glass, it bends:
    A. toward the normal without changing speed
    B. toward the normal and slows down
    C. toward the normal and speeds up
    D. away from the normal and slows down
    E. away from the normal and speeds up

    ans: B

68. A ray of light passes through three media as shown. The speed of light in these media obey:




    A. $v_1 > v_2 > v_3$
    B. $v_3 > v_2 > v_1$
    C. $v_3 > v_1 > v_2$
    D. $v_2 > v_1 > v_3$
    E. $v_1 > v_3 > v_2$

    ans: C

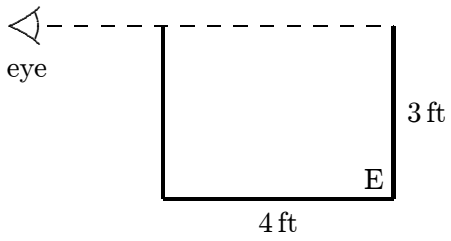
69. As light goes from one medium to another, it is bent away from the normal. Then:
    A. the speed of the light has increased
    B. dispersion must occur
    C. the second medium has a higher index of refraction than the first
    D. no change in speed has occurred
    E. refraction has not occurred because refraction means a bending toward the normal

    ans: A

70. A pole stands in a river, half in and half out of the water. Another pole of the same length stands vertically on the shore at a place where the ground is level. The shadow cast by the pole in the river on the river bottom is:
    A. slightly longer than the shadow of the pole on land
    B. much longer than the shadow of the pole on land
    C. shorter than the shadow of the pole on land
    D. shorter than the shadow of the pole on land if the Sun is high and longer if the sun is low
    E. the same length as the shadow of the pole on land

    ans: C

71. The rectangular metal tank shown is filled with an unknown liquid. The observer, whose eye is level with the top of the tank, can just see corner E. The index of refraction of this liquid is:




A. 1.75
B. 1.67
C. 1.50
D. 1.33
E. 1.25

ans: E

72. The index of refraction of benzene is 1.80. The critical angle for total internal reflection, at a benzene-air interface, is about:

A. 56°
B. 47°
C. 34°
D. 22°
E. 18°

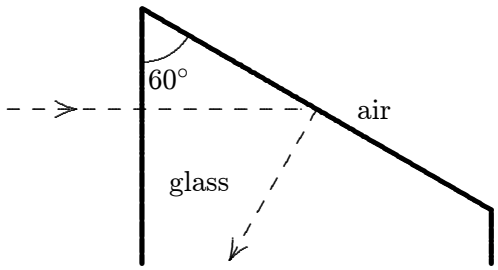
ans: C

73. The index of refraction of a certain glass is 1.50. The sine of the critical angle for total internal reflection at a glass-air interface is:

A. 0.50
B. 0.67
C. 0.75
D. 1.00
E. 1.50

ans: B

74. The illustration shows total internal reflection taking place in a piece of glass. The index of refraction of this glass:




A. is at least 2.0
B. is at most 2.0
C. is at least 1.15
D. is at most 1.15
E. cannot be calculated from the given data

ans: C

75. The critical angle for total internal reflection at a diamond-air interface is 25°. Suppose light is incident at an angle of $\theta$ with the normal. Total internal reflection will occur if the incident medium is:

A. air and $\theta = 25°$
B. air and $\theta > 25°$
C. air and $\theta < 25°$
D. diamond and $\theta < 25°$
E. diamond and $\theta > 25°$

ans: E

76. If $n_{\text{water}} = 1.50$ and $n_{\text{glass}} = 1.33$, then total internal reflection at an interface between this glass and water:

A. occurs whenever the light goes from glass to water
B. occurs whenever the light goes from water to glass
C. may occur when the light goes from glass to water
D. may occur when the light goes from water to glass
E. can never occur at this interface

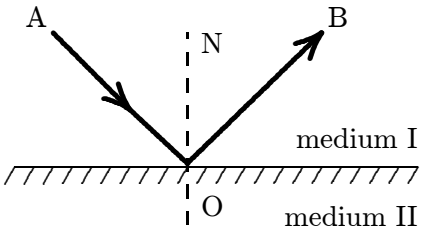
ans: D

77. The separation of white light into colors by a prism is associated with:

A. total internal reflection
B. partial reflection from each surface
C. variation of index of refraction with wavelength
D. a decrease in the speed of light in the glass
E. selective absorption of various colors

ans: C

78. The diagram shows total internal reflection. Which of the following statements is NOT true?




A. Angle AON is the angle of incidence
B. Angle AON = angle BON
C. Angle AON must be the critical angle
D. The speed of light in medium II is greater than that in medium I
E. if angle AON were increased, there would still be total internal reflection

ans: C

79. A ray of light in water (index $n_1$) is incident on its surface (with air) at the critical angle for total internal reflection. Some oil (index $n_2$) is now floated on the water. The angle between the ray in the oil and the normal is:

A. $\sin^{-1}(1.00)$
B. $\sin^{-1}(1/n_1)$
C. $\sin^{-1}(1/n_2)$
D. $\sin^{-1}(n_1/n_2)$
E. $\sin^{-1}(n_2/n_1)$

ans: C

## Chapter 34: IMAGES

1. A virtual image is one:
   A. toward which light rays converge but do not pass through
   B. from which light rays diverge but do not pass through
   C. from which light rays diverge as they pass through
   D. toward which light rays converge and pass through
   E. with a ray normal to a mirror passing through it

   ans: B

2. Which of the following is true of all virtual images?
   A. They can be seen but not photographed
   B. They are ephemeral
   C. They are smaller than the objects
   D. They are larger than the objects
   E. None of the above

   ans: E

3. When you stand in front of a plane mirror, your image is:
   A. real, erect, and smaller than you
   B. real, erect, and the same size as you
   C. virtual, erect, and smaller than you
   D. virtual, erect, and the same size as you
   E. real, inverted, and the same size as you

   ans: D

4. An object is 2 m in front of a plane mirror. Its image is:
   A. virtual, inverted, and 2 m behind the mirror
   B. virtual, inverted, and 2 m in front of the mirror
   C. virtual, erect, and 2 m in front of the mirror
   D. real, erect, and 2 m behind the mirror
   E. none of the above

   ans: E

5. A ball is held 50 cm in front of a plane mirror. The distance between the ball and its image is:
   A. 100 cm
   B. 150 cm
   C. 200 cm
   D. 0
   E. 50 cm

   ans: A

6. A card marked IAHIO8 is standing upright in front of a plane mirror. Which of the following is NOT true?

   A. The image is virtual
   B. The image shifts its position as the observer shifts his position
   C. The image appears as 8OIHAI to a person looking in the mirror
   D. The image is caused mostly by specular rather than diffuse reflection
   E. The image is the same size as the object

   ans: B

7. The angle between a horizontal ruler and a vertical plane mirror is 30°. The angle between the ruler and its image is:

   A. 15°
   B. 30°
   C. 60°
   D. 90°
   E. 180°

   ans: C

8. A 5.0-ft woman wishes to see a full length image of herself in a plane mirror. The minimum length mirror required is:

   A. 5 ft
   B. 10 ft
   C. 2.5 ft
   D. 3.54 ft
   E. variable: the farther away she stands the smaller the required mirror length

   ans: C

9. A man holds a rectangular card in front of and parallel to a plane mirror. In order for him to see the entire image of the card, the least mirror area needed is:

   A. that of the whole mirror, regardless of its size
   B. that of the pupil of his eye
   C. one-half that of the card
   D. one-fourth that of the card
   E. an amount which decreases with his distance from the mirror

   ans: D

10. A light bulb burns in front of the center of a 40-cm wide plane mirror that is hung vertically on a wall. A man walks in front of the mirror along a line that is parallel to the mirror and twice as far from it as the bulb. The greatest distance he can walk and still see the image of the bulb is:

    A. 20 cm
    B. 40 cm
    C. 60 cm
    D. 80 cm
    E. 120 cm

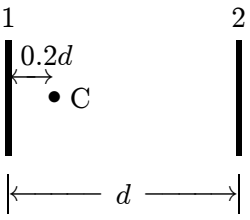
    ans: E

11. A plane mirror is in a vertical plane and is rotating about a vertical axis at 100 rpm. A horizontal beam of light is incident on the mirror. The reflected beam will rotate at:

A. 100 rpm
B. 141 rpm
C. 0 rpm
D. 200 rpm
E. 10,000 rpm

ans: D

12. A candle C sits between two parallel mirrors, a distance $0.2d$ from mirror 1. Here $d$ is the distance between the mirrors. Multiple images of the candle appear in both mirrors. How far behind mirror 1 are the nearest three images of the candle in that mirror?




A. $0.2d$, $1.8d$, $2.2d$
B. $0.2d$, $2.2d$, $4.2d$
C. $0.2d$, $1.8d$, $3.8d$
D. $0.2d$, $0.8d$, $1.4d$
E. $0.2d$, $1.8d$, $3.4d$

ans: A

13. Two plane mirrors make an angle of 120° with each other. The maximum number of images of an object placed between them is:

A. one
B. two
C. three
D. four
E. more than four

ans: B

14. A parallel beam of monochromatic light in air is incident on a plane glass surface. In the glass, the beam:

A. remains parallel
D. undergoes dispersion
B. becomes diverging
E. follows a parabolic path
C. becomes converging

ans: A

15. The focal length of a spherical mirror is $N$ times its radius of curvature where $N$ is:
    A. 1/4
    B. 1/2
    C. 1
    D. 2
    E. 4

    ans: B

16. Real images formed by a spherical mirror are always:
    A. on the side of the mirror opposite the source
    B. on the same side of the mirror as the source but closer to the mirror than the source
    C. on the same side of the mirror as the source but closer to the mirror than the focal point
    D. on the same side of the mirror as the source but further from the mirror than the focal point
    E. none of the above

    ans: E

17. The image produced by a convex mirror of an erect object in front of the mirror is always:
    A. virtual, erect, and larger than the object
    B. virtual, erect, and smaller than the object
    C. real, erect, and larger than the object
    D. real, erect, and smaller than the object
    E. none of the above

    ans: B

18. An erect object is located between a concave mirror and its focal point. Its image is:
    A. real, erect, and larger than the object
    B. real, inverted, and larger than the object
    C. virtual, erect, and larger than the object
    D. virtual, inverted, and larger than the object
    E. virtual, erect, and smaller than the object

    ans: C

19. An erect object is in front of a convex mirror a distance greater than the focal length. The image is:
    A. real, inverted, and smaller than the object
    B. virtual, inverted, and larger than the object
    C. real, inverted, and larger than the object
    D. virtual, erect, and smaller than the object
    E. real, erect, and larger than the object

    ans: D

20. As an object is moved from the center of curvature of a concave mirror toward its focal point its image:

A. remains virtual and becomes larger
B. remains virtual and becomes smaller
C. remains real and becomes larger
D. remains real and becomes smaller
E. remains real and approaches the same size as the object

ans: C

21. As an object is moved from a distant location toward the center of curvature of a concave mirror its image:

A. remains virtual and becomes smaller
B. remains virtual and becomes larger
C. remains real and becomes smaller
D. remains real and becomes larger
E. changes from real to virtual

ans: D

22. The image of an erect candle, formed using a convex mirror, is always:

A. virtual, inverted, and smaller than the candle
B. virtual, inverted, and larger than the candle
C. virtual, erect, and larger than the candle
D. virtual, erect, and smaller than the candle
E. real, erect, and smaller than the candle

ans: D

23. At what distance in front of a concave mirror must an object be placed so that the image and object are the same size?

A. a focal length
B. half a focal length
C. twice a focal length
D. less than half focal length
E. more than twice a focal length

ans: B

24. A point source is to be used with a concave mirror to produce a beam of parallel light. The source should be placed:

A. as close to the mirror as possible
B. at the center of curvature
C. midway between the center of curvature and the focal point
D. midway between the center of curvature and the mirror
E. midway between the focal point and the mirror

ans: D

25. A concave mirror forms a real image that is twice the size of the object. If the object is 20 cm from the mirror, the radius of curvature of the mirror must be about:

A. 13 cm
B. 20 cm
C. 27 cm
D. 40 cm
E. 80 cm

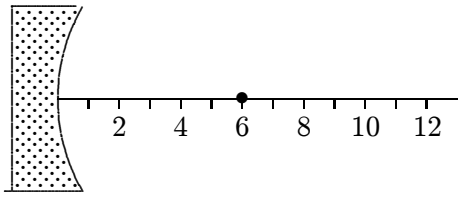
ans: C

26. A man stands with his nose 8 cm from a concave shaving mirror of radius 32 cm The distance from the mirror to the image of his nose is:

A. 8 cm
B. 12 cm
C. 16 cm
D. 24 cm
E. 32 cm

ans: C

27. The figure shows a concave mirror with a small object located at the point marked 6. If the image is also at this point, then the center of curvature of the mirror is at the point marked:




A. 3
B. 4
C. 6
D. 9
E. 12

ans: C

28. A concave spherical mirror has a focal length of 12 cm. If an object is placed 6 cm in front of it the image position is:

A. 4 cm behind the mirror
B. 4 cm in front of the mirror
C. 12 cm behind the mirror
D. 12 cm in front of the mirror
E. at infinity

ans: C

29. A concave spherical mirror has a focal length of 12 cm. If an object is placed 18 cm in front of it the image position is:

A. 7.2 cm behind the mirror
B. 7.2 cm in front of the mirror
C. 36 cm behind the mirror
D. 36 cm in front of the mirror
E. at infinity

ans: D

30. A convex spherical mirror has a focal length of 12 cm. If an object is placed 6 cm in front of it the image position is:

A. 4 cm behind the mirror
B. 4 cm in front of the mirror
C. 12 cm behind the mirror
D. 12 cm in front of the mirror
E. at infinity

ans: A

31. A concave spherical mirror has a focal length of 12 cm. If an erect object is placed 6 cm in front of it:

A. the magnification is 2 and the image is erect
B. the magnification is 2 and the image is inverted
C. the magnification is 0.67 and the image is erect
D. the magnification is 0.67 and the image is inverted
E. the magnification is 0.5 and the image is erect

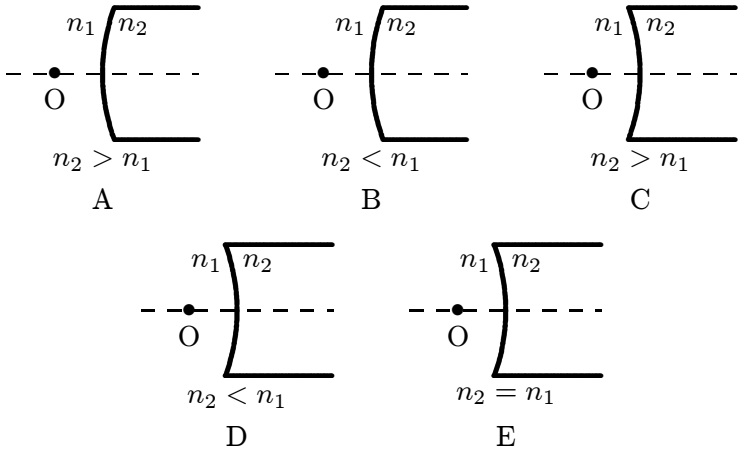
ans: A

32. An erect object is located on the central axis of a spherical mirror. The magnification is $-3$. This means:

A. its image is real, inverted, and on the same side of the mirror
B. its image is virtual, erect, and on the opposite side of the mirror
C. its image is real, erect, and on the same side of the mirror
D. its image is real, inverted, and on the opposite side of the mirror
E. its image is virtual, inverted, and on the opposite side of the mirror

ans: A

33. An object O, in air, is in front of the concave spherical refracting surface of a piece of glass. Which of the general situations depicted below is like this situation?




ans: C

34. A concave refracting surface is one with a center of curvature:

A. to the left of the surface
B. to the right of the surface
C. on the side of the incident light
D. on the side of the refracted light
E. on the side with the higher index of refraction

ans: C

35. A convex refracting surface has a radius of 12 cm. Light is incident in air ($n = 1$) and is refracted into a medium with an index of refraction of 2. Light incident parallel to the central axis is focused at a point:

A. 3 cm from the surface
B. 6 cm from the surface
C. 12 cm from the surface
D. 18 cm from the surface
E. 24 cm from the surface

ans: E

36. A convex refracting surface has a radius of 12 cm. Light is incident in air ($n = 1$) and refracted into a medium with an index of refraction of 2. To obtain light with rays parallel to the central axis after refraction a point source should be placed on the axis:

A. 3 cm from the surface
B. 6 cm from the surface
C. 12 cm from the surface
D. 18 cm from the surface
E. 24 cm from the surface

ans: C

37. A concave refracting surface of a medium with index of refraction $n$ produces a real image no matter where an object is placed outside:

A. always
B. only if the index of refraction of the surrounding medium is less than $n$
C. only if the index of refraction of the surrounding medium is greater than $n$
D. never
E. none of the above

ans: E

38. A convex spherical refracting surface separates a medium with index of refraction 2 from air. The image of an object outside the surface is real:

A. always
B. never
C. only if it is close to the surface
D. only if it is far from the surface
E. only if the radius of curvature is small

ans: D

39. A convex spherical surface with radius $r$ separates a medium with index of refraction 2 from air. As an object is moved toward the surface from far away along the central axis, its image:

A. changes from virtual to real when it is $r/2$ from the surface
B. changes from virtual to real when it is $r$ from the surface
C. changes from real to virtual when it is $r/2$ from the surface
D. changes from real to virtual when it is $r$ from the surface
E. remains real

ans: D

40. A concave spherical surface with radius $r$ separates a medium with index of refraction 2 from air. As an object is moved toward the surface from far away along the central axis, its image:

A. changes from virtual to real when it is $r/2$ from the surface
B. changes from virtual to real when it is $2r$ from the surface
C. changes from real to virtual when it is $r/2$ from the surface
D. changes from real to virtual when it is $2r$ from the surface
E. remains virtual

ans: E

41. An erect object is placed on the central axis of a thin lens, further from the lens than the magnitude of its focal length. The magnification is $+0.4$. This means:

A. the image is real and erect and the lens is a converging lens
B. the image is real and inverted and the lens is a converging lens
C. the image is virtual and erect, and the lens is a diverging lens
D. the image is virtual and erect, and the lens is a converging lens
E. the image is virtual and inverted and the lens is a diverging lens

ans: C

42. Where must an object be placed in front of a converging lens in order to obtain a virtual image?
    A. At the focal point
    B. At twice the focal length
    C. Greater than the focal length
    D. Between the focal point and the lens
    E. Between the focal length and twice the focal length

    ans: D

43. An erect object placed outside the focal point of a converging lens will produce an image that is:
    A. erect and virtual
    B. inverted and virtual
    C. erect and real
    D. inverted and real
    E. impossible to locate

    ans: D

44. An object is 30 cm in front of a converging lens of focal length 10 cm. The image is:
    A. real and larger than the object
    B. real and the same size than the object
    C. real and smaller than the object
    D. virtual and the same size than the object
    E. virtual and smaller than the object

    ans: C

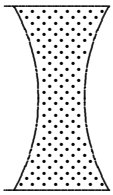
45. Let $p$ denote the object-lens distance and $i$ the image-lens distance. The image produced by a lens of focal length $f$ has a height that can be obtained from the object height by multiplying it by:
    A. $p/i$
    B. $i/p$
    C. $f/p$
    D. $f/i$
    E. $i/f$

    ans: B

46. A camera with a lens of focal length 6.0 cm takes a picture of a 1.4-m tall man standing 11 m away. The height of the image is about:
    A. 0.39 cm
    B. 0.77 cm
    C. 1.5 cm
    D. 3.0 cm
    E. 6.0 cm

    ans: B

47. A hollow lens is made of thin glass, as shown. It can be filled with air, water ($n = 1.3$) or $CS_2$ ($n = 1.6$). The lens will diverge a beam of parallel light if it is filled with:



A. air and immersed in air
B. air and immersed in water
C. water and immersed in CS2
D. $CS_2$ and immersed in water
E. $CS_2$ and immersed in $CS_2$

ans: D

48. The object-lens distance for a certain converging lens is 400 mm. The image is three times the size of the object. To make the image five times the size of the object-lens distance must be changed to:

A. 360 mm
B. 540 mm
C. 600 mm
D. 720 mm
E. 960 mm

ans: A

49. An erect object is $2f$ in front of a converging lens of focal length $f$. The image is:

A. real, inverted, magnified
B. real, erect, same size
C. real, inverted, same size
D. virtual, inverted, reduced
E. real, inverted, reduced

ans: C

50. An ordinary magnifying glass in front of an erect object produces an image that is:

A. real and erect
B. real and inverted
C. virtual and inverted
D. virtual and erect
E. none of these

ans: D

51. The Sun subtends 0.5° as seen from Earth. The diameter of its image, using a 1.0-m focal length lens, is about:

A. 10 cm
B. 2 cm
C. 1 cm
D. 5 mm
E. 1 mm

ans: C

52. An object is in front of a converging lens, at a distance less than the focal length from the lens. Its image is:

A. virtual and larger than the object
B. real and smaller than the object
C. virtual and smaller than the object
D. real and larger than the object
E. virtual and the same size as the object

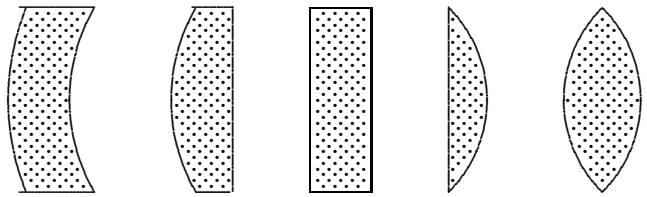
ans: A

53. A plano-convex glass ($n = 1.5$) lens has a curved side whose radius is 50 cm. If the image size is to be the same as the object size, the object should be placed at a distance from the lens of:

A. 50 cm
B. 100 cm
C. 200 cm
D. 400 cm
E. 340 cm

ans: C

54. Which of the following five glass lenses is a diverging lens?




ans: A

55. The bellows of an adjustable camera can be extended so that the largest film to lens distance is one and one-half times the focal length. If the focal length is 12 cm, the nearest object that can be sharply focused on the film must be what distance from the lens?

A. 12 cm
B. 24 cm
C. 36 cm
D. 48 cm
E. 72 cm

ans: C

56. A 3-cm high object is in front of a thin lens. The object distance is 4 cm and the image distance is −8 cm. The image height is:

A. 0.5 cm
B. 1 cm
C. 1.5 cm
D. 6 cm
E. 24 cm

ans: D

57. When a single-lens camera is focused on a distant object, the lens-to-film distance is found to be 40.0 mm. To focus on an object 0.54 m in front of the lens, the film-to-lens distance should be:

A. 40.0 mm
B. 37.3 mm
C. 36.8 mm
D. 42.7 mm
E. 43.2 mm

ans: E

58. In a cinema, a picture 2.5 cm wide on the film is projected to an image 3.0 m wide on a screen that is 18 m away. The focal length of the lens is about:

A. 7.5 cm
B. 10 cm
C. 12.5 cm
D. 15 cm
E. 20 cm

ans: D

59. The term "virtual" as applied to an image made by a mirror means that the image:

A. is on the mirror surface
B. cannot be photographed by a camera
C. is in front of the mirror
D. is the same size as the object
E. cannot be shown directly on a screen

ans: E

60. Which instrument uses a single converging lens with the object placed just inside the focal point?

A. Camera
B. Compound microscope
C. Magnifying glass
D. Overhead projector
E. Telescope

ans: C

61. Let $f_o$ and $f_e$ be the focal lengths of the objective and eyepiece of a compound microscope. In ordinary use, the object:

A. is less than $f_o$ from the objective lens
B. is more that $f_o$ from the objective
C. produces an intermediate image that is slightly more than $f_e$ from the eyepiece
D. produces an intermediate image that is $2f_e$ away from the eyepiece
E. produces an intermediate image that is less than $f_o$ from the objective lens

ans: B

62. Consider the following four statements concerning a compound microscope:

1. Each lens produces an image that is virtual and inverted.
2. The objective lens has a very short focal length.
3. The eyepiece is used as a simple magnifying glass.
4. The objective lens is convex and the eyepiece is concave.

Which two of the four statements are correct?

A. 1, 2
B. 1, 3
C. 1, 4
D. 2, 3
E. 2, 4

ans: D

63. What type of eyeglasses should a nearsighted person wear?

A. diverging lenses
D. bifocal lenses
B. converging lenses
E. plano-convex lenses
C. double convex lenses

ans: A

64. Which of the following is NOT correct for a simple magnifying glass?

A. The image is virtual
B. The image is erect
C. The image is larger than the object
D. The object is inside the focal point
E. The lens is diverging

ans: E

65. A nearsighted person can see clearly only objects within 1.4 m of her eye. To see distant objects, she should wear eyeglasses of what type and focal length?

A. diverging, 2.8 m
B. diverging, 1.4 m
C. converging, 2.8 m
D. converging, 1.4 m
E. diverging, 0.72 m

ans: B

66. A magnifying glass has a focal length of 15 cm. If the near point of the eye is 25 cm from the eye the angular magnification of the glass is about:

A. 0.067
B. 0.33
C. 0.67
D. 1.7
E. 15

ans: D

67. An object is 20 cm to the left of a lens of focal length +10 cm. A second lens, of focal length +12.5 cm, is 30 cm to the right of the first lens. The distance between the original object and the final image is:

A. 28 cm
B. 50 cm
C. 100 cm
D. 0
E. infinity

ans: D

68. A converging lens of focal length 20 cm is placed in contact with a converging lens of focal length 30 cm. The focal length of this combination is:

A. +10 cm
B. −10 cm
C. +60 cm
D. −60 cm
E. +25 cm

ans: A

69. A student sets the cross-hairs of an eyepiece in line with an image that he is measuring. He then notes that when he moves his head slightly to the right, the image moves slightly to the left (with respect to the cross-hairs). Therefore the image is:

A. infinitely far away
B. farther away from him that the cross-hairs
C. nearer to him than the cross-hairs
D. in the focal plane of the eyepiece
E. in the plane of the cross-hairs

ans: C

70. In a two lens microscope, the intermediate image is:

A. virtual, erect, and magnified
B. real, erect, and magnified
C. real, inverted, and magnified
D. virtual, inverted, and reduced
E. virtual, inverted, and magnified

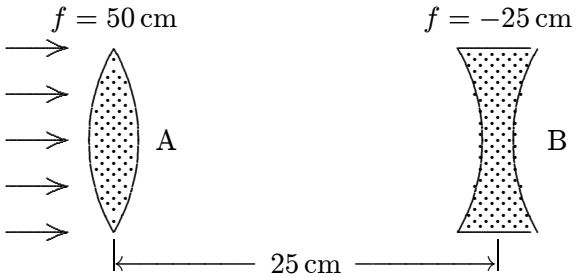
ans: C

71. Two thin lenses (focal lengths $f_1$ and $f_2$) are in contact. Their equivalent focal length is:

A. $f_1 + f_2$
B. $f_1 f_2/(f_1 + f_2)$
C. $1/f_1 + 1/f_2$
D. $f_1 - f_2$
E. $f_1(f_1 - f_2)/f_2$

ans: B

72. The two lenses shown are illuminated by a beam of parallel light from the left. Lens B is then moved slowly toward lens A. The beam emerging from lens B is:




A. initially parallel and then diverging
B. always diverging
C. initially converging and finally parallel
D. always parallel
E. initially converging and finally diverging

ans: A

## Chapter 35: INTERFERENCE

1. A "wave front" is a surface of constant:
   A. phase
   B. frequency
   C. wavelength
   D. amplitude
   E. speed

   ans: A

2. Huygens' construction can be used only:
   A. for light
   B. for an electromagnetic wave
   C. if one of the media is vacuum (or air)
   D. for transverse waves
   E. for all of the above and other situations

   ans: E

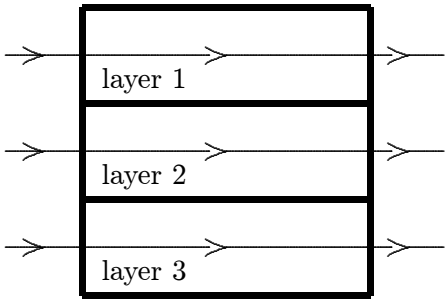
3. Consider (I) the law of reflection and (II) the law of refraction. Huygens' principle can be used to derive:
   A. only I
   B. only II
   C. both I and II
   D. neither I nor II
   E. the question is meaningless because Huygens' principle is for wave fronts whereas both I and II concern rays

   ans: C

4. Units of "optical path length" are:
   A. $m^{-1}$
   B. m
   C. m/s
   D. Hz/m
   E. m/Hz

   ans: B

5. The light waves represented by the three rays shown in the diagram all have the same frequency. 4.7 wavelengths fit into layer 1, 3.2 wavelengths fit into layer 2, and 5.3 wavelengths fit into layer 3. Rank the layers according to the speeds of the waves, least to greatest.




A. 1, 2, 3
B. 2, 1, 3
C. 3, 1, 2
D. 3, 1, 2
E. 1, 3, 2

ans: D

6. Interference of light is evidence that:

A. the speed of light is very large
B. light is a transverse wave
C. light is electromagnetic in character
D. light is a wave phenomenon
E. light does not obey conservation of energy

ans: D

7. The reason there are two slits, rather than one, in a Young's experiment is:

A. to increase the intensity
B. one slit is for frequency, the other for wavelength
C. to create a path length difference
D. one slit is for $\vec{E}$ fields, the other is for $\vec{B}$ fields
E. two slits in parallel offer less resistance

ans: C

8. In a Young's double-slit experiment the center of a bright fringe occurs wherever waves from the slits differ in the distance they travel by a multiple of:

A. a fourth of a wavelength
B. a half a wavelength
C. a wavelength
D. three-fourths of a wavelength
E. none of the above

ans: E

9. In a Young's double-slit experiment the center of a bright fringe occurs wherever waves from the slits differ in phase by a multiple of:

   A. $\pi/4$
   B. $\pi/2$
   C. $\pi$
   D. $3\pi/4$
   E. $2\pi$

   ans: E

10. Waves from two slits are in phase at the slits and travel to a distant screen to produce the third side maximum of the interference pattern. The difference in the distance traveled by the waves is:

   A. half a wavelength
   B. a wavelength
   C. three halves of a wavelength
   D. two wavelengths
   E. three wavelengths

   ans: E

11. Waves from two slits are in phase at the slits and travel to a distant screen to produce the second minimum of the interference pattern. The difference in the distance traveled by the waves is:

   A. half a wavelength
   B. a wavelength
   C. three halves of a wavelength
   D. two wavelengths
   E. five halves of a wavelength

   ans: C

12. A monochromatic light source illuminates a double slit and the resulting interference pattern is observed on a distant screen. Let $d$ = center-to-center slit spacing, $a$ = individual slit width, $D$ = screen-to-slit distance, and $\ell$ = adjacent dark line spacing in the interference pattern. The wavelength of the light is then:

   A. $d\ell/D$
   B. $Ld/a$
   C. $da/D$
   D. $\ell D/a$
   E. $Dd/\ell$

   ans: A

13. Light from a small region of an ordinary incandescent bulb is passed through a yellow filter and then serves as the source for a Young's double-slit experiment. Which of the following changes would cause the interference pattern to be more closely spaced?

A. Use slits that are closer together
B. Use a light source of lower intensity
C. Use a light source of higher intensity
D. Use a blue filter instead of a yellow filter
E. Move the light source further away from the slits.

ans: D

14. In a Young's double-slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance $D$ must be changed to:

A. $D/2$
B. $D/\sqrt{2}$
C. $D\sqrt{2}$
D. $2D$
E. $4D$

ans: D

15. In a Young's double-slit experiment, light of wavelength 500 nm illuminates two slits that are separated by 1 mm. The separation between adjacent bright fringes on a screen 5 m from the slits is:

A. 0.10 cm
B. 0.25 cm
C. 0.50 cm
D. 1.0 cm
E. none of the above

ans: B

16. In a Young's double-slit experiment, the separation between slits is $d$ and the screen is a distance $D$ from the slits. $D$ is much greater than $d$ and $\lambda$ is the wavelength of the light. The number of bright fringes per unit width on the screen is:

A. $Dd/\lambda$
B. $D\lambda/d$
C. $D/d\lambda$
D. $\lambda/Dd$
E. $d/D\lambda$

ans: E

17. In a Young's double-slit experiment, the slit separation is doubled. This results in:

A. an increase in fringe intensity
B. a decrease in fringe intensity
C. a halving of the wavelength
D. a halving of the fringe spacing
E. a doubling of the fringe spacing

ans: D

18. In an experiment to measure the wavelength of light using a double slit, it is found that the fringes are too close together to easily count them. To spread out the fringe pattern, one could:

A. decrease the slit separation
B. increase the slit separation
C. increase the width of each slit
D. decrease the width of each slit
E. none of these

ans: A

19. The phase difference between the two waves that give rise to a dark spot in a Young's double-slit experiment is (where $m$ = integer):

A. zero
B. $2\pi m + \pi/8$
C. $2\pi m + \pi/4$
D. $2\pi m + \pi/2$
E. $2\pi m + \pi$

ans: E

20. In a Young's experiment, it is essential that the two beams:

A. have exactly equal intensity
B. be exactly parallel
C. travel equal distances
D. come originally from the same source
E. be composed of a broad band of frequencies

ans: D

21. A light wave with an electric field amplitude of $E_0$ and a phase constant of zero is to be combined with one of the following waves:

wave A has an amplitude of $E_0$ and a phase constant of zero
wave B has an amplitude of $E_0$ and a phase constant of $\pi$
wave C has an amplitude of $2E_0$ and a phase constant of zero
wave D has an amplitude of $2E_0$ and a phase constant of $\pi$
wave E has an amplitude of $3E_0$ and a phase constant of $\pi$

Which of these combinations produces the greatest intensity?

ans: C

22. A light wave with an electric field amplitude of $2E_0$ and a phase constant of zero is to be combined with one of the following waves:

wave A has an amplitude of $E_0$ and a phase constant of zero
wave B has an amplitude of $E_0$ and a phase constant of $\pi$
wave C has an amplitude of $2E_0$ and a phase constant of zero
wave D has an amplitude of $2E_0$ and a phase constant of $\pi$
wave E has an amplitude of $3E_0$ and a phase constant of $\pi$

Which of these combinations produces the least intensity?

ans: D

23. One of the two slits in a Young's experiment is painted over so that it transmits only one-half the intensity of the other slit. As a result:

A. the fringe system disappears
B. the bright fringes get brighter and the dark ones get darker
C. the fringes just get dimmer
D. the dark fringes just get brighter
E. the dark fringes get brighter and the bright ones get darker

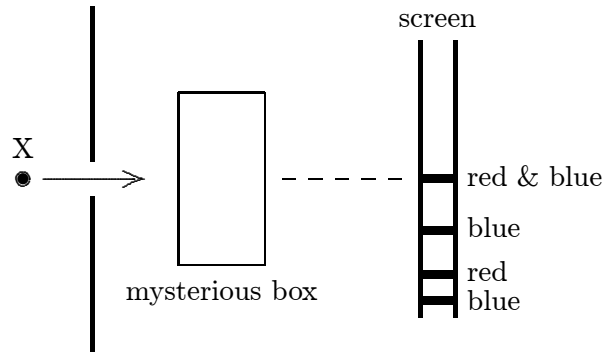
ans: E

24. In a Young's double-slit experiment, a thin sheet of mica is placed over one of the two slits. As a result, the center of the fringe pattern (on the screen) shifts by an amount corresponding to 30 dark bands. The wavelength of the light in this experiment is 480 nm and the index of the mica is 1.60. The mica thickness is:

A. 0.090 mm
B. 0.012 mm
C. 0.014 mm
D. 0.024 mm
E. 0.062 mm

ans: D

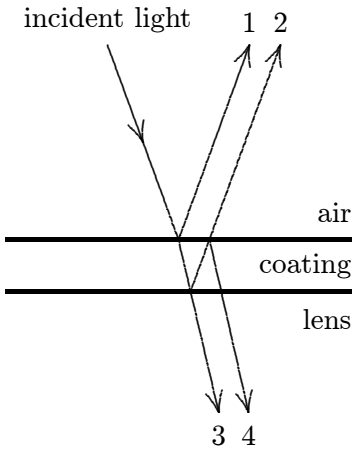
25. Light from a point source X contains only blue and red components. After passing through a mysterious box, the light falls on a screen. Bright red and blue hands are observed as shown. The box must contain:




A. a lens
B. a mirror
C. a prism
D. a double slit
E. a blue and red filter

ans: D

26. Binoculars and microscopes are frequently made with coated optics by adding a thin layer of transparent material to the lens surface as shown. One wants:




A. constructive interference between waves 1 and 2
B. destructive interference between waves 3 and 4
C. constructive interference between 3 and 4
D. the coating to be more transparent than the lens
E. the speed of light in the coating to be less than that in the lens

ans: C

27. Monochromatic light, at normal incidence, strikes a thin film in air. If $\lambda$ denotes the wavelength in the film, what is the thinnest film in which the reflected light will be a maximum?

A. Much less than $\lambda$
B. $\lambda/4$
C. $\lambda/2$
D. $3\lambda/4$
E. $\lambda$

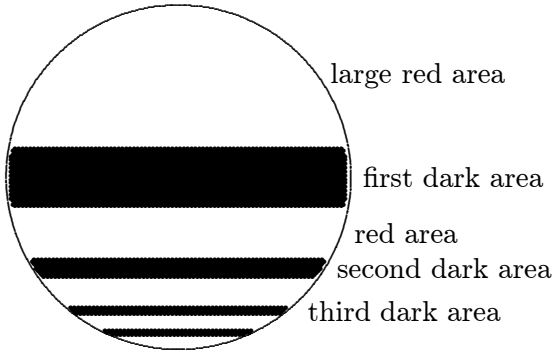
ans: B

28. A soap film is illuminated by white light normal to its surface. The index of refraction of the film is 1.50. Wavelengths of 480 nm and 800 nm and no wavelengths between are be intensified in the reflected beam. The thickness of the film is:

A. $1.5 \times 10^{-5}$ cm
B. $2.4 \times 10^{-5}$ cm
C. $3.6 \times 10^{-5}$ cm
D. $4.0 \times 10^{-5}$ cm
E. $6.0 \times 10^{-5}$ cm

ans: D

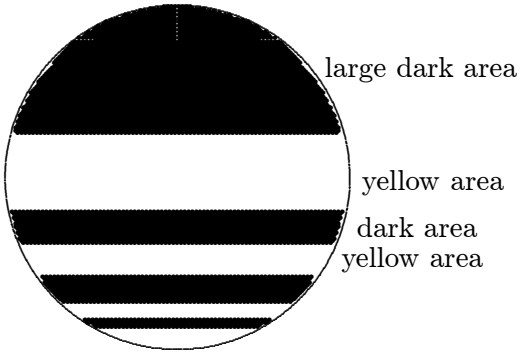
29. Red light is viewed through a thin vertical soap film. At the third dark area shown, the thickness of the film, in terms of the wavelength within the film, is:




A. $\lambda$
B. $3\lambda/4$
C. $\lambda/2$
D. $\lambda/4$
E. $5\lambda/4$

ans: E

30. Yellow light is viewed by reflection from a thin vertical soap film. Let $\lambda$ be the wavelength of the light within the film. Why is there a large dark space at the top of the film?

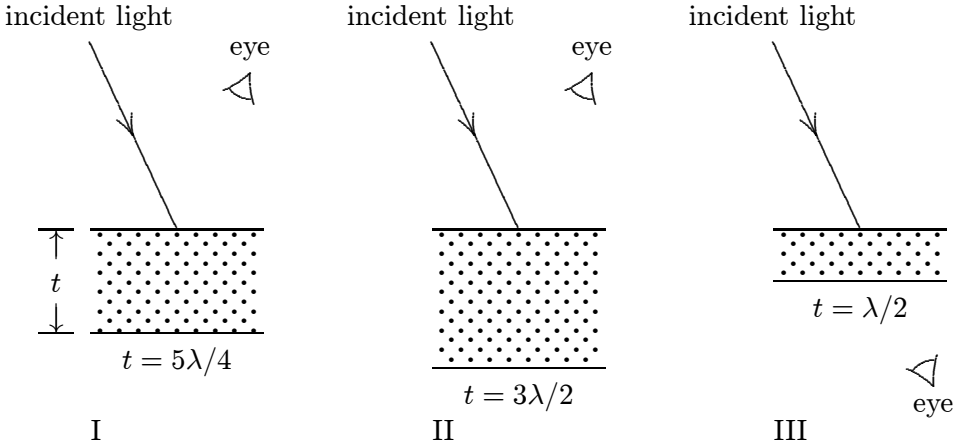



A. no light is transmitted through this part of the film
B. the film thickness there is $\lambda/4$
C. the film thickness there is much less than $\lambda$
D. the film is too thick in this region for thin film formulas to apply
E. the reflected light is in the infrared

ans: C

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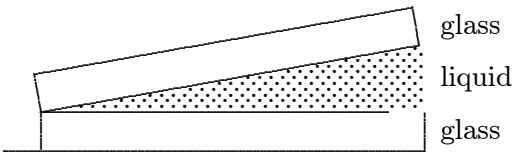
31. Three experiments involving a thin film (in air) are shown. If $t$ denotes the film thickness and $\lambda$ denotes the wavelength of the light in the film, which experiments will produce constructive interference as seen by the observer? The incident light is nearly normal to the surface.




A. I only
B. II only
C. III only
D. I and III only
E. II and III only

ans: D

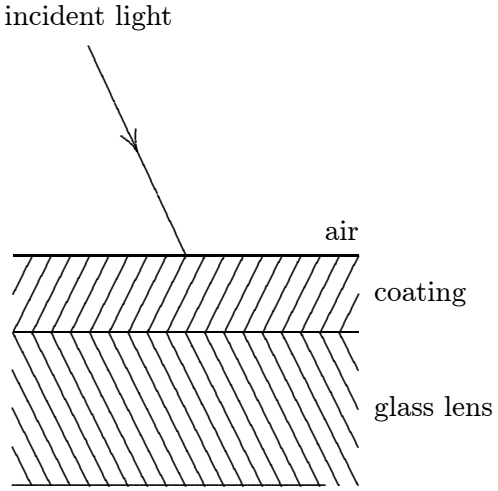
32. A liquid of refractive index $n = 4/3$ replaces the air between a fixed wedge formed from two glass plates as shown. As a result, the spacing between adjacent dark bands in the interference pattern:




A. increases by a factor of 4/3
B. increases by a factor of 3
C. remains the same
D. decreases to 3/4 of its original value
E. decreases to 1/3 of its original value

ans: D

33. A lens with a refractive index of 1.5 is coated with a material of refractive index 1.2 in order to minimize reflection. If $\lambda$ denotes the wavelength of the incident light in air, what is the thinnest possible such coating?




A. $0.5\lambda$
B. $0.416\lambda$
C. $0.3\lambda$
D. $0.208\lambda$
E. $0.25\lambda$

ans: D

34. In a thin film experiment, a wedge of air is used between two glass plates. If the wavelength of the incident light in air is 480 nm, how much thicker is the air wedge at the $16^{\text{th}}$ dark fringe than it is at the $6^{\text{th}}$?

A. 2400 nm
B. 4800 nm
C. 240 nm
D. 480 nm
E. None of these

ans: A

35. An air wedge is formed from two glass plates that are in contact at their left edges. There are ten dark bands when viewed by reflection using monochromatic light. The left edge of the top plate is now slowly lifted until the plates are parallel. During this process:

A. the dark bands crowd toward the right edge
B. the dark bands remain stationary
C. the dark bands crowd toward the left edge
D. the dark bands spread out, disappearing off the right edge
E. the dark bands spread out, disappearing off the left edge

ans: E

36. An air wedge is formed using two glass plates that are in contact along their left edge. When viewed by highly monochromatic light, there are exactly 4001 dark bands in the reflected light. The air is now evacuated (with the glass plates remaining rigidly fixed) and the number of dark bands decreases to exactly 4000. The index of refraction of the air is:

A. 0.00025
B. 0.00050
C. 1.00025
D. 1.00050
E. 1.00000, by definition

ans: C

37. A glass ($n = 1.6$) lens is coated with a thin film ($n = 1.3$) to reduce reflection of certain incident light. If $\lambda$ is the wavelength of the light in the film, the least film thickness is:

A. less than $\lambda/4$
B. $\lambda/4$
C. $\lambda/2$
D. $\lambda$
E. more than $\lambda$

ans: B

38. Two point sources, vibrating in phase, produce an interference pattern in a ripple tank. If the frequency is increased by 20%, the number of nodal lines:

A. is increased by 20%
B. is increased by 40%
C. remains the same
D. is decreased by 20%
E. is decreased by 40%

ans: A

39. If two light waves are coherent:

A. their amplitudes are the same
B. their frequencies are the same
C. their wavelengths are the same
D. their phase difference is constant
E. the difference in their frequencies is constant

ans: D

40. To obtain an observable double-slit interference pattern:

A. the light must be incident normally on the slits
B. the light must be monochromatic
C. the light must consist of plane waves
D. the light must be coherent
E. the screen must be far away from the slits

ans: D

## Chapter 36: DIFFRACTION

1. Sound differs from light in that sound:
   A. is not subject to diffraction
   B. is a torsional wave rather than a longitudinal wave
   C. does not require energy for its origin
   D. is a longitudinal wave rather than a transverse wave
   E. is always monochromatic

   ans: D

2. Radio waves are readily diffracted around buildings whereas light waves are negligibly diffracted around buildings. This is because radio waves:
   A. are plane polarized
   B. have much longer wavelengths than light waves
   C. have much shorter wavelengths than light waves
   D. are nearly monochromatic (single frequency)
   E. are amplitude modulated (AM).

   ans: B

3. Diffraction plays an important role in which of the following phenomena?
   A. The sun appears as a disk rather than a point to the naked eye
   B. Light is bent as it passes through a glass prism
   C. A cheerleader yells through a megaphone
   D. A farsighted person uses eyeglasses of positive focal length
   E. A thin soap film exhibits colors when illuminated with white light

   ans: C

4. The rainbow seen after a rain shower is caused by:
   A. diffraction
   B. interference
   C. refraction
   D. polarization
   E. absorption

   ans: C

5. When a highly coherent beam of light is directed against a very fine wire, the shadow formed behind it is not just that of a single wire but rather looks like the shadow of several parallel wires. The explanation of this involves:
   A. refraction
   B. diffraction
   C. reflection
   D. the Doppler effect
   E. an optical illusion

   ans: B

6. When the atmosphere is not quite clear, one may sometimes see colored circles concentric with the Sun or the Moon. These are generally not more than a few diameters of the Sun or Moon and invariably the innermost ring is blue. The explanation for this phenomena involves:
   A. reflection
   B. refraction
   C. interference
   D. diffraction
   E. the Doppler effect

   ans: D

7. The shimmering or wavy lines that can often be seen near the ground on a hot day are due to:
   A. Brownian movement
   B. reflection
   C. refraction
   D. diffraction
   E. dispersion

   ans: C

8. A point source of monochromatic light is placed in front of a soccer ball and a screen is placed behind the ball. The light intensity pattern on the screen is best described as:
   A. a dark disk on a bright background
   B. a dark disk with bright rings outside
   C. a dark disk with a bright spot at its center
   D. a dark disk with a bright spot at its center and bright rings outside
   E. a bright disk with bright rings outside

   ans: D

9. In the equation $\sin\theta = \lambda/a$ for single-slit diffraction, $\theta$ is:
   A. the angle to the first minimum
   B. the angle to the second maximum
   C. the phase angle between the extreme rays
   D. $N\pi$ where $N$ is an integer
   E. $(N + 1/2)\pi$ where $N$ is an integer

   ans: A

10. In the equation $\phi = (2\pi a/\lambda)\sin\theta$ for single-slit diffraction, $\phi$ is:
   A. the angle to the first minimum
   B. the angle to the second maximum
   C. the phase angle between the extreme rays
   D. $N\pi$ where $N$ is an integer
   E. $(N + 1/2)\pi$ where $N$ is an integer

   ans: C

11. No fringes are seen in a single-slit diffraction pattern if:

A. the screen is far away
B. the wavelength is less than the slit width
C. the wavelength is greater than the slit width
D. the wavelength is less than the distance to the screen
E. the distance to the screen is greater than the slit width

ans: C

12. A student wishes to produce a single-slit diffraction pattern in a ripple tank experiment. He considers the following parameters:

1. frequency
2. wavelength
3. water depth
4. slit width

Which two of the above should be decreased to produce more bending?

A. 1, 3
B. 1, 4
C. 2, 3
D. 2, 4
E. 3, 4

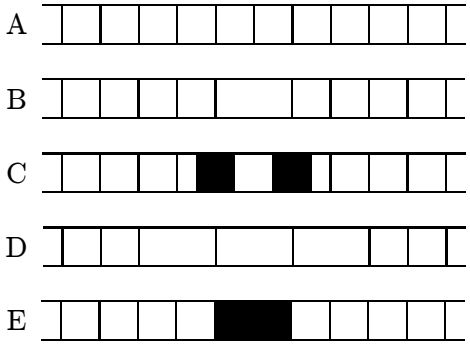
ans: B

13. A parallel beam of monochromatic light is incident on a slit of width 2 cm. The light passing through the slit falls on a screen 2 m away. As the slit width is decreased:

A. the width of the pattern on the screen continuously decreases
B. the width of the pattern on the screen at first decreases but then increases
C. the width of the pattern on the screen increases and then decreases
D. the width of the pattern on the screen remains the same
E. the pattern on the screen changes color going from red to blue

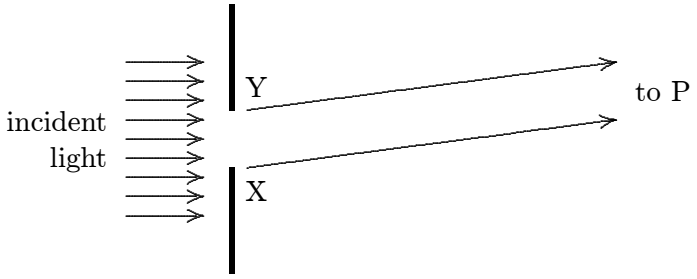
ans: B

14. Monochromatic plane waves of light are incident normally on a single slit. Which one of the five figures below correctly shows the diffraction pattern observed on a distant screen?




ans: B

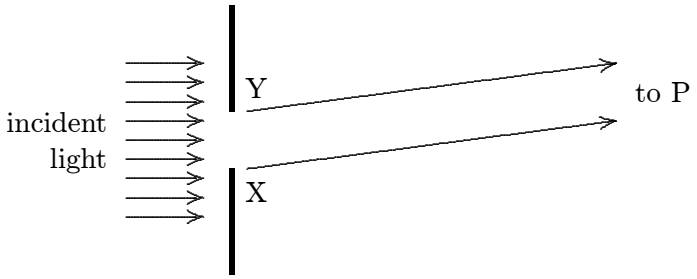
15. The diagram shows a single slit with the direction to a point P on a distant screen shown. At P, the pattern has its second minimum (from its central maximum). If X and Y are the edges of the slit, what is the path length difference (PX) - (PY)?




A. $\lambda/2$
B. $\lambda$
C. $3\lambda/2$
D. $2\lambda$
E. $5\lambda/2$

ans: D

16. The diagram shows a single slit with the direction to a point P on a distant screen shown. At P, the pattern has its maximum nearest the central maximum. If X and Y are the edges of the slit, what is the path length difference (PX) - (PY)?




A. $\lambda/2$
B. $\lambda$
C. $3\lambda/2$
D. $2\lambda$
E. $5\lambda/2$

ans: C

17. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern the phase difference between the Huygens wavelet from the top of the slit and the wavelet from the midpoint of the slit is:

A. $\pi/8$ rad
B. $\pi/4$ rad
C. $\pi/2$ rad
D. $\pi$ rad
E. $3\pi/2$ rad

ans: D

18. At the second minimum adjacent to the central maximum of a single-slit diffraction pattern the Huygens wavelet from the top of the slit is 180° out of phase with the wavelet from:
    A. a point one-fourth of the slit width from the top
    B. the midpoint of the slit
    C. a point one-fourth of the slit width from the bottom of the slit
    D. the bottom of the slit
    E. none of these

    ans: A

19. A plane wave with a wavelength of 500 nm is incident normally on a single slit with a width of $5.0 \times 10^{-6}$ m. Consider waves that reach a point on a far-away screen such that rays from the slit make an angle of 1.0° with the normal. The difference in phase for waves from the top and bottom of the slit is:
    A. 0
    B. 0.55 rad
    C. 1.1 rad
    D. 1.6 rad
    E. 2.2 rad

    ans: C

20. A diffraction pattern is produced on a viewing screen by illuminating a long narrow slit with light of wavelength $\lambda$. If $\lambda$ is increased and no other changes are made:
    A. the intensity at the center of the pattern decreases and the pattern expands away from the bright center
    B. the intensity at the center of the pattern increases and the pattern contracts toward the bright center
    C. the intensity at the center of the pattern does not change and the pattern expands away from the bright center
    D. the intensity at the center of the pattern does not change and the pattern contracts toward the bright center
    E. neither the intensity at the center of the pattern nor the pattern itself change

    ans: C

21. A diffraction pattern is produced on a viewing screen by illuminating a long narrow slit with light of wavelength $\lambda$. If the slit width is decreased and no other changes are made:
    A. the intensity at the center of the pattern decreases and the pattern expands away from the bright center
    B. the intensity at the center of the pattern increases and the pattern contracts toward the bright center
    C. the intensity at the center of the pattern does not change and the pattern expands away from the bright center
    D. the intensity at the center of the pattern does not change and the pattern contracts toward the bright center
    E. neither the intensity at the center of the pattern nor the pattern itself change

    ans: A

22. In order to obtain a good single-slit diffraction pattern, the slit width could be:
    A. $\lambda$
    B. $\lambda/10$
    C. $10\lambda$
    D. $10^4\lambda$
    E. $\lambda/10^4$

    ans: C

23. Consider a single-slit diffraction pattern caused by a slit of width $a$. There is a maximum if $\sin\theta$ is equal to:
    A. slightly more than $3\lambda/2a$
    B. slightly less than $3\lambda/2a$
    C. exactly $3\lambda/2a$
    D. exactly $\lambda/2a$
    E. very nearly $\lambda/2a$

    ans: B

24. Consider a single-slit diffraction pattern caused by a slit of width $a$. There is a minimum if $\sin\theta$ is equal to:
    A. exactly $\lambda/a$
    B. slightly more than $\lambda/a$
    C. slightly less than $\lambda/a$
    D. exactly $\lambda/2a$
    E. very nearly $\lambda/2a$

    ans: A

25. In a single-slit diffraction pattern, the central maximum is about twice as wide as the other maxima. This is because:
    A. half the light is diffracted up and half is diffracted down
    B. the central maximum has both electric and magnetic fields present
    C. the small angle approximation applies only near the central maximum
    D. the screen is flat instead of spherical
    E. none of the above

    ans: E

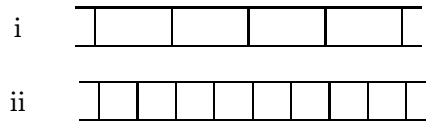
26. The intensity at a secondary maximum of a single-slit diffraction pattern is less than the intensity at the central maximum chiefly because:
    A. some Huygens wavelets sum to zero at the secondary maximum but not at the central maximum
    B. the secondary maximum is further from the slits than the central maximum and intensity decreases as the square of the distance
    C. the Huygens construction is not valid for a secondary maximum
    D. the amplitude of every Huygens wavelet is smaller when it travels to a secondary maximum than when it travels to the central maximum
    E. none of the above

    ans: A

27. Figure (i) shows a double-slit pattern obtained using monochromatic light. Consider the following five possible changes in conditions:

    1. decrease the frequency
    2. increase the frequency
    3. increase the width of each slit
    4. increase the separation between the slits
    5. decrease the separation between the slits

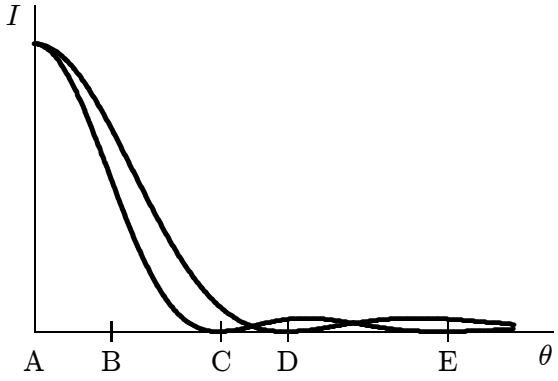
    Which of the above would change Figure (i) into Figure (ii)?




    A. 3 only
    B. 5 only
    C. 1 and 3 only
    D. 1 and 5 only
    E. 2 and 4 only

    ans: E

28. Two wavelengths, 800 nm and 600 nm, are used separately in single-slit diffraction experiments. The diagram shows the intensities on a far-away viewing screen as function of the angle made by the rays with the straight-ahead direction. If both wavelengths are then used simultaneously, at which angle is the light on the screen purely 800-nm light?




    ans: C

29. If we increase the wavelength of the light used to form a double-slit diffraction pattern:
    A. the width of the central diffraction peak increases and the number of bright fringes within the peak increases
    B. the width of the central diffraction peak increases and the number of bright fringes within the peak decreases
    C. the width of the central diffraction peak decreases and the number of bright fringes within the peak increases
    D. the width of the central diffraction peak decreases and the number of bright fringes within the peak decreases
    E. the width of the central diffraction peak increases and the number of bright fringes within the peak stays the same

    ans: E

30. Two slits of width $a$ and separation $d$ are illuminated by a beam of light of wavelength $\lambda$. The separation of the interference fringes on a screen a distance $D$ away is:
    A. $\lambda a/D$
    B. $\lambda d/D$
    C. $\lambda D/d$
    D. $dD/\lambda$
    E. $\lambda D/a$

    ans: C

31. Two slits in an opaque barrier each have a width of 0.020 mm and are separated by 0.050 mm. When coherent monochromatic light passes through the slits the number of interference maxima within the central diffraction maximum:
    A. is 1
    B. is 2
    C. is 4
    D. is 5
    E. cannot be determined unless the wavelength is given

    ans: D

32. When 450-nm light is incident normally on a certain double-slit system the number of interference maxima within the central diffraction maximum is 5. When 900-nm light is incident on the same slit system the number is:
    A. 2
    B. 3
    C. 5
    D. 9
    E. 10

    ans: C

33. In a double-slit diffraction experiment the number of interference fringes within the central diffraction maximum can be increased by:

A. increasing the wavelength
B. decreasing the wavelength
C. decreasing the slit separation
D. increasing the slit width
E. decreasing the slit width

ans: E

34. A diffraction-limited laser of length $\ell$ and aperture diameter $d$ generates light of wavelength $\lambda$. If the beam is directed at the surface of the Moon a distance $D$ away, the radius of the illuminated area on the Moon is approximately:

A. $dD/\ell$
B. $dD/\lambda$
C. $D\lambda/\ell$
D. $D\lambda/d$
E. $\ell\lambda/d$

ans: D

35. Two stars that are close together are photographed through a telescope. The black and white film is equally sensitive to all colors. Which situation would result in the most clearly separated images of the stars?

A. Small lens, red stars
B. Small lens, blue stars
C. Large lens, red stars
D. Large lens, blue stars
E. Large lens, one star red and the other blue

ans: D

36. The resolving power of a telescope can be increased by:

A. increasing the objective focal length and decreasing the eyepiece focal length
B. increasing the lens diameters
C. decreasing the lens diameters
D. inserting a correction lens between objective and eyepiece
E. none of the above

ans: B

37. In the equation $d\sin\theta = m\lambda$ for the lines of a diffraction grating $m$ is:

A. the number of slits
B. the slit width
C. the slit separation
D. the order of the line
E. the index of refraction

ans: D

38. In the equation $d \sin\theta = m\lambda$ for the lines of a diffraction grating $d$ is:
    A. the number of slits
    B. the slit width
    C. the slit separation
    D. the order of the line
    E. the index of refraction

    ans: C

39. As more slits with the same spacing are added to a diffraction grating the lines:
    A. spread farther apart
    B. move closer together
    C. become wider
    D. becomes narrower
    E. do not change in position or width

    ans: D

40. An $N$-slit system has slit separation $d$ and slit width $a$. Plane waves with intensity $I$ and wavelength $\lambda$ are incident normally on it. The angular separation of the lines depends only on:
    A. $a$ and $N$
    B. $a$ and $\lambda$
    C. $N$ and $\lambda$
    D. $d$ and $\lambda$
    E. $I$ and $N$

    ans: D

41. 600-nm light is incident on a diffraction grating with a ruling separation of $1.7 \times 10^{-6}$ m. The second order line occurs at a diffraction angle of:
    A. 0
    B. 10°
    C. 21°
    D. 42°
    E. 45°

    ans: E

42. The widths of the lines produced by monochromatic light falling on a diffraction grating can be reduced by:
    A. increasing the wavelength of the light
    B. increasing the number of rulings without changing their spacing
    C. decreasing the spacing between adjacent rulings without changing the number of rulings
    D. decreasing both the wavelength and the spacing between rulings by the same factor
    E. increasing the number of rulings and decreasing their spacing so the length of the grating remains the same

    ans: B

43. Monochromatic light is normally incident on a diffraction grating that is 1 cm wide and has 10,000 slits. The first order line is deviated at a 30° angle. What is the wavelength, in nm, of the incident light?

A. 300
B. 400
C. 500
D. 600
E. 1000

ans: C

44. A light spectrum is formed on a screen using a diffraction grating. The entire apparatus (source, grating and screen) is now immersed in a liquid of refractive index 1.33. As a result, the pattern on the screen:

A. remains the same
B. spreads out
C. crowds together
D. becomes reversed, with the previously blue end becoming red
E. disappears because the refractive index isn't an integer

ans: C

45. The spacing between adjacent slits on a diffraction grating is $3\lambda$. The deviation $\theta$ of the first order diffracted beam is given by:

A. $\sin(\theta/2) = 1/3$
B. $\sin(\theta/3) = 2/3$
C. $\sin(\theta) = 1/3$
D. $\tan(\theta/2) = 1/3$
E. $\tan(\theta) = 2/3$

ans: C

46. When light of a certain wavelength is incident normally on a certain diffraction grating the line of order 1 is at a diffraction angle of 25°. The diffraction angle for the second order line is:

A. 25°
B. 42°
C. 50°
D. 58°
E. 75°

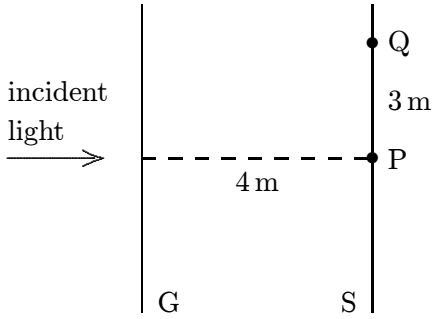
ans D

47. A diffraction grating of width $W$ produces a deviation $\theta$ in second order for light of wavelength $\lambda$. The total number $N$ of slits in the grating is given by:

A. $2W\lambda/\sin\theta$
B. $(W/\lambda)\sin\theta$
C. $\lambda W/2\sin\theta$
D. $(W/2\lambda)\sin\theta$
E. $2\lambda/\sin\theta$

ans: D

48. Light of wavelength $\lambda$ is normally incident on a diffraction grating G. On the screen S, the central line is at P and the first order line is at Q, as shown. The distance between adjacent slits in the grating is:




A. $3\lambda/5$
B. $3\lambda/4$
C. $4\lambda/5$
D. $5\lambda/4$
E. $5\lambda/3$

ans: E

49. 550-nm light is incident normally on a diffraction grating and exactly 6 lines are produced. The ruling separation must be:

A. between $2.75 \times 10^{-7}$ m and $5.50 \times 10^{-7}$ m
B. between $5.50 \times 10^{-7}$ m and $1.10 \times 10^{-6}$ m
C. between $3.30 \times 10^{-6}$ m and $3.85 \times 10^{-6}$ m
D. between $3.85 \times 10^{-6}$ m and $4.40 \times 10^{-6}$ m
E. greater than $4.40 \times 10^{-6}$ m

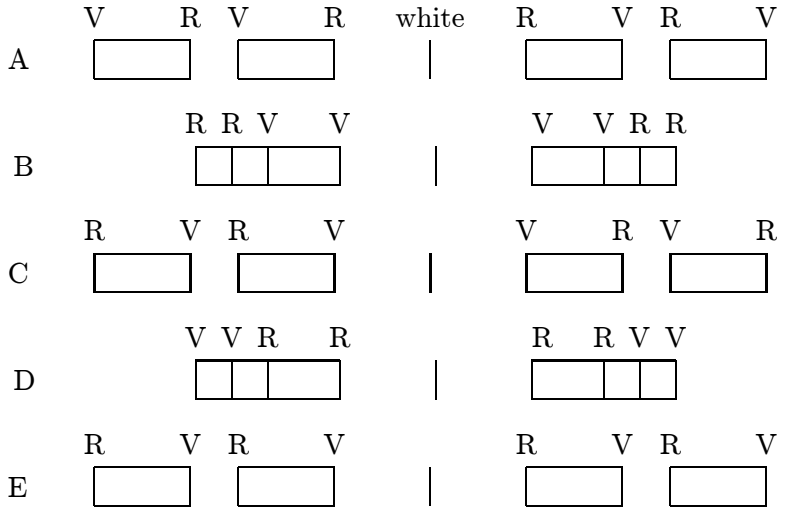
ans: E

50. A mixture of 450-nm and 900-nm light is incident on a diffraction grating. Which of the following is true?

A. all lines of the 900-nm light coincide with even order lines of the 450-nm light
B. all lines of the 450-nm light coincide with even order lines of the 900-nm light
C. all lines of the 900-nm light coincide with odd order lines of the 450-nm light
D. None of the lines of the 450-nm light coincide with lines of the 900-nm light
E. All of the lines of the 450-nm light coincide with lines of the 900-nm light

ans: A

51. A beam of white light (from 400 nm for violet to 700 nm for red) is normally incident on a diffraction grating. It produces two orders on a distant screen. Which diagram below (R = red, V = violet) correctly shows the pattern on the screen?

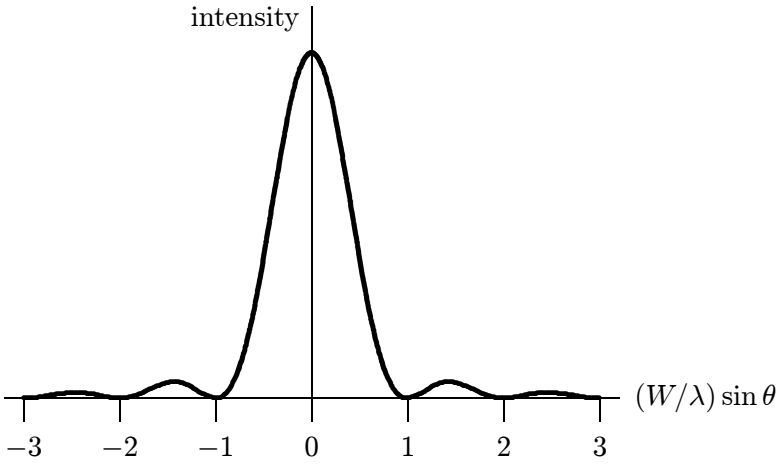



ans: C

52. If white light is incident on a diffraction grating:
    A. the first order lines for all visible wavelengths occur at smaller diffraction angles than any of the second order lines
    B. some first order lines overlap the second order lines if the ruling separation is small but do not if it is large
    C. some first order lines overlap second order lines if the ruling separation is large but do not if it is small
    D. some first order lines overlap second order lines no matter what the ruling separation
    E. first and second order lines have the same range of diffraction angles

    ans: A

53. Light of wavelength is normally incident on some plane optical device. The intensity pattern shown is observed on a distant screen ($\theta$ is the angle measured from the normal of the device). The device could be:




A. a single slit of width $W$
B. a single slit of width $2W$
C. two narrow slits with separation $W$
D. two narrow slits with separation $2W$
E. a diffraction grating with slit separation $W$

ans: A

54. A person with her eye relaxed looks through a diffraction grating at a distant monochromatic point source of light. The slits of the grating are vertical. She sees:

A. one point of light
B. a hazy horizontal strip of light of the same color as the source
C. a hazy strip of light varying from violet to red
D. a sequence of horizontal points of light
E. a sequence of closely spaced vertical lines

ans: D

55. Monochromatic light is normally incident on a diffraction grating. The $m^{\text{th}}$ order line is at a diffraction angle $\theta$ and has width $w$. A wide single slit is now placed in front of the grating and its width is then slowly reduced. As a result:

A. both $\theta$ and $w$ increase
B. both $\theta$ and $w$ decrease
C. $\theta$ remains the same and $w$ increases
D. $\theta$ remains the same and $w$ decreases
E. $\theta$ decreases and $w$ increases

ans: C

56. At a diffraction line phasors associated with waves from the slits of a multiple-slit barrier:
    A. are aligned
    B. form a closed polygon
    C. form a polygon with several sides missing
    D. are parallel but adjacent phasors point in opposite directions
    E. form the arc of a circle

    ans: A

57. For a certain multiple-slit barrier the slit separation is 4 times the slit width. For this system:
    A. the orders of the lines that appear are all multiples of 4
    B. the orders of lines that appear are all multiples of 2
    C. the orders of the missing lines are all multiples of 4
    D. the orders of the missing lines are all multiples of 2
    E. none of the above are true

    ans: C

58. The dispersion $D$ of a grating can have units:
    A. cm
    B. 1/nm
    C. nm/cm
    D. radian
    E. none of these

    ans: B

59. The resolving power $R$ of a grating can have units:
    A. cm
    B. degree/nm
    C. watt
    D. nm/cm
    E. watt/nm

    ans: D

60. The dispersion of a diffraction grating indicates:
    A. the resolution of the grating
    B. the separation of lines of the same order
    C. the number of rulings in the grating
    D. the width of the lines
    E. the separation of lines of different order for the same wavelength

    ans: B

61. The resolving power of a diffraction grating is defined by $R = \lambda/\Delta\lambda$. Here $\lambda$ and $\lambda + \Delta\lambda$ are:
    A. any two wavelengths
    B. any two wavelengths that are nearly the same
    C. two wavelengths for which lines of the same order are separated by $\pi$ radians
    D. two wavelengths for which lines of the same order are separated by $2\pi$ radians
    E. two wavelengths for which lines of the same order are separated by half the width of a maximum

    ans: E

62. A light beam incident on a diffraction grating consists of waves with two different wavelengths. The separation of the two first order lines is great if:
    A. the dispersion is great
    B. the resolution is great
    C. the dispersion is small
    D. the resolution is small
    E. none of the above (line separation does not depend on either dispersion or resolution)

    ans: A

63. To obtain greater dispersion by a diffraction grating:
    A. the slit width should be increased
    B. the slit width should be decreased
    C. the slit separation should be increased
    D. the slit separation should be decreased
    E. more slits with the same width and separation should be added to the system

    ans: D

64. Two nearly equal wavelengths of light are incident on an $N$-slit grating. The two wavelengths are not resolvable. When $N$ is increased they become resolvable. This is because:
    A. more light gets through the grating
    B. the lines get more intense
    C. the entire pattern spreads out
    D. there are more orders present
    E. the lines become more narrow

    ans: E

65. A diffraction grating just resolves the wavelengths 400.0 nm and 400.1 nm in first order. The number of slits in the grating is:
    A. 400
    B. 1000
    C. 2500
    D. 4000
    E. not enough information is given

    ans: D

66. What is the minimum number of slits required in a diffraction grating to just resolve light with wavelengths of 471.0 nm and 471.6 nm?

A. 99
B. 197
C. 393
D. 786
E. 1179

ans: C

67. X rays are:

A. electromagnetic waves
B. negatively charged ions
C. rapidly moving electrons
D. rapidly moving protons
E. rapidly moving neutrons

ans: A

68. Bragg's law for x-ray diffraction is $2d\sin\theta = m\lambda$, where $\theta$ is the angle between the incident beam and:

A. a reflecting plane of atoms
B. the normal to a reflecting plane of atoms
C. the scattered beam
D. the normal to the scattered beam
E. the refracted beam

ans: A

69. Bragg's law for x-ray diffraction is $2d\sin\theta = m\lambda$, where the quantity $d$ is:

A. the height of a unit cell
B. the smallest interatomic distance
C. the distance from detector to sample
D. the distance between planes of atoms
E. the usual calculus symbol for a differential

ans: D

70. Which of the following is true for Bragg diffraction but not for diffraction from a grating?

A. Two different wavelengths may be used
B. For a given wavelength, a maximum may exist in several directions
C. Long waves are deviated more than short ones
D. There is only one grating spacing
E. Maxima occur only for particular angles of incidence

ans: E

71. The longest x-ray wavelength that can be diffracted by crystal planes with a separation of 0.316 nm is:

A. 0.158 nm
B. 0.316 nm
C. 0.474 nm
D. 0.632 nm
E. 1.26 nm

ans: D

72. A beam of x rays of wavelength 0.20 nm is diffracted by a set of planes in a crystal whose separation is $3.1 \times 10^{-8}$ cm. The smallest angle between the beam and the crystal planes for which a reflection occurs is:

A. 0.70 rad
B. 0.33 rad
C. 0.033 rad
D. 0.066 rad
E. no such angle exists

ans: C

73. An x-ray beam of wavelength $3 \times 10^{-11}$ m is incident on a calcite crystal of lattice spacing 0.3 nm. The smallest angle between crystal planes and the x-ray beam that will result in constructive interference is:

A. 2.87°
B. 5.73°
C. 11.63°
D. 23.27°
E. none of these

ans: A

74. A beam of x rays of wavelength 0.10 nm is found to diffract in second order from the face of a LiF crystal at a Bragg angle of 30°. The distance between adjacent crystal planes, in nm, is about:

A. 0.15
B. 0.20
C. 0.25
D. 0.30
E. 0.40

ans: B

## Chapter 37: SPECIAL THEORY OF RELATIVITY

1. A basic postulate of Einstein's theory of relativity is:
   A. moving clocks run more slowly than when they are at rest
   B. moving rods are shorter than when they are at rest
   C. light has both wave and particle properties
   D. the laws of physics must be the same for observers moving with uniform velocity relative to each other
   E. everything is relative

   ans: D

2. A consequence of Einstein's theory of relativity is:
   A. moving clocks run more slowly than when they are at rest
   B. moving rods are longer than when they are at rest
   C. light has both wave and particle properties
   D. the laws of physics must appear the same to all observers moving with uniform velocity relative to each other
   E. everything is relative

   ans: A

3. A consequence of Einstein's theory of relativity is:
   A. moving clocks run faster than when they are at rest
   B. moving rods are shorter than when they are at rest
   C. light has both wave and particle properties
   D. the laws of physics must appear the same to all observers moving with uniform velocity relative to each other
   E. everything is relative

   ans: B

4. According to the theory of relativity:
   A. moving clocks run fast
   B. energy is not conserved in high speed collisions
   C. the speed of light must be measured relative to the ether
   D. momentum is not conserved in high speed collisions
   E. none of the above are true

   ans: E

5. Two events occur simultaneously on the $x$ axis of reference frame S, one at $x = -a$ and the other at $x = +a$. According to an observer moving in the positive $x$ direction:
   A. the event at $x = +a$ occurs first
   B. the event at $x = -a$ occurs first
   C. either event might occur first, depending on the value of $a$ and the observer's speed
   D. the events are simultaneous
   E. none of the above

   ans: A

6. Two events occur simultaneously at separated points on the $y$ axis of reference frame S. According to an observer moving in the positive $x$ direction:
   A. the event with the greater $y$ coordinate occurs first
   B. the event with the greater $y$ coordinate occurs last
   C. either event might occur first, depending on the observer's speed
   D. the events are simultaneous
   E. none of the above

   ans: D

7. A train traveling very fast ($v = 0.6c$) has an engineer (E) at the front, a guard (G) at the rear, and an observer (S′) exactly half way between them. Both E and G are equipped with yellow signaling lamps. The train passes a station, closely observed by the station master (S). Both E and G use their lamps to send signals. According to both S and S′ these signals arrive simultaneously at the instant S′ is passing S. According to S′:
   A. E and G sent their signals simultaneously from different distances
   B. G sent his signal before E and from farther away
   C. G sent his signal before E but was the same distance away
   D. E sent his signal before G and from farther away
   E. none of the above

   ans: E

8. The proper time between two events is measured by clocks at rest in a reference frame in which the two events:
   A. occur at the same time
   B. occur at the same coordinates
   C. are separated by the distance a light signal can travel during the time interval
   D. occur in Boston
   E. satisfy none of the above

   ans: B

9. The spaceship U.S.S. Enterprise, traveling through the galaxy, sends out a smaller explorer craft that travels to a nearby planet and signals its findings back. The proper time for the trip to the planet is measured by clocks:
   A. on board the Enterprise
   B. on board the explorer craft
   C. on Earth
   D. at the center of the galaxy
   E. none of the above

   ans: B

10. Spaceship A, traveling past us at $0.7c$, sends a message capsule to spaceship B, which is in front of A and is traveling in the same direction as A at $0.8c$ relative to us. The capsule travels at $0.95c$ relative to us. A clock that measures the proper time between the sending and receiving of the capsule travels:

A. in the same direction as the spaceships at $0.7c$ relative to us
B. in the opposite direction from the spaceships at $0.7c$ relative to us
C. in the same direction as the spaceships at $0.8c$ relative to us
D. in the same direction as the spaceships at $0.95c$ relative to us
E. in the opposite direction from the spaceships at $0.95c$ relative to us

ans: D

11. Two events occur on the $x$ axis separated in time by $\Delta t$ and in space by $\Delta x$. A reference frame, traveling at less than the speed of light, in which the two events occur at the same time:

A. exists no matter what the values of $\Delta x$ and $\Delta t$
B. exists only if $\Delta x/\Delta t < c$
C. exists only if $\Delta x/\Delta t > c$
D. exists only if $\Delta x/\Delta t = c$
E. does not exist under any condition

ans: C

12. Two events occur on the $x$ axis separated in time by $\Delta t$ and in space by $\Delta x$. A reference frame, traveling at less than the speed of light, in which the two events occur at the same coordinate:

A. exists no matter what the values of $\Delta x$ and $\Delta t$
B. exists only if $\Delta x/\Delta t < c$
C. exists only if $\Delta x/\Delta t > c$
D. exists only if $\Delta x/\Delta t = c$
E. does not exist under any condition

ans: B

13. As we watch, a spaceship passes us in time $\Delta t$. The crew of the spaceship measures the passage time and finds it to be $\Delta t'$. Which of the following statements is true?

A. $\Delta t$ is the proper time for the passage and it is smaller than $\Delta t'$
B. $\Delta t$ is the proper time for the passage and it is greater than $\Delta t'$
C. $\Delta t'$ is the proper time for the passage and it is smaller than $\Delta t$
D. $\Delta t'$ is the proper time for the passage and it is greater than $\Delta t$
E. None of the above statements are true.

ans: C

14. A millionairess was told in 1992 that she had exactly 15 years to live. However, if she travels away from the Earth at $0.8c$ and then returns at the same speed, the last New Year's day the doctors expect her to celebrate is:

A. 2001
B. 2003
C. 2007
D. 2010
E. 2017

ans: E

15. Two events occur 100 m apart with an intervening time interval of 0.60 $\mu$s. The speed of a reference frame in which they occur at the same coordinate is:

A. 0
B. $0.25c$
C. $0.56c$
D. $1.1c$
E. $1.8c$

ans: C

16. Two independent events occur 100 m apart with an intervening time interval of 0.42 $\mu$s. The proper time in $\mu$s between the events is:

A. 0
B. 0.16
C. 0.26
D. 0.42
E. 0.69

ans: C

17. Two events occur 100 m apart with an intervening time interval of 0.37 $\mu$s. The speed of a clock that measures the proper time between the events is:

A. 0
B. $0.45c$
C. $0.56c$
D. $0.90c$
E. $1.8c$

ans: D

18. A rocket traveling with constant velocity makes a $8.4 \times 10^{15}$ m trip in exactly one year. The proper time in years between events that mark the beginning and end of the trip is:

A. 0.21
B. 0.46
C. 1.0
D. 2.2
E. 4.7

ans: B

19. An observer notices that a moving clock runs slow by a factor of exactly 10. The speed of the clock is:

A. $0.100c$
B. $0.0100c$
C. $0.990c$
D. $0.900c$
E. $0.995c$

ans: E

20. A meson when at rest decays $2\,\mu$s after it is created. If moving in the laboratory at $0.99c$, its lifetime according to laboratory clocks would be:

A. the same
B. 0.28 si
C. $14\,\mu$s
D. 4.6 s
E. none of these

ans: C

21. Pi mesons at rest have a half-life of $T$. If a beam of pi mesons is traveling at a speed of $v = \beta c$, the distance in which the intensity of the beam is halved is:

A. $c\beta T(1-\beta^2)^{-1/2}$
B. $c\beta T[(1+\beta)/(1-\beta)]^{1/2}$
C. $\beta vT$
D. $(1-\beta^2)^{1/2}vT$
E. none of the above

ans: A

22. A meson moving through a laboratory of length $x$ at a speed $v$ decays after a lifetime $T$ as measured by an observer at rest in the laboratory. If the meson were at rest in the laboratory its lifetime would be:

A. $T(1-v/c)$
B. $T(1-v/c)^{-1}$
C. $T(1-v^2/c^2)^{-1/2}$
D. $T(1-v^2/c^2)^{1/2}$
E. $(T-vx/c^2)(1-v^2/c^2)^{-1/2}$

ans: D

23. A measurement of the length of an object that is moving relative to the laboratory consists of noting the coordinates of the front and back:

A. at different times according to clocks at rest in the laboratory
B. at the same time according to clocks that move with the object
C. at the same time according to clocks at rest in the laboratory
D. at the same time according to clocks at rest with respect to the fixed stars
E. none of the above

ans: C

24. A meter stick moves in the direction of its length through a laboratory. According to measurements taken in the laboratory, its length is 0.31 m The speed of the meter stick relative to the laboratory is:

A. $0.096c$
B. $0.31c$
C. $0.69c$
D. $0.83c$
E. $0.95c$

ans: E

25. A meter stick moves sideways at $0.95c$. According to measurements taken in the laboratory, its length is:

A. 0
B. 0.098 m
C. 0.31 m
D. 3.2 m
E. 1.0 m

ans: E

26. A rocket ship of rest length 100 m is moving at speed $0.8c$ past a timing device that records the time interval between the passage of the front and back ends of the ship. This time interval is:

A. 0.20 $\mu$s
B. 0.25 $\mu$s
C. 0.33 $\mu$s
D. 0.52 $\mu$s
E. 0.69 $\mu$s

ans: B

27. The length of a meter stick moving at $0.95c$ in the direction of its length is measured by simultaneously marking its ends on a stationary axis. As measured by clocks moving with the stick, the time interval between the making of the back mark and the making of the front mark is:

A. 0
B. $3.1 \times 10^{-10}$ s
C. $1.0 \times 10^{-9}$ s
D. $3.2 \times 10^{-9}$ s
E. $1.0 \times 10^{-8}$ s

ans: D

28. A certain automobile is 6 m long if at rest. If it is measured to be 4/5 as long, its speed is:

A. $0.1c$
B. $0.3c$
C. $0.6c$
D. $0.8c$
E. $> 0.95c$

ans: C

29. As a rocket ship moves by at $0.95c$ a mark is made on a stationary axis at the front end of the rocket and $9 \times 10^{-8}$ s later a mark is made on the axis at the back end. The marks are found to be 100 m apart. The rest length of the rocket is:

A. 31 m
B. 78 m
C. 100 m
D. 240 m
E. 320 m

ans: D

30. A clock is moving along the $x$ axis at $0.6c$. It reads zero as it passes the origin ($x = 0$). When it passes the $x = 180\,\text{m}$ mark on the $x$ axis the clock reads:

A. $0.60\,\mu\text{s}$
B. $0.80\,\mu\text{s}$
C. $1.00\,\mu\text{s}$
D. $1.25\,\mu\text{s}$
E. $1.67\,\mu\text{s}$

ans: B

31. Relative to reference frame 1, reference frame 2 moves with speed $v$ in the negative $x$ direction. When the origins of the two frames coincide the clocks in both frames are set to zero. An event occurs at coordinate $x_1$ and time $t_1$ as measured in reference frame 1 and at coordinate $x_2$ and time $t_2$ as measured in frame 2. If $\gamma = 1/\sqrt{1 - v^2/c^2}$, then the coordinates and times of the event are related by:

A. $x_2 = \gamma[x_1 - vt_1]$ and $t_2 = \gamma[t_1 - vx_1/c^2]$
B. $x_2 = \gamma[x_1 - vt_1]$ and $t_2 = \gamma[t_1 + vx_1/c^2]$
C. $x_2 = \gamma[x_1 + vt_1]$ and $t_2 = \gamma[t_1 - vx_1/c^2]$
D. $x_2 = \gamma[x_1 + vt_1]$ and $t_2 = \gamma[t_1 + vx_1/c^2]$
E. none of the above are correct

ans: D

32. An event occurs at $x = 500\,\text{m}$, $t = 0.90\,\mu\text{s}$ in one frame of reference. Another frame is moving at $0.90c$ in the positive $x$ direction. The origins coincide at $t = 0$ and clocks in the second frame are zeroed when the origins coincide. The coordinate and time of the event in the second frame are:

A. $500\,\text{m}$, $0.90\,\mu\text{s}$
B. $1700\,\text{m}$, $5.5\,\mu\text{s}$
C. $740\,\text{m}$, $2.4\,\mu\text{s}$
D. $260\,\text{m}$, $-0.60\,\mu\text{s}$
E. $590\,\text{m}$, $-1.4\,\mu\text{s}$

ans: E

33. An event occurs at $x = 500\,\text{m}$, $t = 0.90\,\mu\text{s}$ in one frame of reference. Another frame is moving at $0.90c$ in the negative $x$ direction. The origins coincide at $t = 0$ and clocks in the second frame are zeroed when the origins coincide. The coordinate and time of the event in the second frame are:

A. $500\,\text{m}$, $0.90\,\mu\text{s}$
B. $1700\,\text{m}$, $5.5\,\mu\text{s}$
C. $740\,\text{m}$, $2.4\,\mu\text{s}$
D. $260\,\text{m}$, $-0.60\,\mu\text{s}$
E. $590\,\text{m}$, $-1.4\,\mu\text{s}$

ans: B

34. Two flashes of light occur simultaneously at $t = 0$ in reference frame S, one at $x = 0$ and the other at $x = 600\,\text{m}$. They are observed in reference frame S′, which is moving at $0.95c$ in the positive $x$ direction. The origins of the two frames coincide at $t = 0$ and the clocks of S′ are zeroed when the origins coincide. In S′ the coordinate where the leading edges of the two light flashes meet and the time when they meet are:

A. 300 m, 1.0 $\mu$s
B. 15 m, 0.050 $\mu$s
C. 585 m, 1.95 $\mu$s
D. 49 m, 0.16 $\mu$s
E. 1900 m, 0.16 $\mu$s

ans: D

35. Frame S′ moves in the positive $x$ direction at $0.6c$ with respect to frame S. A particle moves in the positive $x$ direction at $0.4c$ as measured by an observer in S′. The speed of the particle as measured by an observer in S is:

A. $c/5$
B. $5c/19$
C. $8c/25$
D. $25c/31$
E. $c$

ans: D

36. Star S1 is moving away from us at a speed of $0.8c$. Star S2 is moving away from us in the opposite direction at a speed of $0.5c$. The speed of S1 as measured by an observer on S2 is:

A. $0.21c$
B. $0.5c$
C. $0.93c$
D. $1.3c$
E. $2.17c$

ans: C

37. Observer A measures the velocity of a rocket as $\vec{v}$ and a comet as $\vec{u}$. Here $\vec{u}$ and $\vec{v}$ are parallel and in the direction of the observer's positive $x$ axis. The speed of the comet as measured by observer B on the rocket is:

A. $(u - v)/(1 - uv/c^2)$
B. $(u - v)/(1 - v^2/c^2)$
C. $(u - v)/(1 - v^2/c^2)^{1/2}$
D. $(u - v)/(1 + uv/c^2)$
E. $(u + v)/(1 - uv/c^2)$

ans: A

38. Two electrons move in opposite directions at $0.70c$ as measured in the laboratory. The speed of one electron as measured from the other is:

A. $0.35c$
B. $0.70c$
C. $0.94c$
D. $1.00c$
E. $1.40c$

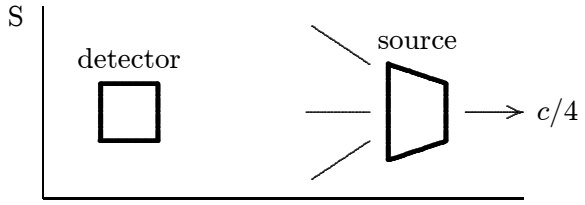
ans: C

39. Light from some stars shows an apparent change in frequency because of:

A. interference
B. refraction by layers of air
C. diffraction
D. reflection
E. relative motion

ans: E

40. While emitting light of proper frequency $f_0$, a source moves to the right with speed $c/4$ relative to reference frame S. A detector, to the left of the source, measures the frequency to be $f$, which is greater than $f_0$. This means:




A. the detector is moving to the right with a speed that is greater than $c/4$ relative to S
B. the detector is moving to the right with a speed that is less than $c/4$ relative to S
C. the detector is moving to the left with a speed that is greater than $c/4$ relative to S
D. the detector is moving to the left with a speed that is less than $c/4$ relative to S
E. the detector is not moving

ans: A

41. Light from a stationary spaceship is observed, then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

A. a higher frequency and a longer wavelength than before
B. a lower frequency and a shorter wavelength than before
C. a higher frequency and a shorter wavelength than before
D. a lower frequency and a longer wavelength than before
E. the same frequency and wavelength as before

ans: D

42. A train traveling very fast ($v = 0.6c$) has an engineer (E) at the front, a guard (G) at the rear, and a passenger (S′) exactly halfway between them. Both E and G are equipped with yellow signaling lamps. The train passes a station, closely observed by the station master (S). Both E and G use their lamps to send signals. According to both S and S′ these signals arrive simultaneously at the instant S′ is passing S. According to S, the signal from E will look _____ and that from G will look _____.

A. red, blue
B. yellow, yellow
C. blue, red
D. blue, blue
E. red, red

ans: A

43. A console lamp in the cabin of a spaceship appears green when the ship and observer are both at rest on Earth. When the ship is moving at $0.90c$ away from Earth, passengers on board see:

A. nothing (the frequency is too high to be seen)
B. nothing (the frequency is too low to be seen)
C. a red lamp
D. a violet lamp
E. a green lamp

ans: E

44. A spectral line of a certain star is observed to be "red shifted" from a wavelength of 500 nm to a wavelength of 1500 nm. Interpreting this as a Doppler effect, the speed of recession of this star is:

A. $0.33c$
B. $0.50c$
C. $0.71c$
D. $0.8c$
E. $c$

ans: D

45. A source at rest emits light of wavelength 500 nm. When it is moving at $0.90c$ toward an observer, the observer detects light of wavelength:

A. 26 nm
B. 115 nm
C. 500 nm
D. 2200 nm
E. 9500 nm

ans: B

46. A source at rest emits light of wavelength 500 nm. When it is moving at $0.90c$ away from an observer, the observer detects light of wavelength:

A. 26 nm
B. 115 nm
C. 500 nm
D. 2200 nm
E. 9500 nm

ans: D

47. Visible light, with a frequency of $6.0 \times 10^{14}$ Hz, is reflected from a spaceship moving directly away at a speed of $0.90c$. The frequency of the reflected waves observed at the source is:

A. $3.2 \times 10^{13}$ Hz
B. $1.4 \times 10^{14}$ Hz
C. $6.0 \times 10^{14}$ Hz
D. $2.6 \times 10^{15}$ Hz
E. $1.1 \times 10^{16}$ Hz

ans: A

48. How fast should you move away from a $6.0 \times 10^{14}$ Hz light source to observe waves with a frequency of $4.0 \times 10^{14}$ Hz?

A. $0.20c$
B. $0.38c$
C. $0.45c$
D. $0.51c$
E. $0.76c$

ans: B

49. A particle with rest mass m moves with speed $0.6c$. Its kinetic energy is:

A. $0.18mc^2$
B. $0.22mc^2$
C. $0.25mc^2$
D. $mc^2$
E. $1.25mc^2$

ans: C

50. An electron is moving at $0.6c$. If we calculate its kinetic energy using $(1/2)mv^2$, we get a result that is:

A. just right
B. just half enough
C. twice the correct value
D. about 1% too low
E. about 28% too low

ans: E

51. The velocity of an electron is changed from $c/2$ in the $-x$ direction to $c/2$ in the $+x$ direction. As a result, its kinetic energy changes by:

A. $mc^2$
B. $0.5mc^2$
C. $\sqrt{2}mc^2$
D. $2mc^2$
E. zero

ans: E

52. The work that must be done to increase the speed of an electron ($m = 9.11 \times 10^{-31}$ kg) from $0.90c$ to $0.95c$ is:

A. $2.6 \times 10^{-13}$ J
B. $8.2 \times 10^{-13}$ J
C. $3.2 \times 10^{-13}$ J
D. $7.4 \times 10^{-14}$ J
E. $3.8 \times 10^{-15}$ J

ans: D

53. An electron ($m = 9.11 \times 10^{-31}$ kg) has a speed of $0.95c$. Its kinetic energy is:

A. $8.2 \times 10^{-14}$ J
B. $1.8 \times 10^{-13}$ J
C. $2.0 \times 10^{-13}$ J
D. $2.2 \times 10^{-13}$ J
E. $2.6 \times 10^{-13}$ J

ans: B

54. An electron ($m = 9.11 \times 10^{-31}$ kg) has a speed of $0.95c$. The magnitude of its momentum is:

A. $2.6 \times 10^{-22}$ kg $\cdot$ m/s
B. $2.9 \times 10^{-22}$ kg $\cdot$ m/s
C. $6.0 \times 10^{-22}$ kg $\cdot$ m/s
D. $8.3 \times 10^{-22}$ kg $\cdot$ m/s
E. $8.8 \times 10^{-22}$ kg $\cdot$ m/s

ans: D

55. According to the special theory of relativity:

A. all forms of energy have mass-like properties
B. moving particles lose mass
C. momentum is not conserved in high speed collisions
D. a rod moving rapidly sideways is shorter along its length
E. a rod moving rapidly sideways is longer along its length

ans: A

56. The mass of a particle is $m$. In order for its total energy to be twice its rest energy, its momentum must be:

A. $mc/2$
B. $mc/\sqrt{2}$
C. $mc$
D. $\sqrt{3}mc$
E. $2mc$

ans: D

57. If the kinetic energy of a particle is equal to its rest energy then its speed must be:

A. $0.25c$
B. $0.50c$
C. $0.87c$
D. $c$
E. unknown unless its mass is given

ans: C

58. If the mass of a particle is zero its speed must be:

A. $c$
B. infinite
C. 0
D. any speed less than $c$
E. any speed greater than $c$

ans: A

59. A particle with zero mass and energy $E$ carries momentum:

A. $Ec$
B. $Ec^2$
C. $\sqrt{Ec}$
D. $E/c$
E. $E/c^2$

ans: D

60. According to relativity theory a particle of mass $m$ with a momentum of $2mc$ has a speed of:

A. $2c$
B. $4c$
C. $c$
D. $c/2$
E. $0.89c$

ans: E

61. If the kinetic energy of a free particle is much greater than its rest energy then its kinetic energy is proportional to:

A. the magnitude of its momentum
B. the square of the magnitude of its momentum
C. the square root of the magnitude of its momentum
D. the reciprocal of the magnitude of its momentum
E. none of the above

ans: A

62. If the kinetic energy of a free particle is much less than its rest energy then its kinetic energy is proportional to:

A. the magnitude of its momentum
B. the square of the magnitude of its momentum
C. the square root of the magnitude of its momentum
D. the reciprocal of the magnitude of its momentum
E. none of the above

ans: B

63. The magnitude of the momentum of a particle can never exceed:

A. $mc$, where $m$ is its mass
B. $E/c$, where $E$ is its total energy
C. $K/c$, where $K$ is its kinetic energy
D. none of the above, but there is an upper limit
E. none of the above; there is no upper limit

ans: B

64. An electron ($m = 9.11 \times 10^{-31}\,\text{kg}$) has a momentum of $4.0 \times 10^{-22}\,\text{kg} \cdot \text{m/s}$. Its kinetic energy is:

A. $6.3 \times 10^{-14}\,\text{J}$
B. $8.2 \times 10^{-14}\,\text{J}$
C. $1.2 \times 10^{-13}\,\text{J}$
D. $1.5 \times 10^{-13}\,\text{J}$
E. $2.7 \times 10^{-13}\,\text{J}$

ans: A

65. A certain particle has a kinetic energy of $3.2 \times 10^{-10}\,\text{J}$ and a momentum of $1.7 \times 10^{-18}\,\text{kg} \cdot \text{m/s}$. Its mass is:

A. $9.11 \times 10^{-31}\,\text{kg}$
B. $2.7 \times 10^{-27}\,\text{kg}$
C. $4.5 \times 10^{-27}\,\text{kg}$
D. $6.3 \times 10^{-27}\,\text{kg}$
E. $8.6 \times 10^{-27}\,\text{kg}$

ans: B

66. An electron ($m = 9.11 \times 10^{-31}$ kg, $q = 1.60 \times 10^{-19}$ C) travels at $0.95c$ around a circular orbit perpendicular to a uniform 1.8-T magnetic field. The radius of its orbit is:

A. 0.28 mm
B. 0.90 mm
C. 1.1 mm
D. 2.9 mm
E. 4.7 mm

ans: D

67. An electron ($m = 9.11 \times 10^{-31}$ kg, $q = 1.60 \times 10^{-19}$ C) travels around a 1.7-mm radius circular orbit perpendicular to a 2.8-T magnetic field. Its speed is:

A. $0.16c$
B. $0.36c$
C. $0.94c$
D. $c$
E. $2.8c$

ans: C

## Chapter 38: PHOTONS AND MATTER WAVES

1. The units of the Planck constant $h$ are those of:
   A. energy
   B. power
   C. momentum
   D. angular momentum
   E. frequency

   ans: D

2. If $h$ is the Planck constant, then $\hbar$ is:
   A. $2\pi h$
   B. $2h$
   C. $h/2$
   D. $h/2\pi$
   E. $2h/\pi$

   ans: D

3. The quantization of energy, $E = nhf$, is not important for an ordinary pendulum because:
   A. the formula applies only to mass-spring oscillators
   B. the allowed energy levels are too closely spaced
   C. the allowed energy levels are too widely spaced
   D. the formula applies only to atoms
   E. the value of $h$ for a pendulum is too large

   ans: B

4. The frequency of light beam A is twice that of light beam B. The ratio $E_A/E_B$ of photon energies is:
   A. 1/2
   B. 1/4
   C. 1
   D. 2
   E. 4

   ans: D

5. The wavelength of light beam B is twice the wavelength of light beam B. The energy of a photon in beam A is:
   A. half the energy of a photon in beam B
   B. one-fourth the energy of a photon in beam B
   C. equal to the energy of a photon in beam B
   D. twice the energy of a photon in beam B
   E. four times the energy of a photon in beam B

   ans: A

6. A photon in light beam A has twice the energy of a photon in light beam B. The ratio $p_A/p_B$ of their momenta is:

   A. 1/2
   B. 1/4
   C. 1
   D. 2
   E. 4

   ans: D

7. Which of the following electromagnetic radiations has photons with the greatest energy?

   A. blue light
   B. yellow light
   C. x rays
   D. radio waves
   E. microwaves

   ans: C

8. Which of the following electromagnetic radiations has photons with the greatest momentum?

   A. blue light
   B. yellow light
   C. x rays
   D. radio waves
   E. microwaves

   ans: C

9. Rank following electromagnetic radiations according to the energies of their photons, from least to greatest.

   1. blue light
   2. yellow light
   3. x rays
   4. radio waves

   A. 1, 2, 3, 4
   B. 4, 2, 1, 3
   C. 4, 1, 2, 3
   D. 3, 2, 1, 4
   E. 3, 1, 2, 4

   ans: B

10. The intensity of a uniform light beam with a wavelength of 500 nm is 2000 $\mathrm{W/m^2}$. The photon flux (in number/$\mathrm{m^2 \cdot}$ s) is about:

    A. $5 \times 10^{17}$
    B. $5 \times 10^{19}$
    C. $5 \times 10^{21}$
    D. $5 \times 10^{23}$
    E. $5 \times 10^{25}$

    ans: C

11. The concentration of photons in a uniform light beam with a wavelength of 500 nm is $1.7 \times 10^{13}\,\mathrm{m}^{-3}$. The intensity of the beam is:
    A. $6.7 \times 10^{-6}\,\mathrm{W/m^2}$
    B. $1.0 \times 10^{3}\,\mathrm{W/m^2}$
    C. $2.0 \times 10^{3}\,\mathrm{W/m^2}$
    D. $4.0 \times 10^{3}\,\mathrm{W/m^2}$
    E. $3.2 \times 10^{2}\,\mathrm{W/m^2}$

    ans: C

12. Light beams A and B have the same intensity but the wavelength associated with beam A is longer than that associated with beam B. The photon flux (number crossing a unit area per unit time) is:
    A. greater for A than for B
    B. greater for B than for A
    C. the same for A and B
    D. greater for A than for B only if both have short wavelengths
    E. greater for B than for A only if both have short wavelengths

    ans: A

13. In a photoelectric effect experiment the stopping potential is:
    A. the energy required to remove an electron from the sample
    B. the kinetic energy of the most energetic electron ejected
    C. the potential energy of the most energetic electron ejected
    D. the photon energy
    E. the electric potential that causes the electron current to vanish

    ans: E

14. In a photoelectric effect experiment at a frequency above cut off, the stopping potential is proportional to:
    A. the energy of the least energetic electron before it is ejected
    B. the energy of the least energetic electron after it is ejected
    C. the energy of the most energetic electron before it is ejected
    D. the energy of the most energetic electron after it is ejected
    E. the electron potential energy at the surface of the sample

    ans: D

15. In a photoelectric effect experiment at a frequency above cut off, the number of electrons ejected is proportional to:
    A. their kinetic energy
    B. their potential energy
    C. the work function
    D. the frequency of the incident light
    E. the number of photons that hit the sample

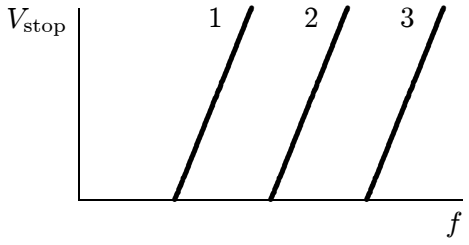
    ans: E

16. In a photoelectric effect experiment no electrons are ejected if the frequency of the incident light is less than $A/h$, where $h$ is the Planck constant and $A$ is:

A. the maximum energy needed to eject the least energetic electron
B. the minimum energy needed to eject the least energetic electron
C. the maximum energy needed to eject the most energetic electron
D. the minimum energy needed to eject the most energetic electron
E. the intensity of the incident light

ans: D

17. The diagram shows the graphs of the stopping potential as a function of the frequency of the incident light for photoelectric experiments performed on three different materials. Rank the materials according to the values of their work functions, from least to greatest.




A. 1, 2, 3
B. 3, 2, 1
C. 2, 3, 1
D. 2, 1, 3
E. 1, 3, 2

ans: A

18. The work function for a certain sample is 2.3 eV. The stopping potential for electrons ejected from the sample by $7.0 \times 10^{14}$-Hz electromagnetic radiation is:

A. 0
B. 0.60 V
C. 2.3 V
D. 2.9 V
E. 5.2 V

ans: B

19. The stopping potential for electrons ejected by $6.8 \times 10^{14}$-Hz electromagnetic radiation incident on a certain sample is 1.8 V. The kinetic energy of the most energetic electrons ejected and the work function of the sample, respectively, are:

A. 1.8 eV, 2.8 eV
B. 1.8 eV, 1.0 eV
C. 1.8 eV, 4.6 eV
D. 2.8 eV, 1.0 eV
E. 1.0 eV, 4.6 eV

ans: B

20. Separate Compton effect experiments are carried out using visible light and x rays. The scattered radiation is observed at the same scattering angle. For these experiments:
    A. the x rays have the greater shift in wavelength and the greater change in photon energy
    B. the two radiations have the same shift in wavelength and the x rays have the greater change in photon energy
    C. the two radiations have the same shift in wavelength and the visible light has the greater change in photon energy
    D. the two radiations have the same shift in wavelength and the same change in photon energy
    E. the visible light has the greater shift in wavelength and the greater shift in photon energy

    ans: B

21. In Compton scattering from stationary particles the maximum change in wavelength can be made smaller by using:
    A. higher frequency radiation
    B. lower frequency radiation
    C. more massive particles
    D. less massive particles
    E. particles with greater charge

    ans: C

22. Of the following, Compton scattering from electrons is most easily observed for:
    A. microwaves
    B. infrared light
    C. visible light
    D. ultraviolet light
    E. x rays

    ans: E

23. In Compton scattering from stationary electrons the largest change in wavelength occurs when the photon is scattered through:
    A. 0°
    B. 22.5°
    C. 45°
    D. 90°
    E. 180°

    ans: E

24. In Compton scattering from stationary electrons the frequency of the emitted light is independent of:
    A. the frequency of the incident light
    B. the speed of the electron
    C. the scattering angle
    D. the electron recoil energy
    E. none of the above

    ans: E

25. In Compton scattering from stationary electrons the largest change in wavelength that can occur is:

A. $2.43 \times 10^{-15}$ m
B. $2.43 \times 10^{-12}$ m
C. $2.43 \times 10^{-9}$ m
D. dependent on the frequency of the incident light
E. dependent on the work function

ans: B

26. Electromagnetic radiation with a wavelength of $5.7 \times 10^{-12}$ m is incident on stationary electrons. Radiation that has a wavelength of $6.57 \times 10^{-12}$ m is detected at a scattering angle of:

A. $10^\circ$
B. $121^\circ$
C. $40^\circ$
D. $50^\circ$
E. $69^\circ$

ans: D

27. Electromagnetic radiation with a wavelength of $3.5 \times 10^{-12}$ m is scattered from stationary electrons and photons that have been scattered through $50^\circ$ are detected. An electron from which one of these photons was scattered receives an energy of:

A. 0
B. $1.1 \times 10^{-14}$ J
C. $1.9 \times 10^{-14}$ J
D. $2.3 \times 10^{-14}$ J
E. $1.3 \times 10^{-13}$ J

ans: B

28. Electromagnetic radiation with a wavelength of $3.5 \times 10^{-12}$ m is scattered from stationary electrons and photons that have been scattered through $50^\circ$ are detected. After a scattering event the magnitude of the electron's momentum is:

A. 0
B. $1.5 \times 10^{-22}$ kg · m/s
C. $2.0 \times 10^{-22}$ kg · m/s
D. $2.2 \times 10^{-22}$ kg · m/s
E. $8.7 \times 10^{-23}$ kg · m/s

ans: B

29. Consider the following:

1. a photoelectric process in which some emitted electrons have kinetic energy greater than $hf$, where $f$ is the frequency of the incident light.
2. a photoelectric process in which all emitted electrons have energy less than $hf$.
3. Compton scattering from stationary electrons for which the emitted light has a wavelength that is greater than that of the incident light.
4. Compton scattering from stationary electrons for which the emitted light has a wavelength that is less than that of the incident light.

The only possible processes are:

A. 1
B. 3
C. 1 and 3
D. 2 and 3
E. 2 and 4

ans: D

30. J. J. Thompson's measurement of $e/m$ for electrons provides evidence of the:

A. wave nature of matter
B. particle nature of matter
C. wave nature of radiation
D. particle nature of radiation
E. transverse wave nature of light

ans: B

31. Evidence for the wave nature of matter is:

A. electron diffraction experiments of Davisson and Germer
B. Thompson's measurement of $e/m$
C. Young's double slit experiment
D. the Compton effect
E. Lenz's law

ans: A

32. Which of the following is NOT evidence for the wave nature of matter?

A. The photoelectric effect
B. The diffraction pattern obtained when electrons pass through a slit
C. Electron tunneling
D. The validity of the Heisenberg uncertainty principle
E. The interference pattern obtained when electrons pass through a two-slit system

ans: A

33. Of the following which is the best evidence for the wave nature of matter?

A. The photoelectric effect
B. The Compton effect
C. The spectral radiancy of cavity radiation
D. The relationship between momentum and energy for an electron
E. The reflection of electrons by crystals

ans: E

34. Monoenergetic electrons are incident on a single slit barrier. If the energy of each incident electron is increased the central maximum of the diffraction pattern:

A. widens
B. narrows
C. stays the same width
D. widens for slow electrons and narrows for fast electrons
E. narrows for slow electrons and widens for fast electrons

ans: B

35. A free electron and a free proton have the same kinetic energy. This means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has:

A. a shorter wavelength and a greater frequency
B. a longer wavelength and a greater frequency
C. a shorter wavelength and the same frequency
D. a longer wavelength and the same frequency
E. a shorter wavelength and a smaller frequency

ans: D

36. A free electron and a free proton have the same momentum. This means that, compared to the matter wave associated with the proton, the matter wave associated with the electron:

A. has a shorter wavelength and a greater frequency
B. has a longer wavelength and a greater frequency
C. has the same wavelength and the same frequency
D. has the same wavelength and a greater frequency
E. has the same wavelength and a smaller frequency

ans: D

37. A free electron and a free proton have the same speed. This means that, compared to the matter wave associated with the proton, the matter wave associated with the electron:

A. has a shorter wavelength and a greater frequency
B. has a longer wavelength and a greater frequency
C. has the same wavelength and the same frequency
D. has the same wavelength and a greater frequency
E. has a longer wavelength and a smaller frequency

ans: E

38. Consider the following three particles:

1. a free electron with speed $v_0$
2. a free proton with speed $v_0$
3. a free proton with speed $2v_0$

Rank them according to the wavelengths of their matter waves, least to greatest.

A. 1, 2, 3
B. 3, 2, 1
C. 2, 3, 1
D. 1, 3, 2
E. 1, then 2 and 3 tied

ans: B

39. Consider the following three particles:
    1. a free electron with kinetic energy $K_0$
    2. a free proton with kinetic energy $K_0$
    3. a free proton with kinetic energy $2K_0$

    Rank them according to the wavelengths of their matter waves, least to greatest.

    A. 1, 2, 3
    B. 3, 2, 1
    C. 2, 3, 1
    D. 1, 3, 2
    E. 1, then 2 and 3 tied

    ans: B

40. A free electron has a momentum of $5.0 \times 10^{-24}\,\text{kg} \cdot \text{m/s}$. The wavelength of its wave function is:

    A. $1.3 \times 10^{-8}\,\text{m}$
    B. $1.3 \times 10^{-10}\,\text{m}$
    C. $2.1 \times 10^{-11}\,\text{m}$
    D. $2.1 \times 10^{-13}\,\text{m}$
    E. none of these

    ans: B

41. The frequency and wavelength of the matter wave associated with a 10-eV free electron are:

    A. $1.5 \times 10^{34}\,\text{Hz}$, $3.9 \times 10^{-10}\,\text{m}$
    B. $1.5 \times 10^{34}\,\text{Hz}$, $1.3 \times 10^{-34}\,\text{m}$
    C. $2.4 \times 10^{15}\,\text{Hz}$, $1.2 \times 10^{-9}\,\text{m}$
    D. $2.4 \times 10^{15}\,\text{Hz}$, $3.9 \times 10^{-10}\,\text{m}$
    E. $4.8 \times 10^{15}\,\text{Hz}$, $1.9 \times 10^{-10}\,\text{m}$

    ans: D

42. If the kinetic energy of a non-relativistic free electron doubles, the frequency of its wave function changes by the factor:

    A. $1/\sqrt{2}$
    B. $1/2$
    C. $1/4$
    D. $\sqrt{2}$
    E. 2

    ans: E

43. A non-relativistic free electron has kinetic energy $K$. If its wavelength doubles, its kinetic energy is:

    A. $4K$
    B. $2K$
    C. still $K$
    D. $K/2$
    E. $K/4$

    ans: E

44. The probability that a particle is in a given small region of space is proportional to:
    A. its energy
    B. its momentum
    C. the frequency of its wave function
    D. the wavelength of its wave function
    E. the square of the magnitude of its wave function

    ans: E

45. $\psi(x)$ is the wave function for a particle moving along the $x$ axis. The probability that the particle is in the interval from $x = a$ to $x = b$ is given by:
    A. $\psi(b) - \psi(a)$
    B. $|\psi(b)|/|\psi(a)|$
    C. $|\psi(b)|^2/|\psi(a)|^2$
    D. $\int_a^b \psi(x)\,dx$
    E. $\int_a^b |\psi(x)|^2\,dx$

    ans: E

46. The significance of $|\psi|^2$ is:
    A. probability
    B. energy
    C. probability density
    D. energy density
    E. wavelength

    ans: C

47. Maxwell's equations are to electric and magnetic fields as _____ equation is to the wave function for a particle.
    A. Einstein's
    B. Fermi's
    C. Newton's
    D. Schrödinger's
    E. Bohr's

    ans: D

48. A free electron in motion along the $x$ axis has a localized wave function. The uncertainty in its momentum is decreased if:
    A. the wave function is made more narrow
    B. the wave function is made less narrow
    C. the wave function remains the same but the energy of the electron is increased
    D. the wave function remains the same but the energy of the electron is decreased
    E. none of the above

    ans: B

49. The uncertainty in position of an electron in a certain state is $5 \times 10^{-10}$ m. The uncertainty in its momentum might be:

A. $5.0 \times 10^{-24}$ kg · m/s
B. $4.0 \times 10^{-24}$ kg · m/s
C. $3.0 \times 10^{-24}$ kg · m/s
D. all of the above
E. none of the above

ans: D

50. The reflection coefficient $R$ for a certain barrier tunneling problem is 0.80. The corresponding transmission coefficient $T$ is:

A. 0.80
B. 0.60
C. 0.50
D. 0.20
E. 0

ans: D

51. An electron with energy $E$ is incident upon a potential energy barrier of height $E_{\text{pot}} > E$ and thickness $L$. The transmission coefficient $T$:

A. is zero
B. decreases exponentially with $L$
C. is proportional to $1/L$
D. is proportional to $1/L^2$
E. is non-zero and independent of $L$

ans: B

52. In order to tunnel through a potential barrier a particle must:

A. have energy greater than the barrier height
B. have spin
C. be massive
D. have a wavelength longer than the barrier width
E. none of the above

ans: E

53. An electron with energy $E$ is incident on a potential energy barrier of height $E_{\text{pot}}$ and thickness $L$. The probability of tunneling increases if:

A. $E$ decreases without any other changes
B. $E_{\text{pot}}$ increases without any other changes
C. $L$ decreases without any other changes
D. $E$ and $E_{\text{pot}}$ increase by the same amount
E. $E$ and $E_{\text{pot}}$ decrease by the same amount

ans: C

54. Identical particles, each with energy $E$, are incident on the following four potential energy barriers:

    1. barrier height $= 5E$, barrier width $= 2L$
    2. barrier height $= 10E$, barrier width $= L$
    3. barrier height $= 17E$, barrier width $= L/2$
    4. barrier height $= 26E$, barrier width $= L/3$

    Rank the barriers in terms of the probability that the particles tunnel through them, from least probability to greatest probability.

    A. 1, 2, 3, 4
    B. 4, 3, 2, 1
    C. 1 and 2 tied, then 3, then 4
    D. 2, then 1 and 3 tied, then 4
    E. 3, 2, 1, 4

    ans: A

## Chapter 39: MORE ABOUT MATTER WAVES

1. If a wave function $\psi$ for a particle moving along the $x$ axis is normalized, then:
   A. $\int |\psi|^2 \, dt = 1$
   B. $\int |\psi|^2 \, dx = 1$
   C. $\partial\psi/\partial x = 1$
   D. $\partial\psi/\partial t = 1$
   E. $|\psi|^2 = 1$

   ans: B

2. The energy of a particle in a one-dimensional trap with zero potential energy in the interior and infinite potential energy at the walls is proportional to ($n$ = quantum number):
   A. $n$
   B. $1/n$
   C. $1/n^2$
   D. $\sqrt{n}$
   E. $n^2$

   ans: E

3. The ground state energy of an electron in a one-dimensional trap with zero potential energy in the interior and infinite potential energy at the walls is 2.0 eV. If the width of the well is doubled, the ground state energy will be:
   A. 0.5 eV
   B. 1.0 eV
   C. 2.0 eV
   D. 4.0 eV
   E. 8.0 eV

   ans: A

4. An electron is in a one-dimensional trap with zero potential energy in the interior and infinite potential energy at the walls. The ratio $E_3/E_1$ of the energy for $n = 3$ to that for $n = 1$ is:
   A. 1/3
   B. 1/9
   C. 3/1
   D. 9/1
   E. 1/1

   ans: D

5. A particle is trapped in a one-dimensional well with infinite potential energy at the walls. Three possible pairs of energy levels are

    1. $n = 3$ and $n = 1$
    2. $n = 3$ and $n = 2$
    3. $n = 4$ and $n = 3$

    Order these pairs according to the difference in energy, least to greatest.

    A. 1, 2, 3
    B. 3, 2, 1
    C. 2, 3, 1
    D. 1, 3, 2
    E. 3, 1, 2

    ans: C

6. Identical particles are trapped in one-dimensional wells with infinite potential energy at the walls. The widths $L$ of the traps and the quantum numbers $n$ of the particles are

    1. $L = 2L_0$, $n = 2$
    2. $L = 2L_0$, $n = 4$
    3. $L = 3L_0$, $n = 3$
    4. $L = 4L_0$, $n = 2$

    Rank them according to the kinetic energies of the particles, least to greatest.

    A. 1, 2, 3, 4
    B. 4, 3, 2, 1
    C. 1 and 3 tied, then 2, 4
    D. 4, 2, then 1 and 3 tied
    E. 1, 3, then 2 and 4 tied

    ans: D

7. Four different particles are trapped in one-dimensional wells with infinite potential energy at their walls. The masses of the particles and the width of the wells are

    1. mass $= 4m_0$, width $= 2L_0$
    2. mass $= 2m_0$, width $= 2L_0$
    3. mass $= 4m_0$, width $= L_0$
    4. mass $= m_0$, width $= 2L_0$

    Rank them according to the kinetic energies of the particles when they are in their ground states.

    A. 1, 2, 3, 4
    B. 1, 2, 3 and 4 tied
    C. 1 and 2 tied, then 3, 4
    D. 4, 3, 2, 1
    E. 3, 1, 2, 4

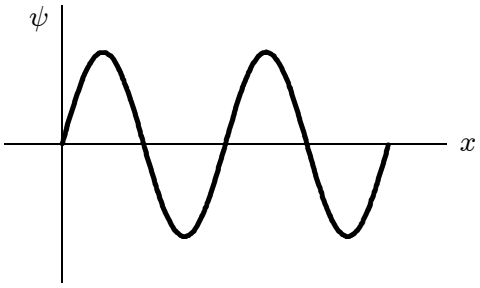
    ans: B

8. The ground state energy of an electron in a one-dimensional trap with zero potential energy in the interior and infinite potential energy at the walls:

A. is zero
B. decreases with temperature
C. increases with temperature
D. is independent of temperature
E. oscillates with time

ans: D

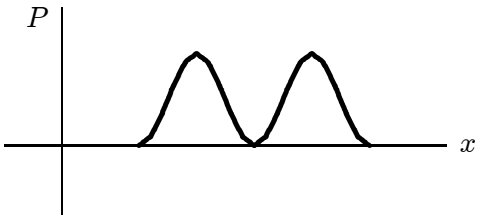
9. An electron is in a one-dimensional trap with zero potential energy in the interior and infinite potential energy at the walls. A graph of its wave function $\psi(x)$ versus $x$ is shown. The value of quantum number $n$ is:




A. 0
B. 2
C. 4
D. 6
E. 8

ans: C

10. An electron is in a one-dimensional trap with zero potential energy in the interior and infinite potential energy at the walls. A graph of its probability density $P(x)$ versus $x$ is shown. The value of the quantum number $n$ is:




A. 0
B. 1
C. 2
D. 3
E. 4

ans: C

11. A particle is trapped in an infinite potential energy well. It is in the state with quantum number $n = 14$. How many nodes does the probability density have (counting the nodes at the ends of the well)?

A. none
B. 7
C. 13
D. 14
E. 15

ans: E

12. A particle is trapped in an infinite potential energy well. It is in the state with quantum number $n = 14$. How many maxima does the probability density have?

A. none
B. 7
C. 13
D. 14
E. 15

ans: D

13. A particle is confined to a one-dimensional trap by infinite potential energy walls. Of the following states, designed by the quantum number $n$, for which one is the probability density greatest near the center of the well?

A. $n = 2$
B. $n = 3$
C. $n = 4$
D. $n = 5$
E. $n = 6$

ans: B

14. Two one-dimensional traps have infinite potential energy at their walls Trap A has width $L$ and trap B has width $2L$. For which value of the quantum number $n$ does a particle in trap B have the same energy as a particle in the ground state of trap A?

A. $n = 1$
B. $n = 2$
C. $n = 3$
D. $n = 4$
E. $n = 5$

ans: B

15. An electron is trapped in a deep well with a width of 0.3 nm. If it is in the state with quantum number $n = 3$ its kinetic energy is:

A. $6.0 \times 10^{-28}$ J
B. $1.8 \times 10^{-27}$ J
C. $6.7 \times 10^{-19}$ J
D. $2.0 \times 10^{-18}$ J
E. $6.0 \times 10^{-18}$ J

ans: E

16. An electron is in a one-dimensional well with finite potential energy barriers at the walls. The matter wave:

A. is zero at the barriers
B. is zero everywhere within each barrier
C. is zero in the well
D. extends into the barriers
E. is discontinuous at the barriers

ans: D

17. A particle is confined by finite potential energy walls to a one-dimensional trap from $x = 0$ to $x = L$. Its wave function in the region $x > L$ has the form:

A. $\psi(x) = A\sin(kx)$
B. $\psi(x) = Ae^{kx}$
C. $\psi(x) = Ae^{-kx}$
D. $\psi(x) = Ae^{ikx}$
E. $\psi(x) = 0$

ans: C

18. A particle is trapped in a finite potential energy well that is deep enough so that the electron can be in the state with $n = 4$. For this state how many nodes does the probability density have?

A. none
B. 1
C. 3
D. 5
E. 7

ans: C

19. A particle in a certain finite potential energy well can have any of five quantized energy values and no more. Which of the following would allow it to have any of six quantized energy levels?

A. Increase the momentum of the particle
B. Decrease the momentum of the particle
C. Decrease the well width
D. Increase the well depth
E. Decrease the well depth

ans: D

20. A particle in a certain finite potential energy well can have any of five quantized energy values and no more. Which of the following would allow it to have any of six quantized energy levels?

A. Increase the energy of the particle
B. Decrease the energy of the particle
C. Make the well shallower
D. Make the well deeper
E. Make the well narrower

ans: D

21. An electron in an atom initially has an energy 5.5 eV above the ground state energy. It drops to a state with energy 3.2 eV above the ground state energy and emits a photon in the process. The wave associated with the photon has a wavelength of:

A. $5.4 \times 10^{-7}$ m
B. $3.0 \times 10^{-7}$ m
C. $1.7 \times 10^{-7}$ m
D. $1.15 \times 10^{-7}$ m
E. $1.0 \times 10^{-7}$ m

ans: A

22. An electron in an atom drops from an energy level at $-1.1 \times 10^{-18}$ J to an energy level at $-2.4 \times 10^{-18}$ J. The wave associated with the emitted photon has a frequency of:

A. $2.0 \times 10^{17}$ Hz
B. $2.0 \times 10^{15}$ Hz
C. $2.0 \times 10^{13}$ Hz
D. $2.0 \times 10^{11}$ Hz
E. $2.0 \times 10^{9}$ Hz

ans: B

23. An electron in an atom initially has an energy 7.5 eV above the ground state energy. It drops to a state with an energy of 3.2 eV above the ground state energy and emits a photon in the process. The momentum of the photon is:

A. $1.7 \times 10^{-27}$ kg · m/s
B. $2.3 \times 10^{-27}$ kg · m/s
C. $4.0 \times 10^{-27}$ kg · m/s
D. $5.7 \times 10^{-27}$ kg · m/s
E. $8.0 \times 10^{-27}$ kg · m/s

ans: B

24. The quantum number $n$ is most closely associated with what property of the electron in a hydrogen atom?

A. Energy
B. Orbital angular momentum
C. Spin angular momentum
D. Magnetic moment
E. $z$ component of angular momentum

ans: A

25. Take the potential energy of a hydrogen atom to be zero for infinite separation of the electron and proton. Then, according to quantum theory the energy $E_n$ of a state with principal quantum number $n$ is proportional to:

A. $n$
B. $n^2$
C. $1/n$
D. $1/n^2$
E. none of the above

ans: D

26. The binding energy of an electron in the ground state in a hydrogen atom is about:
A. 13.6 eV
B. 3.4 eV
C. 10.2 eV
D. 1.0 eV
E. 27.2 eV
ans: A

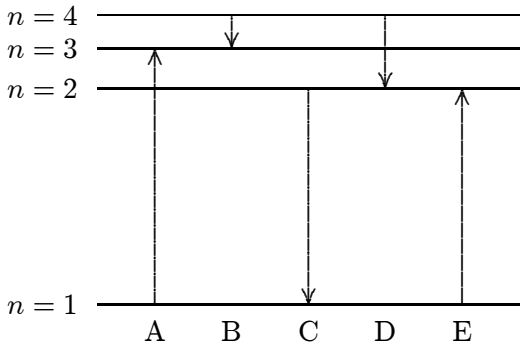
27. Take the potential energy of a hydrogen atom to be zero for infinite separation of the electron and proton. Then the ground state energy is −13.6 eV. The energy of the first excited state is:
A. 0
B. −3.4 eV
C. −6.8 eV
D. −9.6 eV
E. −27 eV
ans: B

28. Take the potential energy of a hydrogen atom to be zero for infinite separation of the electron and proton. Then the ground state energy is −13.6 eV. The negative sign indicates:
A. the kinetic energy is negative
B. the potential energy is positive
C. the electron might escape from the atom
D. the electron and proton are bound together
E. none of the above
ans: D

29. Take the potential energy of a hydrogen atom to be zero for infinite separation of the electron and proton. Then the ground state energy of a hydrogen atom is −13.6 eV. When the electron is in the first excited state its excitation energy is:
A. 0
B. 3.4 eV
C. 6.8 eV
D. 10.2 eV
E. 13.6 eV
ans: D

30. Take the potential energy of a hydrogen atom to be zero for infinite separation of the electron and proton. Then the ground state energy is −13.6 eV. When the electron is in the first excited state the ionization energy is:
A. 0
B. 3.4 eV
C. 6.8 eV
D. 10.2 eV
E. 13.6 eV
ans: B

31. The diagram shows the energy levels for an electron in a certain atom. Of the transitions shown, which represents the emission of a photon with the most energy?




ans: C

32. When a hydrogen atom makes the transition from the second excited state to the ground state (at $-13.6\,\text{eV}$) the energy of the photon emitted is:

A. 0
B. 1.5 eV
C. 9.1 eV
D. 12.1 eV
E. 13.6 eV

ans: D

33. The series limit for the Balmer series represents a transition $m \to n$, where $(m, n)$ is

A. $(2, 1)$
B. $(3, 2)$
C. $(\infty, 0)$
D. $(\infty, 1)$
E. $(\infty, 2)$

ans: E

34. The Balmer series of hydrogen is important because it:

A. is the only one for which the quantum theory can be used
B. is the only series that occurs for hydrogen
C. is in the visible region
D. involves the lowest possible quantum number n
E. involves the highest possible quantum number n

ans: C

35. The principle of complementarity is due to:

A. Einstein
B. Maxwell
C. Newton
D. Bohr
E. Schrodinger

ans: D

36. Which of the following sets of quantum numbers is possible for an electron in a hydrogen atom?
A. $n = 4$, $\ell = 3$, $m_\ell = -3$
B. $n = 4$, $\ell = 4$, $m_\ell = -2$
C. $n = 5$, $\ell = -1$, $m_\ell = 2$
D. $n = 3$, $\ell = 1$, $m_\ell = -2$
E. $n = 2$, $\ell = 3$, $m_\ell = -2$
ans: A

37. The wave function for an electron in a state with zero angular momentum:
A. is zero everywhere
B. is spherically symmetric
C. depends on the angle from the $z$ axis
D. depends on the angle from the $x$ axis
E. is spherically symmetric for some shells and depends on the angle from the $z$ axis for others
ans: B

38. Consider the following:
1. the probability density for an $\ell = 0$ state
2. the probability density for a state with $\ell \neq 0$
3. the average of the probability densities for all states in an $\ell \neq 0$ subshell

Of these which are spherically symmetric?
A. only 1
B. only 2
C. only 1 and 2
D. only 1 and 3
E. 1, 2, and 3
ans: D

39. If the wave function $\psi$ is spherically symmetric then the radial probability density is given by:
A. $4\pi r^2 \psi$
B. $|\psi|^2$
C. $4\pi r^2 |\psi|^2$
D. $4\pi |\psi|^2$
E. $4\pi r |\psi|^2$
ans: C

40. If $P(r)$ is the radial probability density then the probability that the separation of the electron and proton is between $r$ and $r + dr$ is:
A. $P\,dr$
B. $|P|^2\,dr$
C. $4\pi r^2 P\,dr$
D. $4\pi r^2 |P|\,dr$
E. $4\pi |P|^2\,dr$
ans: A

41. The radial probability density for the electron in the ground state of a hydrogen atom has a peak at about:

A. 0.5 pm
B. 5 pm
C. 50 pm
D. 500 pm
E. 5000 pm

ans: C

## Chapter 40: ALL ABOUT ATOMS

1. The magnitude of the orbital angular momentum of an electron in an atom is what multiple of $\hbar$? ($\ell$ is a positive integer.)

   A. 1
   B. 1/2
   C. $\sqrt{\ell(\ell+1)}$
   D. $2\ell+1$
   E. $\ell^2$

   ans: C

2. The magnetic quantum number $m_\ell$ is most closely associated with what property of an electron in an atom?

   A. Magnitude of the orbital angular momentum
   B. Energy
   C. $z$ component of the spin angular momentum
   D. $z$ component of the orbital angular momentum
   E. Radius of the orbit

   ans: D

3. The quantum number $m_s$ is most closely associated with what property of the electron in an atom?

   A. Magnitude of the orbital angular momentum
   B. Energy
   C. $z$ component of the spin angular momentum
   D. $z$ component of the orbital angular momentum
   E. Radius of the orbit

   ans: C

4. Possible values of the principal quantum number $n$ for an electron in an atom are:

   A. only 0 and 1
   B. only 0, 1, 2, ..., $\infty$
   C. only 0, 1, ..., $\ell-1$
   D. only 1/2 and $-1/2$
   E. only 1, 2, 3, ..., $\infty$

   ans: E

5. The number of values of the orbital quantum number $\ell$ associated with the principal quantum number $n=3$ is:

   A. 1
   B. 2
   C. 3
   D. 4
   E. 7

   ans: C

6. The number of possible values of the magnetic quantum number $m_\ell$ associated with a given value of the orbital quantum number $\ell$ is:

A. 1
B. 2
C. $\ell$
D. $2\ell$
E. $2\ell + 1$

ans: E

7. An atom is in a state with orbital quantum number $\ell = 2$. Possible values of the magnetic quantum number $m_\ell$ are:

A. 1, 2
B. 0, 1, 2
C. 0, 1
D. $-1$, 0, 1
E. $-2$, $-1$, 0, 1, 2

ans: E

8. An electron is in a quantum state for which the magnitude of the orbital angular momentum is $6\sqrt{2}\hbar$. How many allowed values of the $z$ component of the angular momentum are there?

A. 4
B. 5
C. 7
D. 8
E. 9

ans: D

9. An electron is in a quantum state for which there are seven allowed values of the $z$ component of the angular momentum. The magnitude of the angular momentum is:

A. $\sqrt{3}\hbar$
B. $\sqrt{7}\hbar$
C. $\sqrt{9}\hbar$
D. $\sqrt{12}\hbar$
E. $\sqrt{14}\hbar$

ans: D

10. The number of states in a subshell with orbital quantum number $\ell = 3$ is:

A. 2
B. 3
C. 7
D. 9
E. 14

ans: E

11. The number of states in a shell with principal quantum number $n = 3$ is:

A. 3
B. 9
C. 15
D. 18
E. 25

ans: D

12. An electron in an atom is in a state with principal quantum number $n = 4$. The possible values of the orbital quantum number $\ell$ are:

A. 1, 2, 3
B. 1, 2, 3, 4
C. $-3$, $-2$, $-1$, 0, 1, 2, 3
D. 0, 1, 2, 3
E. 0, 1, 2

ans: D

13. Space quantization means that:

A. space is quantized
B. $L_z$ can have only certain discrete values
C. $\vec{L}$ and $\vec{\boldsymbol{\mu}}$ are in the same direction
D. $\vec{L}$ and $\vec{\boldsymbol{\mu}}$ are in opposite directions
E. an electron has a magnetic dipole moment

ans: B

14. An electron in an atom is in a state with $\ell = 3$ and $m_\ell = 2$. The angle between $\vec{L}$ and the $z$ axis is:

A. 48.2°
B. 60°
C. 30°
D. 35.3°
E. 54.7°

ans: E

15. An electron in an atom is in a state with $\ell = 5$. The minimum angle between $\vec{L}$ and the $z$ axis is:

A. 0°
B. 18.0°
C. 24.1°
D. 36.7°
E. 33.6°

ans: C

16. In the relation $\mu_z = -m_\ell \mu_B$, the quantity $\mu_B$ is:
    A. the Bohr magneton
    B. the component of the dipole moment along the magnetic field
    C. the permeability of the material
    D. a friction coefficient
    E. none of the above

    ans: A

17. The electron states that constitute a single shell for an atom all have:
    A. the same value of $n$
    B. the same value of $\ell$
    C. the same value of $n$ and the same value of $\ell$
    D. the same value of $\ell$ and the same value of $m_\ell$
    E. the same set of all four quantum numbers

    ans: A

18. The electron states that constitute a single subshell for an atom all have:
    A. only the same value of $n$
    B. only the same value of $\ell$
    C. only the same value of $n$ and the same value of $\ell$
    D. only the same value of $\ell$ and the same value of $m_\ell$
    E. the same set of all four quantum numbers

    ans: C

19. The total number of electron states with $n = 2$ and $\ell = 1$ for an atom is:
    A. two
    B. four
    C. six
    D. eight
    E. ten

    ans: C

20. The possible values for the magnetic quantum number $m_s$ of an electron in an atom:
    A. depend on $n$
    B. depend on $\ell$
    C. depend on both $n$ and $\ell$
    D. depend on whether there is an external magnetic field present
    E. are $\pm 1/2$

    ans: E

21. The Stern-Gerlach experiment makes use of:
    A. a strong uniform magnetic field
    B. a strong non-uniform magnetic field
    C. a strong uniform electric field
    D. a strong non-uniform electric field
    E. strong perpendicular electric and magnetic fields

    ans: B

22. The magnetic field $\vec{B}$ is along the $z$ axis in a Stern-Gerlach experiment. The force it exerts on a magnetic dipole with dipole moment $\vec{\mu}$ is proportional to:

A. $\mu_z^2$
B. $B^2$
C. $dB/dz$
D. $d^2B/dz^2$
E. $\int B\,dz$

ans: C

23. A magnetic dipole $\vec{\mu}$ is placed in a strong uniform magnetic field $\vec{B}$. The associated force exerted on the dipole is:

A. along $\vec{\mu}$
B. along $-\vec{\mu}$
C. along $\vec{B}$
D. along $\vec{\mu} \times \vec{B}$
E. zero

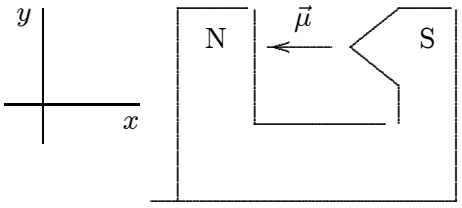
ans: E

24. The force exerted on a magnetic dipole as it moves with velocity $\vec{v}$ through a Stern-Gerlach apparatus is:

A. proportional to $v$
B. proportional to $1/v$
C. zero
D. proportional to $v^2$
E. independent of $v$

ans: E

25. A magnetic dipole is placed between the poles of a magnet as shown. The direction of the associated force exerted on the dipole is:




A. positive $x$
B. positive $y$
C. negative $x$
D. negative $y$
E. into or out of the page

ans: C

26. To observe the Zeeman effect one uses:
    A. a strong uniform magnetic field
    B. a strong non-uniform magnetic field
    C. a strong uniform electric field
    D. a strong non-uniform electric field
    E. mutually perpendicular electric and magnetic fields

    ans: B

27. An electron in a K shell of an atom has the principal quantum number:
    A. $n = 0$
    B. $n = 1$
    C. $n = 2$
    D. $n = 3$
    E. $n = \infty$

    ans: B

28. An electron in an L shell of an atom has the principal quantum number:
    A. $n = 0$
    B. $n = 1$
    C. $n = 2$
    D. $n = 3$
    E. $n = \infty$

    ans: C

29. The most energetic photon in a continuous x-ray spectrum has an energy approximately equal to:
    A. the energy of all the electrons in a target atom
    B. the kinetic energy of an incident-beam electron
    C. the rest energy, $mc^2$, of an electron
    D. the total energy of a K-electron in the target atom
    E. the kinetic energy of a K-electron in the target atom

    ans: B

30. Two different electron beams are incident on two different targets and both produce x rays. The cutoff wavelength for target 1 is shorter than the cutoff wavelength for target 2. We can conclude that:
    A. target 2 has a higher atomic number than target 1
    B. target 2 has a lower atomic number than target 1
    C. the electrons in beam 1 have greater kinetic energy than those in beam 2
    D. the electrons in beam 1 have less kinetic energy than those in beam 2
    E. target 1 is thicker than target 2

    ans: C

31. A photon with the smallest wavelength in the continuous z-ray spectrum is emitted when:
    A. an electron is knocked from a K shell
    B. a valence electron is knocked from the atom
    C. the incident electron becomes bound to the atom
    D. the atom has the greatest recoil energy
    E. the incident electron loses all its energy in a single decelerating event

    ans: E

32. Radiation with the minimum wavelength as well as the K x-ray lines are detected for a certain target. The energy of the incident electrons is then doubled, with the result that
    A. the minimum wavelength increases and the wavelengths of the K lines remain the same
    B. the minimum wavelength decreases and the wavelengths of the K lines remain the same
    C. the minimum wavelength and the wavelengths of the K lines all increase
    D. the minimum wavelength and the wavelengths of the K lines all decrease
    E. the minimum wavelength increases and the wavelengths of the K lines all decrease

    ans: B

33. Characteristic K x-radiation of an element occurs when:
    A. the incident electron is absorbed by a target nucleus
    B. the incident electron is scattered by a target atom without an energy loss
    C. an electron is ejected from an outer shell of a target atom
    D. an electron in a target atom makes a transition to the lowest energy state
    E. the incident electron goes into the lowest energy state

    ans: D

34. The $K_\alpha$ x rays arising from a cobalt ($Z = 27$) target have a wavelength of about 179 pm. The atomic number of a target that gives rise to $K_\alpha$ x rays with a wavelength one-third as great ($\approx 60$ pm) is:
    A. $Z = 9$
    B. $Z = 10$
    C. $Z = 12$
    D. $Z = 16$
    E. $Z = 46$

    ans: E

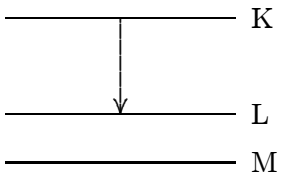
35. In connection with x-ray emission the symbol $K_\alpha$ refers to:
    A. an alpha particle radiation
    B. an effect of the dielectric constant on energy levels
    C. x-ray radiation from potassium
    D. x-ray radiation associated with an electron going from $n = \infty$ to $n = 1$
    E. x-ray radiation associated with an electron going from $n = 2$ to $n = 1$

    ans: E

36. In connection with x-ray emission the symbol $L_\beta$ refers to:
    A. a beta particle radiation
    B. an atomic state of angular momentum $h/2\pi$
    C. the inductance associated with an orbiting electron
    D. x-radiation associated with an electron going from $n = 4$ to $n = 2$
    E. none of the above

    ans: D

37. The transition shown gives rise to an x-ray. The correct label for this is:




    A. $K_\alpha$
    B. $K_\beta$
    C. $L_\alpha$
    D. $L_\beta$
    E. KL

    ans: A

38. In a Moseley graph:
    A. the x-ray frequency is plotted as a function of atomic number
    B. the square of the x-ray frequency is plotted as a function of atomic number
    C. the square root of the x-ray frequency is plotted as a function of atomic number
    D. the x-ray frequency is plotted as a function of the square root of atomic number
    E. the square root of the x-ray frequency is plotted as a function of atomic mass

    ans: C

39. In calculating the x-ray energy levels the effective charge of the nucleus is taken to be $Z - b$, where $Z$ is the atomic number. The parameter $b$ enters because:
    A. an electron is removed from the inner shell
    B. a proton is removed from the nucleus
    C. the quantum mechanical force between two charges is less than the classical force
    D. the nucleus is screened by electrons
    E. the Pauli exclusion principle must be obeyed

    ans: D

40. The ratio of the wavelength of the $K_\alpha$ x-ray line for Nb ($Z = 41$) to that of Ga ($Z = 31$) is:
    A. 9/16
    B. 16/9
    C. 3/4
    D. 4/3
    E. 1.15

    ans: A

41. The Pauli exclusion principle is obeyed by:

A. all particles
B. all charged particles
C. all particles with spin quantum numbers of 1/2
D. all particles with spin quantum numbers of 1
E. all particles with mass

ans: C

42. No state in an atom can be occupied by more than one electron. This is most closely related to:

A. the wave nature of matter
B. the finite value for the speed of light
C. the Bohr magneton
D. the Pauli exclusion principle
E. the Einstein-de Haas effect

ans: D

43. Electrons are in a two-dimensional square potential energy well with sides of length $L$. The potential energy is infinite at the sides and zero inside. The single-particle energies are given by $(h^2/8mL^2)(n_x^2 + n_y^2)$, where $n_x$ and $n_y$ are integers. At most the number of electrons that can have energy $8(h^2/8mL^2)$ is:

A. 1
B. 2
C. 3
D. 4
E. any number

ans: B

44. Five electrons are in a two-dimensional square potential energy well with sides of length $L$. The potential energy is infinite at the sides and zero inside. The single-particle energies are given by $(h^2/8mL^2)(n_x^2 + n_y^2)$, where $n_x$ and $n_y$ are integers. In units of $(h^2/8mL^2)$ the energy of the ground state of the system is:

A. 0
B. 10
C. 19
D. 24
E. 48

ans: C

45. Five electrons are in a two-dimensional square potential energy well with sides of length $L$. The potential energy is infinite at the sides and zero inside. The single-particle energies are given by $(h^2/8mL^2)(n_x^2+n_y^2)$, where $n_x$ and $n_y$ are integers. In units of $(h^2/8mL^2)$ the energy of the first excited state of the system is:

A. 13
B. 22
C. 24
D. 25
E. 27

ans: B

46. Electrons are in a two-dimensional square potential energy well with sides of length $L$. The potential energy is infinite at the sides and zero inside. The single-particle energies are given by $(h^2/8mL^2)(n_x^2+n_y^2)$, where $n_x$ and $n_y$ are integers. The number of single-particle states with energy $5(h^2/8mL^2)$ is:

A. 1
B. 2
C. 3
D. 4
E. 5

ans: B

47. Six electrons are in a two-dimensional square potential energy well with sides of length $L$. The potential energy is infinite at the sides and zero inside. The single-particle energies are given by $(h^2/8mL^2)(n_x^2+n_y^2)$, where $n_x$ and $n_y$ are integers. If a seventh electron is added to the system when it is in its ground state the least energy the additional electron can have is:

A. $2(h^2/8mL^2)$
B. $5(h^2/8mL^2)$
C. $10(h^2/8mL^2)$
D. $13(h^2/8mL^2)$
E. $18(h^2/8mL^2)$

ans: C

48. When a lithium atom is made from a helium atom by adding a proton (and neutron) to the nucleus and an electron outside, the electron goes into an $n = 2$, $\ell = 0$ state rather than an $n = 1$, $\ell = 0$ state. This is an indication that electrons:

A. obey the Pauli exclusion principle
B. obey the minimum energy principle
C. undergo the Zeeman effect
D. are diffracted
E. and protons are interchangeable

ans: A

49. When a lithium atom in its ground state is made from a helium atom by adding a proton (and neutron) to the nucleus and an electron outside, the electron goes into an $n = 2$, $\ell = 0$ state rather than an $n = 3$, $\ell = 0$ state. This is an indication that electrons:

A. obey the Pauli exclusion principle
B. obey the minimum energy principle
C. undergo the Zeeman effect
D. are diffracted
E. and protons are interchangeable

ans: B

50. If electrons did not have intrinsic angular momentum (spin) but still obeyed the Pauli exclusion principle, the states occupied by electrons in the ground state of helium would be:

A. $(n = 1, \ell = 0)$; $(n = 1, \ell = 0)$
B. $(n = 1, \ell = 0)$; $(n = 1, \ell = 1)$
C. $(n = 1, \ell = 0)$; $(n = 2, \ell = 0)$
D. $(n = 2, \ell = 0)$; $(n = 2, \ell = 1)$
E. $(n = 2, \ell = 1)$; $(n = 2, \ell = 1)$

ans: C

51. The minimum energy principle tells us that:

A. the energy of an atom with a high atomic number is less than the energy of an atom with a low atomic number
B. the energy of an atom with a low atomic number is less than the energy of an atom with a high atomic number
C. when an atom makes an upward transition the energy of the absorbed photon is the least possible
D. the ground state configuration of any atom is the one with the least energy
E. the ground state configuration of any atom is the one with the least ionization energy

ans: D

52. Which of the following $(n, \ell, m_\ell, m_s)$ combinations is impossible for an electron in an atom?

A. 3, 1, 1, $-1/2$
B. 6, 2, 0, $1/2$
C. 3, 2, $-2$, $-1/2$
D. 3, 1, $-2$, $1/2$
E. 1, 0, 0, $-1/2$

ans: D

53. Which of the following subshells cannot exist?

A. 3p
B. 2p
C. 4d
D. 3d
E. 2d

ans: E

54. For any atom other than hydrogen and helium all electrons in the same shell have:

A. the same energy
B. the same magnitude of angular momentum
C. the same magnetic quantum number
D. the same spin quantum number
E. none of the above

ans: E

55. The states being filled from the beginning to end of the lanthanide series of atoms are:

A. $n = 3$, $\ell = 2$ states
B. $n = 4$, $\ell = 1$ states
C. $n = 4$, $\ell = 2$ states
D. $n = 4$, $\ell = 3$ states
E. $n = 5$, $\ell = 2$ states

ans: D

56. The most energetic electron in any atom at the beginning of a period of the periodic table is in:

A. an $\ell = 0$ state
B. an $\ell = 1$ state
C. an $\ell = 2$ state
D. an $n = 0$ state with unspecified angular momentum
E. an $n = 1$ state with unspecified angular momentum

ans: A

57. The most energetic electron in any atom at the end of a period of the periodic table is in:

A. an $\ell = 0$ state
B. an $\ell = 1$ state
C. an $\ell = 2$ state
D. an $n = 0$ state with unspecified angular momentum
E. an $n = 1$ state with unspecified angular momentum

ans: B

58. The group of atoms at the ends of periods of the periodic table are called:

A. alkali metals
B. rare earths
C. transition metal atoms
D. alkaline atoms
E. inert gas atoms

ans: E

59. The group of atoms at the beginning of periods of the periodic table are called:
    A. alkali metal atoms
    B. rare earth atoms
    C. transition metal atoms
    D. alkaline atoms
    E. inert gas atoms

    ans: A

60. Suppose the energy required to ionize an argon atom is $i$, the energy to excite it is $e$, and its thermal energy at room temperature is $t$. In increasing order, these three energies are:
    A. $i$, $e$, $t$
    B. $t$, $i$, $e$
    C. $e$, $t$, $i$
    D. $i$, $t$, $e$
    E. $t$, $e$, $i$

    ans: C

61. The ionization energy of an atom in its ground state is:
    A. the energy required to remove the least energetic electron
    B. the energy required to remove the most energetic electron
    C. the energy difference between the most energetic electron and the least energetic electron
    D. the same as the energy of a $K_\alpha$ photon
    E. the same as the excitation energy of the most energetic electron

    ans: B

62. The effective charge acting on a single valence electron outside a closed shell is about $Ne$, where $N$ is:
    A. the atomic number of the nucleus
    B. the atomic mass of the atom
    C. usually between 1 and 3
    D. half the atomic number
    E. less than 1

    ans: C

63. In a laser:
    A. excited atoms are stimulated to emit photons by radiation external to the laser
    B. the transitions for laser emission are directly to the ground state
    C. the states which give rise to laser emission are usually very unstable states that decay rapidly
    D. the state in which an atom is initially excited is never between two states that are involved in the stimulated emission
    E. a minimum of two energy levels are required.

    ans: D

64. Photons in a laser beam have the same energy, wavelength, polarization direction, and phase because:

A. each is produced in an emission that is stimulated by another
B. all come from the same atom
C. the lasing material has only two quantum states
D. all photons are alike, no matter what their source
E. none of the above

ans: A

65. A laser must be pumped to achieve:

A. a metastable state
B. fast response
C. stimulated emission
D. population inversion
E. the same wavelength for all photons

ans: D

66. Photons in a laser beam are produced by:

A. transitions from a metastable state
B. transitions to a metastable state
C. transitions from a state that decays rapidly
D. splitting of other photons
E. pumping

ans: A

67. Which of the following is essential for laser action to occur between two energy levels of an atom?

A. the lower level is metastable
B. the upper level is metastable
C. the lower level is the ground state
D. there are more atoms in the lower level than in the upper level
E. the lasing material is a gas

ans: B

68. Which of the following is essential for laser action to occur between two energy levels of an atom?

A. the lower level is metastable
B. there are more atoms the upper level than in the lower level
C. there are more atoms in the lower level than in the upper level
D. the lower level is the ground state
E. the lasing material is a gas

ans: B

69. Population inversion is important for the generation of a laser beam because it assures that:
A. spontaneous emission does not occur more often than stimulated emission
B. photons do not split too rapidly
C. more photons are emitted than are absorbed
D. photons do not collide with each other
E. photons do not make upward transitions

ans: C

70. A metastable state is important for the generation of a laser beam because it assures that:
A. spontaneous emission does not occur more often than stimulated emission
B. photons do not split too rapidly
C. more photons are emitted than are absorbed
D. photons do not collide with each other
E. photons do not make upward transitions

ans: A

71. Electrons in a certain laser make transitions from a metastable state to the ground state. Initially there are $6 \times 10^{20}$ atoms in the metastable state and $2 \times 10^{20}$ atoms in the ground state. The number of photons that can be produced in a single burst is about:
A. $2 \times 10^{20}$
B. $3 \times 10^{20}$
C. $4 \times 10^{20}$
D. $6 \times 10^{20}$
E. $8 \times 10^{20}$

ans: C

72. In a helium-neon laser, the laser light arises from a transition from a _____ state to a _____ state.
A. He, He
B. Ne, Ne
C. He, Ne
D. Ne, He
E. N, He

ans: B

73. The purpose of the mirrors at the ends of a helium-neon laser is:
A. to assure that no laser light leaks out
B. to increase the number of stimulated emissions
C. to absorb some of the photons
D. to keep the light used for pumping inside the laser
E. to double the effective length of the laser

ans: B

74. A group of electromagnetic waves might

I. be monochromatic
II. be coherent
III. have the same polarization direction

Which of these describe the waves from a laser?

A. I only
B. II only
C. III only
D. I and II only
E. I, II, and III

ans: E

75. A laser beam can be sharply focused because it is:

A. highly coherent
B. plane polarized
C. intense
D. circularly polarized
E. highly directional

ans: E

76. The "e" in laser stands for:

A. electric
B. emf
C. energy
D. emission
E. entropy

ans: D

## Chapter 41: CONDUCTION OF ELECTRICITY IN SOLIDS

1. In a pure metal the collisions that are characterized by the mean free time $\tau$ in the expression for the resistivity are chiefly between:
   A. electrons and other electrons
   B. electrons with energy about equal to the Fermi energy and atoms
   C. all electrons and atoms
   D. electrons with energy much less than the Fermi energy and atoms
   E. atoms and other atoms

   ans: B

2. A certain metal has $5.3 \times 10^{29}$ conduction electrons/$\mathrm{m}^3$ and an electrical resistivity of $1.9 \times 10^{-9}\,\Omega \cdot \mathrm{m}$. The average time between collisions of electrons with atoms in the metal is:
   A. $5.6 \times 10^{-33}\,\mathrm{s}$
   B. $1.3 \times 10^{-31}\,\mathrm{s}$
   C. $9.9 \times 10^{-22}\,\mathrm{s}$
   D. $4.6 \times 10^{-15}\,\mathrm{s}$
   E. $3.5 \times 10^{-14}\,\mathrm{s}$

   ans: C

3. Which one of the following statements concerning electron energy bands in solids is true?
   A. The bands occur as a direct consequence of the Fermi-Dirac occupancy probability function
   B. Electrical conduction arises from the motion of electrons in completely filled bands
   C. Within a given band, all electron energy levels are equal to each other
   D. An insulator has a large energy separation between the highest filled band and the lowest empty band
   E. Only insulators have energy bands

   ans: D

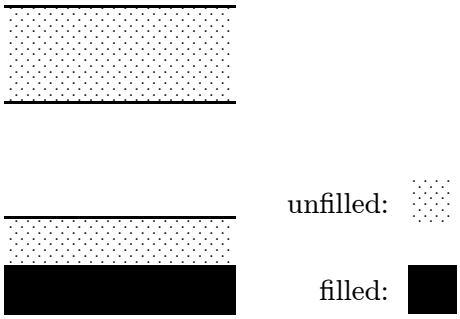
4. If $E_0$ and $E_T$ are the average energies of the "free" electrons in a metal at 0 K and room temperature, respectively, then the ratio $E_T/E_0$ is approximately:
   A. 0
   B. 1
   C. 100
   D. $10^6$
   E. infinity

   ans: B

5. The energy gap (in eV) between the valence and conduction bands of an insulator is of the order of:
   A. $10^{-19}$
   B. 0.001
   C. 0.1
   D. 10
   E. 1000

   ans: D

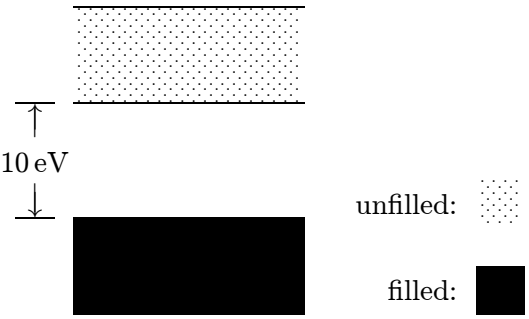
6. The energy level diagram shown applies to:




A. a conductor
B. an insulator
C. a semiconductor
D. an isolated molecule
E. an isolated atom

ans: A

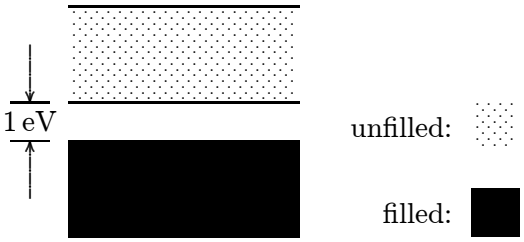
7. The energy level diagram shown applies to:




A. a conductor
B. an insulator
C. a semiconductor
D. an isolated atom
E. a free-electron gas

ans: B

8. The energy level diagram shown applies to:




A. a conductor
B. an insulator
C. a semiconductor
D. an isolated molecule
E. an isolated atom

ans: C

9. Possible units for the density of states function $N(E)$ are:

A. $\mathrm{J/m^3}$
B. $\mathrm{1/J}$
C. $\mathrm{m^{-3}}$
D. $\mathrm{J^{-1}{\cdot}m^{-3}}$
E. $\mathrm{kg/m^3}$

ans: D

10. The density of states for a metal depends primarily on:

A. the temperature
B. the energy
C. the density of the metal
D. the volume of the sample
E. none of these

ans: B

11. The Fermi-Dirac occupancy probability $P(E)$ varies between:

A. 0 and 1
B. 0 and infinity
C. 1 and infinity
D. $-1$ and 1
E. 0 and $E_F$

ans: A

12. For a metal at absolute temperature $T$, with Fermi energy $E_F$, the occupancy probability is given by:
    A. $e^{(E-E_F)/kT}$
    B. $e^{-(E-E_F)/kT}$
    C. $\dfrac{1}{e^{(E-E_F)/kT}+1}$
    D. $\dfrac{1}{e^{-(E-E_F)/kT}+1}$
    E. $\dfrac{1}{e^{(E-E_F)/kT}-1}$
    ans: C

13. In a metal at 0 K, the Fermi energy is:
    A. the highest energy of any electron
    B. the lowest energy of any electron
    C. the mean thermal energy of the electrons
    D. the energy of the top of the valence band
    E. the energy at the bottom of the conduction band
    ans: A

14. The occupancy probability for a state with energy equal to the Fermi energy is:
    A. 0
    B. 0.5
    C. 1
    D. 1.5
    E. 2
    ans: B

15. The Fermi energy of a metal depends primarily on:
    A. the temperature
    B. the volume of the sample
    C. the mass density of the metal
    D. the size of the sample
    E. the number density of conduction electrons
    ans: E

16. The speed of an electron with energy equal to the Fermi energy for copper is on the order of:
    A. $10^{6}$ m/s
    B. $10^{-6}$ m/s
    C. 10 m/s
    D. $10^{-1}$ m/s
    E. $10^{9}$ m/s
    ans: A

17. At $T = 0\,\text{K}$ the probability that a state $0.50\,\text{eV}$ below the Fermi level is occupied is about:
    A. 0
    B. $5.0 \times 10^{-9}$
    C. $5.0 \times 10^{-6}$
    D. $5.0 \times 10^{-3}$
    E. 1

    ans: E

18. At $T = 0\,\text{K}$ the probability that a state $0.50\,\text{eV}$ above the Fermi level is occupied is about:
    A. 0
    B. $5.0 \times 10^{-9}$
    C. $5.0 \times 10^{-6}$
    D. $5.0 \times 10^{-3}$
    E. 1

    ans: A

19. At room temperature $kT$ is about $0.0259\,\text{eV}$. The probability that a state $0.50\,\text{eV}$ above the Fermi level is occupied at room temperature is:
    A. 1
    B. 0.05
    C. 0.025
    D. $5.0 \times 10^{-6}$
    E. $4.1 \times 10^{-9}$

    ans: E

20. At room temperature $kT$ is about $0.0259\,\text{eV}$. The probability that a state $0.50\,\text{eV}$ below the Fermi level is unoccupied at room temperature is:
    A. 1
    B. 0.05
    C. 0.025
    D. $5.0 \times 10^{-6}$
    E. $4.1 \times 10^{-9}$

    ans: E

21. If the density of states is $N(E)$ and the occupancy probability is $P(E)$, then the density of occupied states is:
    A. $N(E) + P(E)$
    B. $N(E)/P(E)$
    C. $N(E) - P(E)$
    D. $N(E)P(E)$
    E. $P(E)/N(E)$

    ans: D

22. A hole refers to:
    A. a proton
    B. a positively charged electron
    C. an electron that has somehow lost its charge
    D. a microscopic defect in a solid
    E. the absence of an electron in an otherwise filled band

    ans: E

23. Electrons in a full band do not contribute to the current when an electric field exists in a solid because:
    A. the field cannot exert a force on them
    B. the individual contributions cancel each other
    C. they are not moving
    D. they make transitions to other bands
    E. they leave the solid

    ans: B

24. For a pure semiconductor the Fermi level is:
    A. in the conduction band
    B. well above the conduction band
    C. in the valence band
    D. well below the valence band
    E. near the center of the gap between the valence and conduction bands

    ans: E

25. The number density $n$ of conduction electrons, the resistivity $\rho$, and the temperature coefficient of resistivity $\alpha$ are given below for five materials. Which is a semiconductor?
    A. $n = 10^{29}\,\mathrm{m}^{-3}$, $\rho = 10^{-8}\,\Omega\cdot\mathrm{m}$, $\alpha = +10^{-3}\,\mathrm{K}^{-1}$
    B. $n = 10^{28}\,\mathrm{m}^{-3}$, $\rho = 10^{-9}\,\Omega\cdot\mathrm{m}$, $\alpha = -10^{-3}\,\mathrm{K}^{-1}$
    C. $n = 10^{28}\,\mathrm{m}^{-3}$, $\rho = 10^{-9}\,\Omega\cdot\mathrm{m}$, $\alpha = +10^{-3}\,\mathrm{K}^{-1}$
    D. $n = 10^{15}\,\mathrm{m}^{-3}$, $\rho = 10^{3}\,\Omega\cdot\mathrm{m}$, $\alpha = -10^{-2}\,\mathrm{K}^{-1}$
    E. $n = 10^{15}\,\mathrm{m}^{-3}$, $\rho = 10^{-7}\,\Omega\cdot\mathrm{m}$, $\alpha = +10^{-3}\,\mathrm{K}^{-1}$

    ans: D

26. A pure semiconductor at room temperature has:
    A. more electrons/$\mathrm{m}^3$ in its conduction band than holes/$\mathrm{m}^3$ in its valence band
    B. more electrons/$\mathrm{m}^3$ in its conduction band than a typical metal
    C. more electrons/$\mathrm{m}^3$ in its valence band than at $T = 0\,\mathrm{K}$
    D. more holes/$\mathrm{m}^3$ in its valence band than electrons/$\mathrm{m}^3$ in its valence band
    E. none of the above

    ans: E

27. For a metal at room temperature the temperature coefficient of resistivity is determined primarily by:

A. the number of electrons in the conduction band
B. the number of impurity atoms
C. the binding energy of outer shell electrons
D. collisions between conduction electrons and atoms
E. none of the above

ans: D

28. For a pure semiconductor at room temperature the temperature coefficient of resistivity is determined primarily by:

A. the number of electrons in the conduction band
B. the number of replacement atoms
C. the binding energy of outer shell electrons
D. collisions between conduction electrons and atoms
E. none of the above

ans: A

29. A certain material has a resistivity of $7.8 \times 10^3\,\Omega \cdot \text{m}$ at room temperature and it increases as the temperature is raised by 100° C. The material is most likely:

A. a metal
B. a pure semiconductor
C. a heavily doped semiconductor
D. an insulator
E. none of the above

ans: C

30. A certain material has a resistivity of $7.8 \times 10^3\,\Omega \cdot \text{m}$ at room temperature and it decreases as the temperature is raised by 100° C. The material is most likely:

A. a metal
B. a pure semiconductor
C. a heavily doped semiconductor
D. an insulator
E. none of the above

ans: B

31. A certain material has a resistivity of $7.8 \times 10^{-8}\,\Omega \cdot \text{m}$ at room temperature and it increases as the temperature is raised by 100° C. The material is most likely:

A. a metal
B. a pure semiconductor
C. a heavily doped semiconductor
D. an insulator
E. none of the above

ans: A

32. Donor atoms introduced into a pure semiconductor at room temperature:
    A. increase the number of electrons in the conduction band
    B. increase the number of holes in the valence band
    C. lower the Fermi level
    D. increase the electrical resistivity
    E. none of the above

    ans: A

33. Acceptor atoms introduced into a pure semiconductor at room temperature:
    A. increase the number of electrons in the conduction band
    B. increase the number of holes in the valence band
    C. raise the Fermi level
    D. increase the electrical resistivity
    E. none of the above

    ans: B

34. An acceptor replacement atom in silicon might have _____ electrons in its outer shell.
    A. 3
    B. 4
    C. 5
    D. 6
    E. 7

    ans: A

35. A donor replacement atom in silicon might have _____ electrons in its outer shell.
    A. 1
    B. 2
    C. 3
    D. 4
    E. 5

    ans: E

36. A given doped semiconductor can be identified as $p$ or $n$ type by:
    A. measuring its electrical conductivity
    B. measuring its magnetic susceptibility
    C. measuring its coefficient of resistivity
    D. measuring its heat capacity
    E. performing a Hall effect experiment

    ans: E

37. The contact electric field in the depletion region of a $p$-$n$ junction is produced by:
    A. electrons in the conduction band alone
    B. holes in the valence band alone
    C. electrons and holes together
    D. charged replacement atoms
    E. an applied bias potential difference

    ans: D

38. For an unbiased $p$-$n$ junction, the energy at the bottom of the conduction band on the $n$ side is:
    A. higher than the energy at the bottom of the conduction band on the $p$ side
    B. lower than the energy at the bottom of the conduction band on the $p$ side
    C. lower than the energy at the top of the valence band on the $n$ side
    D. lower than the energy at the top of the valence band on the $p$ side
    E. the same as the energy at the bottom of the conduction band on the $p$ side

    ans: B

39. In an unbiased $p$-$n$ junction:
    A. the electric potential vanishes everywhere
    B. the electric field vanishes everywhere
    C. the drift current vanishes everywhere
    D. the diffusion current vanishes everywhere
    E. the diffusion and drift currents cancel each other

    ans: E

40. Application of a forward bias to a $p$-$n$ junction:
    A. narrows the depletion zone
    B. increases the electric field in the depletion zone
    C. increases the potential difference across the depletion zone
    D. increases the number of donors on the $n$ side
    E. decreases the number of donors on the $n$ side

    ans: A

41. Application of a forward bias to a $p$-$n$ junction:
    A. increases the drift current in the depletion zone
    B. increases the diffusion current in the depletion zone
    C. decreases the drift current on the $p$ side outside the depletion zone
    D. decreases the drift current on the $n$ side outside the depletion zone
    E. does not change the current anywhere

    ans: B

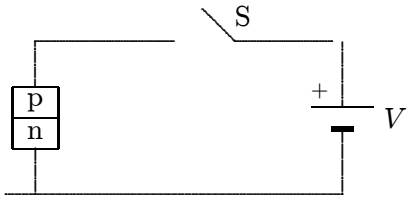
42. When a forward bias is applied to a $p$-$n$ junction the concentration of electrons on the $p$ side:
    A. increases slightly
    B. increases dramatically
    C. decreases slightly
    D. decreases dramatically
    E. does not change

    ans: B

43. Which of the following is NOT true when a back bias is applied to a $p$-$n$ junction?
A. Electrons flow from the $p$ to the $n$ side
B. Holes flow from the $p$ to the $n$ side
C. The electric field in the depletion zone increases
D. The potential difference across the depletion zone increases
E. The depletion zone narrows

ans: B

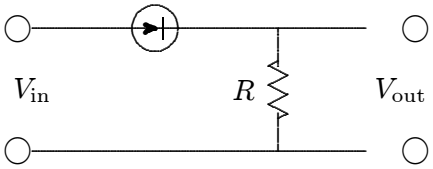
44. Switch S is closed to apply a potential difference $V$ across a $p$-$n$ junction as shown. Relative to the energy levels of the $n$-type material, with the switch open, the electron levels of the $p$-type material are:

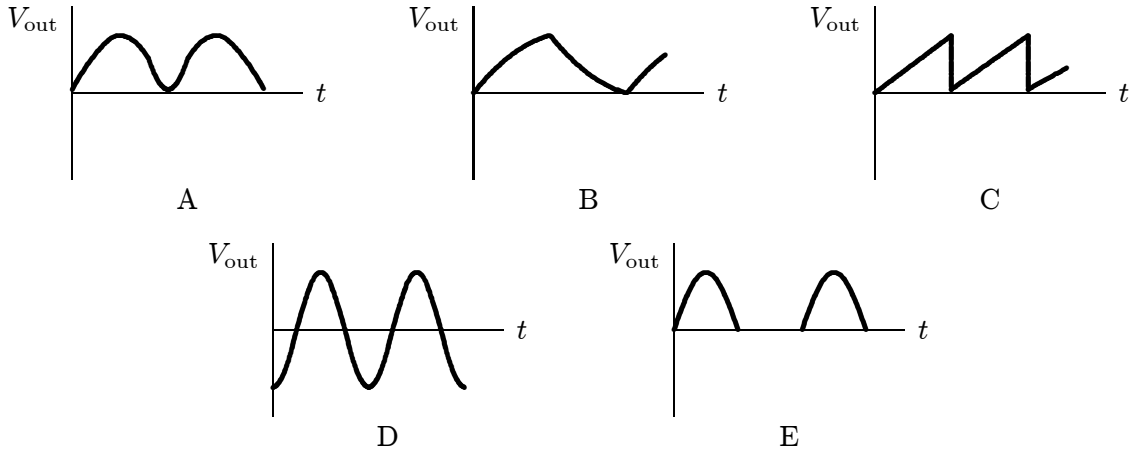



A. unchanged
B. lowered by the amount $e^{-Ve/kT}$
C. lowered by the amount $Ve$
D. raised by the amount $e^{-Ve/kT}$
E. raised by the amount $Ve$

ans: C

45. A sinusoidal potential difference $V_{\text{in}} = V_m \sin(\omega t)$ is applied to the $p$-$n$ junction as shown. Which graph correctly shows $V_{\text{out}}$ as a function of time?




ans: E

46. In normal operation the current in a MOSFIT device is controlled by changing:
    A. the number of donors and acceptors
    B. the width of the depletion zone
    C. the size of the sample
    D. the density of electron states
    E. the temperature

    ans: B

47. "LED" stands for:
    A. Less Energy Donated
    B. Light Energy Degrader
    C. Luminescent Energy Developer
    D. Laser Energy Detonator
    E. none of the above

    ans: E

48. A light emitting diode emits light when:
    A. electrons are excited from the valence to the conduction band
    B. electrons from the conduction band recombine with holes from the valence band
    C. electrons collide with atoms
    D. electrons are accelerated by the electric field in the depletion region
    E. the junction gets hot

    ans: B

49. The gap between the valence and conduction bands of a certain semiconductor is 0.85 eV. When this semiconductor is used to form a light emitting diode, the wavelength of the light emitted:
    A. is in a range above $1.5 \times 10^{-6}$ m
    B. is in a range below $1.5 \times 10^{-6}$ m
    C. is always $1.5 \times 10^{-6}$ m
    D. is in a range centered on $1.5 \times 10^{-6}$ m
    E. has nothing to do with the gap

    ans: B

## Chapter 42: NUCLEAR PHYSICS

1. The smallest particle of any chemical element that can exist by itself and yet retain the qualities that distinguish it as that element is:
   A. an electron
   B. a proton
   C. a neutron
   D. an atom
   E. a molecule

   ans: D

2. Of the following, which has the smallest rest energy?
   A. A neutron
   B. An electron
   C. An ion
   D. A proton
   E. An atom

   ans: B

3. The mass of an electron:
   A. is almost the same as that of a neutron
   B. is negative
   C. equals that of a proton
   D. is zero if the electron is at rest
   E. is much less than that of a proton

   ans: E

4. The mass of a neutron:
   A. equals that of an electron
   B. equals that of a proton
   C. is a little more than that of a proton
   D. is exactly that of a proton plus an electron
   E. is as yet unmeasured

   ans: C

5. The mass of a hydrogen atom, in kilograms, is approximately:
   A. $10^{-27}$
   B. $10^{-31}$
   C. $10^{-24}$
   D. $10^{-13}$
   E. $10^{-8}$

   ans: A

6. 1 atomic mass unit is about:
   A. $1.66 \times 10^{-31}$ kg
   B. $9.11 \times 10^{-31}$ kg
   C. $1.66 \times 10^{-27}$ kg
   D. $9.11 \times 10^{-27}$ kg
   E. $1.66 \times 10^{-25}$ kg

   ans: C

7. The atomic number of an element is:
   A. the whole number nearest to its mass
   B. the number of protons in its nucleus
   C. the nearest whole number of hydrogen atoms having the same mass as a single atom of the given element
   D. the number of neutrons in its nucleus
   E. its order of discovery

   ans: B

8. Iron has atomic number 26. Naturally mined iron contains isotopes of mass numbers 54, 56, 57, and 58. Which of the following statements is FALSE?
   A. Every atom of iron has 26 protons
   B. Some iron atoms have 30 neutrons
   C. Some iron atoms have 54 neutrons
   D. The isotopes may be separated in a mass spectrometer
   E. There are four kinds of naturally occurring iron atoms with the same chemical properties

   ans: C

9. Let $Z$ denote the atomic number and $A$ denote the mass number of a nucleus. The number of neutrons in this nucleus is:
   A. $Z$
   B. $A - Z$
   C. $A - 2Z$
   D. $A$
   E. $2A - Z$

   ans: B

10. The isotopes of an element:
   A. cannot be separated at all
   B. occur well separated in nature
   C. have similar chemical behavior
   D. cannot be separated by physical methods
   E. have equal masses

   ans: C

11. Bromine, with atomic mass 79.942 u, is composed of nearly equal amounts of two isotopes, one of which contains 79 nucleons per atom. The mass number of the other isotope is:
    A. 78
    B. 79
    C. 80
    D. 81
    E. 82

    ans: D

12. The mass density of an atomic nucleus is:
    A. about $10^{15}\,\mathrm{kg/m^3}$
    B. about $10^{12}\,\mathrm{kg/m^3}$
    C. increases with increasing nuclear mass
    D. increases with decreasing nuclear radius
    E. about the same as that of all other nuclei

    ans: E

13. Volumes of atomic nuclei are proportional to:
    A. the mass number
    B. the atomic number
    C. the total nuclear spin
    D. the number of neutrons
    E. none of these

    ans: A

14. A femtometer is:
    A. larger than $10^{-9}\,\mathrm{m}$
    B. $10^{-9}\,\mathrm{m}$
    C. $10^{-12}\,\mathrm{m}$
    D. $10^{-15}\,\mathrm{m}$
    E. $10^{-18}\,\mathrm{m}$

    ans: D

15. A nucleus with a mass number of 64 has a mean radius of about:
    A. 4.8 fm
    B. 19 fm
    C. 77 fm
    D. 260 fm
    E. $2.6 \times 10^5$ fm

    ans: A

16. A proton in a large nucleus:
    A. attracts all other protons
    B. repels all other protons
    C. repels all neutrons
    D. attracts some protons and repels others
    E. attracts some neutrons and repels others

    ans: D

17. Two protons are separated by $10^{-16}$ m. The nuclear (N), electrostatic (E), and gravitational (G) forces between these protons, in order of increasing strength, are:
    A. E, N, G
    B. N, G, E
    C. G, E, N
    D. G, N, E
    E. E, G, N

    ans: C

18. Two protons are about $10^{-10}$ m apart. Their relative motion is chiefly determined by:
    A. gravitational forces
    B. electrical forces
    C. nuclear forces
    D. magnetic forces
    E. torque due to electric dipole moments

    ans: B

19. The binding energy of a nucleus is the energy that must be supplied to:
    A. remove a nucleon
    B. remove an alpha particle
    C. remove a beta particle
    D. separate the nucleus into its constituent nucleons
    E. separate the nucleus into a collection of alpha particles

    ans: D

20. If a nucleus has mass $M$, $Z$ protons (mass $m_p$), and $N$ neutrons (mass $m_n$), its binding energy is equal to:
    A. $Mc^2$
    B. $(M - Zm_p - Nm_n)c^2$
    C. $(Zm_p + Nm_n - M)c^2$
    D. $(Zm_p + Nm_n)c^2$
    E. $(Zm_p - M)c^2$

    ans: C

21. Stable nuclei generally:

A. have a greater number of protons than neutrons
B. have low mass numbers
C. have high mass numbers
D. are beta emitters
E. none of the above

ans: E

22. Let $A$ be the mass number and $Z$ be the atomic number of a nucleus. Which of the following is approximately correct for light nuclei?

A. $Z = 2A$
B. $Z = A$
C. $Z = A/2$
D. $Z = \sqrt{A}$
E. $Z = A^2$

ans: C

23. The greatest binding energy per nucleon occurs for nuclides with masses near that of:

A. helium
B. sodium
C. iron
D. mercury
E. uranium

ans: C

24. Which of the following nuclides is least likely to be detected?

A. $^{52}$Fe ($Z = 26$)
B. $^{115}$Nd ($Z = 60$)
C. $^{175}$Lu ($Z = 71$)
D. $^{208}$Pb ($Z = 82$)
E. $^{238}$U ($Z = 92$)

ans: B

25. The half-life of a radioactive substance is:

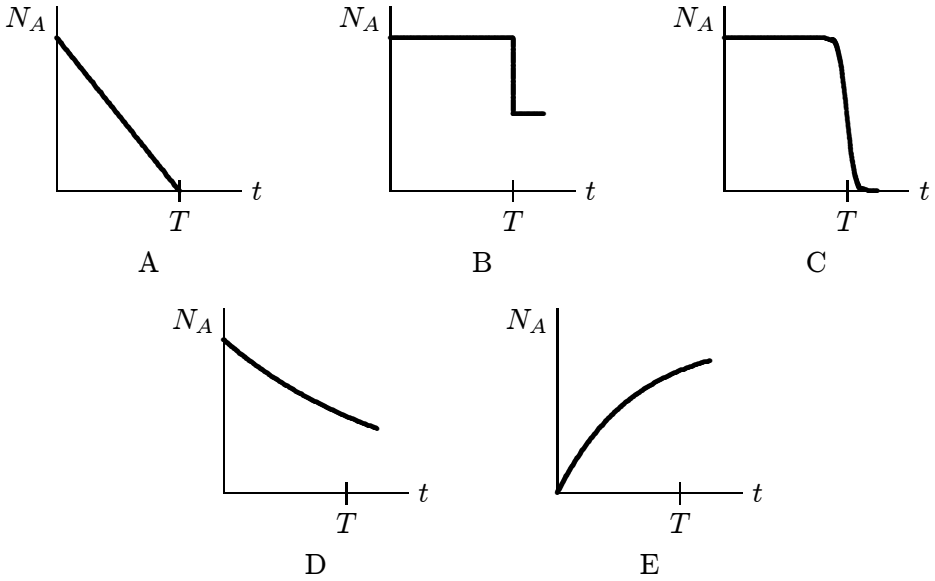
A. half the time it takes for the entire substance to decay
B. usually about 50 years
C. the time for radium to change into lead
D. calculated from $E = mc^2$
E. the time for half the substance to decay

ans: E

26. Which expression correctly describes the radioactive decay of a substance whose half-life is $T$?
A. $N(t) = N_0 e^{-(t \ln 2)/T}$
B. $N(t) = N_0 e^{-t/T}$
C. $N(t) = N_0 e^{-tT}$
D. $N(t) = N_0 e^{-tT \ln 2}$
E. $N(t) = N_0 e^{-t/T \ln 2}$

ans: A

27. Radioactive element A decays to the stable element B with a half-life $T$. Starting with a sample of pure A and no B, which graph below correctly shows the number of A atoms, $N_A$, as a function of time $t$?




ans: D

28. A large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:
A. the number of undecayed nuclei present at that instant
B. the time since the decays started
C. the time remaining before all have decayed
D. the half-life of the decay
E. the average time between decays

ans: A

29. The relation between the disintegration constant $\lambda$ and the half-life $T$ of a radioactive substance is:
A. $\lambda = 2T$
B. $\lambda = 1/T$
C. $\lambda = 2/T$
D. $\lambda T = \ln 2$
E. $\lambda T = \ln(1/2)$

ans: D

30. Possible units for the disintegration constant $\lambda$ are:

A. kg/s
B. s/kg
C. hour
D. $\text{day}^{-1}$
E. $\text{cm}^{-1}$

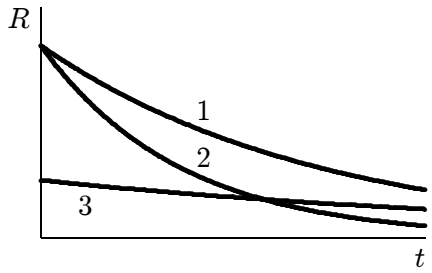
ans: D

31. The half-life of a given nuclear disintegration A $\rightarrow$ B:

A. depends on the initial number of A atoms
B. depends on the initial number of B atoms
C. is an exponentially increasing function of time
D. is an exponentially decreasing function of time
E. none of the above

ans: E

32. The graph shows the activity $R$ as a function of time $t$ for three radioactive samples. Rank the samples according to their half-lives, shortest to longest.




A. 1, 2, 3
B. 1, 3, 2
C. 2, 1, 3
D. 2, 3, 1
E. 3, 1, 2

ans: C

33. The half-life of radium is about 1600 years. If a rock initially contains 1 g of radium, the amount left after 6400 years will be about:

A. 938 mg
B. 62 mg
C. 31 mg
D. 16 mg
E. less than 16 mg

ans: C

34. Starting with a sample of pure $^{66}$Cu, 7/8 of it decays into Zn in 15 minutes. The corresponding half-life is:

A. 15 minutes
B. 5 minutes
C. 7 minutes
D. 3.75 minutes
E. 10 minutes

ans: B

35. $^{210}$Bi (an isotope of bismuth) has a half-life of 5.0 days. The time for three-quarters of a sample of $^{210}$Bi to decay is:

A. 2.5 days
B. 10 days
C. 15 days
D. 20 days
E. 3.75 days

ans: B

36. Radioactive $^{90}$Sr has a half-life of 30 years. What percent of a sample of $^{90}$Sr will remain after 60 years?

A. 0%
B. 25%
C. 50%
D. 75%
E. 14%

ans: B

37. The half-life of a radioactive isotope is 6.5 h. If there are initially $48 \times 10^{32}$ atoms of this isotope, the number of atoms of this isotope remaining after 26 h is:

A. $12 \times 10^{32}$
B. $6 \times 10^{32}$
C. $3 \times 10^{32}$
D. $6 \times 10^{4}$
E. $3 \times 10^{2}$

ans: C

38. At the end of 14 min, 1/16 of a sample of radioactive polonium remains. The corresponding half-life is:

A. (7/8) min
B. (8/7) min
C. (7/4) min
D. (7/2) min
E. (14/3) min

ans: D

39. The half-life of a radioactive isotope is 140 days. In how many days does the decay rate of a sample of this isotope decrease to one-fourth of its initial decay rate?

A. 35
B. 105
C. 187
D. 210
E. 280

ans: E

40. Of the three common types of radiation (alpha, beta, gamma) from radioactive sources, electric charge is carried by:

A. only beta and gamma
B. only beta
C. only alpha and gamma
D. only alpha
E. only alpha and beta

ans: E

41. An alpha particle is:

A. a helium atom with two electrons removed
B. an aggregate of two or more electrons
C. a hydrogen atom
D. the ultimate unit of positive charge
E. sometimes negatively charged

ans: A

42. A nucleus with mass number $A$ and atomic number $Z$ emits an alpha particle. The mass number and atomic number, respectively, of the daughter nucleus are:

A. $A$, $Z-2$
B. $A-2$, $Z-2$
C. $A-2$, $Z$
D. $A-4$, $Z$
E. $A-4$, $Z-2$

ans: E

43. Radioactive polonium, $^{214}$Po ($Z = 84$), decays by alpha emission to:

A. $^{214}$Po ($Z = 84$)
B. $^{210}$Pb ($Z = 82$)
C. $^{214}$At ($Z = 85$)
D. $^{218}$Po ($Z = 84$)
E. $^{210}$Bi ($Z = 83$)

ans: B

44. A radium atom, $^{226}$Ra ($Z = 86$) emits an alpha particle. The number of protons in the resulting atom is:

A. 84
B. 85
C. 86
D. 88
E. some other number

ans: A

45. Some alpha emitters have longer half-lives than others because:

A. their alpha particles have greater mass
B. their alpha particles have less mass
C. their barriers to decay are higher and wider
D. their barriers to decay are lower and narrower
E. their decays include the emission of a photon

ans: C

46. In an alpha decay the disintegration energy appears chiefly as:

A. photon energies
B. the kinetic energies of the alpha and the daughter nucleus
C. the excitation energy of the daughter nucleus
D. the excitation energy of the alpha particle
E. heat

ans: B

47. Rank the following collections of particles according to the total binding energy of all the particles in each collection, least to greatest.

collection 1: $^{244}$Pu ($Z = 94$) nucleus alone
collection 2: $^{240}$U ($Z = 92$) nucleus, $\alpha$ particle
collection 3: $^{240}$U ($Z = 92$) nucleus, two separated protons, two separated neutrons

A. 1, 2, 3
B. 3, 2, 1
C. 2, 1, 3
D. 1, 3, 2
E. 2, 3, 1

ans: D

48. A beta particle is:

A. a helium nucleus
B. an electron or a positron
C. a radioactive element
D. any negative particle
E. a hydrogen atom

ans: B

49. Beta particles from various radioactive sources all have:

A. the same mass
D. the same speed
B. the same charge
E. the same deflection
C. the same energy in a magnetic field

ans: A

50. A radioactive atom X emits a $\beta^-$ particle. The resulting atom:

A. must be very reactive chemically
B. has an atomic number that is one more than that of X
C. has a mass number that is one less than that of X
D. must be radioactive
E. is the same chemical element as X

ans: B

51. A nucleus with mass number $A$ and atomic number $Z$ undergoes $\beta^-$ decay. The mass number and atomic number, respectively, of the daughter nucleus are:

A. $A$, $Z-1$
B. $A-1$, $Z$
C. $A+1$, $Z-1$
D. $A$, $Z+1$
E. $A$, $Z-1$

ans: D

52. A nucleus with mass number $A$ and atomic number $Z$ undergoes $\beta^+$ decay. The mass number and atomic number, respectively, of the daughter nucleus are:

A. $A-1$, $Z-1$
B. $A-1$, $Z+1$
C. $A+1$, $Z-1$
D. $A$, $Z+1$
E. $A$, $Z-1$

ans: E

53. In addition to the daughter nucleus and an electron or positron, the products of a beta decay include:

A. a neutron
B. a neutrino
C. a proton
D. an alpha particle
E. no other particle

ans: B

54. The energies of electrons emitted in $\beta^-$ decays have a continuous spectrum because:

A. the original neutron has a continuous spectrum
B. a neutrino can carry off energy
C. the emitted electron is free
D. energy is not conserved
E. the daughter nucleus may have any energy

ans: B

55. If $^{204}$Tl ($Z = 81$) emits a $\beta^-$ particle from its nucleus:

A. stable Tl is formed
B. $^{202}$Hg ($Z = 80$) is formed
C. $^{204}$Pb ($Z = 82$) is formed
D. radioactive Tl is formed
E. $^{197}$Au ($Z = 79$) is formed

ans: C

56. An atom of $^{235}$U ($Z = 92$) disintegrates to $^{207}$Pb ($Z = 82$) with a half-life of about a billion years by emitting seven alpha particles and _____ $\beta^-$ particles:

A. 3
B. 4
C. 5
D. 6
E. 7

ans: B

57. When ordinary sodium ($^{23}$Na, $Z = 11$) is bombarded with deuterons, the products are a neutron and:

A. $^{27}$Al, $Z = 13$
B. $^{24}$Na, $Z = 11$
C. $^{24}$Mg, $Z = 12$
D. $^{25}$Mg, $Z = 12$
E. $^{20}$Ne, $Z = 10$

ans: D

58. $^{65}$Cu can be turned into $^{66}$Cu, with no accompanying product except a gamma, if bombarded with:

A. protons
B. neutrons
C. deuterons
D. electrons
E. alpha particles

ans: B

59. Magnesium has atomic number 12, hydrogen has atomic number 1, and helium has atomic number 2. In the nuclear reaction $^{24}\text{Mg} + {}^{2}\text{H} \rightarrow (\ ) + {}^{4}\text{He}$ the missing quantity is:

A. $^{23}\text{Na}$ $(Z = 11)$
B. $^{22}\text{Ne}$ $(Z = 10)$
C. $^{21}\text{Na}$ $(Z = 11)$
D. $^{21}\text{Ne}$ $(Z = 10)$
E. $^{22}\text{Na}$ $(Z = 11)$

ans: E

60. Aluminum has atomic number 13, helium has atomic number 2, and silicon has atomic number 14. In the nuclear reaction $^{27}\text{Al} + {}^{4}\text{He} \rightarrow {}^{30}\text{Si} + (\ )$ the missing particle is:

A. an $\alpha$ particle
B. a positron
C. an electron
D. a proton
E. a neutron

ans: D

61. The $^{66}\text{Cu}$ $(Z = 29)$ produced in a nuclear bombardment is unstable, changing to $^{66}\text{Zn}$ $(Z = 30)$ by the emission of:

A. a proton
B. a gamma ray photon
C. a positron
D. an electron
E. an alpha particle

ans: D

62. When ordinary sulfur, $^{32}\text{S}$ $(Z = 16)$, is bombarded with neutrons, the products are $^{32}\text{P}$ $(Z = 15)$ and:

A. alpha particles
B. protons
C. deuterons
D. gamma ray particles
E. electrons

ans: B

63. A certain nucleus, after absorbing a neutron, emits a $\beta^-$ and then splits into two alpha particles. The $(A, Z)$ of the original nucleus must have been:

A. 6, 2
B. 6, 3
C. 7, 2
D. 7, 3
E. 8, 4

ans: D

64. When $^{23}$Na ($Z = 11$) is bombarded with protons, the products are $^{20}$Ne ($Z = 10$) and:
    A. a neutron
    B. an alpha particle
    C. a deuteron
    D. a gamma ray particle
    E. two beta particles

    ans: B

65. Bombardment of $^{28}$Si ($Z = 14$) with alpha particles may produce:
    A. a proton and $^{31}$P ($Z = 15$)
    B. hydrogen and $^{32}$S ($Z = 16$)
    C. a deuteron and $^{27}$Al ($Z = 13$)
    D. helium and $^{31}$P ($Z = 15$)
    E. $^{35}$Cl ($Z = 17$)

    ans: A

66. The becquerel is the correct unit to use in reporting the measurement of:
    A. the rate of decay of a radioactive source
    B. the ability of a beam of gamma ray photons to produce ions in a target
    C. the energy delivered by radiation to a target
    D. the biological effect of radiation
    E. none of the above

    ans: A

67. The gray is the correct unit to use in reporting the measurement of:
    A. the rate of decay of a radioactive source
    B. the ability of a beam of gamma ray photons to produce ions in a target
    C. the energy per unit mass of target delivered by radiation to a target
    D. the biological effect of radiation
    E. none of the above

    ans: C

68. The sievert is the correct unit to use in reporting the measurement of:
    A. the rate of decay of a radioactive source
    B. the ability of a beam of gamma ray photons to produce ions in a target
    C. the energy delivered by radiation to a target
    D. the biological effect of radiation
    E. none of the above

    ans: D

## Chapter 43: ENERGY FROM THE NUCLEUS

1. If the nucleus of a lead atom were broken into two identical nuclei, the total mass of the resultant nuclei would be:

   A. the same as before
   B. greater than before
   C. less than before
   D. converted into radiation
   E. converted into kinetic energy

   ans: C

2. Consider the following energies:

   1. minimum energy needed to excite a hydrogen atom
   2. energy needed to ionize a hydrogen atom
   3. energy released in $^{235}$U fission
   4. energy needed to remove a neutron from a $^{12}$C nucleus

   Rank them in order of increasing value.

   A. 1, 2, 3, 4
   B. 1, 3, 2, 4
   C. 1, 2, 4, 3
   D. 2, 1, 4, 3
   E. 2, 4, 1, 3

   ans: C

3. The binding energy per nucleon:

   A. increases for all fission events
   B. increases for some, but not all, fission events
   C. decreases for all fission events
   D. decreases for some, but not all, fission events
   E. remains the same for all fission events

   ans: A

4. When uranium undergoes fission as a result of neutron bombardment, the energy released is due to:

   A. oxidation of the uranium
   B. kinetic energy of the bombarding neutrons
   C. radioactivity of the uranium nucleus
   D. radioactivity of the fission products
   E. a reduction in binding energy

   ans: E

5. The energy supplied by a thermal neutron in a fission event is essentially its:
   A. excitation energy
   B. binding energy
   C. kinetic energy
   D. rest energy
   E. electric potential energy

   ans: B

6. The barrier to fission comes about because the fragments:
   A. attract each other via the strong nuclear force
   B. repel each other electrically
   C. produce magnetic fields
   D. have large masses
   E. attract electrons electrically

   ans: A

7. $^{235}$U is readily made fissionable by a thermal neutron but $^{238}$U is not because:
   A. the neutron has a smaller binding energy in $^{236}$U
   B. the neutron has a smaller excitation energy in $^{236}$U
   C. the potential barrier for the fragments is less in $^{239}$U
   D. the neutron binding energy is greater than the barrier height for $^{236}$U and less than the barrier height for $^{239}$U
   E. the neutron binding energy is less than the barrier height for $^{236}$U and greater than the barrier height for $^{239}$U

   ans: D

8. An explosion does not result from a small piece of $^{235}$U because:
   A. it does not fission
   B. the neutrons released move too fast
   C. $^{238}$U is required
   D. too many neutrons escape, preventing a chain reaction from starting
   E. a few neutrons must be injected to start the chain reaction

   ans: D

9. When $^{236}$U fissions the fragments are:
   A. always $^{140}$Xe and $^{94}$Sr
   B. always identical
   C. never $^{140}$Xe and $^{94}$Sr
   D. never identical
   E. none of the above

   ans: E

10. Fission fragments usually decay by emitting:
    A. alpha particles
    B. electrons and neutrinos
    C. positrons and neutrinos
    D. only neutrons
    E. only electrons

    ans: B

11. When $^{236}$U fissions, the products might be:
    A. $^{146}$Ba, $^{89}$Kr, and a proton
    B. $^{146}$Ba, $^{89}$Kr, and a neutron
    C. $^{148}$Cs and $^{85}$Br
    D. $^{133}$I, $^{92}$Sr, and an alpha particle
    E. two uranium nuclei

    ans: B

12. Consider all possible fission events. Which of the following statements is true?
    A. Light initial fragments have more protons than neutrons and heavy initial fragments have fewer protons than neutrons
    B. Heavy initial fragments have more protons than neutrons and light initial fragments have fewer protons than neutrons
    C. All initial fragments have more protons than neutrons
    D. All initial fragments have about the same number of protons and neutrons
    E. All initial fragments have more neutrons than protons

    ans: E

13. Which one of the following represents a fission reaction that can be activated by slow neutrons?
    A. $^{238}\mathrm{U}_{92} + {}^{1}\mathrm{n}_{0} \rightarrow {}^{90}\mathrm{Kr}_{36} + {}^{146}\mathrm{Cs}_{55} + {}^{2}\mathrm{H}_{1} + {}^{1}\mathrm{n}_{0}$
    B. $^{239}\mathrm{Pu}_{94} + {}^{1}\mathrm{n}_{0} \rightarrow {}^{96}\mathrm{Sr}_{38} + {}^{141}\mathrm{Ba}_{56} + 3\,{}^{1}\mathrm{n}_{0}$
    C. $^{238}\mathrm{U}_{92} \rightarrow {}^{234}\mathrm{Th}_{90} + {}^{4}\mathrm{He}_{2}$
    D. $^{3}\mathrm{H}_{1} + {}^{2}\mathrm{H}_{1} \rightarrow {}^{4}\mathrm{He}_{2} + {}^{1}\mathrm{n}_{0}$
    E. $^{107}\mathrm{Ag}_{47} + {}^{1}\mathrm{n}_{0} \rightarrow {}^{108}\mathrm{Ag}_{47} \rightarrow {}^{108}\mathrm{Cd}_{48} + {}^{0}e_{-1}$

    ans: B

14. In the uranium disintegration series:
    A. the emission of a $\beta^-$ particle increases the mass number $A$ by one and decreases the atomic number $Z$ by one
    B. the disintegrating element merely ejects atomic electrons
    C. the emission of an $\alpha$ particle decreases the mass number $A$ by four and decreases the atomic number $Z$ by two
    D. the nucleus always remains unaffected
    E. the series of disintegrations continues until an element having eight outermost orbital electrons is obtained

    ans: C

15. Separation of the isotopes of uranium requires a physical, rather than chemical, method because:

A. mixing other chemicals with uranium is too dangerous
B. the isotopes are chemically the same
C. the isotopes have exactly the same number of neutrons per nucleus
D. natural uranium contains only 0.7% $^{235}$U
E. uranium is the heaviest element in nature

ans: B

16. Which one of the following is NOT needed in a nuclear fission reactor?

A. Moderator
B. Fuel
C. Coolant
D. Control device
E. Accelerator

ans: E

17. The function of the control rods in a nuclear reactor is to:

A. increase fission by slowing down the neutrons
B. decrease the energy of the neutrons without absorbing them
C. increase the ability of the neutrons to cause fission
D. decrease fission by absorbing neutrons
E. provide the critical mass for the fission reaction

ans: D

18. A nuclear reactor is operating at a certain power level, with its multiplication factor adjusted to unity. The control rods are now used to reduce the power output to one-half its former value. After the reduction in power the multiplication factor is maintained at:

A. 1/2
B. 1/4
C. 2
D. 4
E. 1

ans: E

19. The purpose of a moderator in a nuclear reactor is to:

A. provide neutrons for the fission process
B. slow down fast neutrons to increase the probability of capture by uranium
C. absorb dangerous gamma radiation
D. shield the reactor operator from dangerous radiation
E. none of the above

ans: B

20. In a neutron-induced fission process, delayed neutrons come from:
    A. the fission products
    B. the original nucleus just before it absorbs the neutron
    C. the original nucleus just after it absorbs the neutron
    D. the moderator material
    E. the control rods

    ans: A

21. In a nuclear reactor the fissionable fuel is formed into pellets rather than finely ground and the pellets are mixed with the moderator. This reduces the probability of:
    A. non-fissioning absorption of neutrons
    B. loss of neutrons through the reactor container
    C. absorption of two neutrons by single fissionable nucleus
    D. loss of neutrons in the control rods
    E. none of the above

    ans: A

22. In a subcritical nuclear reactor:
    A. the number of fission events per unit time decreases with time
    B. the number of fission events per unit time increases with time
    C. each fission event produces fewer neutrons than when the reactor is critical
    D. each fission event produces more neutrons than when the reactor is critical
    E. none of the above

    ans: A

23. In the normal operation of a nuclear reactor:
    A. control rods are adjusted so the reactor is subcritical
    B. control rods are adjusted so the reactor is critical
    C. the moderating fluid is drained
    D. the moderating fluid is continually recycled
    E. none of the above

    ans: B

24. In a nuclear power plant, the power discharged to the environment:
    A. can be made zero by proper design
    B. must be less than the electrical power generated
    C. must be greater than the electrical power generated
    D. can be entirely recycled to produce an equal amount of electrical power
    E. is not any of the above

    ans: E

25. The binding energy per nucleon:
    A. increases for all fusion events
    B. increases for some, but not all, fusion events
    C. remains the same for some fusion events
    D. decreases for all fusion events
    E. decreases for some, but not all, fusion events

    ans: A

26. To produce energy by fusion of two nuclei, the nuclei must:
    A. have at least several thousand electron volts of kinetic energy
    B. both be above iron in mass number
    C. have more neutrons than protons
    D. be unstable
    E. be magic number nuclei

    ans: A

27. Which one of the following represents a fusion reaction that yields large amounts of energy?
    A. $^{238}\mathrm{U}_{92} + {}^{1}\mathrm{n}_{0} \rightarrow {}^{90}\mathrm{Kr}_{36} + {}^{146}\mathrm{Cs}_{55} + {}^{2}\mathrm{H}_{1} + {}^{1}\mathrm{n}_{0}$
    B. $^{239}\mathrm{Pu}_{92} + {}^{1}\mathrm{n}_{0} \rightarrow {}^{96}\mathrm{Sr}_{38} + {}^{141}\mathrm{Ba}_{56} + 3{}^{1}\mathrm{n}_{0}$
    C. $^{238}\mathrm{U}_{92} \rightarrow {}^{234}\mathrm{Th}_{90} + {}^{4}\mathrm{He}_{2}$
    D. $^{3}\mathrm{H}_{1} + {}^{2}\mathrm{H}_{1} \rightarrow {}^{4}\mathrm{He}_{2} + {}^{1}\mathrm{n}_{0}$
    E. $^{107}\mathrm{Ag}_{47} + {}^{1}\mathrm{n}_{0} \rightarrow {}^{108}\mathrm{Ag}_{47} \rightarrow {}^{108}\mathrm{Cd}_{48} + {}^{0}e_{-1}$

    ans: D

28. The barrier to fusion comes about because protons:
    A. attract each other via the strong nuclear force
    B. repel each other electrically
    C. produce magnetic fields
    D. attract neutrons via the strong nuclear force
    E. attract electrons electrically

    ans: B

29. High temperatures are required in thermonuclear fusion so that:
    A. some nuclei are moving fast enough to overcome the barrier to fusion
    B. there is a high probability some nuclei will strike each other head on
    C. the atoms are ionized
    D. thermal expansion gives the nuclei more room
    E. the uncertainty principle can be circumvented

    ans: A

30. For a controlled nuclear fusion reaction, one needs:
    A. high number density $n$ and high temperature $T$
    B. high number density $n$ and low temperature $T$
    C. low number density $n$ and high temperature $T$
    D. low number density $n$ and low temperature $T$
    E. high number density $n$ and temperature $T = 0\,\mathrm{K}$

    ans: A

31. Most of the energy produced by the Sun is due to:
    A. nuclear fission
    B. nuclear fusion
    C. chemical reaction
    D. gravitational collapse
    E. induced emfs associated with the Sun's magnetic field

    ans: B

32. Nuclear fusion in stars produces all the chemical elements with mass numbers less than:
    A. 56
    B. 66
    C. 70
    D. 82
    E. 92

    ans: A

33. Nuclear fusion in the Sun is increasing its supply of:
    A. hydrogen
    B. helium
    C. nucleons
    D. positrons
    E. neutrons

    ans: B

34. Which of the following chemical elements is not produced by thermonuclear fusion in stars?
    A. Carbon ($Z = 6$, $A \approx 12$)
    B. Silicon ($Z = 14$, $A \approx 28$)
    C. Oxygen ($Z = 8$, $A \approx 16$)
    D. Mercury ($Z = 80$, $A \approx 200$)
    E. Chromium ($Z = 24$, $A \approx 52$)

    ans: D

35. The first step of the proton-proton cycle is:
    A. ${}^1\mathrm{H} + {}^1\mathrm{H} \rightarrow {}^2\mathrm{H}$
    B. ${}^1\mathrm{H} + {}^1\mathrm{H} \rightarrow {}^2\mathrm{H} + e^+ + \nu$
    C. ${}^1\mathrm{H} + {}^1\mathrm{H} \rightarrow {}^2\mathrm{H} + e^- + \nu$
    D. ${}^1\mathrm{H} + {}^1\mathrm{H} \rightarrow {}^2\mathrm{H} + \gamma$
    E. ${}^1\mathrm{H} + {}^1\mathrm{H} \rightarrow {}^3\mathrm{H} + e^- + \nu$

    ans: B

36. The overall proton-proton cycle is equivalent to:
    A. $2\,{}^1\mathrm{H} \rightarrow {}^2\mathrm{H}$
    B. $4\,{}^1\mathrm{H} \rightarrow {}^4\mathrm{H}$
    C. $4\,{}^1\mathrm{H} \rightarrow {}^4\mathrm{H} + 4\mathrm{n}$
    D. $4\,{}^1\mathrm{H} + 2e^- \rightarrow {}^4\mathrm{He} + 2\nu + 6\gamma$
    E. $4\,{}^1\mathrm{H} + 2e^+ \rightarrow {}^4\mathrm{He} + 2\nu + 3\gamma$

    ans: D

37. The energy released in a complete proton-proton cycle is about:
    A. 3 keV
    B. 30 keV
    C. 3 MeV
    D. 30 MeV
    E. 300 MeV

    ans: D

38. For purposes of a practical (energy producing) reaction one wants a disintegration energy $Q$ that is:
    A. positive for fusion reactions and negative for fission reactions
    B. negative for fusion reactions and positive for fission reactions
    C. negative for both fusion and fission reactions
    D. positive for both fusion and fission reactions
    E. as close to zero as possible for both fusion and fission reactions

    ans: D

39. Lawson's number is $10^{20}\,\mathrm{s}\cdot\mathrm{m}^{-3}$. If the density of deuteron nuclei is $2\times10^{21}\,\mathrm{m}^{-3}$ what should the confinement time be to achieve sustained fusion?
    A. 16 ms
    B. 50 ms
    C. 160 ms
    D. 250 ms
    E. 500 ms

    ans: B

40. Tokamaks confine deuteron plasmas using:
    A. thick steel walls
    B. magnetic fields
    C. laser beams
    D. vacuum tubes
    E. electric fields

    ans: B

41. Most magnetic confinement projects attempt:
    A. proton-proton fusion
    B. proton-deuteron fusion
    C. deuteron-deuteron fusion
    D. deuteron-triton fusion
    E. triton-triton fusion

    ans: C

42. Compared to fusion in a tokamak, laser fusion makes use of:
    A. smaller particle number densities
    B. greater particle number densities
    C. longer confinement times
    D. higher temperatures
    E. lower temperatures

    ans: B

43. Most laser fusion projects attempt:
    A. proton-proton fusion
    B. proton-deuteron fusion
    C. deuteron-deuteron fusion
    D. deuteron-triton fusion
    E. triton-triton fusion

    ans: D

44. In laser fusion, the laser light is:
    A. emitted by the reacting nuclei
    B. used to cause transitions between nuclear energy levels
    C. used to cause transitions between atomic energy levels
    D. used to replace the emitted gamma rays
    E. used to heat the fuel pellet

    ans: E

## Chapter 44: QUARKS, LEPTONS, AND THE BIG BANG

1. Which of the following particles is stable?
   A. Neutron
   B. Proton
   C. Pion
   D. Muon
   E. Kaon

   ans: B

2. The stability of the proton is predicted by the laws of conservation of energy and conservation of:
   A. momentum
   B. angular momentum
   C. baryon number
   D. lepton number
   E. strangeness

   ans: C

3. When a kaon decays via the strong interaction the products must include a:
   A. baryon
   B. lepton
   C. strange particle
   D. electron
   E. neutrino

   ans: C

4. A particle with spin angular momentum $\hbar/2$ is called a:
   A. lepton
   B. hadron
   C. fermion
   D. boson
   E. electron

   ans: C

5. A particle with spin angular momentum $\hbar$ is called a:
   A. lepton
   B. hadron
   C. fermion
   D. boson
   E. electron

   ans: D

6. An example of a fermion is a:
   A. photon
   B. pion
   C. neutrino
   D. kaon
   E. none of these

   ans: C

7. An example of a boson is a:
   A. photon
   B. electron
   C. neutrino
   D. proton
   E. neutron

   ans: A

8. All particles with spin angular momentum $\hbar/2$:
   A. interact via the strong force
   B. travel at the speed of light
   C. obey the Pauli exclusion principle
   D. have non-zero rest mass
   E. are charged

   ans: C

9. All leptons interact with each other via the:
   A. strong force
   B. weak force
   C. electromagnetic force
   D. strange force
   E. none of these

   ans: B

10. An electron participates in:
   A. the strong force only
   B. the strong and weak forces only
   C. the electromagnetic and gravitational forces only
   D. the electromagnetic, gravitational, and weak forces only
   E. the electromagnetic, gravitational, and strong forces only

   ans: D

11. Which of the following particles has a lepton number of zero?
   A. $e^+$
   B. $\mu^+$
   C. $\nu_e$
   D. $\overline{\nu}_\mu$
   E. p

   ans: E

12. Which of the following particles has a lepton number of +1?
    A. $e^+$
    B. $\mu^+$
    C. $\mu^-$
    D. $\overline{\nu}_e$
    E. p

    ans: C

13. $\pi^+$ represents a pion (a meson), $\mu^-$ represents a muon (a lepton), $\nu_e$ represents an electron neutrino (a lepton), $\nu_\mu$ and p represents a proton represents a muon neutrino (a lepton). Which of the following decays might occur?
    A. $\pi^+ \longrightarrow \mu^- + \nu_\mu$
    B. $\pi^+ \longrightarrow p + \nu_e$
    C. $\pi^+ \longrightarrow \mu^+ + \overline{\nu}_e$
    D. $\pi^+ \longrightarrow p + \overline{\nu}_\mu$
    E. $\pi^+ \longrightarrow \mu^+ + \nu_\mu$

    ans: E

14. A particle can decay to particles with greater total rest mass:
    A. only if antiparticles are produced
    B. only if photons are also produced
    C. only if neutrinos are also produced
    D. only if the original particle has kinetic energy
    E. never

    ans: E

15. The interaction $\pi^- + p \rightarrow \pi^- + \Sigma^+$ violates the principle of conservation of:
    A. baryon number
    B. lepton number
    C. strangeness
    D. angular momentum
    E. none of these

    ans: C

16. The interaction $\pi^- + p \rightarrow K^- + \Sigma^+$ violates the principle of conservation of:
    A. baryon number
    B. lepton number
    C. strangeness
    D. angular momentum
    E. none of these

    ans: E

17. A neutral muon cannot decay into two neutrinos. Of the following conservation laws, which would be violated if it did?

A. Energy
B. Baryon number
C. Charge
D. Angular momentum
E. None of the above

ans: D

18. A positron cannot decay into three neutrinos. Of the following conservation laws, which would be violated if it did?

A. Energy
B. Baryon number
C. Lepton number
D. Linear momentum
E. Angular momentum

ans: C

19. Two particles interact to produce only photons, with the original particles disappearing. The particles must have been:

A. mesons
B. strange particles
C. strongly interacting
D. leptons
E. a particle, antiparticle pair

ans: E

20. Two baryons interact to produce pions only, the original baryons disappearing. One of the baryons must have been:

A. a proton
B. an omega minus
C. a sigma
D. an antiparticle
E. none of these

ans: D

21. A baryon with strangeness $-1$ decays via the strong interaction into two particles, one of which is a baryon with strangeness 0. The other might be:

A. a baryon with strangeness 0
B. a baryon with strangeness $+1$
C. a meson with strangeness $-1$
D. a meson with strangeness $+1$
E. a meson with strangeness 0

ans: C

22. A baryon with strangeness 0 decays via the strong interaction into two particles, one of which is a baryon with strangeness +1. The other might be:

A. a baryon with strangeness 0
B. a baryon with strangeness +1
C. a baryon with strangeness −1
D. a meson with strangeness +1
E. a meson with strangeness −1

ans: E

23. In order of increasing strength the four basic interactions are:

A. gravitational, weak, electromagnetic, and strong
B. gravitational, electromagnetic, weak, and strong
C. weak, gravitational, electromagnetic, and strong
D. weak, electromagnetic, gravitational, and strong
E. weak, electromagnetic, strong, and gravitational

ans: A

24. The two basic interactions that have finite ranges are:

A. electromagnetic and gravitational
B. electromagnetic and strong
C. electromagnetic and weak
D. gravitational and weak
E. weak and strong

ans: E

25. A certain process produces baryons that decay with a lifetime of $4 \times 10^{-24}$ s. The decay is a result of:

A. the gravitational interaction
B. the weak interaction
C. the electromagnetic interaction
D. the strong interaction
E. some combination of the above

ans: D

26. A certain process produces mesons that decay with a lifetime of $6 \times 10^{-10}$ s. The decay is a result of:

A. the gravitational interaction
B. the weak interaction
C. the electromagnetic interaction
D. the strong interaction
E. some combination of the above

ans: B

27. Compared to the lifetimes of particles that decay via the weak interaction, the lifetimes of particles that decay via the strong interaction are:

A. $10^{-12}$ times as long
B. $10^{-23}$ times as long
C. $10^{24}$ times as long
D. $10^{12}$ times as long
E. about the same

ans: A

28. Strangeness is conserved in:

A. all particle decays
B. no particle decays
C. all weak particle decays
D. all strong particle decays
E. some strong particle decays

ans: D

29. Different types of neutrinos can be distinguished from each other by:

A. the directions of their spins
B. the leptons with which they interact
C. the baryons with which they interact
D. the number of photons that accompany them
E. their baryon numbers

ans: B

30. All known quarks have:

A. charges that are multiples of $e$ and integer baryon numbers
B. charges that are multiples of $e$ and baryon numbers that are either $+1/3$ or $-1/3$
C. charges that are multiples of $e/3$ and integer baryon numbers
D. charges that are multiples of $e/3$ and baryon numbers that are either $+1/3$ or $-1/3$
E. charges that are multiples of $2e/3$ and baryon numbers that are either $+1/3$ or $-1/3$

ans: D

31. The baryon number of a quark is:

A. 0
B. 1/2
C. 1/3
D. 2/3
E. 1

ans: C

32. Quarks are the constituents of:

A. all particles
B. all leptons
C. all strongly interacting particles
D. only strange particles
E. only mesons

ans: C

33. Any meson is a combination of:

A. three quarks
B. two quarks and an antiquark
C. one quark and two antiquarks
D. one quark and one antiquark
E. two quarks

ans: D

34. Any baryon is a combination of:

A. three quarks
B. two quarks and an antiquark
C. one quark and two antiquarks
D. one quark and one antiquark
E. two quarks

ans: A

35. The quark content of a proton is:

A. uuu
B. uud
C. udd
D. ddd
E. uds

ans: B

36. The quark content of a $\pi^+$ meson is:

A. uu
B. $u\overline{u}$
C. ud
D. $u\overline{d}$
E. $\overline{d}d$

ans: D

37. In terms of quark content a beta decay can be written:

A. $udd \rightarrow uud + e^- + \nu$
B. $udd \rightarrow udd + d\overline{d} + \nu$
C. $udd \rightarrow udd + d\overline{d} + e^-$
D. $udd \rightarrow uud + u\overline{d} + \nu$
E. $udd \rightarrow uud + u\overline{d} + e^- + \nu$

ans: A

38. The up quark u has charge $+2e/3$ and strangeness 0; the down quark d has charge $-e/3$ and strangeness 0; the strange quark s has charge $-e/3$ and strangeness $-1$. This means there can be no baryon with:

A. charge 0 and strangeness 0
B. charge $-e$ and strangeness $-1$
C. charge $+e$ and strangeness $-1$
D. charge $+e$ and strangeness $-2$
E. charge 0 and strangeness $+2$

ans: C

39. The up quark u has charge $+2e/3$ and strangeness 0; the down quark d has charge $-e/3$ and strangeness 0; the strange quark s has charge $-e/3$ and strangeness $-1$. This means there can be no meson with:

A. charge 0 and strangeness $-1$
B. charge $-e$ and strangeness $-1$
C. charge $+e$ and strangeness $-1$
D. charge $+e$ and strangeness $+1$
E. charge 0 and strangeness $+1$

ans: C

40. Messenger particles of the electromagnetic interaction are called:

A. gluons
B. photons
C. W and Z
D. gravitons
E. pions

ans: B

41. Messenger particles of the strong interaction are called:

A. gluons
B. photons
C. W and Z
D. gravitons
E. pions

ans: A

42. Messenger particles of the weak interaction are called:

A. gluons
B. photons
C. W and Z
D. gravitons
E. pions

ans: C

43. A down quark can be changed into an up quark (plus other particles perhaps) by
    A. the gravitational interaction
    B. the electromagnetic interaction
    C. the weak interaction
    D. the strong interaction
    E. none of these

    ans: C

44. The color theory explains why quarks:
    A. form particles in pairs and triplets
    B. have charge that is a multiple of $e/3$
    C. have spin
    D. have mass
    E. none of the above

    ans: A

45. Color is carried by:
    A. only quarks
    B. only leptons
    C. only quarks and leptons
    D. only quarks and gluons
    E. only photons and gluons

    ans: D

46. Hubble's law is evidence that:
    A. the speed of light is increasing
    B. the universe is expanding
    C. the Earth is slowing down in its orbit
    D. galaxies have rotational motion
    E. none of the above

    ans: B

47. Objects in the universe are receding from us with a speed that is proportional to:
    A. the reciprocal of their distance from us
    B. the reciprocal of the square of their distance from us
    C. their distance from us
    D. the square of their distance from us
    E. their distance from the center of the universe

    ans: C

48. The velocities of distant objects in the universe indicate that the time elapsed since the big bang is about:
    A. $10^5$ y
    B. $10^{10}$ y
    C. $10^{15}$ y
    D. $10^{20}$ y
    E. $10^{25}$ y

    ans: B

49. The intensity of the microwave background radiation, a remnant of the big bang:
    A. is greatest in directions toward the center of the galaxy
    B. is least in directions toward the center of the galaxy
    C. is proportional to the reciprocal of the distance from us
    D. is proportional to the square of the distance from us
    E. is nearly the same in all directions

    ans: E

50. As a result of the big bang there is, in addition to the microwave background radiation, a uniform distribution of background:
    A. electrons
    B. quarks
    C. gluons
    D. neutrinos
    E. atoms

    ans: D

51. Dark matter is suspected to exist in the universe because:
    A. the night sky is dark between stars
    B. the orbital period of stars in the outer parts of a galaxy is greater than the orbital period of stars near the galactic center
    C. the orbital period of stars in the outer parts of a galaxy is less than the orbital period of stars near the galactic center
    D. the orbital period of stars in the outer parts of a galaxy is about the same as the orbital period of stars near the galactic center
    E. all galaxies have about the same mass

    ans: D

52. If dark matter did not exist it is likely that:
    A. the universe would expand forever
    B. the universe would begin contracting soon
    C. the night sky would be brighter
    D. the night sky would be darker
    E. we would be able to see the center of the universe

    ans: A